

8

7

6

5

4

3

2

1

1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.

2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.

3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

REV

ECN

DESCRIPTION OF REVISION

CK APPD

2010-10-12

SCHEM, FLYING_DUTCHMAN, MLB, K91F

REV B RELEASE, 01/31/11

Page

(.csa)

Contents

Sync

Date

1

1

Table of Contents

MASTER

MASTER

2

2

System Block Diagram

K17_REF

06/30/2009

3

3

Power Block Diagram

K17_REF

06/30/2009

4

4

Revision History

MASTER

MASTER

5

5

BOM Configuration

K17_REF

05/28/2009

6

7

Functional / ICT Test

K18_MLB

04/27/2010

7

8

Power Aliases

K18_MLB

04/27/2010

8

9

Signal Aliases

K18_MLB

04/27/2010

9

10

CPU DMI/PEG/FDI/RSVD

K92_SUMA

06/21/2010

10

11

CPU CLOCK/MISC/JTAG

K92_MLB

08/03/2010

11

12

CPU DDR3 INTERFACES

K92_SUMA

06/15/2010

12

13

CPU POWER

K92_MLB

08/03/2010

13

14

CPU POWER AND GND

K92_SUMA

06/15/2010

14

16

CPU DECOUPLING-I

K92_MLB

08/19/2010

15

17

CPU DECOUPLING-II

K92_MLB

08/19/2010

16

18

PCH SATA/PCIE/CLK/LPC/SPI

K91_MLB

10/19/2010

17

19

PCH DMI/FDI/GRAPHICS

K92_MLB

07/06/2010

18

20

PCH PCI/FLASHCACHE/USB

K92_MLB

07/06/2010

19

21

PCH MISC

K91_MLB

10/20/2010

20

22

PCH POWER

K92_MLB

07/06/2010

21

23

PCH GROUNDS

K92_MLB

04/30/2010

22

24

PCH DECOUPLING

K92_MLB

07/06/2010

23

25

CPU & PCH XDP

K91_MLB

10/17/2010

24

26

USB HUBS

K91_ERIC

10/08/2010

25

28

Chipset Support

K92_MLB

07/06/2010

26

29

DDR3 S0-DIMM Connector A

K92_SUMA

06/23/2010

27

30

DDR3 Byte/Bit Swaps

K92_SUMA

05/10/2010

28

31

DDR3 S0-DIMM Connector B

K92_SUMA

06/23/2010

29

32

CPU Memory S3 Support

K18_MLB

04/27/2010

30

33

FSB/DDR3/FRAMEBUF Vref Margining

K18_MLB

04/27/2010

31

34

X19/ALS/CAMERA CONNECTOR

K91_MARY

10/08/2010

32

35

SD READER CONNECTOR

K91_ERIC

10/08/2010

33

36

T29 Host (1 of 2)

T29_REF

10/12/2010

34

37

T29 Host (2 of 2)

T29_REF

10/12/2010

35

38

T29 Power Support

T29_REF

10/12/2010

36

39

ETHERNET PHY (CAESAR IV)

K91_ERIC

10/11/2010

37

40

ETHERNET Connector

K91_TRINHNI

05/26/2010

38

41

FireWire LLC/PHY (FW643)

K18_MLB

04/27/2010

39

42

FireWire Port & PHY Power

T27_REF

06/10/2010

40

43

FireWire Connector

T27_REF

06/10/2010

41

45

SATA/IR/SIL Connectors

K91_ERIC

11/08/2010

42

46

External USB Connectors

K91_ERIC

10/08/2010

43

48

Front Flex Support

K18_MLB

04/27/2010

44

49

SMC

K91_BEN

07/12/2010

45

50

SMC Support

K91_BEN

07/12/2010

Page

(.csa)

Contents

Sync

Date

46

51

LPC+SPI Debug Connector

K18_MLB

04/27/2010

47

52

SMBus Connections

K18_MLB

04/27/2010

48

53

Voltage & Load Side Current Sensing

K91_DINESH

08/16/2010

49

54

High Side and CPU/AXG Current Sensing

K91_DINESH

10/29/2010

50

55

Thermal Sensors

K91_DINESH

09/22/2010

51

56

Fan Connectors

K18_MLB

04/27/2010

52

57

WELLSPRING 1

K91_ERIC

10/08/2010

53

58

WELLSPRING 2

K91_ERIC

07/14/2010

54

59

Digital Accelerometer

K91_DINESH

08/06/2010

55

61

SPI ROM

K91_BEN

06/08/2010

56

62

AUDIO: CODEC/REGULATOR

K91_AUDIO

09/30/2010

57

63

AUDIO: LINE INPUT FILTER

K91_AUDIO

07/12/2010

58

65

AUDIO: HEADPHONE FILTER

K91_AUDIO

07/12/2010

59

66

AUDIO: SPEAKER AMP

K91_AUDIO

07/12/2010

60

67

AUDIO: JACKS

K91_AUDIO

09/30/2010

61

68

AUDIO: JACK TRANSLATORS

K91_AUDIO

09/21/2010

62

69

DC-In & Battery Connectors

K91_ERIC

10/08/2010

63

70

PBus Supply & Battery Charger

K91_CHANG

07/20/2010

64

71

System Agent Supply

K91_ERIC

10/08/2010

65

72

5V / 3.3V Power Supply

K91_ERIC

10/08/2010

66

73

1.5V DDR3 Supply

K91_ERIC

10/08/2010

67

74

CPU IMVP7 & AXG VCore Regulator

K91_ERIC

10/08/2010

68

75

CPU IMVP7 & AXG VCore Output

K91_ERIC

09/22/2010

69

76

CPU VCCIO (1.05V) Power Supply

K91_ERIC

10/08/2010

70

77

Misc Power Supplies

K91_ERIC

11/01/2010

71

78

Power FETs

K91_MARY

10/14/2010

72

79

Power Control 1/ENABLE

K91_MARY

07/22/2010

73

80

Whistler PCI-E

K92_SUMA

06/15/2010

74

81

Whistler CORE/FB POWER

K92_SUMA

06/15/2010

75

82

Whistler FRAME BUFFER I/F

K92_MLB

08/03/2010

76

84

GDDR5 Frame Buffer A

K92_MLB

08/19/2010

77

85

GDDR5 Frame Buffer B

K92_MLB

08/19/2010

78

86

Whistler LVDS/DP/GPIO

K92_MLB

12/01/2010

79

87

Whistler GPIOs & STRAPs

K92_MLB

11/23/2010

80

88

Whistler DP PWR/GNDs

K92_SUMA

06/15/2010

81

89

GPU (Whistler) CORE SUPPLY

K91_ERIC

12/21/2010

82

90

LVDS Display Connector

K18_MLB

04/27/2010

83

92

Muxed Graphics Support

K92_MLB

11/21/2010

84

93

DisplayPort/T29 A MUXing

T29_REF

10/16/2010

85

94

DisplayPort/T29 A Connector

T29_REF

10/16/2010

86

95

1V0 GPU / 1V5 FB Power Supply

K91_ERIC

10/08/2010

87

96

Graphics MUX (GMUX)

K91_MARY

08/03/2010

88

97

LCD Backlight Driver

K90I_KIRAN

06/25/2010

89

99

Power Sequencing EG/PCH S0

K91_MARY

08/03/2010

90

100

CPU Constraints

K92_MLB

08/09/2010

Page

(.csa)

Contents

Sync

Date

91

101

Memory Constraints

K18_MLB

04/27/2010

92

102

PCH Constraints 1

K92_MLB

08/09/2010

93

103

PCH Constraints 2

K92_MLB

08/09/2010

94

104

Ethernet/FW Constraints

K91_ERIC

08/03/2010

95

105

T29 Constraints

T29_REF

10/16/2010

96

106

SMC Constraints

K18_MLB

04/27/2010

97

107

GPU (Whistler) CONSTRAINTS

K92_MLB

08/09/2010

98

108

Project Specific Constraints

K18_MLB

04/27/2010

99

109

PCB Rule Definitions

K18_MLB

04/27/2010

100

130

DEBUG SENSORS AND ADC

K91_DINESH

08/06/2010

101

132

Power Supplies BIST

K91_DINESH

08/18/2010

ALIASES

RESOLVED

Schematic / PCB #'s

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
051-8620	1	SCHEM, MLB, K91	SCH	CRITICAL	
820-2915	1	PCBF, MLB, K91	PCB	CRITICAL	

DRAWING

TITLE=MLB

ABBREV=DRAWING

LAST_MODIFIED=Mon Jan 31 12:49:37 2011

SCHEM, MLB, K91

Apple Inc.

NOTICE OF PROPRIETARY PROPERTY:
THE INFORMATION CONTAINED HEREIN IS THE
PROPRIETARY PROPERTY OF APPLE COMPUTER, INC.
THE POSSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

DRAWING NUMBER

REVISION

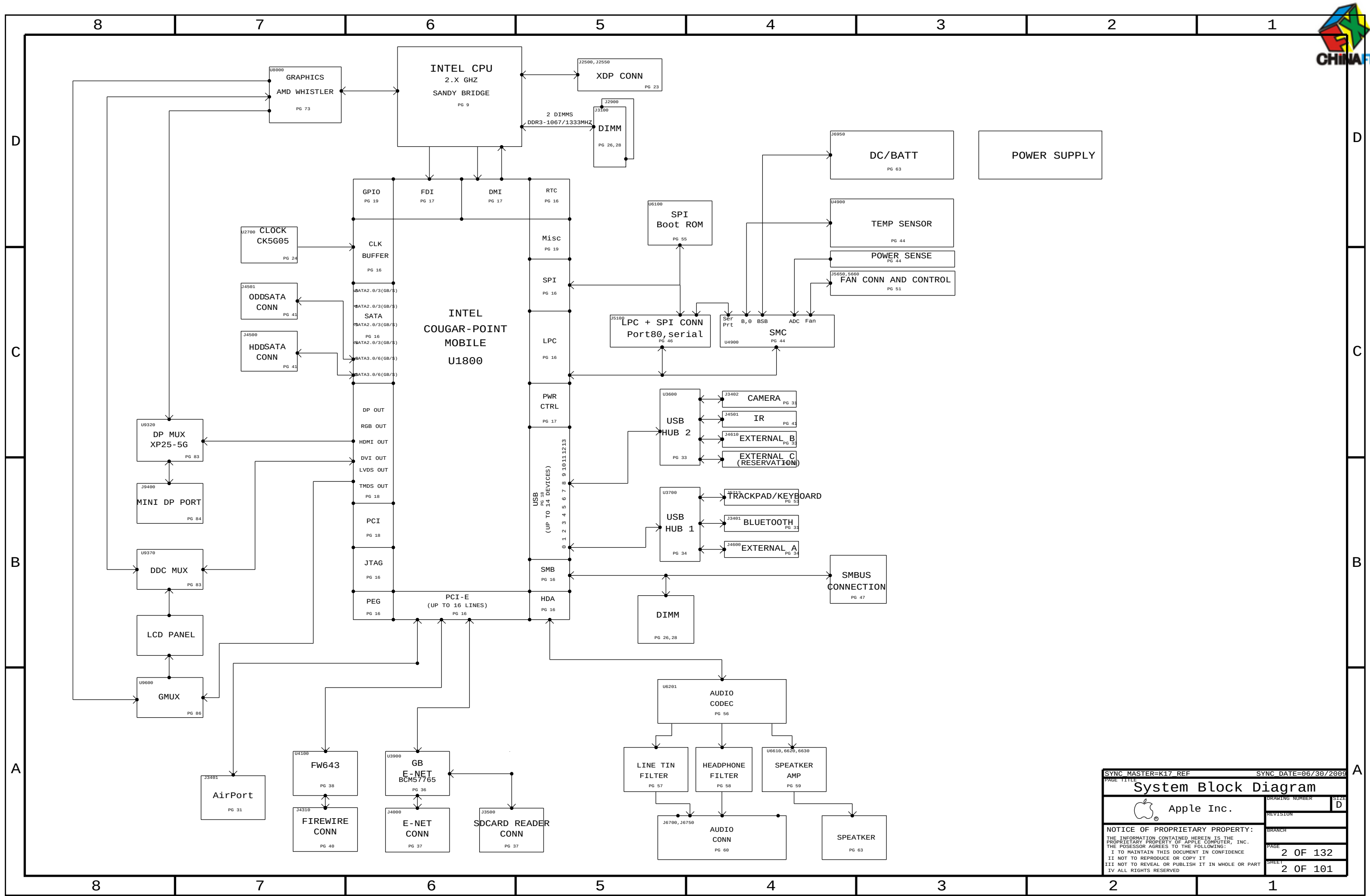
BRANCH

PAGE

SHEET

1 OF 132

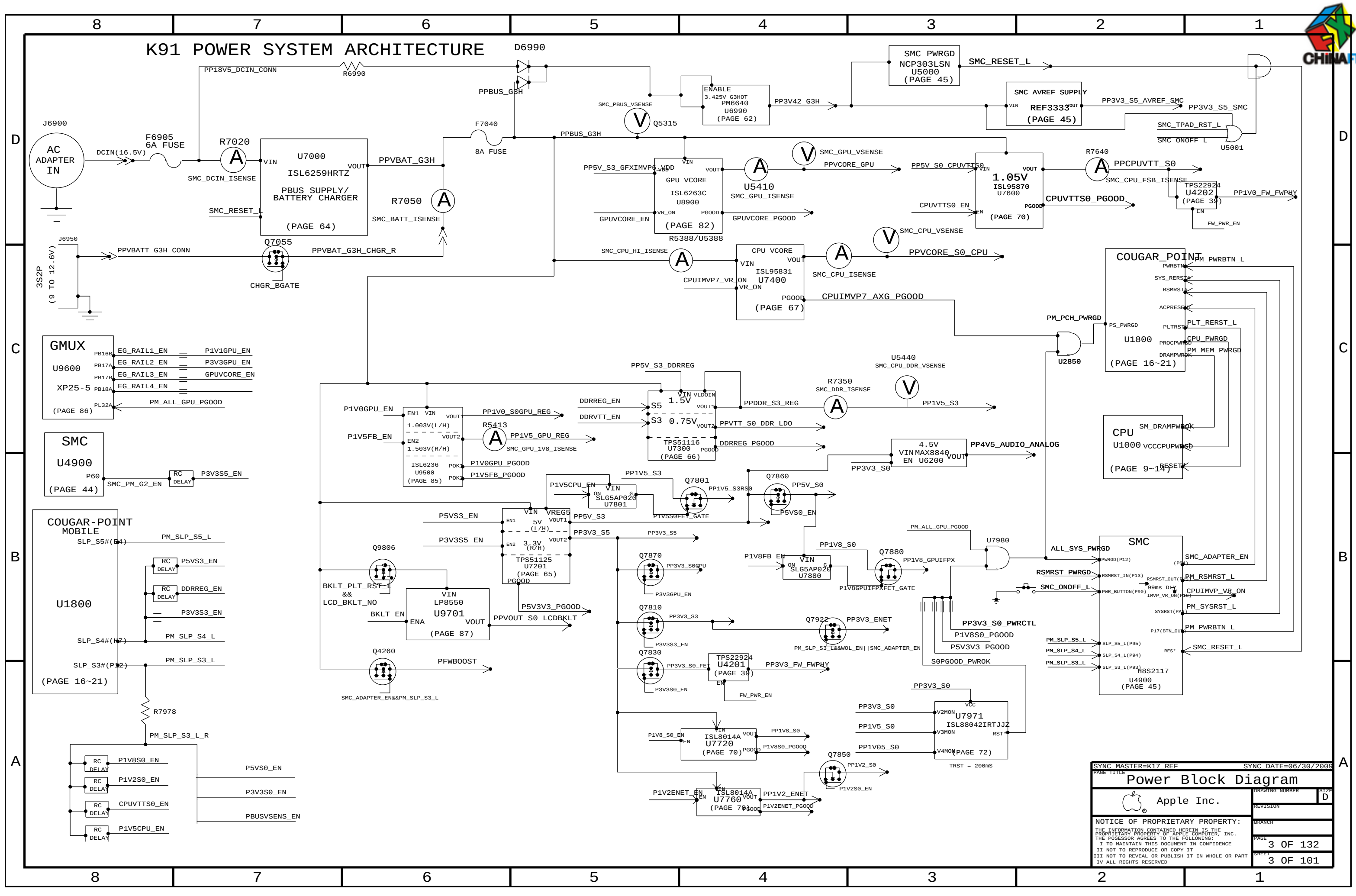
1 OF 101



SYNC MASTER=K17 REF		SYNC DATE=06/30/2009	
PAGE 1111			
System Block Diagram			
 Apple Inc.		DRAWING NUMBER	SIZE D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		REVISION	
		BRANCH	
		PAGE	2 OF 132
		SHEET	2 OF 101



K91 POWER SYSTEM ARCHITECTURE



PAGE TITLE		PAGE 1	
Power Block Diagram		SIZE D	
Apple Inc.		REVISION	
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		3 OF 132	
II NOT TO REPRODUCE OR COPY IT		SHEET	
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART		3 OF 101	
IV ALL RIGHTS RESERVED			



8	7	6	5	4	3	2	1
D							D
C							C
B							B
A							A
8	7	6	5	4	3	2	1

SYNC_MASTER=MASTER		SYNC_DATE=MASTER	
PAGE TITLE		Revision History	
 Apple Inc.		DRAWING NUMBER	SIZE
		REVISION	D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		BRANCH	
		PAGE	4 OF 132
		SHEET	4 OF 101



D

7

C

B

A

8	7	6	5	4	3	2	1
---	---	---	---	---	---	---	---

Alternate Parts

EFI ROM

PSOC

Bar Code Labels / EEEE #'s

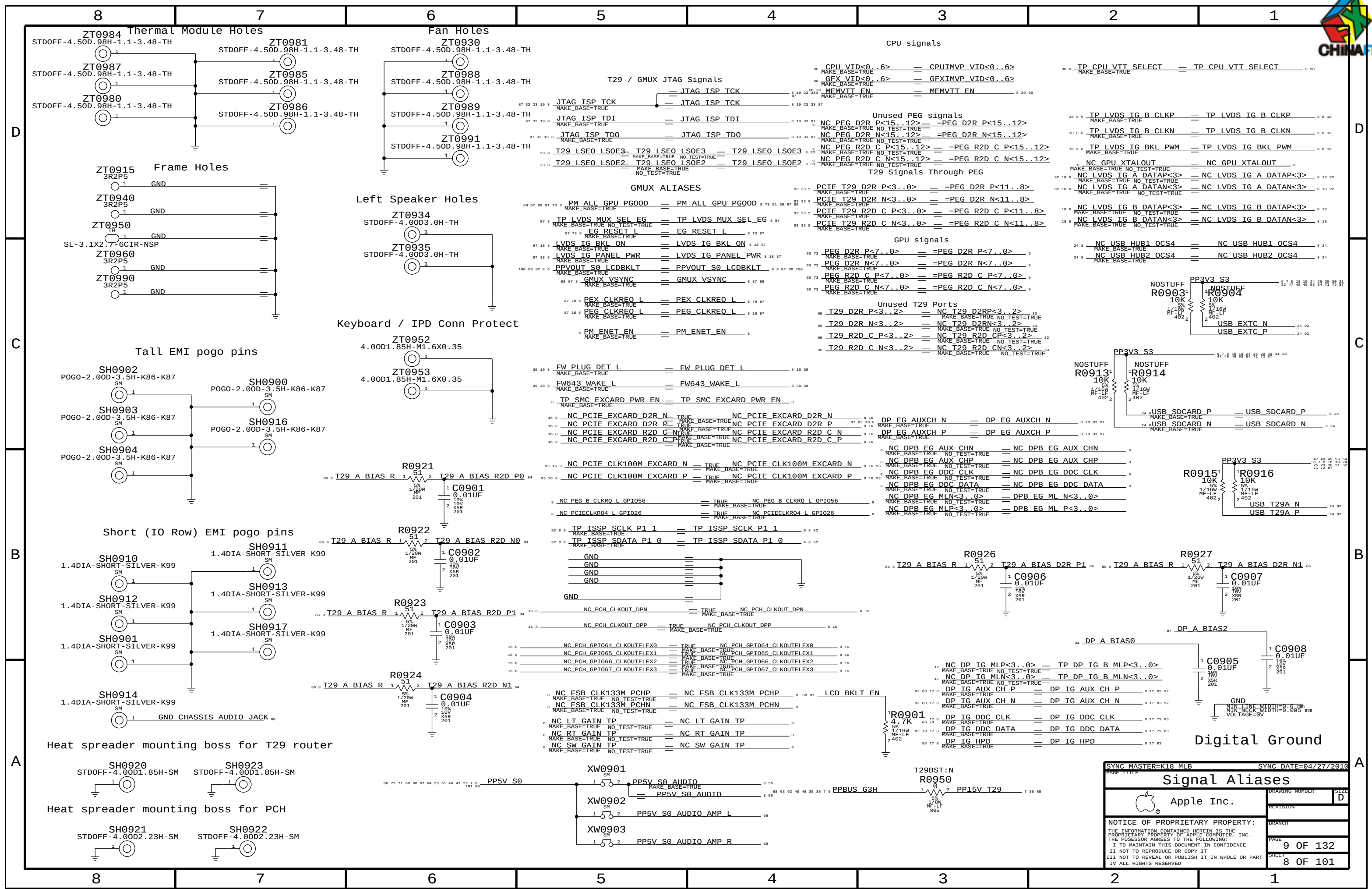
D

C

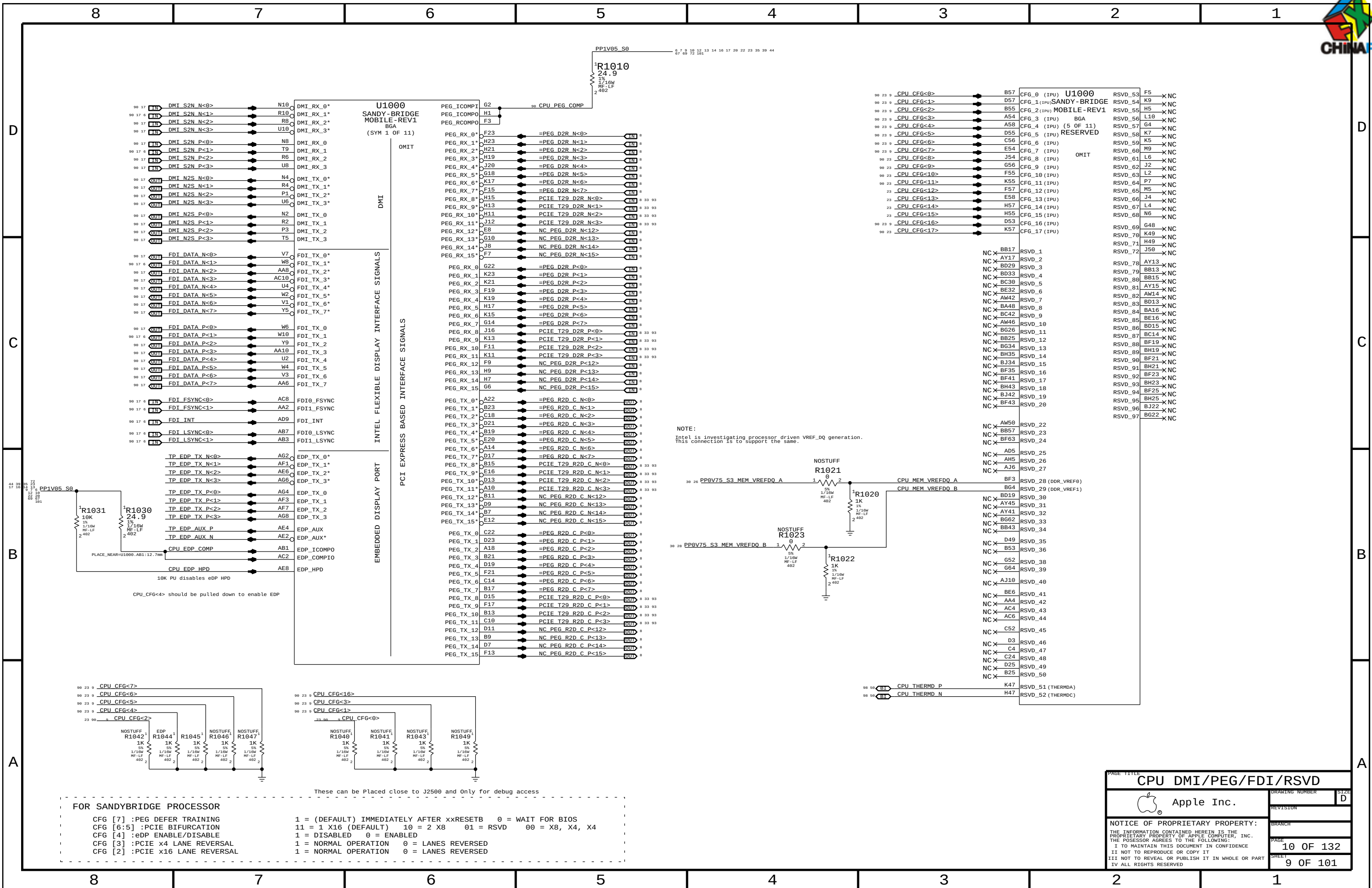
B

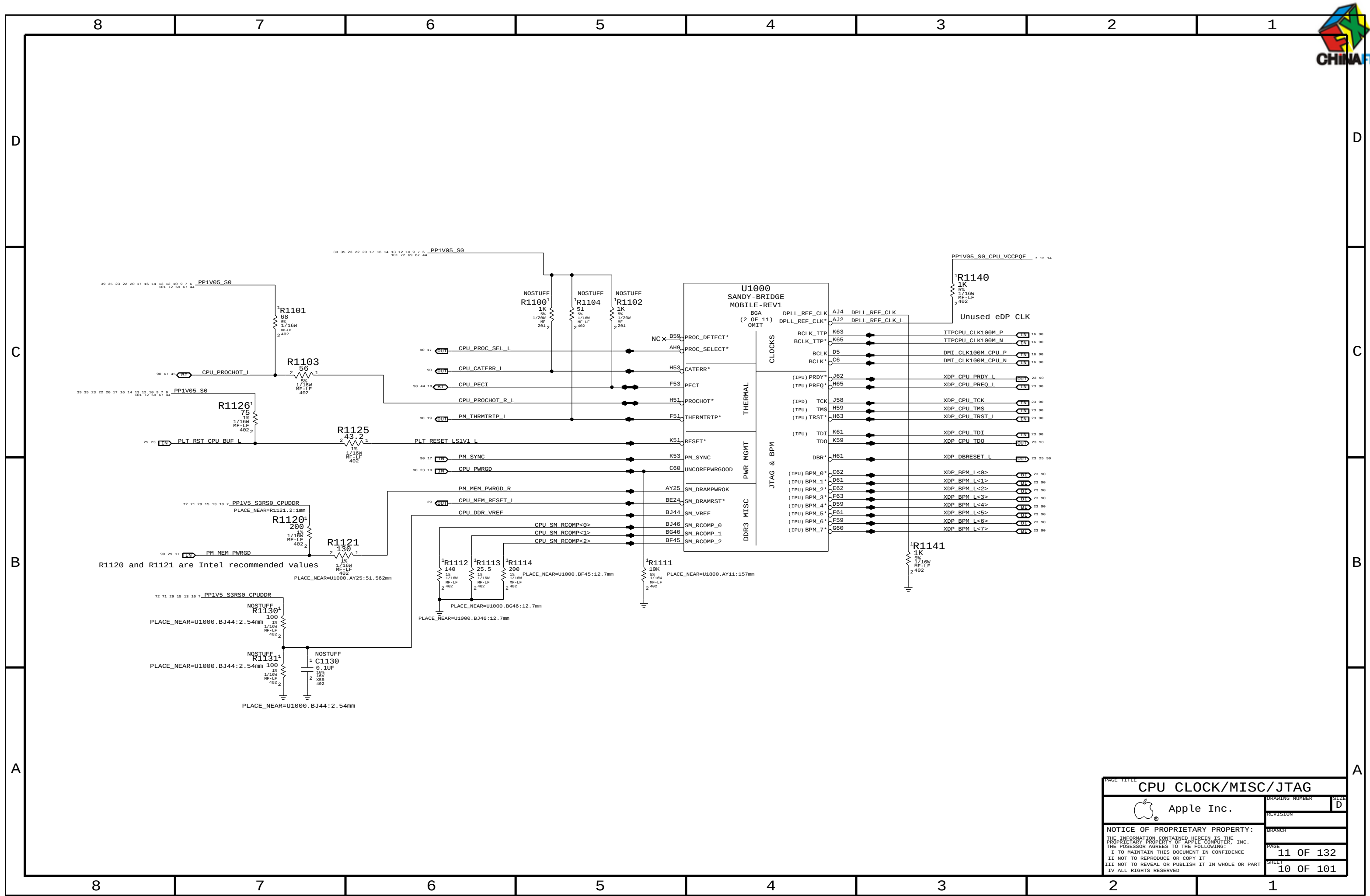
A

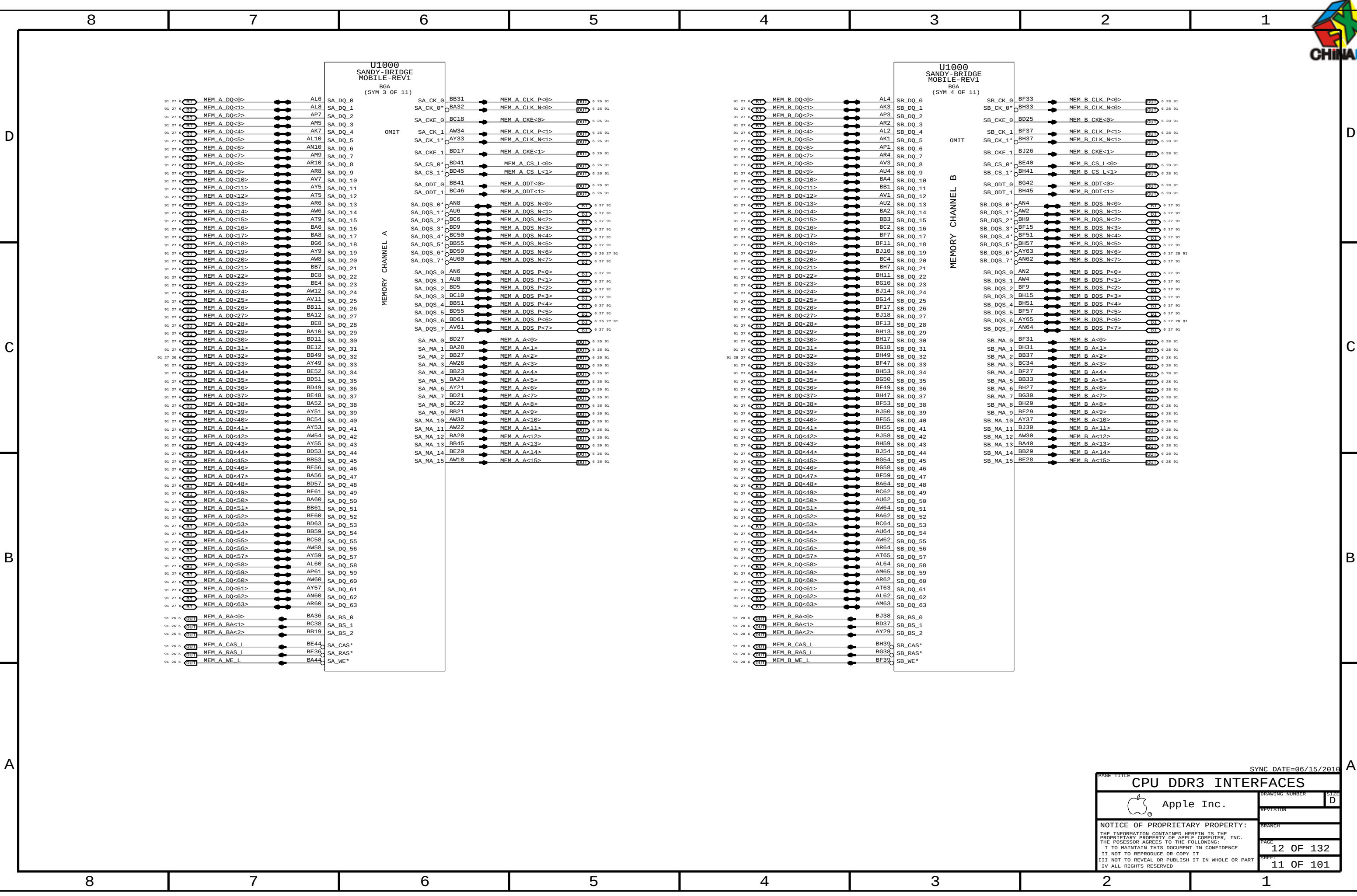
1

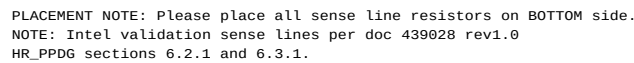


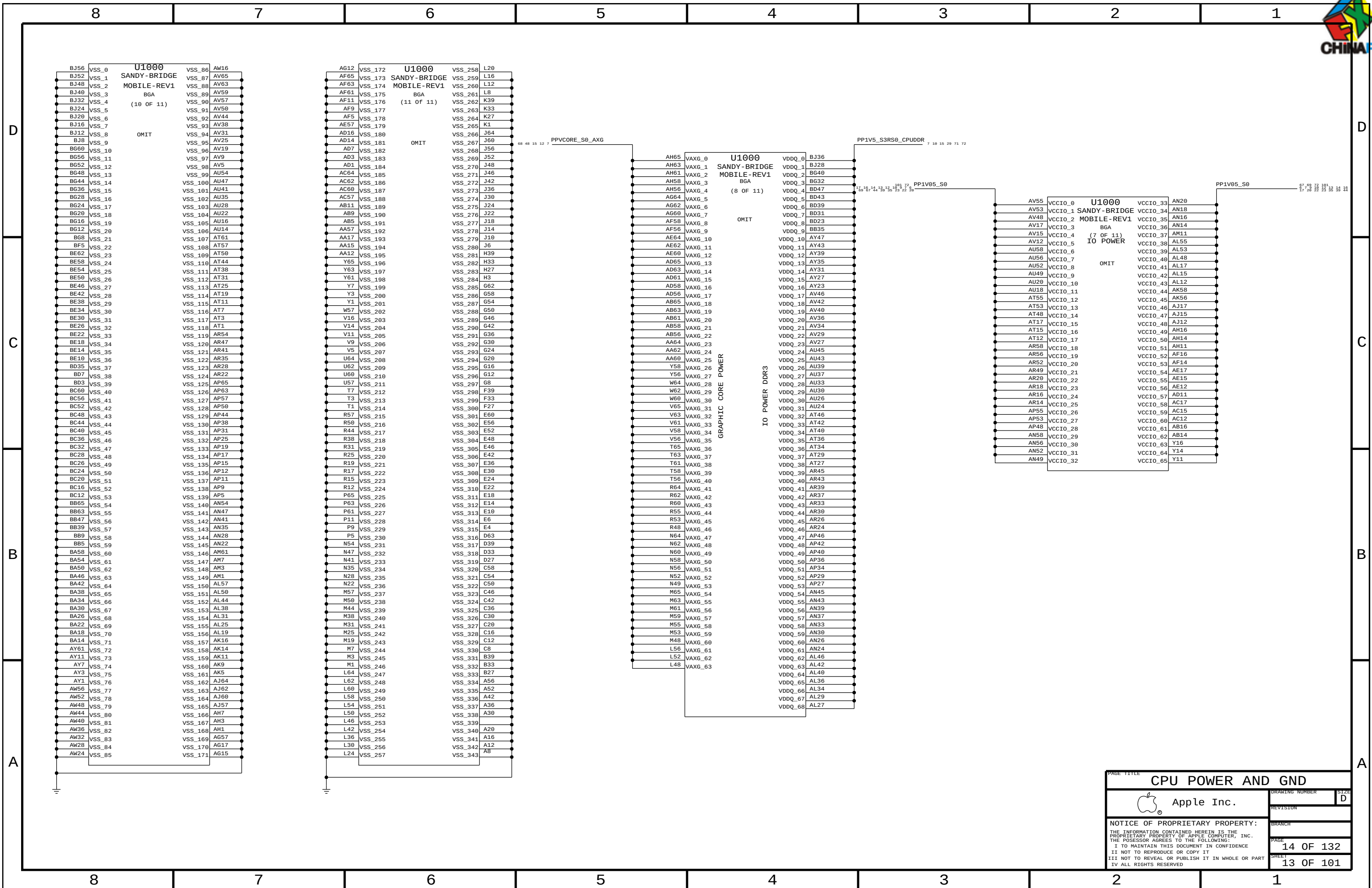
SYNC MASTER=K18 MLB		SYNC DATE=04/27/2016	
PAGE TITLE		SIZE	
Signal Aliases		D	
Apple Inc.		REVISION	
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		9 OF 132	
II NOT TO REPRODUCE OR COPY IT		SHEET	
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART		8 OF 101	
IV ALL RIGHTS RESERVED			











PAGE TITLE		CPU POWER AND GND	
Apple Inc.		DRAWING NUMBER	SIZE
NOTICE OF PROPRIETARY PROPERTY:		REVISION	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:		BRANCH	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		PAGE	14 OF 132
II NOT TO REPRODUCE OR COPY IT		SHEET	13 OF 101
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			

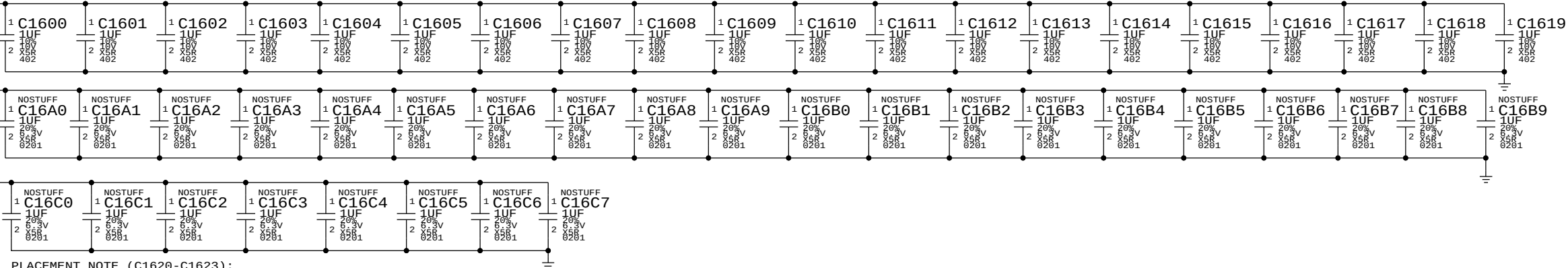


CPU VCORE DECOUPLING

Intel recommendation: 4x 470uF 4mOhm, 2x 470uF 4mOhm (NOSTUFF), 16x 22uF 0805, 4x 10uF 0603, 20x 1uF 0402, 28x 1uF 0402 (NOSTUFF)
Apple Implementation: 4x 470uF 4mOhm, 1x 470uF 4mOhm (NOSTUFF), 16x 22uF 0603, 4x 10uF 0402, 20x 1uF 0402, 28x 1uF 0201 (NOSTUFF), 4x 22uF 0603 (NOSTUFF)

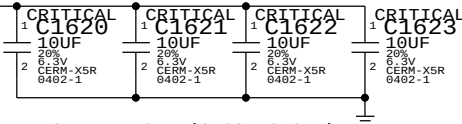
PLACEMENT_NOTE (C1600-C16C7):

Place on bottom side of U1000



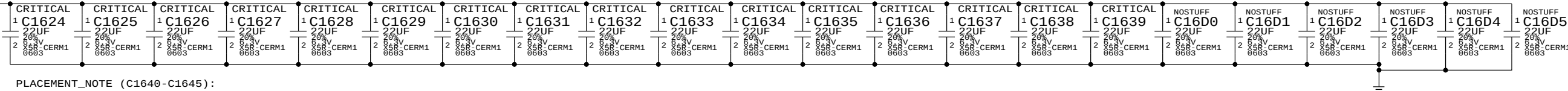
PLACEMENT_NOTE (C1620-C1623):

Place near U1000 on bottom side



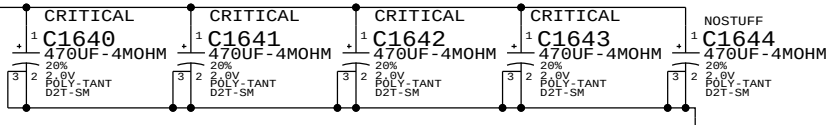
PLACEMENT_NOTE (C1624-C16D5):

Place near inductors on bottom side.



PLACEMENT_NOTE (C1640-C1645):

Place near inductors on bottom side.

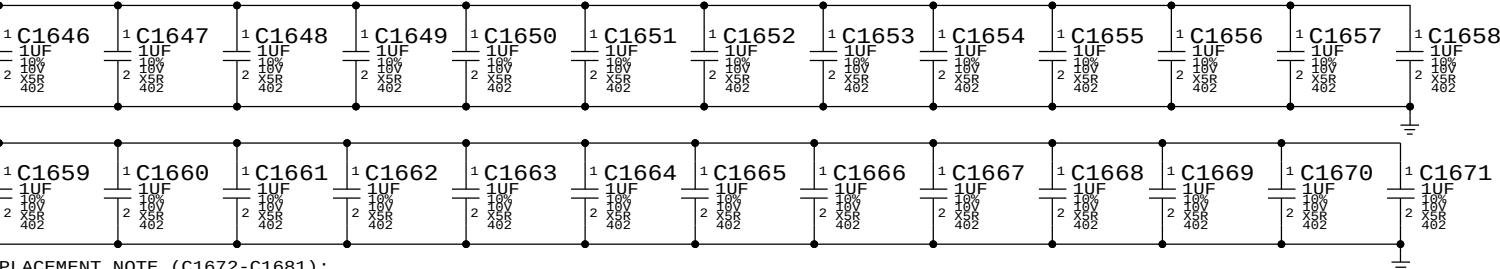


CPU VCCIO/VCCPQ DECOUPLING

Intel recommendation: 2x 330uF, 10x 10uF 0603, 26x 1uF 0402
Apple Implementation: 2x 330uF, 10x 10uF 0603, 26x 1uF 0402

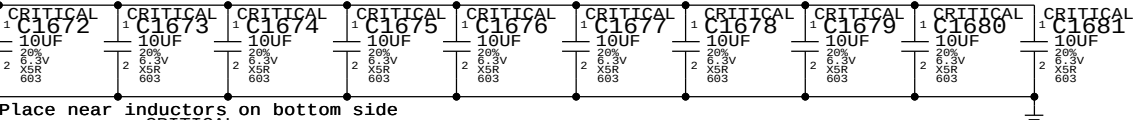
PLACEMENT_NOTE (C1646-C1671):

Place on bottom side of U1000

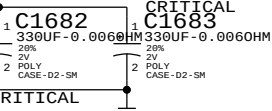


PLACEMENT_NOTE (C1672-C1681):

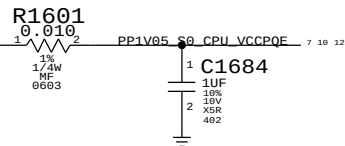
Place near U1000 on bottom side



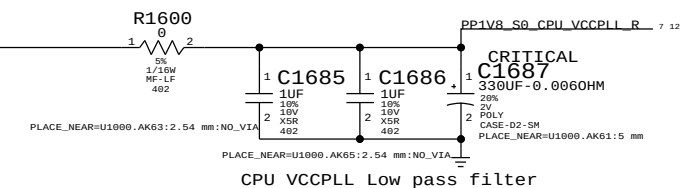
Place near inductors on bottom side



Intel recommendation: 1x 10mOhm resistor, 1x 1uF 0402



CPU VCCPLL DECOUPLING



CPU VCCPLL Low pass filter

SYNC_MASTER=K92_MLB		SYNC_DATE=08/19/2016	
PAGE TITLE		CPU DECOUPLING-I	
Apple Inc.		DRAWING NUMBER	SIZE
NOTICE OF PROPRIETARY PROPERTY:		REVISION	BRANCH
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE	16 OF 132
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		SHEET	14 OF 101
II NOT TO REPRODUCE OR COPY IT			
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			

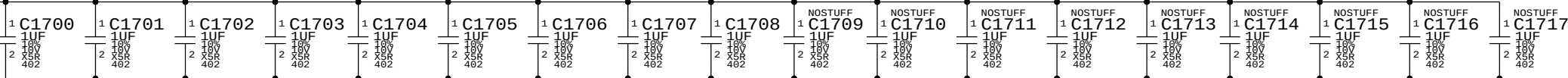


VAXG DECOUPLING

Intel recommendation: 2x 470uF 4mOhm, 2x 470uF 4mOhm (NOSTUFF), 6x 22uF 0805, 2x 22uF 0805 (NOSTUFF), 6x 10uF 0603, 2x 10uF 0603 (NOSTUFF), 9x 1uF 0402, 9x 1uF 0402 (NOSTUFF)
Apple Implementation: 2x 470uF 4mOhm, 1x 470uF 4mOhm (NOSTUFF), 6x 22uF 0603, 2x 22uF 0603 (NOSTUFF), 6x 10uF 0402, 2x 10uF 0402 (NOSTUFF), 9x 1uF 0402, 9x 1uF 0402 (NOSTUFF)

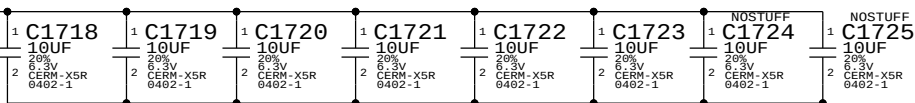
PLACEMENT_NOTE (C1700-C1708):

Place on bottom side of U1000



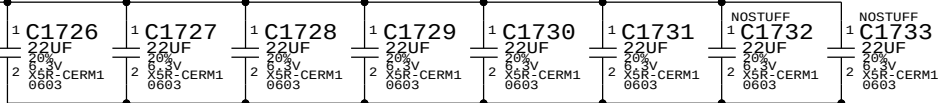
PLACEMENT_NOTE (C1718-C1723):

Place close to U1000 on bottom side



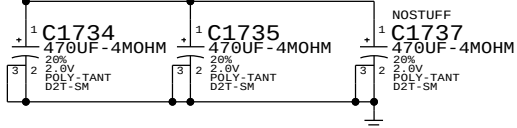
PLACEMENT_NOTE (C1726-C1731):

Place near inductors on bottom side.



PLACEMENT_NOTE (C1734-C1735):

Place near inductors on bottom side.

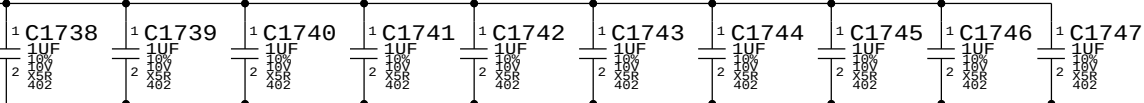


CPU VDDQ/VCCDQ DECOUPLING

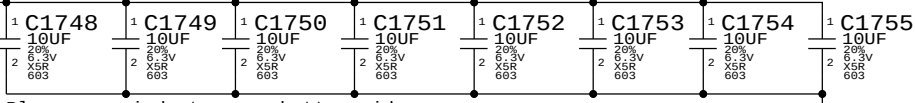
Intel recommendation: 1x 330uF, 8x 10uF 0603, 10x 1uF 0402
Apple Implementation: 1x 330uF, 8x 10uF 0603, 10x 1uF 0402

PLACEMENT_NOTE (C1738-C1747):

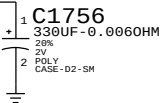
Place on bottom side of U1000



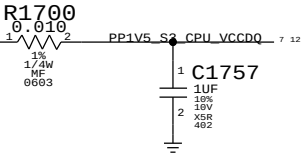
Place close to U1000 on bottom side



Place near inductors on bottom side



Intel recommendation: 1x 10mOhm resistor, 1x 1uF 0402

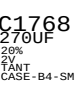
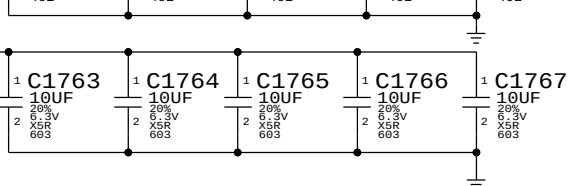
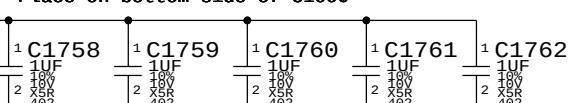


CPU VCCSA DECOUPLING

Intel recommendation: 1x 330uF, 5x 10uF 0603, 5x 1uF 0402
Apple Implementation: 1x 330uF, 5x 10uF 0603, 5x 1uF 0402

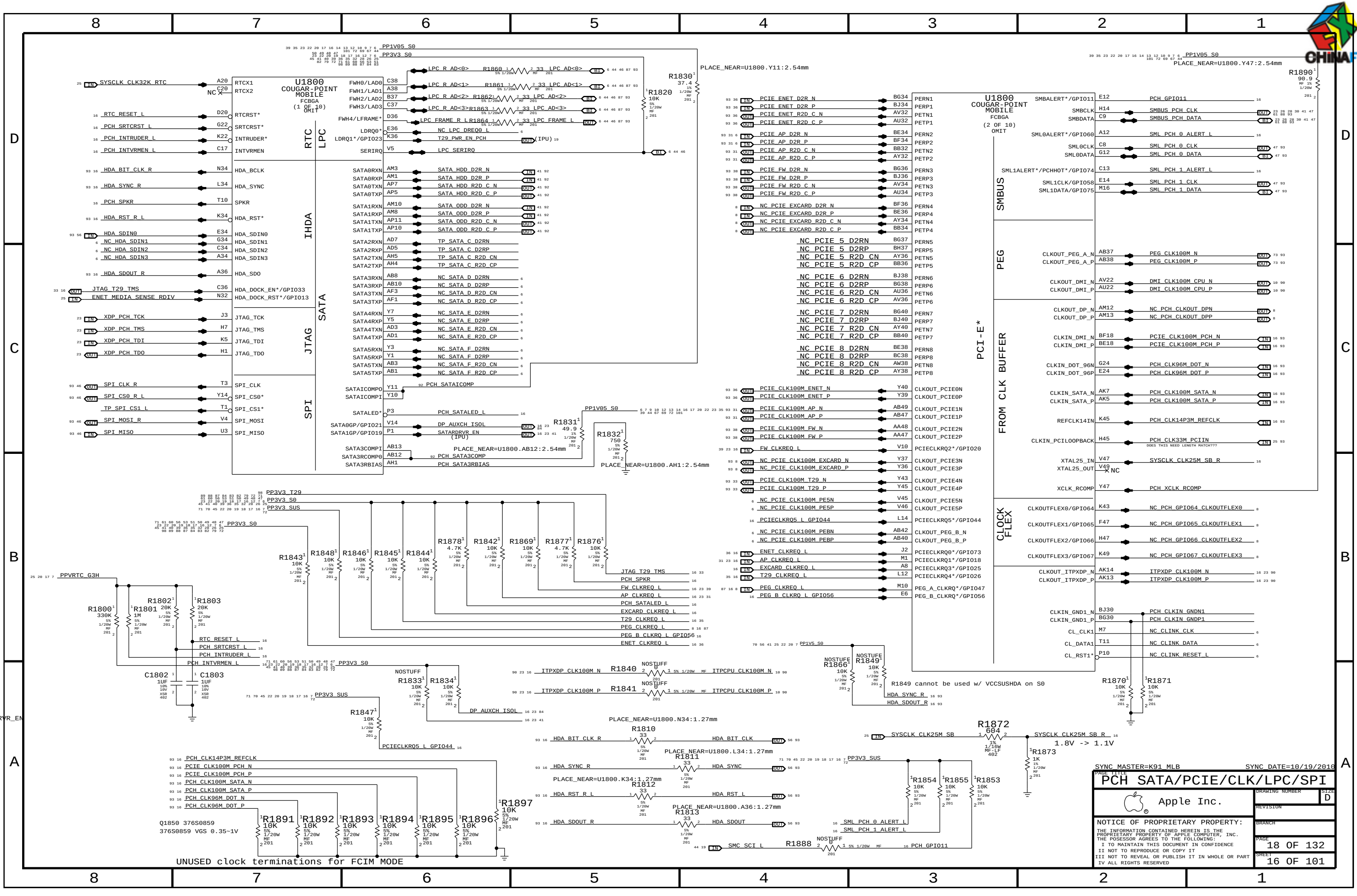
PLACEMENT_NOTE (C1758-C1762):

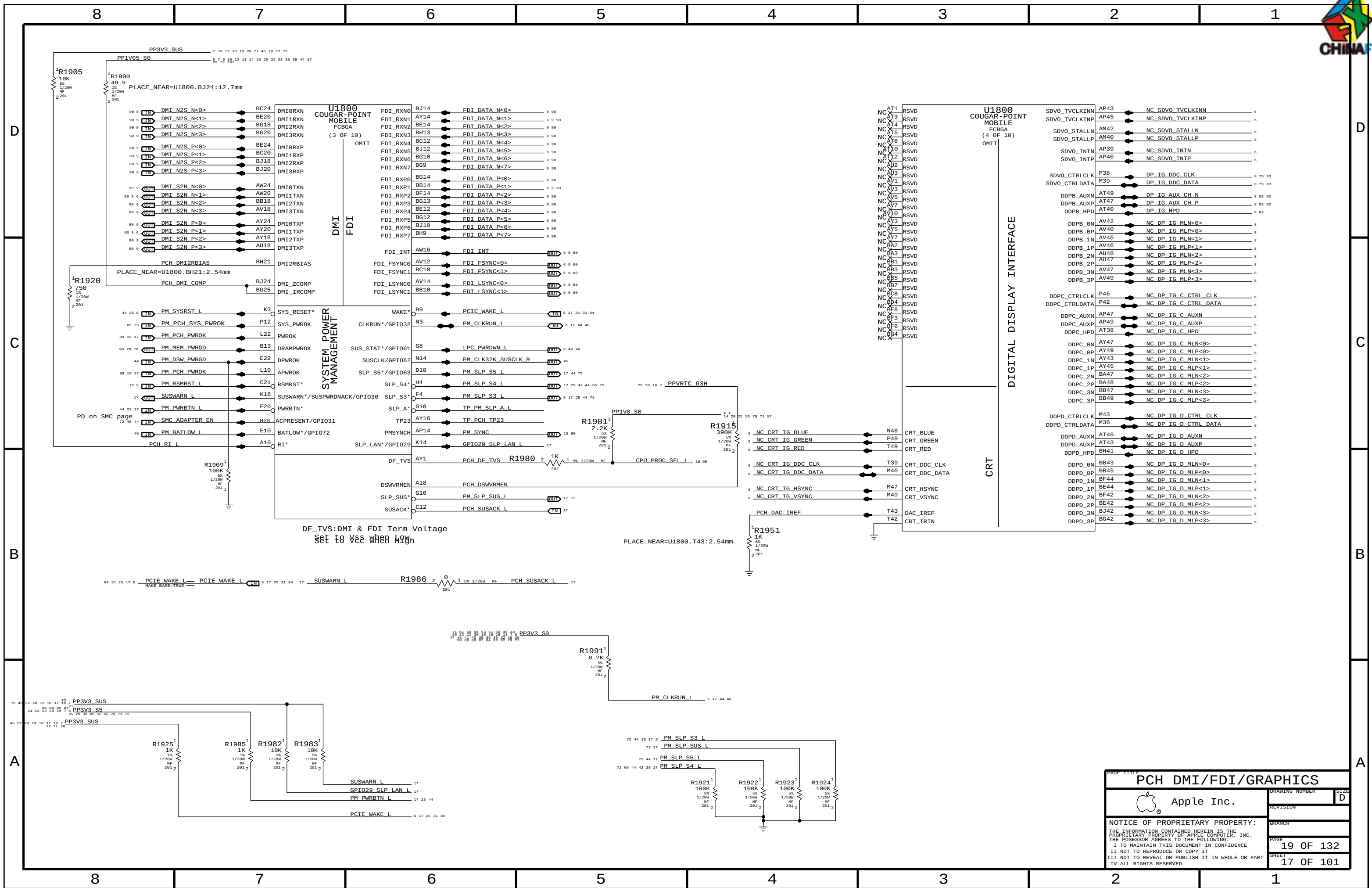
Place on bottom side of U1000

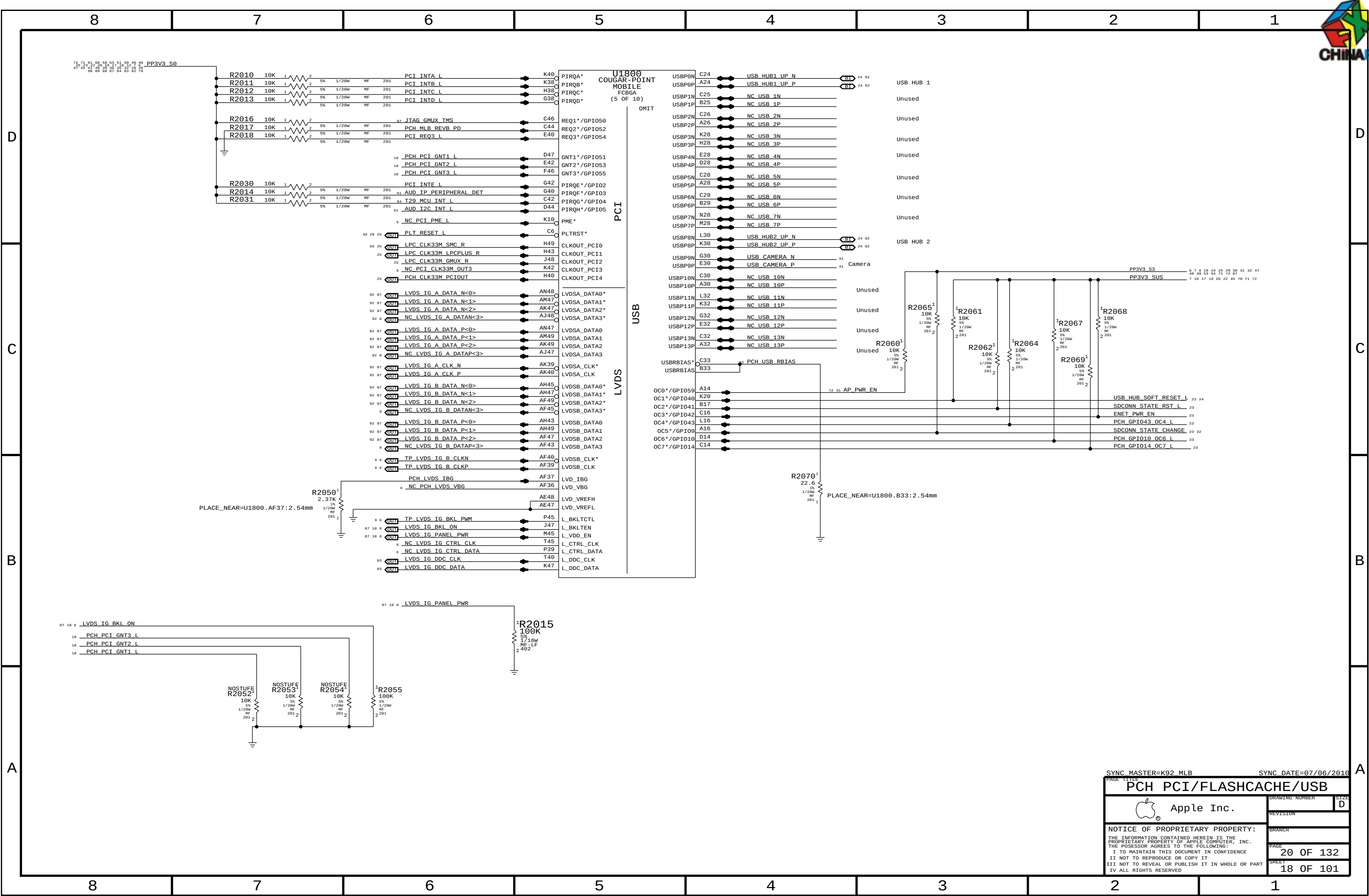


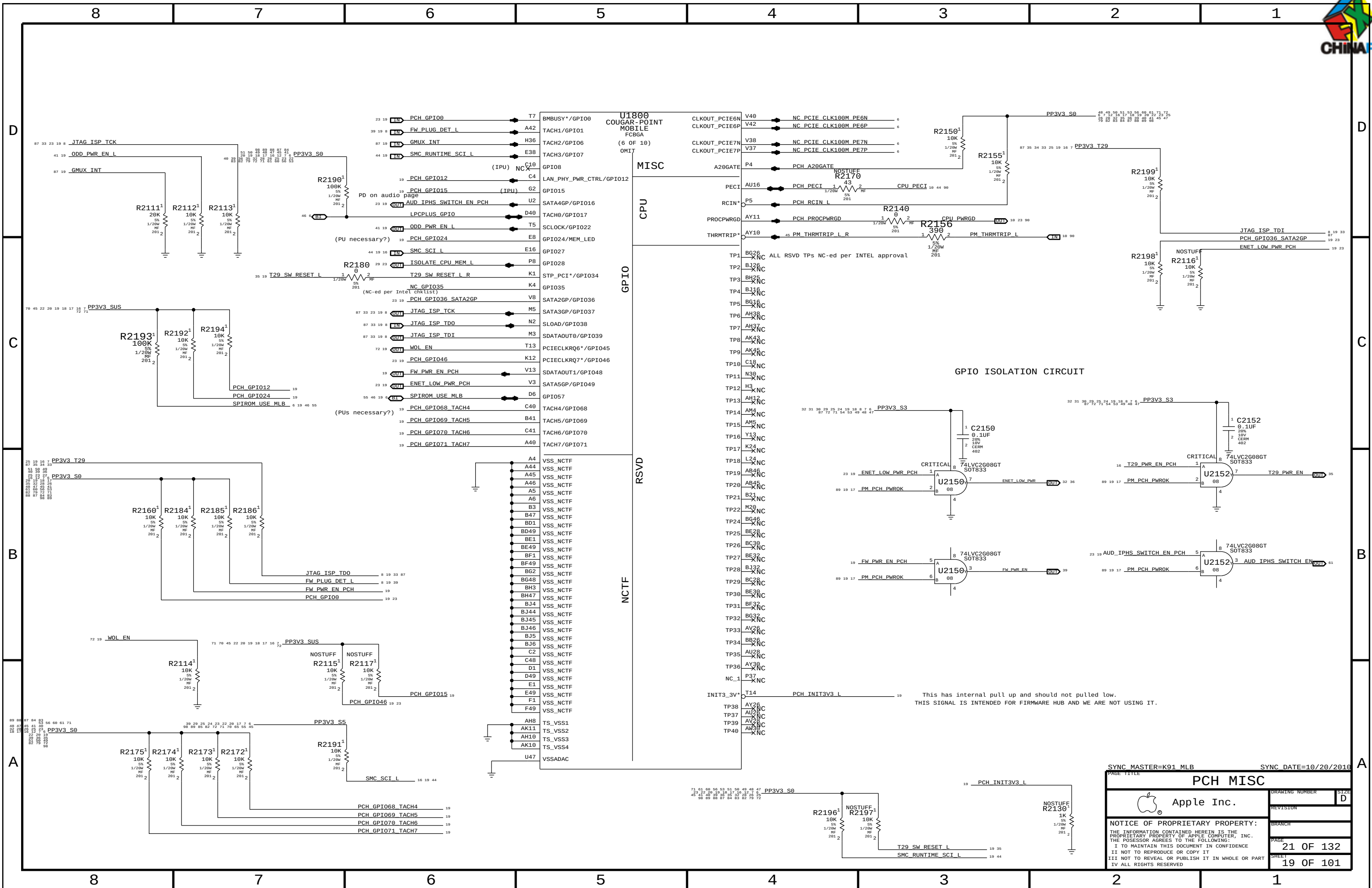
SYNC_MASTER=K92_MLB SYNC_DATE=08/19/2016

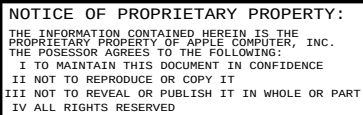
PAGE TITLE		DRAWING NUMBER		SIZE
CPU DECOUPLING-II		D		D
Apple Inc.		REVISION		
NOTICE OF PROPRIETARY PROPERTY:		BRANCH		
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE		17 OF 132
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		SHEET		15 OF 101
II NOT TO REPRODUCE OR COPY IT				
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART				
IV ALL RIGHTS RESERVED				











DRAWING NUMBER		SIZE
		D
REVISION		
BRANCH		
PAGE		
22 OF 132		
SHEET		
20 OF 101		



D

C

B

A

D

C

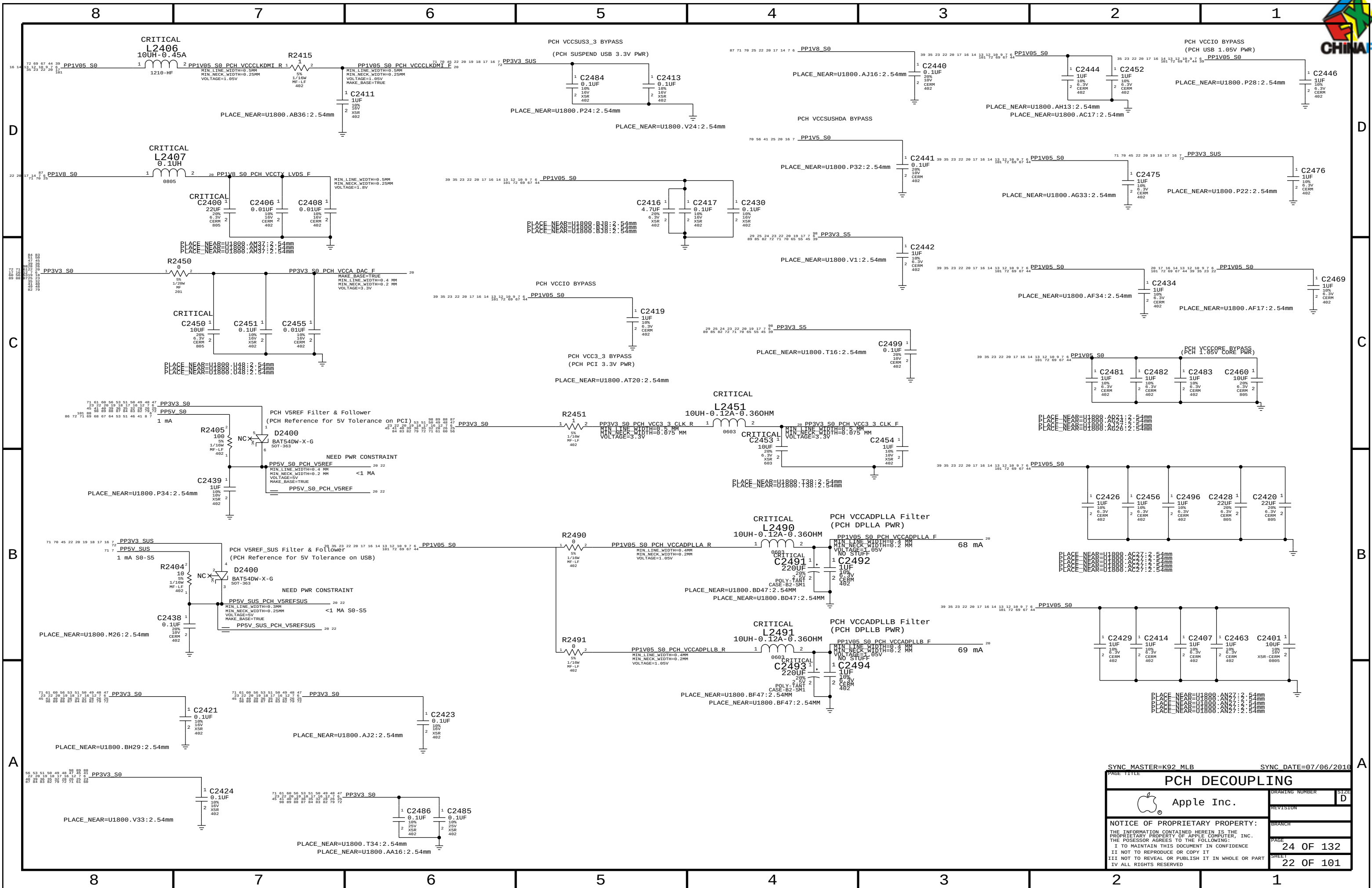
B

A

AJ3	VSS	U1800	VSS	AK46
N24	VSS	COUGAR-POINT	VSS	AK8
BG29	VSS	MOBILE	VSS	AL16
H5	VSS	FCBGA	VSS	AL17
AA17	VSS	(9 OF 10)	VSS	AL19
AA2	VSS	VSS	VSS	AL2
AA3	VSS	OMIT	VSS	AL21
AA33	VSS		VSS	AL23
AA34	VSS		VSS	AL26
AB11	VSS		VSS	AL27
AB14	VSS		VSS	AL31
AB39	VSS		VSS	AL33
AB4	VSS		VSS	AL34
AB43	VSS		VSS	AL48
AB5	VSS		VSS	AM11
AB7	VSS		VSS	AM14
AC19	VSS		VSS	AM36
AC2	VSS		VSS	AM39
AC21	VSS		VSS	AM43
AC24	VSS		VSS	AM45
AC33	VSS		VSS	AM46
AC34	VSS		VSS	AM7
AC48	VSS		VSS	AN2
AD10	VSS		VSS	AN29
AD11	VSS		VSS	AN3
AD12	VSS		VSS	AN31
AD13	VSS		VSS	AP12
AD14	VSS		VSS	AP13
AD16	VSS		VSS	AP19
AD19	VSS		VSS	AP28
AD24	VSS		VSS	AP39
AD26	VSS		VSS	AP32
AD27	VSS		VSS	AP38
AD33	VSS		VSS	AP4
AD34	VSS		VSS	AP42
AD36	VSS		VSS	AP46
AD37	VSS		VSS	AP8
AD38	VSS		VSS	AR2
AD39	VSS		VSS	AR48
AD4	VSS		VSS	AT11
AD40	VSS		VSS	AT13
AD42	VSS		VSS	AT18
AD43	VSS		VSS	AT22
AD45	VSS		VSS	AT26
AD46	VSS		VSS	AT28
AD47	VSS		VSS	AT30
AD8	VSS		VSS	AT32
AE2	VSS		VSS	AT34
AE3	VSS		VSS	AT39
AF10	VSS		VSS	AT42
AF12	VSS		VSS	AT46
AF16	VSS		VSS	AT7
AF19	VSS		VSS	AU24
AF24	VSS		VSS	AU30
AF26	VSS		VSS	AV11
AF27	VSS		VSS	AV16
AF29	VSS		VSS	AV20
AF31	VSS		VSS	AV24
AF38	VSS		VSS	AV30
AF4	VSS		VSS	AV38
AF42	VSS		VSS	AV4
AF46	VSS		VSS	AV43
AF5	VSS		VSS	AV8
AF7	VSS		VSS	Aw14
AF8	VSS		VSS	Aw18
AG19	VSS		VSS	Aw2
AG2	VSS		VSS	Aw22
AG31	VSS		VSS	Aw26
AG48	VSS		VSS	Aw28
AH11	VSS		VSS	Aw32
AH3	VSS		VSS	Aw34
AH36	VSS		VSS	Aw36
AH39	VSS		VSS	Aw40
AH40	VSS		VSS	Aw48
AH42	VSS		VSS	AY12
AH46	VSS		VSS	AY22
AH7	VSS		VSS	AY28
AJ19	VSS		VSS	AY4
AJ21	VSS		VSS	
AJ24	VSS		VSS	AY46
AJ33	VSS		VSS	AY8
AJ34	VSS		VSS	B11
AK12	VSS		VSS	B15
AK3	VSS		VSS	B19
AK38	VSS		VSS	B23
AK4	VSS		VSS	B27
AK42	VSS		VSS	B31

B35	VSS	U1800	VSS	G48
B39	VSS	COUGAR-POINT	VSS	H10
B43	VSS	MOBILE	VSS	H12
B7	VSS	FCBGA	VSS	H16
BB12	VSS	(10 OF 10)	VSS	H18
BB16	VSS	VSS	VSS	H22
BB20	VSS	OMIT	VSS	H24
BB22	VSS		VSS	H26
BB24	VSS		VSS	H30
BB28	VSS		VSS	H32
BB30	VSS		VSS	H34
BB38	VSS		VSS	H46
BB4	VSS		VSS	K18
BB46	VSS		VSS	K26
BC14	VSS		VSS	K39
BC18	VSS		VSS	K46
BC2	VSS		VSS	K7
BC22	VSS		VSS	L18
BC26	VSS		VSS	L2
BC32	VSS		VSS	L20
BC34	VSS		VSS	L26
BC36	VSS		VSS	L28
BC40	VSS		VSS	L36
BC42	VSS		VSS	L48
BC48	VSS		VSS	M12
BD3	VSS		VSS	M14
BD46	VSS		VSS	M18
B05	VSS		VSS	M22
BE10	VSS		VSS	M24
BE22	VSS		VSS	M30
BE26	VSS		VSS	M32
BE40	VSS		VSS	M34
BF10	VSS		VSS	M38
BF12	VSS		VSS	M4
BF16	VSS		VSS	M42
BF20	VSS		VSS	M46
BF22	VSS		VSS	M8
BF24	VSS		VSS	N18
BF26	VSS		VSS	N47
BF28	VSS		VSS	P11
BF30	VSS		VSS	P16
BF38	VSS		VSS	P18
BF40	VSS		VSS	P30
BF8	VSS		VSS	P40
BG17	VSS		VSS	P43
BG21	VSS		VSS	P47
BG22	VSS		VSS	P7
BG24	VSS		VSS	R2
BG33	VSS		VSS	R48
BG41	VSS		VSS	T12
BG44	VSS		VSS	T31
B68	VSS		VSS	T33
BH11	VSS		VSS	T36
BH15	VSS		VSS	T37
BH17	VSS		VSS	T4
BH19	VSS		VSS	T46
BH27	VSS		VSS	T47
BH31	VSS		VSS	T8
BH33	VSS		VSS	V11
BH35	VSS		VSS	V26
BH39	VSS		VSS	V27
BH43	VSS		VSS	V29
BH7	VSS		VSS	V31
C22	VSS		VSS	V36
D12	VSS		VSS	V39
D16	VSS		VSS	V43
D18	VSS		VSS	V7
D22	VSS		VSS	W17
D24	VSS		VSS	W19
D26	VSS		VSS	W2
D3	VSS		VSS	W27
D30	VSS		VSS	W34
D32	VSS		VSS	W48
D34	VSS		VSS	Y12
D38	VSS		VSS	Y38
D42	VSS		VSS	Y4
D8	VSS		VSS	Y42
E18	VSS		VSS	Y46
E26	VSS		VSS	Y8
F3	VSS		VSS	V17
F45	VSS		VSS	AP3
G14	VSS		VSS	AP1
G18	VSS		VSS	BE16
G20	VSS		VSS	BC16
G26	VSS		VSS	BG28
G28	VSS		VSS	BJ28
G36	VSS		VSS	

PAGE TITLE		SYNC MASTER=K92 MLB		SYNC DATE=04/30/2016	
PCH GROUNDS				DRAWING NUMBER	
				SIZE D	
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED				REVISION	
				BRANCH	
				PAGE	
				23 OF 132	
				SHEET	
				21 OF 101	





PROCESSOR MINI XDP

D

C

B

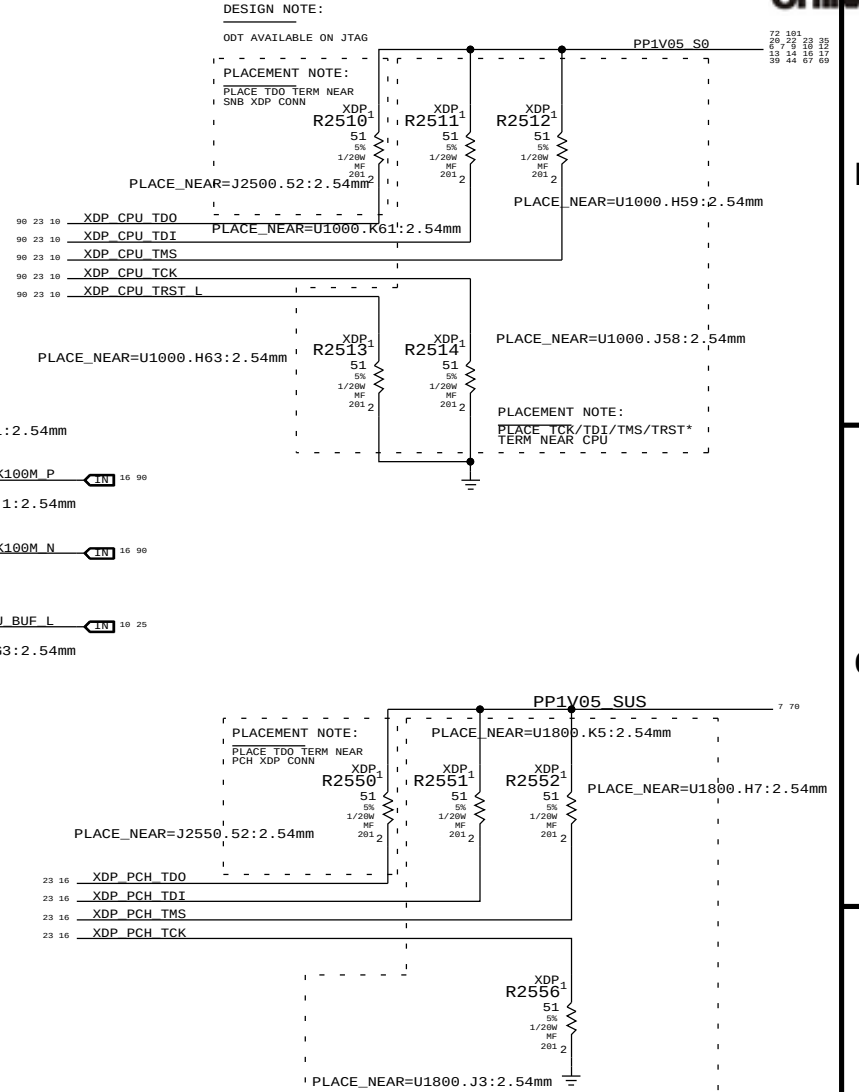
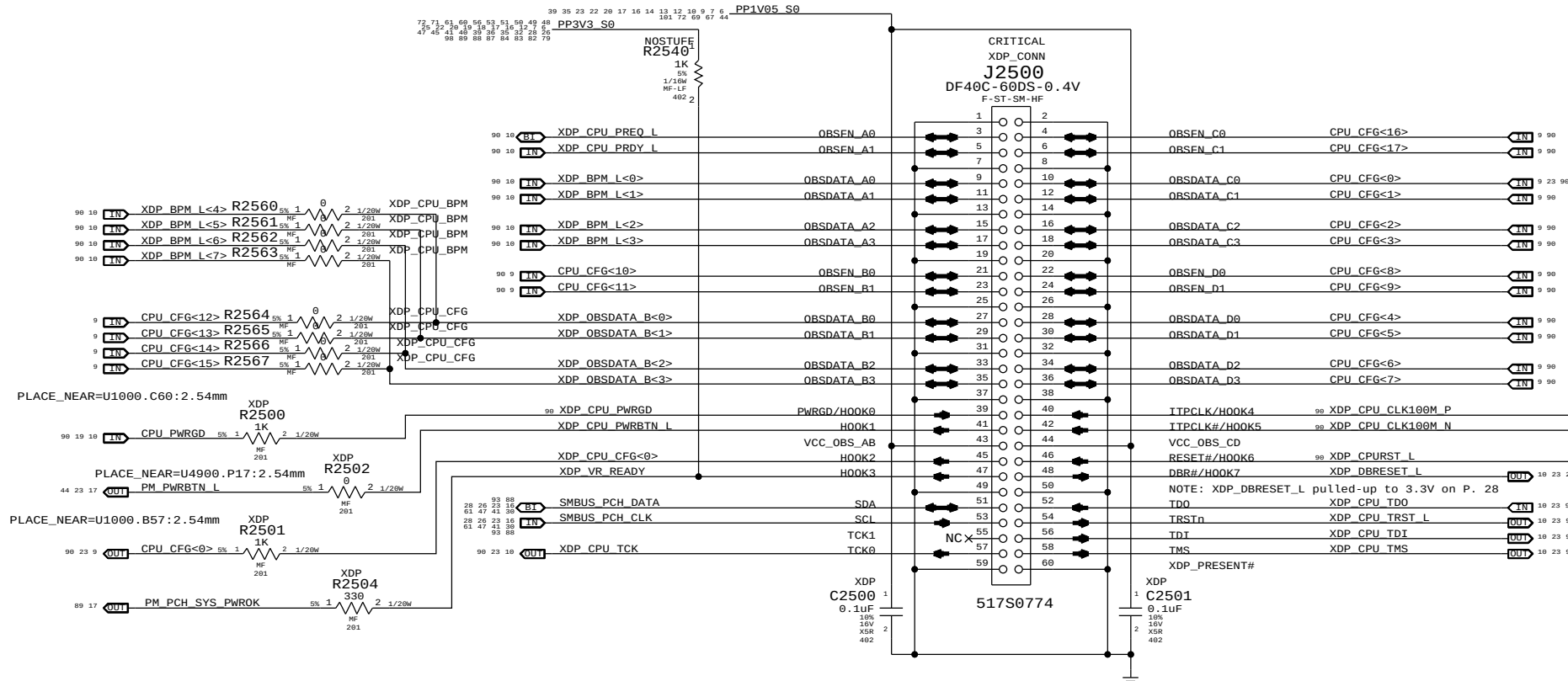
A

D

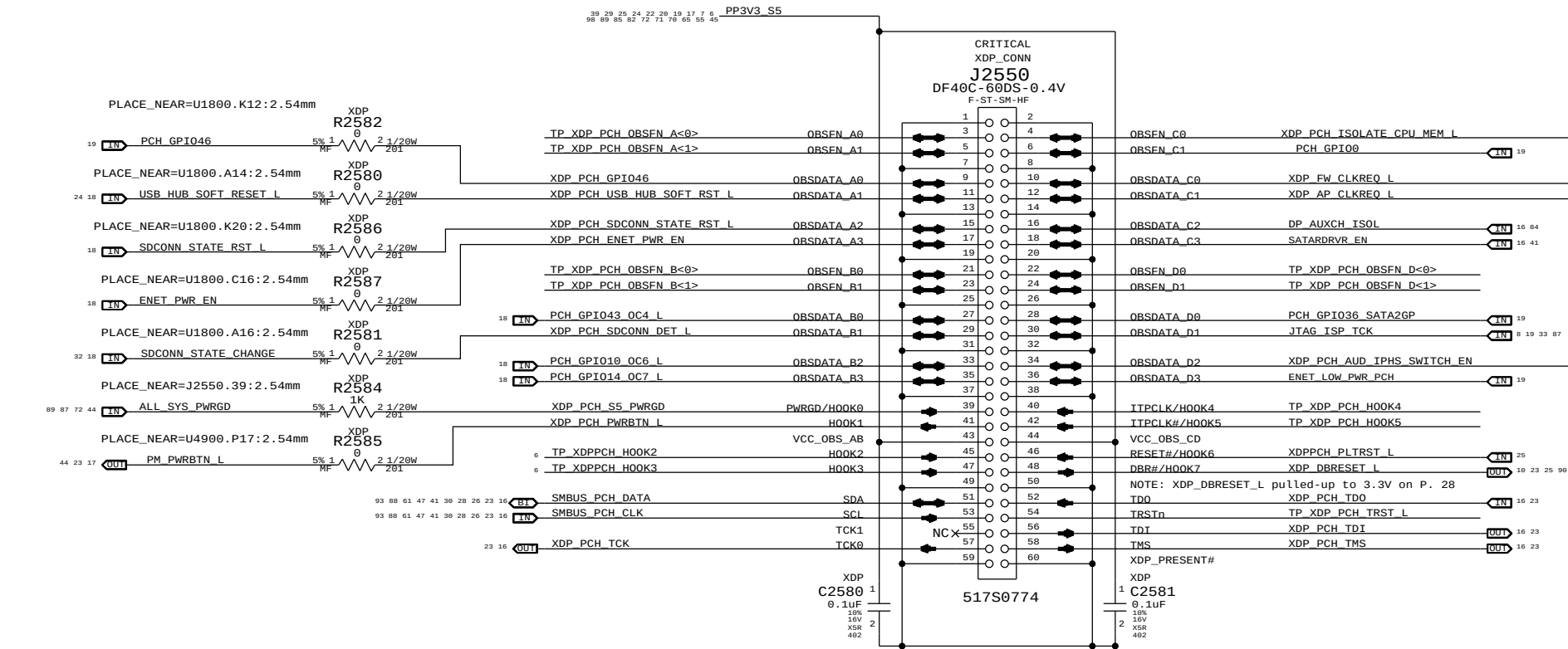
C

B

A



PCH MINI XDP



PAGE TITLE		SYNC DATE=10/17/2016	
CPU & PCH XDP		DRAWING NUMBER	
Apple Inc.		SIZE D	
NOTICE OF PROPRIETARY PROPERTY:		REVISION	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:		BRANCH	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		PAGE	
II NOT TO REPRODUCE OR COPY IT		25 OF 132	
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART		SHEET	
IV ALL RIGHTS RESERVED		23 OF 101	

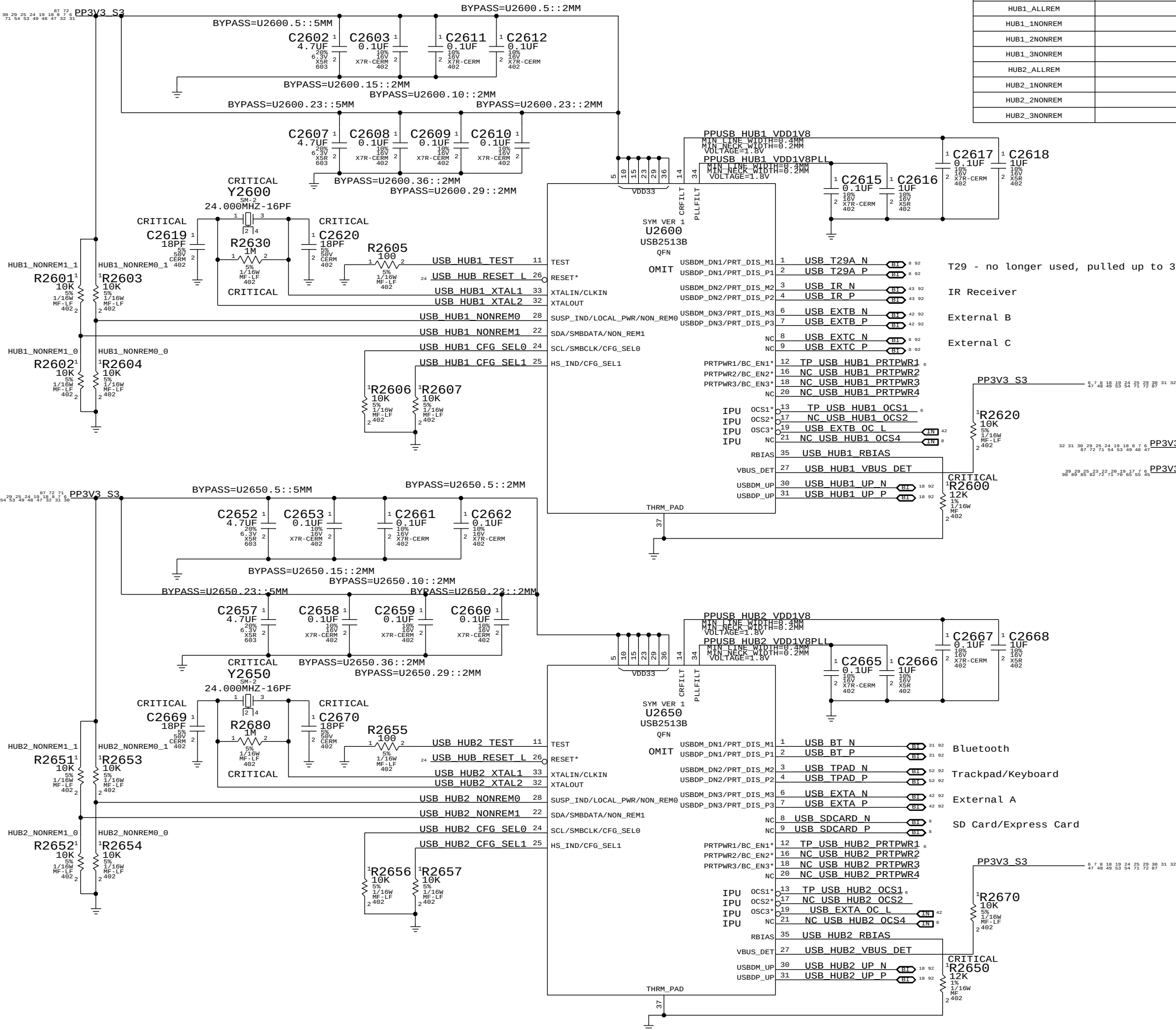


BOM GROUP	BOM OPTIONS
HUB1_ALLREM	HUB1_NONREM1_0, HUB1_NONREM0_0
HUB1_1NONREM	HUB1_NONREM1_0, HUB1_NONREM0_1
HUB1_2NONREM	HUB1_NONREM1_1, HUB1_NONREM0_0
HUB1_3NONREM	HUB1_NONREM1_1, HUB1_NONREM0_1
HUB2_ALLREM	HUB2_NONREM1_0, HUB2_NONREM0_0
HUB2_1NONREM	HUB2_NONREM1_0, HUB2_NONREM0_1
HUB2_2NONREM	HUB2_NONREM1_1, HUB2_NONREM0_0
HUB2_3NONREM	HUB2_NONREM1_1, HUB2_NONREM0_1

NON_REM1	NON_REM0	DESCRIPTION
0	0	All ports are removable
0	1	Port 1 is non removable
1	0	Port 1 and 2 are non removable
1	1	Port 1, 2, and 3 are non removable

BOM TABLE

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
338S0720	2	SMSC USB2514	U2600, U2650	CRITICAL	USBHUB_2514
338S0824	2	SMSC USB2514B	U2600, U2650	CRITICAL	USBHUB_2514B
338S0721	2	SMSC USX2061	U2600, U2650	CRITICAL	USBHUB_2061
338S0923	2	SMSC USX2513B	U2600, U2650	CRITICAL	USBHUB_2513B

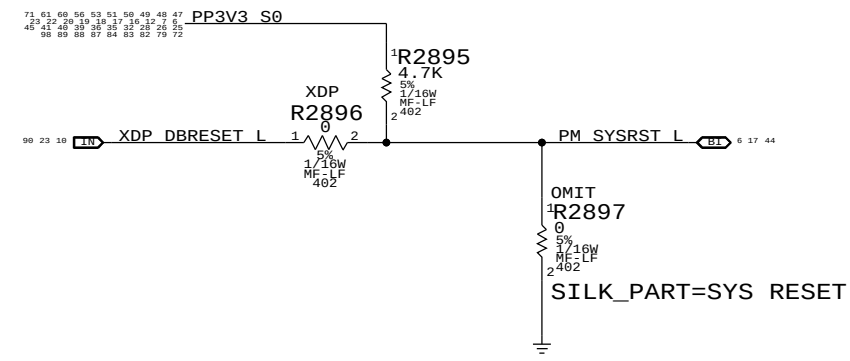


SYNC MASTER=K91 ERIC SYNC DATE=10/08/2010

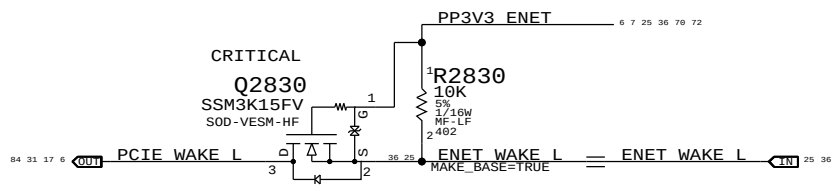
PAGE TITLE		USB HUBS	
Apple Inc.		DRAWING NUMBER	SIZE D
NOTICE OF PROPRIETARY PROPERTY:		REVISION	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:		BRANCH	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		PAGE	26 OF 132
II NOT TO REPRODUCE OR COPY IT		SHEET	24 OF 101
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			



PCH Reset Button

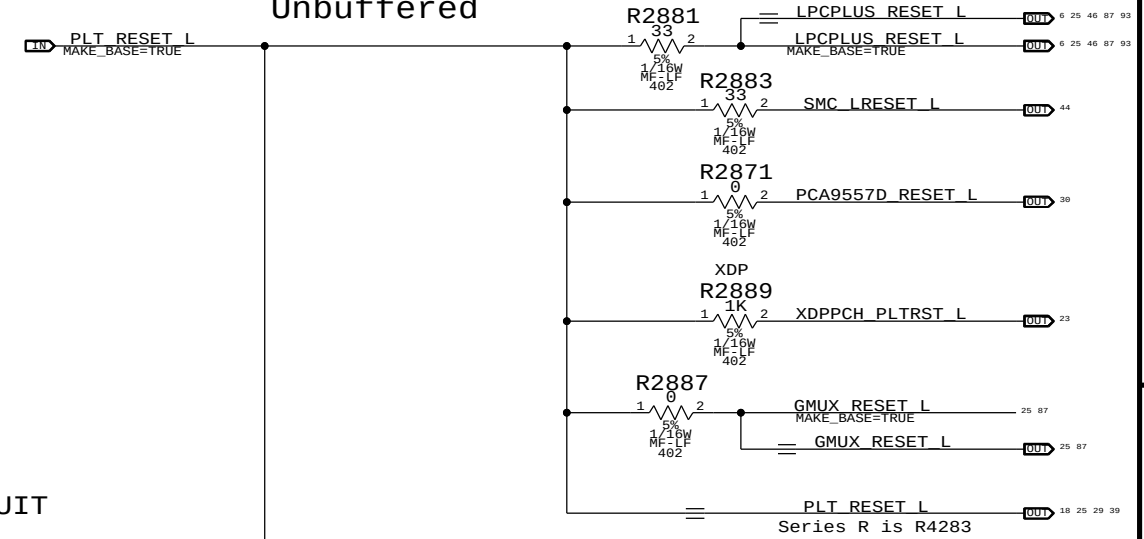


Ethernet WAKE# Isolation

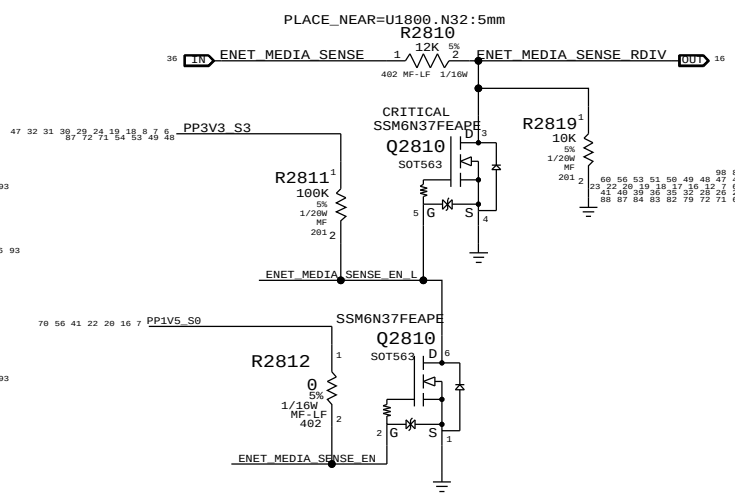


Platform Reset Connections

Unbuffered

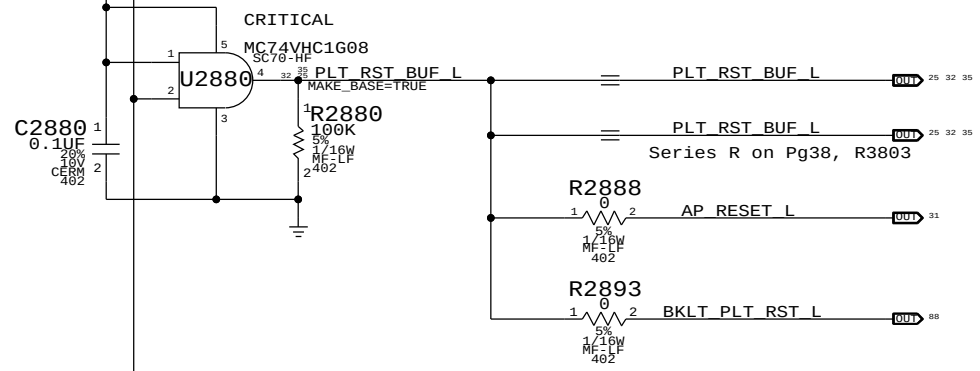


ENET_MEDIA_SENSE ISOLATION CIRCUIT



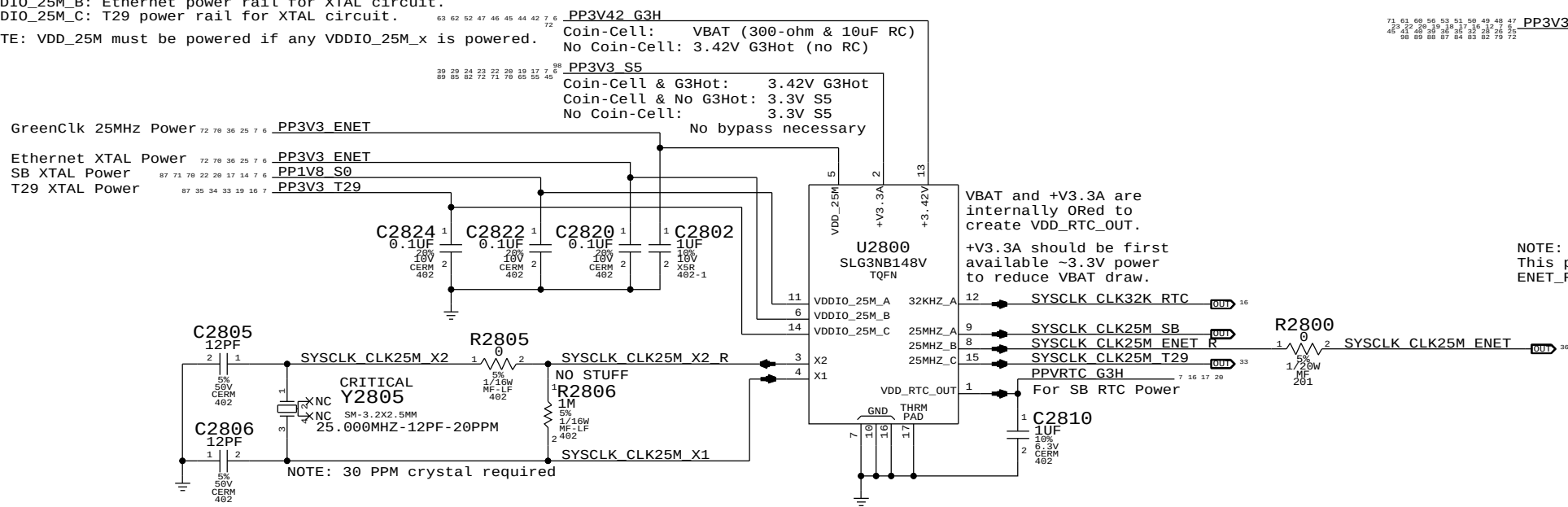
Buffered

Note: Based on K91/K92 layout, ENET,AP and BKLT are moved to Buffered reset.

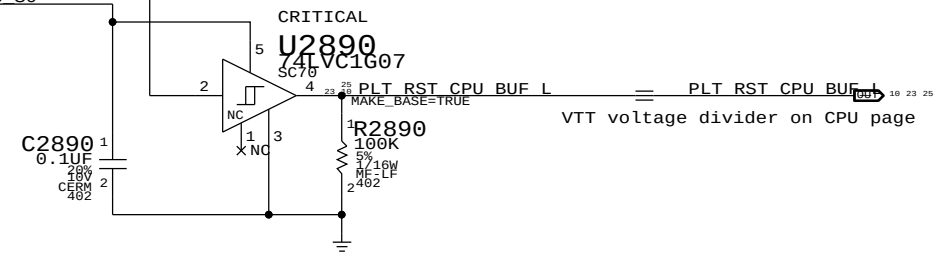


System RTC Power Source & 32kHz / 25MHz Clock Generator

VDDIO_25M_A: SB power rail for XTAL circuit.
VDDIO_25M_B: Ethernet power rail for XTAL circuit.
VDDIO_25M_C: T29 power rail for XTAL circuit.
NOTE: VDD_25M must be powered if any VDDIO_25M_x is powered.



Buffered CPU reset



NOTE:
This page is different for K92.
ENET_RESET_L hooked up differently on both the projects.

SYNC MASTER=K92 MLB		SYNC DATE=07/06/2016	
PAGE TITLE		Chipset Support	
Apple Inc.		DRAWING NUMBER	SIZE D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		REVISION	BRANCH
		PAGE	28 OF 132
		SHEET	25 OF 101



Page Notes

Power aliases required by this page:
- =PP1V5_S0_MEM_A
- =PP1V5_S3_MEM_A
- =PP0V75_S0_MEM_VTT_A
- =PPSPD_S0_MEM_A (2.5 - 3.3V)

Signal aliases required by this page:
- =I2C_S0D1MMA_SCL
- =I2C_S0D1MMA_SDA

BOM options provided by this page:
(NONE)

DDR3 DECOUPLING AND GND RETURN CAPS (SPACE EVENLY AT CONNECTOR)

D

D

C

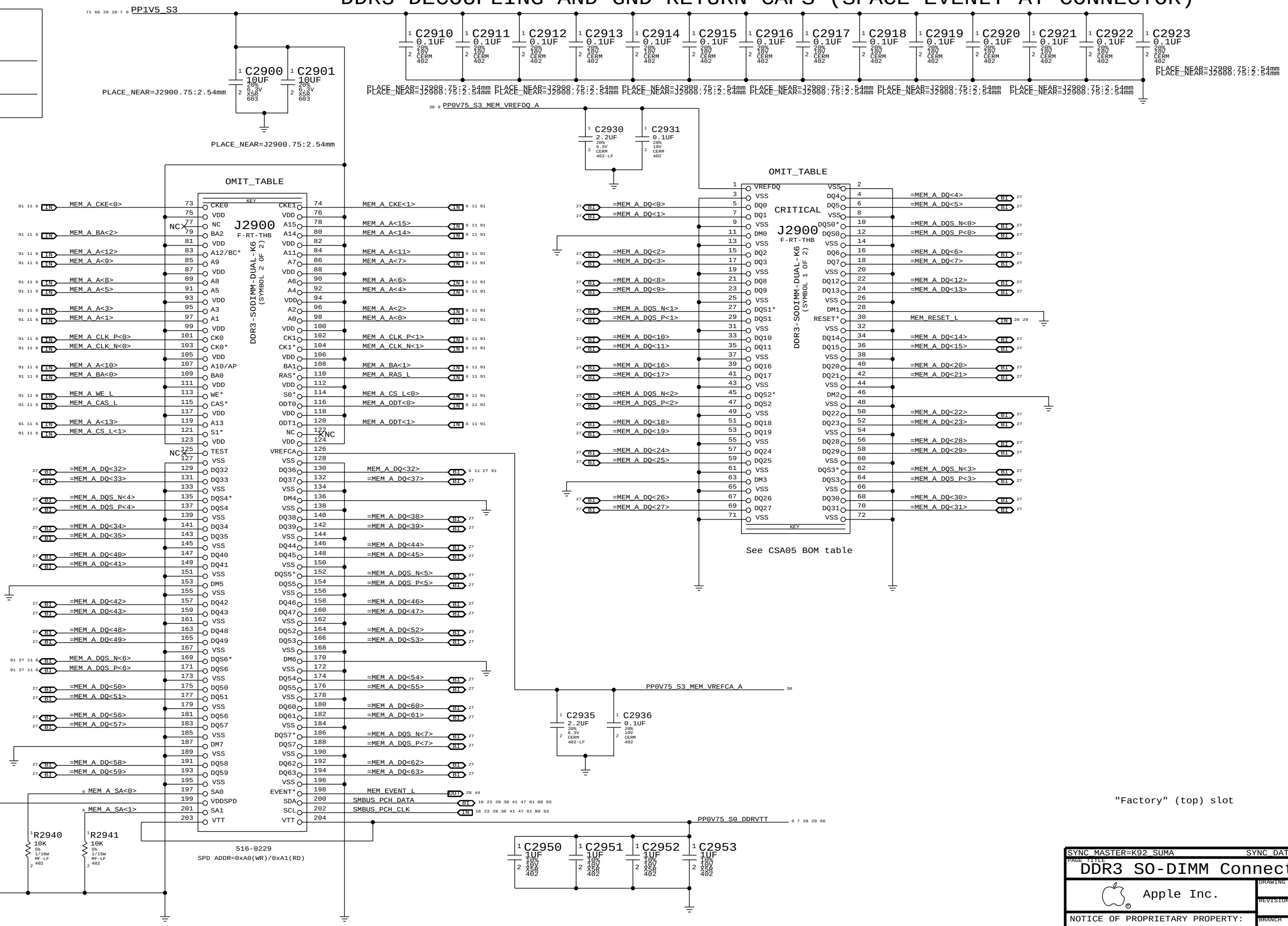
C

B

B

A

A



"Factory" (top) slot

SYNC MASTER=K92 SUMA		SYNC DATE=06/23/2016	
DDR3 S0-DIMM Connector A			
Apple Inc.		DRAWING NUMBER	SIZE D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		REVISION	BRANCH
		PAGE 29 OF 132	SHEET 26 OF 101



8	7	6	5	4	3	2	1
D	CPU CHANNEL A DQS 0 -> DIMM A DQS 0		CPU CHANNEL B DQS 0 -> DIMM B DQS 0				
	91 11 6 MEM A DQS N<0>	=MEM A DQS N<0>	26	91 11 6 MEM B DQS N<0>	=MEM B DQS N<0>	28	
	91 11 6 MEM A DQS P<0>	=MEM A DQS P<0>	26	91 11 6 MEM B DQS P<0>	=MEM B DQS P<0>	28	
	91 11 6 MEM A DQ<7>	=MEM A DQ<3>	26	91 11 6 MEM B DQ<7>	=MEM B DQ<6>	28	
	91 11 6 MEM A DQ<6>	=MEM A DQ<6>	26	91 11 6 MEM B DQ<6>	=MEM B DQ<3>	28	
	91 11 6 MEM A DQ<5>	=MEM A DQ<5>	26	91 11 6 MEM B DQ<5>	=MEM B DQ<5>	28	
	91 11 6 MEM A DQ<4>	=MEM A DQ<4>	26	91 11 6 MEM B DQ<4>	=MEM B DQ<4>	28	
	91 11 6 MEM A DQ<3>	=MEM A DQ<7>	26	91 11 6 MEM B DQ<3>	=MEM B DQ<1>	28	
	91 11 6 MEM A DQ<2>	=MEM A DQ<0>	26	91 11 6 MEM B DQ<2>	=MEM B DQ<7>	28	
C	CPU CHANNEL A DQS 1 -> DIMM A DQS 1		CPU CHANNEL B DQS 1 -> DIMM B DQS 1				
	91 11 6 MEM A DQS N<1>	=MEM A DQS N<1>	26	91 11 6 MEM B DQS N<1>	=MEM B DQS N<1>	28	
	91 11 6 MEM A DQS P<1>	=MEM A DQS P<1>	26	91 11 6 MEM B DQS P<1>	=MEM B DQS P<1>	28	
	91 11 6 MEM A DQ<15>	=MEM A DQ<15>	26	91 11 6 MEM B DQ<15>	=MEM B DQ<15>	28	
	91 11 6 MEM A DQ<14>	=MEM A DQ<14>	26	91 11 6 MEM B DQ<14>	=MEM B DQ<14>	28	
	91 11 6 MEM A DQ<13>	=MEM A DQ<12>	26	91 11 6 MEM B DQ<13>	=MEM B DQ<13>	28	
	91 11 6 MEM A DQ<12>	=MEM A DQ<13>	26	91 11 6 MEM B DQ<12>	=MEM B DQ<12>	28	
	91 11 6 MEM A DQ<11>	=MEM A DQ<10>	26	91 11 6 MEM B DQ<11>	=MEM B DQ<11>	28	
	91 11 6 MEM A DQ<10>	=MEM A DQ<11>	26	91 11 6 MEM B DQ<10>	=MEM B DQ<10>	28	
B	CPU CHANNEL A DQS 2 -> DIMM A DQS 2		CPU CHANNEL B DQS 2 -> DIMM B DQS 2				
	91 11 6 MEM A DQS N<2>	=MEM A DQS N<2>	26	91 11 6 MEM B DQS N<2>	=MEM B DQS N<2>	28	
	91 11 6 MEM A DQS P<2>	=MEM A DQS P<2>	26	91 11 6 MEM B DQS P<2>	=MEM B DQS P<2>	28	
	91 11 6 MEM A DQ<23>	=MEM A DQ<23>	26	91 11 6 MEM B DQ<23>	=MEM B DQ<23>	28	
	91 11 6 MEM A DQ<22>	=MEM A DQ<22>	26	91 11 6 MEM B DQ<22>	=MEM B DQ<22>	28	
	91 11 6 MEM A DQ<21>	=MEM A DQ<17>	26	91 11 6 MEM B DQ<21>	=MEM B DQ<21>	28	
	91 11 6 MEM A DQ<20>	=MEM A DQ<20>	26	91 11 6 MEM B DQ<20>	=MEM B DQ<20>	28	
	91 11 6 MEM A DQ<19>	=MEM A DQ<19>	26	91 11 6 MEM B DQ<19>	=MEM B DQ<19>	28	
	91 11 6 MEM A DQ<18>	=MEM A DQ<18>	26	91 11 6 MEM B DQ<18>	=MEM B DQ<18>	28	
A	CPU CHANNEL A DQS 3 -> DIMM A DQS 3		CPU CHANNEL B DQS 3 -> DIMM B DQS 3				
	91 11 6 MEM A DQS N<3>	=MEM A DQS N<3>	26	91 11 6 MEM B DQS N<3>	=MEM B DQS N<3>	28	
	91 11 6 MEM A DQS P<3>	=MEM A DQS P<3>	26	91 11 6 MEM B DQS P<3>	=MEM B DQS P<3>	28	
	91 11 6 MEM A DQ<31>	=MEM A DQ<31>	26	91 11 6 MEM B DQ<31>	=MEM B DQ<31>	28	
	91 11 6 MEM A DQ<30>	=MEM A DQ<30>	26	91 11 6 MEM B DQ<30>	=MEM B DQ<30>	28	
	91 11 6 MEM A DQ<29>	=MEM A DQ<29>	26	91 11 6 MEM B DQ<29>	=MEM B DQ<29>	28	
	91 11 6 MEM A DQ<28>	=MEM A DQ<28>	26	91 11 6 MEM B DQ<28>	=MEM B DQ<28>	28	
	91 11 6 MEM A DQ<27>	=MEM A DQ<27>	26	91 11 6 MEM B DQ<27>	=MEM B DQ<27>	28	
	91 11 6 MEM A DQ<26>	=MEM A DQ<26>	26	91 11 6 MEM B DQ<26>	=MEM B DQ<26>	28	
	CPU CHANNEL A DQS 4 -> DIMM A DQS 4		CPU CHANNEL B DQS 4 -> DIMM B DQS 4				
	91 11 6 MEM A DQS N<4>	=MEM A DQS N<4>	26	91 11 6 MEM B DQS N<4>	=MEM B DQS N<4>	28	
	91 11 6 MEM A DQS P<4>	=MEM A DQS P<4>	26	91 11 6 MEM B DQS P<4>	=MEM B DQS P<4>	28	
	91 11 6 MEM A DQ<39>	=MEM A DQ<38>	26	91 11 6 MEM B DQ<39>	=MEM B DQ<39>	28	
	91 11 6 MEM A DQ<38>	=MEM A DQ<37>	26	91 11 6 MEM B DQ<38>	=MEM B DQ<38>	28	
	91 11 6 MEM A DQ<37>	=MEM A DQ<39>	26	91 11 6 MEM B DQ<37>	=MEM B DQ<37>	28	
	91 11 6 MEM A DQ<36>	=MEM A DQ<33>	26	91 11 6 MEM B DQ<36>	=MEM B DQ<36>	28	
	91 11 6 MEM A DQ<35>	=MEM A DQ<35>	26	91 11 6 MEM B DQ<35>	=MEM B DQ<35>	28	
	91 11 6 MEM A DQ<34>	=MEM A DQ<34>	26	91 11 6 MEM B DQ<34>	=MEM B DQ<34>	28	
	CPU CHANNEL A DQS 5 -> DIMM A DQS 5		CPU CHANNEL B DQS 5 -> DIMM B DQS 5				
	91 11 6 MEM A DQS N<5>	=MEM A DQS N<5>	26	91 11 6 MEM B DQS N<5>	=MEM B DQS N<5>	28	
	91 11 6 MEM A DQS P<5>	=MEM A DQS P<5>	26	91 11 6 MEM B DQS P<5>	=MEM B DQS P<5>	28	
	91 11 6 MEM A DQ<47>	=MEM A DQ<47>	26	91 11 6 MEM B DQ<47>	=MEM B DQ<47>	28	
	91 11 6 MEM A DQ<46>	=MEM A DQ<41>	26	91 11 6 MEM B DQ<46>	=MEM B DQ<46>	28	
	91 11 6 MEM A DQ<45>	=MEM A DQ<43>	26	91 11 6 MEM B DQ<45>	=MEM B DQ<45>	28	
	91 11 6 MEM A DQ<44>	=MEM A DQ<44>	26	91 11 6 MEM B DQ<44>	=MEM B DQ<44>	28	
	91 11 6 MEM A DQ<43>	=MEM A DQ<40>	26	91 11 6 MEM B DQ<43>	=MEM B DQ<43>	28	
	91 11 6 MEM A DQ<42>	=MEM A DQ<46>	26	91 11 6 MEM B DQ<42>	=MEM B DQ<42>	28	
	CPU CHANNEL A DQS 6 -> DIMM A DQS 6		CPU CHANNEL B DQS 6 -> DIMM B DQS 6				
	91 27 26 11 6 MEM A DQS N<6>	=MEM A DQS N<6>	6 11 26 27 91	27 11 6 MEM B DQS N<6>	=MEM B DQS N<6>	6 11 27 28 91	
	91 27 26 11 6 MEM A DQS P<6>	=MEM A DQS P<6>	6 11 26 27 91	27 11 6 MEM B DQS P<6>	=MEM B DQS P<6>	6 11 27 28 91	
	91 11 6 MEM A DQ<55>	=MEM A DQ<49>	26	91 11 6 MEM B DQ<55>	=MEM B DQ<55>	28	
	91 11 6 MEM A DQ<54>	=MEM A DQ<54>	26	91 11 6 MEM B DQ<54>	=MEM B DQ<54>	28	
	91 11 6 MEM A DQ<53>	=MEM A DQ<55>	26	91 11 6 MEM B DQ<53>	=MEM B DQ<53>	28	
	91 11 6 MEM A DQ<52>	=MEM A DQ<52>	26	91 11 6 MEM B DQ<52>	=MEM B DQ<52>	28	
	91 11 6 MEM A DQ<51>	=MEM A DQ<51>	26	91 11 6 MEM B DQ<51>	=MEM B DQ<51>	28	
	91 11 6 MEM A DQ<50>	=MEM A DQ<50>	26	91 11 6 MEM B DQ<50>	=MEM B DQ<50>	28	
	CPU CHANNEL A DQS 7 -> DIMM A DQS 7		CPU CHANNEL B DQS 7 -> DIMM B DQS 7				
	91 11 6 MEM A DQS N<7>	=MEM A DQS N<7>	26	91 11 6 MEM B DQS N<7>	=MEM B DQS N<7>	28	
	91 11 6 MEM A DQS P<7>	=MEM A DQS P<7>	26	91 11 6 MEM B DQS P<7>	=MEM B DQS P<7>	28	
	91 11 6 MEM A DQ<63>	=MEM A DQ<59>	26	91 11 6 MEM B DQ<63>	=MEM B DQ<63>	28	
	91 11 6 MEM A DQ<62>	=MEM A DQ<58>	26	91 11 6 MEM B DQ<62>	=MEM B DQ<62>	28	
	91 11 6 MEM A DQ<61>	=MEM A DQ<56>	26	91 11 6 MEM B DQ<61>	=MEM B DQ<61>	28	
	91 11 6 MEM A DQ<60>	=MEM A DQ<61>	26	91 11 6 MEM B DQ<60>	=MEM B DQ<60>	28	
	91 11 6 MEM A DQ<59>	=MEM A DQ<63>	26	91 11 6 MEM B DQ<59>	=MEM B DQ<59>	28	
	91 11 6 MEM A DQ<58>	=MEM A DQ<62>	26	91 11 6 MEM B DQ<58>	=MEM B DQ<58>	28	

SYNC MASTER=K92 SUMA		SYNC DATE=05/10/2010	
PAGE TITLE			
DDR3 Byte/Bit Swaps			
Apple Inc.		DRAWING NUMBER	SIZE
		REVISION	D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		BRANCH	
		PAGE	30 OF 132
		SHEET	27 OF 101



Page Notes

Power aliases required by this page:

- =PP1V5_S0_MEM_B
- =PP1V5_S3_MEM_B
- =PP0V75_S0_MEM_VTT_B
- =PPSPD_S0_MEM_B (2.5 - 3.3V)

Signal aliases required by this page:

- =I2C_S0DIMM_SCL
- =I2C_S0DIMM_SDA

BOM options provided by this page:

(NONE)

DDR3 DECOUPLING AND GND RETURN CAPS (SPACE EVENLY AT CONNECTOR)

71 66 29 26 7 6 PP1V5_S3

PLACE_NEAR=J3100.75:2.54mm

PLACE_NEAR=J3100.75:2.54mm

PLACE_NEAR=J3100.75:2.54mm

PLACE_NEAR=J3100.75:2.54mm

PLACE_NEAR=J3100.75:2.54mm

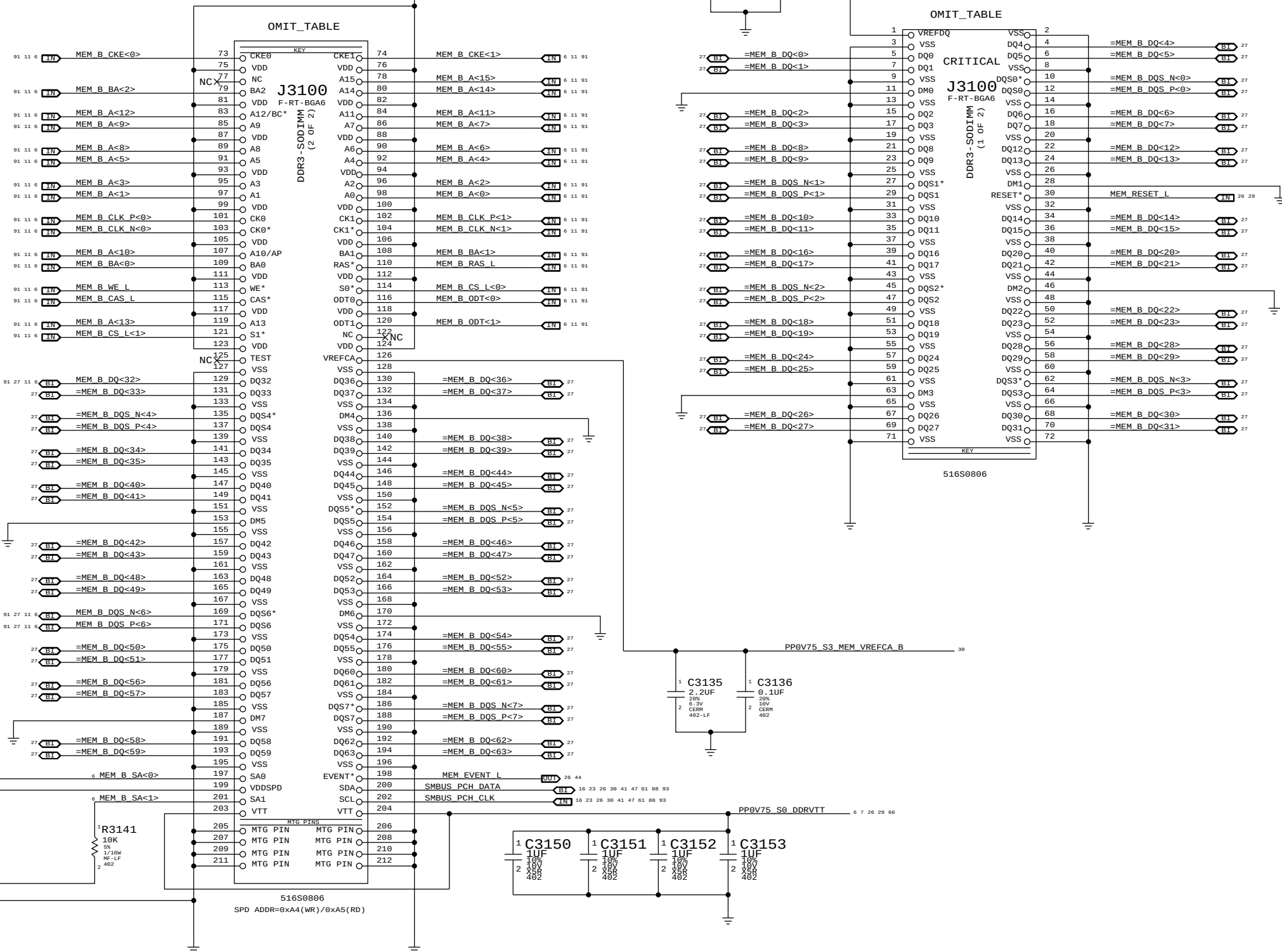
PLACE_NEAR=J3100.75:2.54mm

PLACE_NEAR=J3100.75:2.54mm

PLACE_NEAR=J3100.75:2.54mm

PLACE_NEAR=J3100.75:2.54mm

PLACE_NEAR=J3100.75:2.54mm



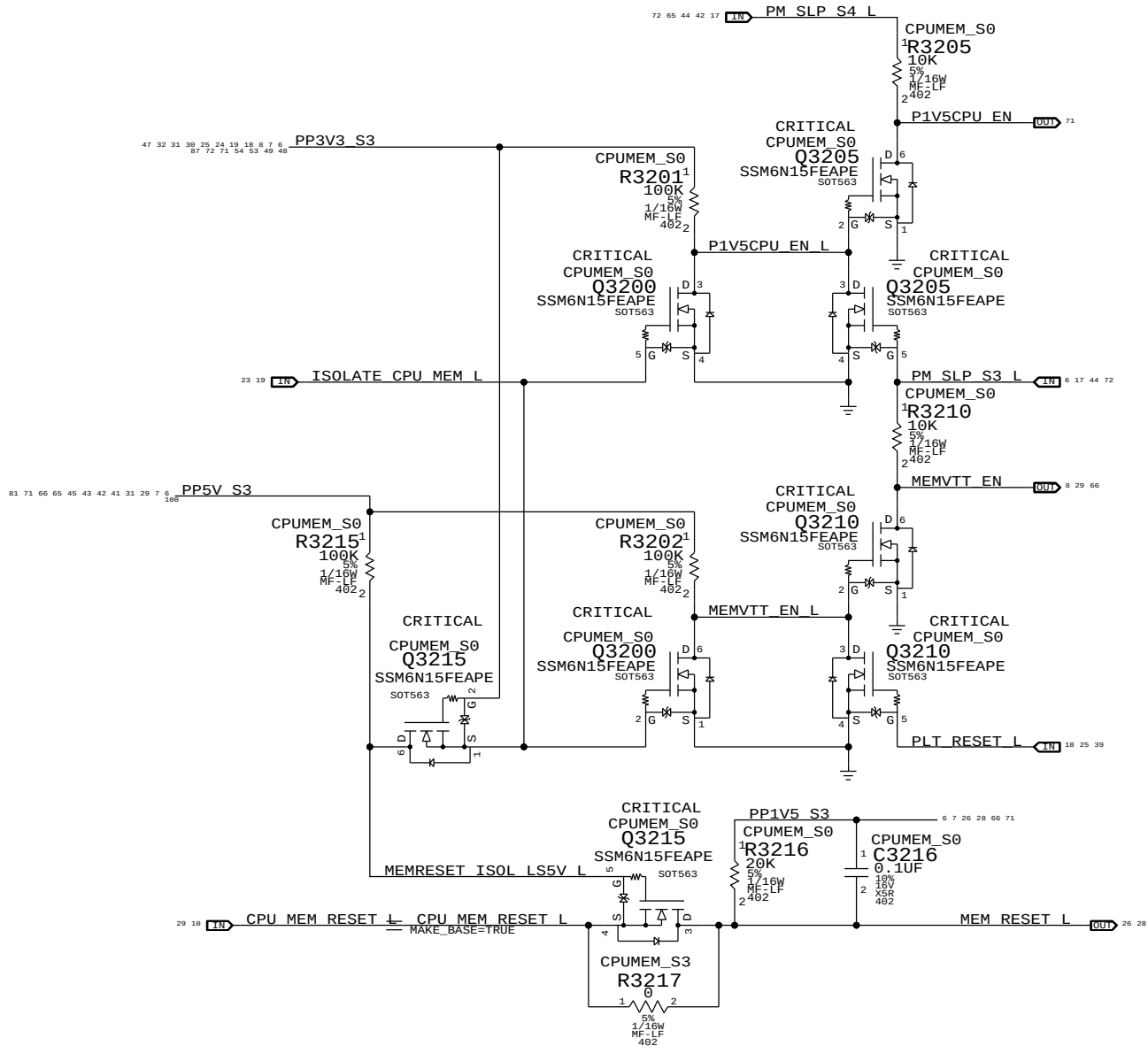
"Expansion" (bottom) slot

SYNC MASTER=K92 SUMA		SYNC DATE=06/23/2016	
DDR3 S0-DIMM Connector B			
Apple Inc.		DRAWING NUMBER	SIZE D
NOTICE OF PROPRIETARY PROPERTY:		REVISION	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:		BRANCH	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		PAGE	31 OF 132
II NOT TO REPRODUCE OR COPY IT		SHEET	28 OF 101
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			

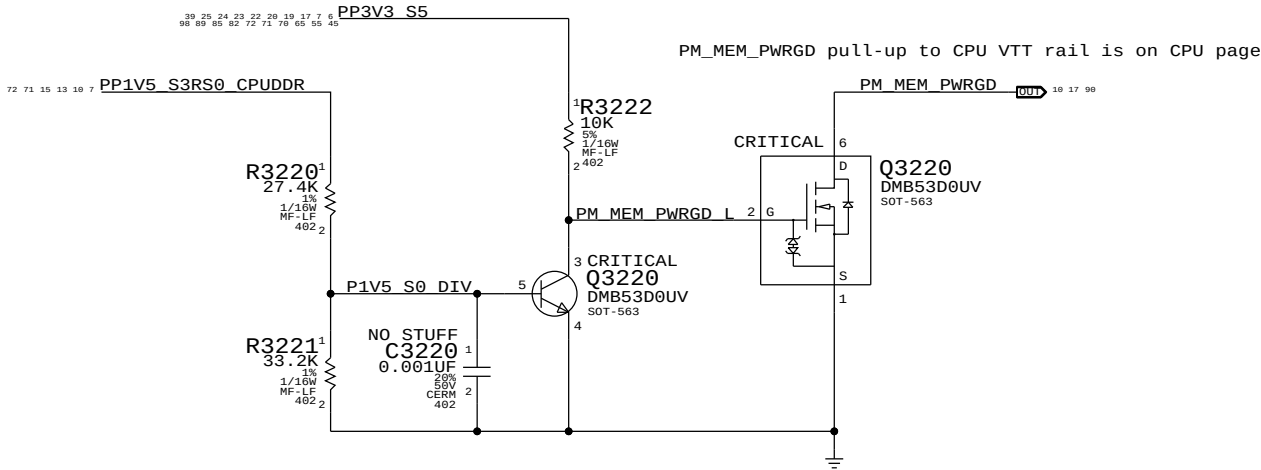
The circuit below handles CPU and VTT power during S0->S3->S0 transitions, as well as isolating the CPU's SM_DRAMRST# output from the S0-DIMMs when necessary.

ISOLATE_CPU_MEM_L GPIO state during S3->S0 transitions determines behavior of signals.
 WHEN HIGH: CPU 1.5V remains powered in S3, VTT follows S0 rails, MEM_RESET_L not isolated.
 WHEN LOW: CPU 1.5V follows S0 rails, VTT ensures clean CKE transition, MEM_RESET_L isolated.

$P1V5CPU_EN = (ISOLATE_CPU_MEM_L + PM_SLP_S3_L) * PM_SLP_S4_L$
 $MEMVTT_EN = (ISOLATE_CPU_MEM_L + PLT_RST_L) * PM_SLP_S3_L$
 $MEM_RESET_L = !ISOLATE_CPU_MEM_L + CPU_MEM_RESET_L$

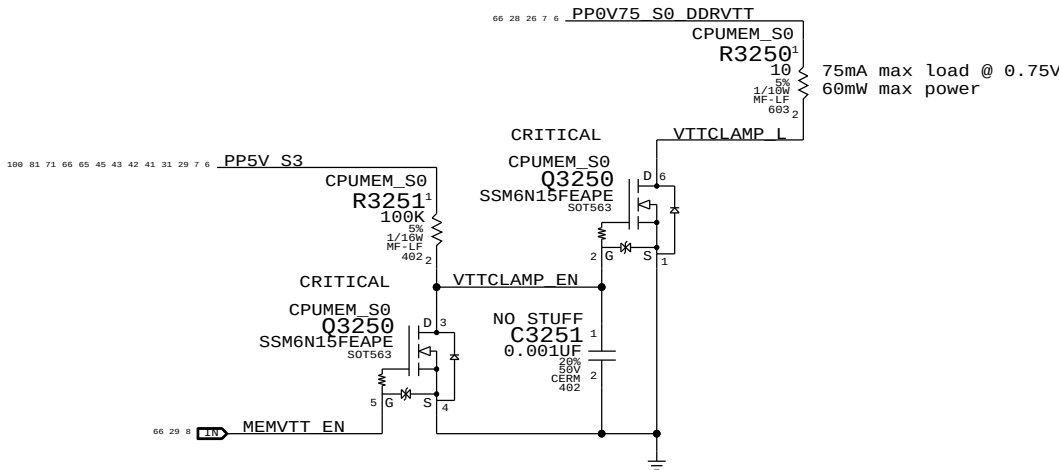


1V5 S0 "PGOOD" for CPU



MEMVTT Clamp

Ensures CKE signals are held low in S3



Step	ISOLATE_CPU_MEM_L	PLT_RESET_L	PM_SLP_S3_L	PM_SLP_S4_L	CPU_MEM_RESET_L	MEM_RESET_L	MEMVTT_EN	P1V5CPU_EN
S0	0	1	1	1	1	CPU_MEM_RESET_L	1	1
to	1	0	1	1	1	1	1	1
2	0	0	1	1	1	1	0	1
3	0	0	0	1	X	1	0	0
4	0	0	1	1	X	1	0	1
5	0	1	1	1	0 (*)	1	1	1
6	0	1	1	1	1	1	1	1
to	7	1	1	1	1	CPU_MEM_RESET_L	1	1
S0								

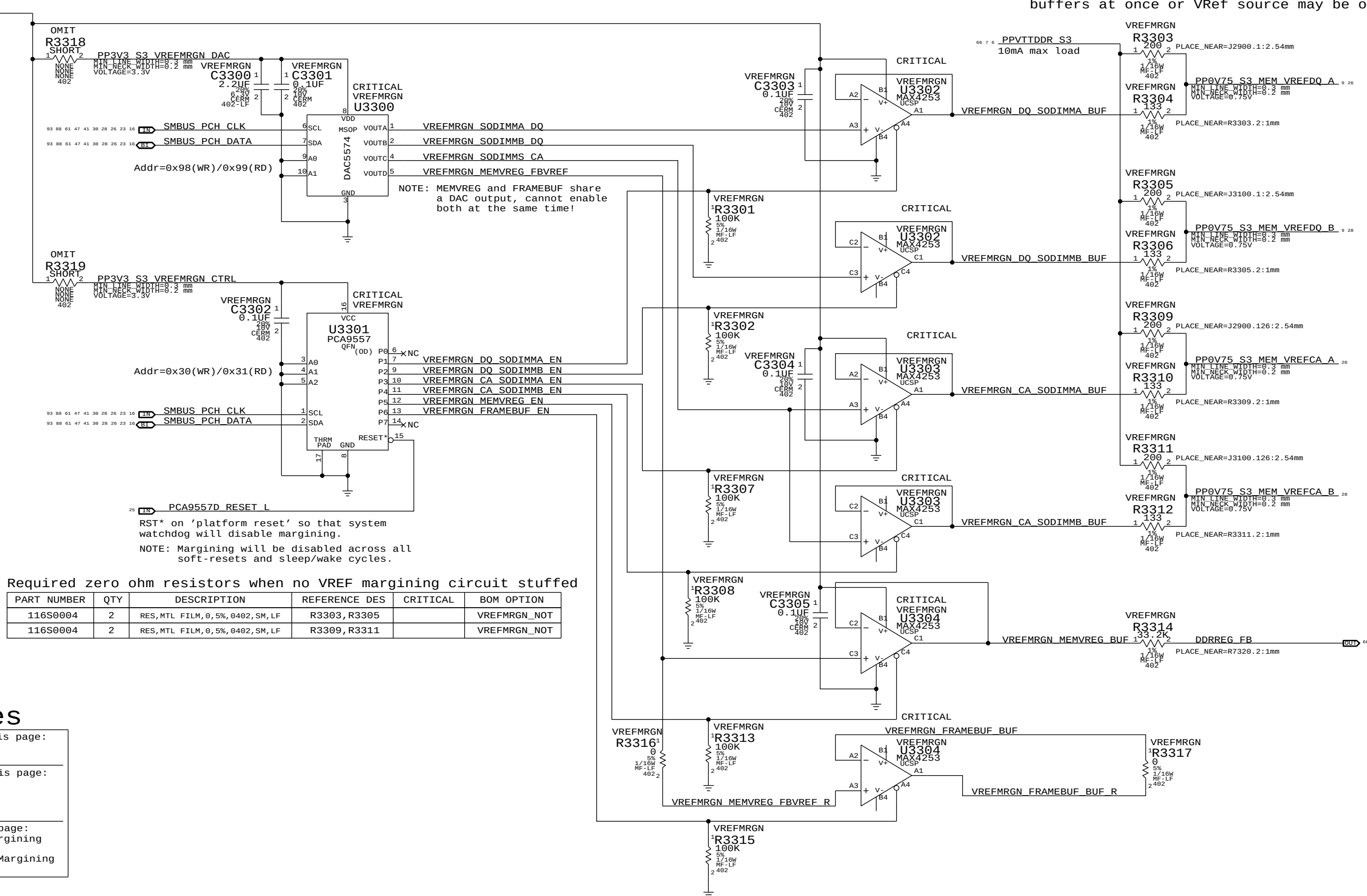
(*) CPU_MEM_RESET_L asserts due to loss of PM_MEM_PWRGD, must wait for software to clear before deasserting ISOLATE_CPU_MEM_L GPIO.

NOTE: In the event of a S3->S5 transition ISOLATE_CPU_MEM_L will still be asserted on next S5->S0 transition. Rails will power-up as if from S3, but MEM_RESET_L will not properly assert. Software must deassert ISOLATE_CPU_MEM_L and then generate a valid reset cycle on CPU_MEM_RESET_L.

PAGE TITLE		SYNC DATE=04/27/2016	
CPU Memory S3 Support			
Apple Inc.		DRAWING NUMBER	SIZE
NOTICE OF PROPRIETARY PROPERTY:		REVISION	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:		BRANCH	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		PAGE	
II NOT TO REPRODUCE OR COPY IT		32 OF 132	
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART		SHEET	
IV ALL RIGHTS RESERVED		29 OF 101	



NOTE: Must not enable more than two S0-DIMM margining buffers at once or VRef source may be overloaded.



Required zero ohm resistors when no VREF margining circuit stuffed

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
116S0004	2	RES, MTL FILM, 0, 5%, 0402, SM, LF	R3303, R3305		VREFMRGN_NOT
116S0004	2	RES, MTL FILM, 0, 5%, 0402, SM, LF	R3309, R3311		VREFMRGN_NOT

Page Notes

Power aliases required by this page:
- =PP3V3_S3_VREFMRGN
- =PPVTT_S3_DDR_BUF

Signal aliases required by this page:
- =I2C_VREFDACS_SCL
- =I2C_VREFDACS_SDA
- =I2C_PCA9557D_SCL
- =I2C_PCA9557D_SDA

BOM options provided by this page:
VREFMRGN - Stuffs VREF Margining Circuitry.
VREFMRGN_NOT - Bypasses VREF Margining Circuitry.

	MEM A VREF DQ	MEM B VREF DQ	MEM A VREF CA	MEM B VREF CA	MEM VREG	GPU Frame Buffer (1.8V, 70% Vref)
DAC Channel:	A	B	C	C	D	D
PCA9557D Pin:	1	2	3	4	5	6
Nominal value		0.75V (DAC: 0x3A)			1.5V (DAC: 0x3A)	1.267V (DAC: 0x8B)
Margined target:		0.300V - 1.200V (+/- 450mV)			1.000V - 2.000V (+/- 500mV)	1.056V - 1.442V (+/- 180mV)
DAC range:		0.000V - 1.501V (0x00 - 0x74)			0.000V - 3.000V (0x00 - 0x74)	0.000V - 3.300V (0x00 - 0xFF)
VRef current:		+3.4mA - -3.4mA (- = sourced)			+61uA - -61uA (- = sourced)	+6.0mA - -5.0mA (- = sourced)
DAC step size:		7.69mV / step @ output			8.59mV / step @ output	1.51mV / step @ output

SYNC MASTER=K18 MLB

SYNC DATE=04/27/2016

PAGE TITLE

FSB/DDR3/FRAMEBUF Vref Margining

Apple Inc.

DRAWING NUMBER

SIZE

D

NOTICE OF PROPRIETARY PROPERTY:

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:

I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE

II NOT TO REPRODUCE OR COPY IT

III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART

IV ALL RIGHTS RESERVED

REVISION

BRANCH

PAGE

33 OF 132

SHEET

30 OF 101



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
117S0002	4	RES, 600M, 0201	L3470, L3471, L3473, L3474		

3V S3 WLAN FET	
MOSFET	TPCP8102
CHANNEL	P-TYPE
RDS(ON)	20-30 MOHM @2.5V
LOADING	1 A (EDP)

AIRPORT

BLUETOOTH

Supervisor & CLKFREG # Isolation
Delay = 60 ms +/- 20%

518S0816
CRITICAL
J3402
CCR20-6K710S

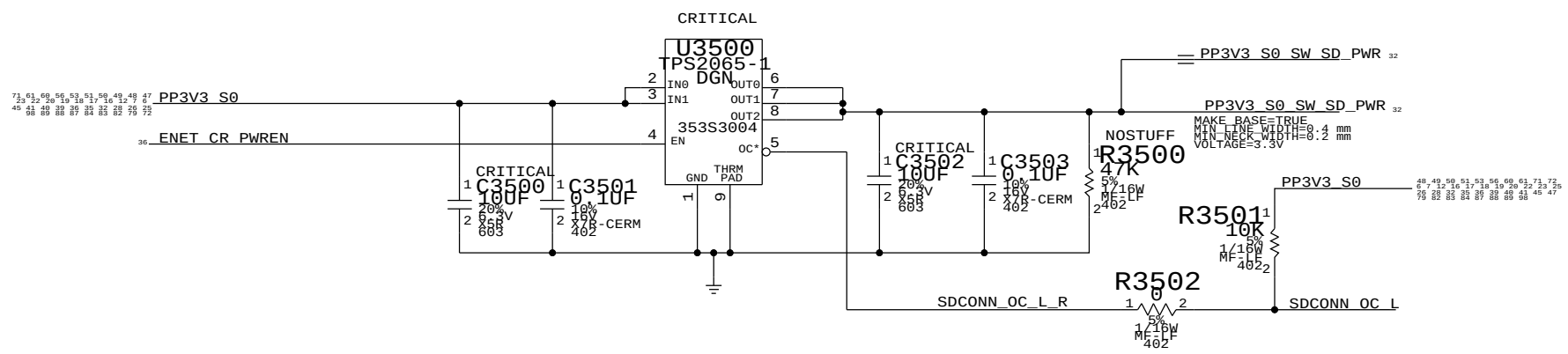
ALS
CAMERA

SYNC MASTER=K91 MARY		SYNC DATE=10/08/2016	
PAGE TITLE		X19/ALS/CAMERA CONNECTOR	
Apple Inc.		DRAWING NUMBER	SIZE D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		REVISION	PAGE 34 OF 132
		BRANCH	SHEET 31 OF 101

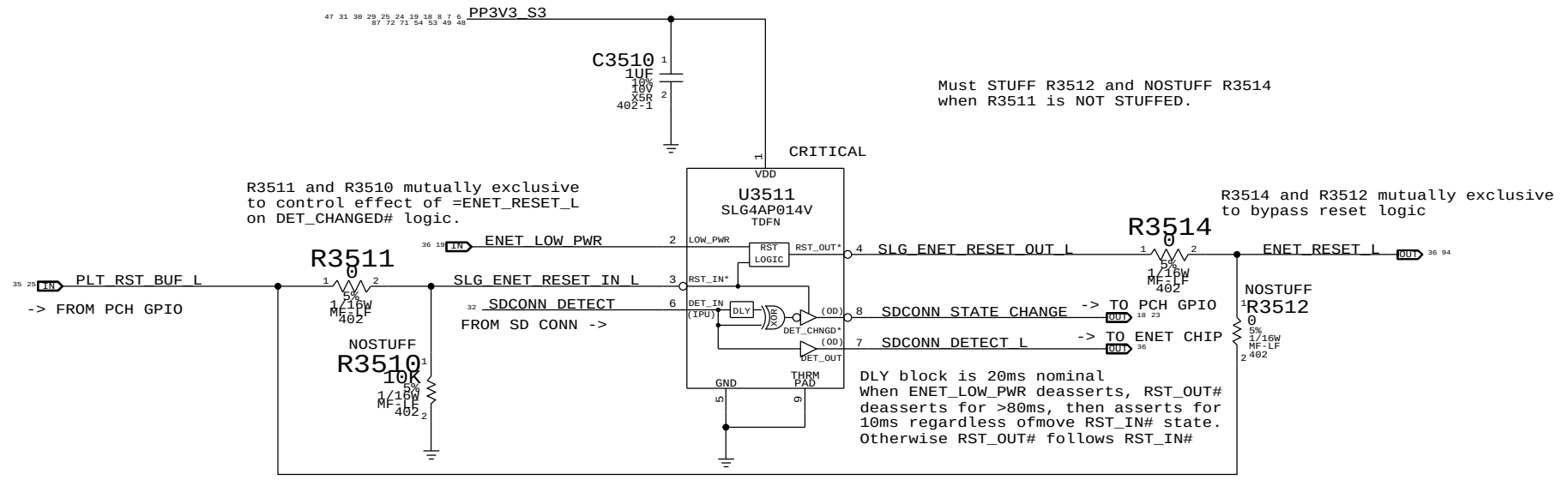


SD CARD 3.3V OVERCURRENT PROTECTION CHIP WITH ACTIVE LOAD DISCHARGE

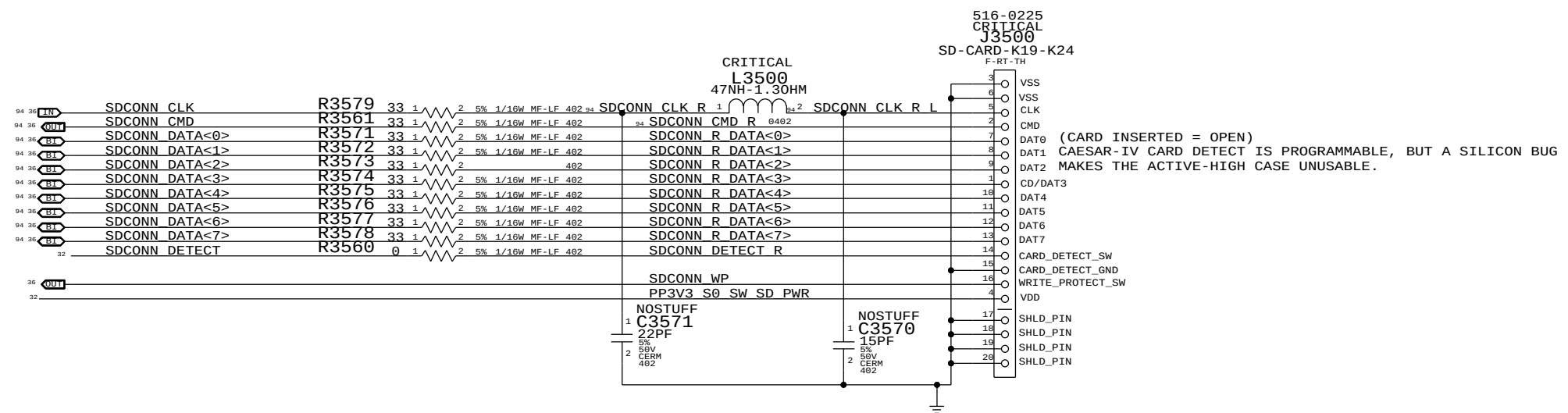
TPS2065-1 (1.0A limit) has active load discharge so R4810 is NOSTUFF.



SDCONN DETECT DEBOUNCE, INVERSION, AND DETECT-CHANGED PCH GPIO LATCH CIRCUIT



SD CARD CONNECTOR



SD READER CONNECTOR

Apple Inc.

NOTICE OF PROPRIETARY PROPERTY:
THE INFORMATION CONTAINED HEREIN IS THE
PROPRIETARY PROPERTY OF APPLE COMPUTER, INC.
THE POSSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

DRAWING NUMBER

REVISION

BRANCH

PAGE

SHEET

35 OF 132

32 OF 101

SYNC_MASTER=K91-ERIC SYNC DATE=10/08/2010

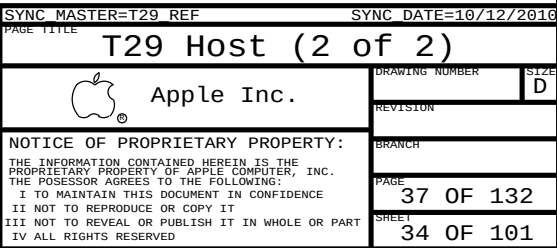
SD READER CONNECTOR

Apple Inc.

NOTICE OF PROPRIETARY PROPERTY:
THE INFORMATION CONTAINED HEREIN IS THE
PROPRIETARY PROPERTY OF APPLE COMPUTER, INC.
THE POSSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

1/16W MF-LF





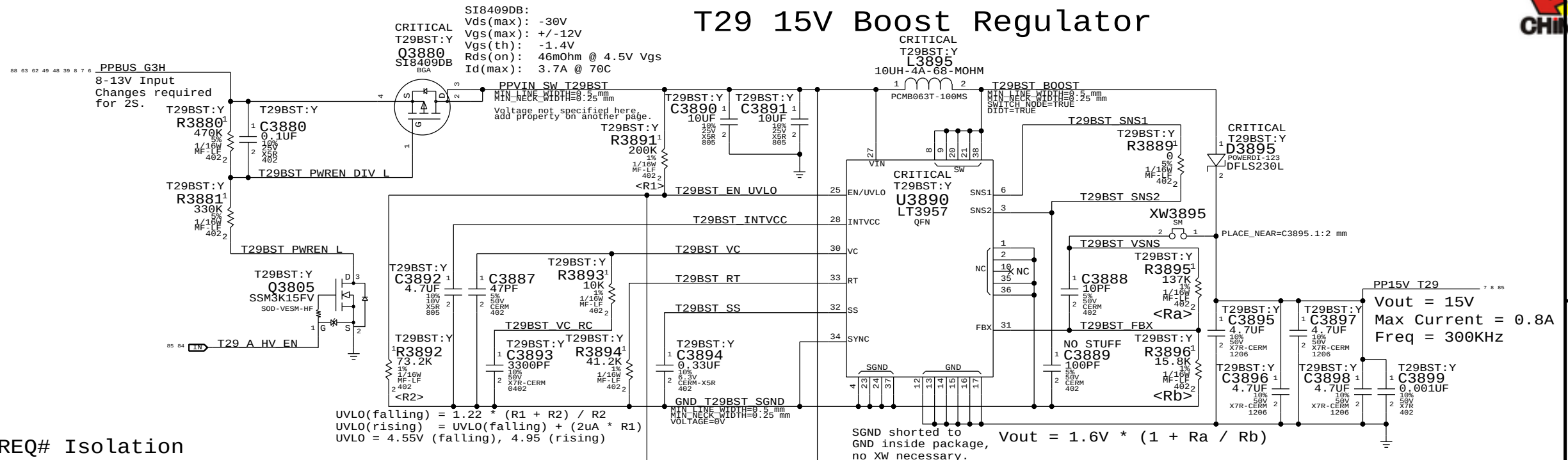
Current numbers from Vendor slide (<REDACTED> power measure 1.ppt), emailed 6/21/2010, TDP @ 90C.



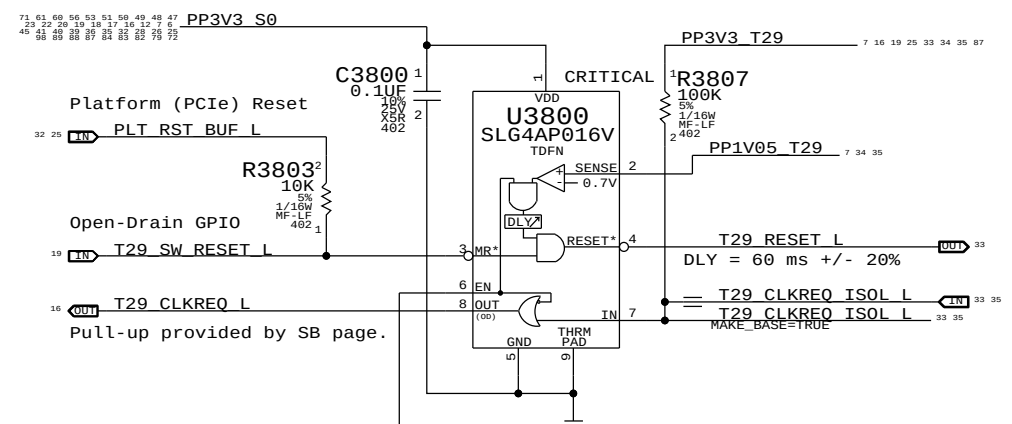
Page Notes

Power aliases required by this page:
- =PPVIN_SW_T29BST (8-13V Boost Input)
- =PP18V_T29_REG (18V Boost Output)
- =PP3V3_T29_P3V3T29FET (3.3V FET Input)
- =PP3V3_T29_FET (3.3V FET Output)
- =PP3V3_S0_T29PWRTL (1.05V FET Input)
- =PP1V05_T29_P1V05T29FET (1.05V FET Input)
- =PP1V05_T29_FET (1.05V FET Output)
Signal aliases required by this page:
- =T29_CLKREQ_L
- =T29_RESET_L
BOM options provided by this page:
T29BST:Y - Stuffs 18V boost circuitry.

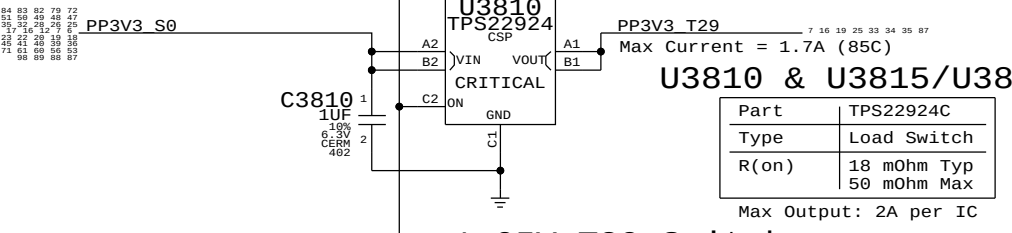
T29 15V Boost Regulator



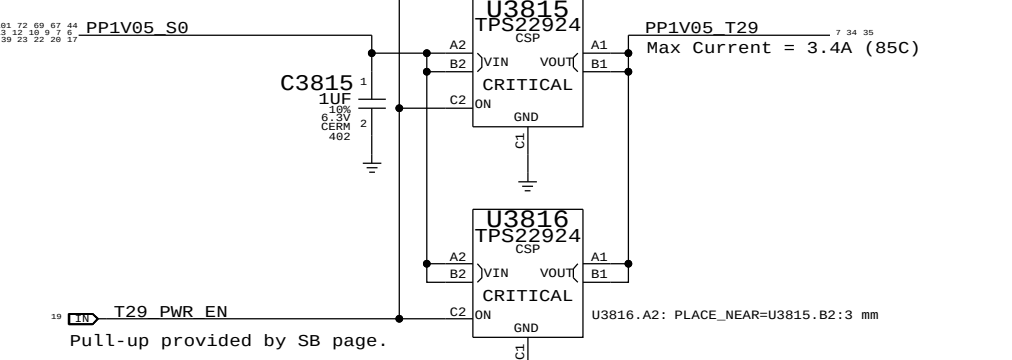
Supervisor & CLKREQ# Isolation



3.3V T29 Switch



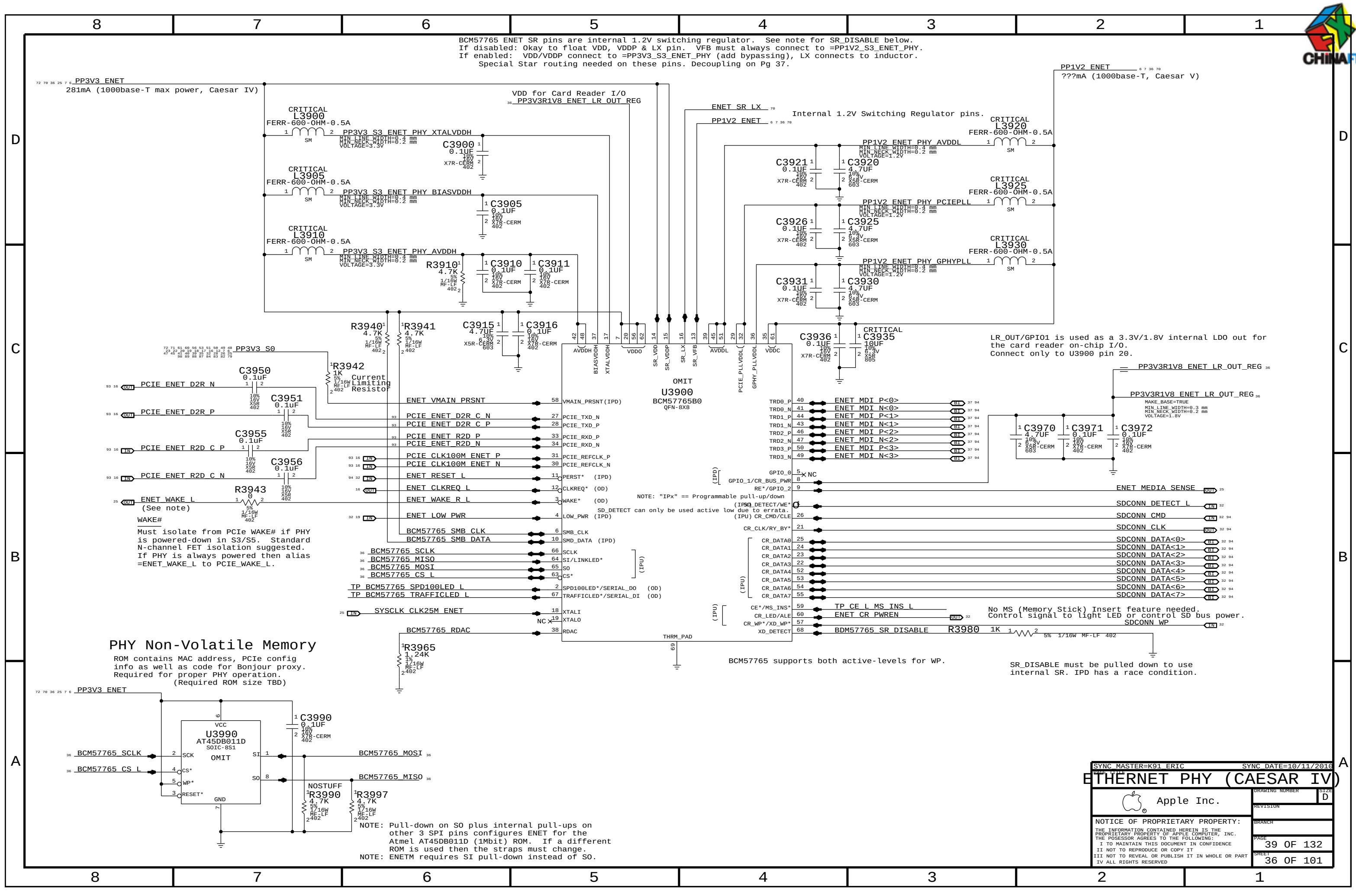
1.05V T29 Switch



SYNC MASTER=T29 REF		SYNC DATE=10/12/2016	
PAGE TITLE		T29 Power Support	
Apple Inc.		DRAWING NUMBER	SIZE D
NOTICE OF PROPRIETARY PROPERTY:		REVISION	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:		BRANCH	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		PAGE	38 OF 132
II NOT TO REPRODUCE OR COPY IT		SHEET	35 OF 101
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			



BCM57765 ENET SR pins are internal 1.2V switching regulator. See note for SR_DISABLE below.
If disabled: Okay to float VDD, VDDP & LX pin. VFB must always connect to =PP1V2_S3_ENET_PHY.
If enabled: VDD/VDDP connect to =PP3V3_S3_ENET_PHY (add bypassing), LX connects to inductor.
Special Star routing needed on these pins. Decoupling on Pg 37.



PHY Non-Volatile Memory

ROM contains MAC address, PCIe config info as well as code for Bonjour proxy. Required for proper PHY operation.
(Required ROM size TBD)

SYNC MASTER=K91 ERIC		SYNC DATE=10/11/2016	
PAGE 111111		DRAWING NUMBER	
ETHERNET PHY (CAESAR IV)		SIZE	
Apple Inc.		REVISION	
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		39 OF 132	
II NOT TO REPRODUCE OR COPY IT		SHEET	
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART		36 OF 101	
IV ALL RIGHTS RESERVED			

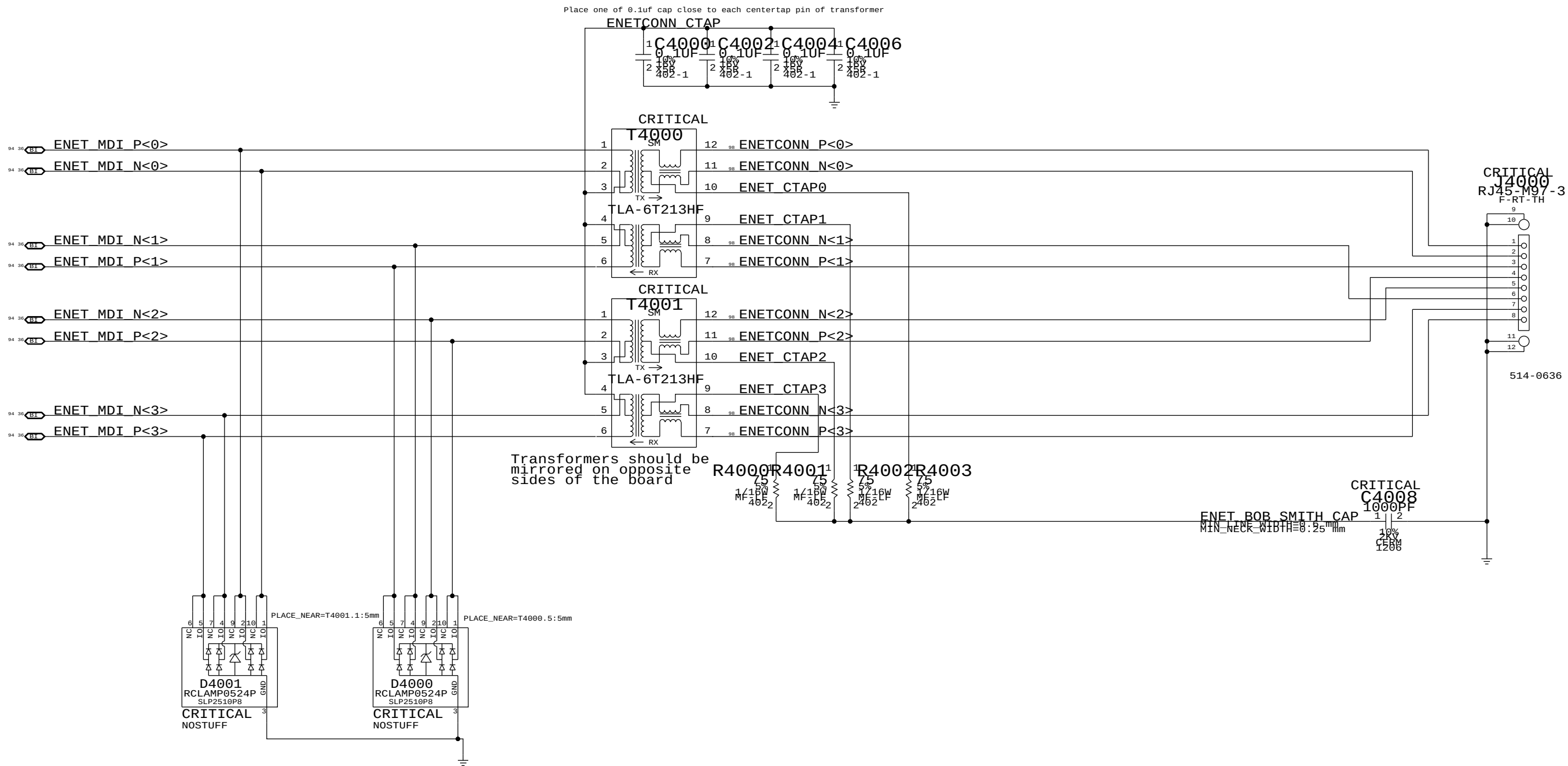


Page Notes

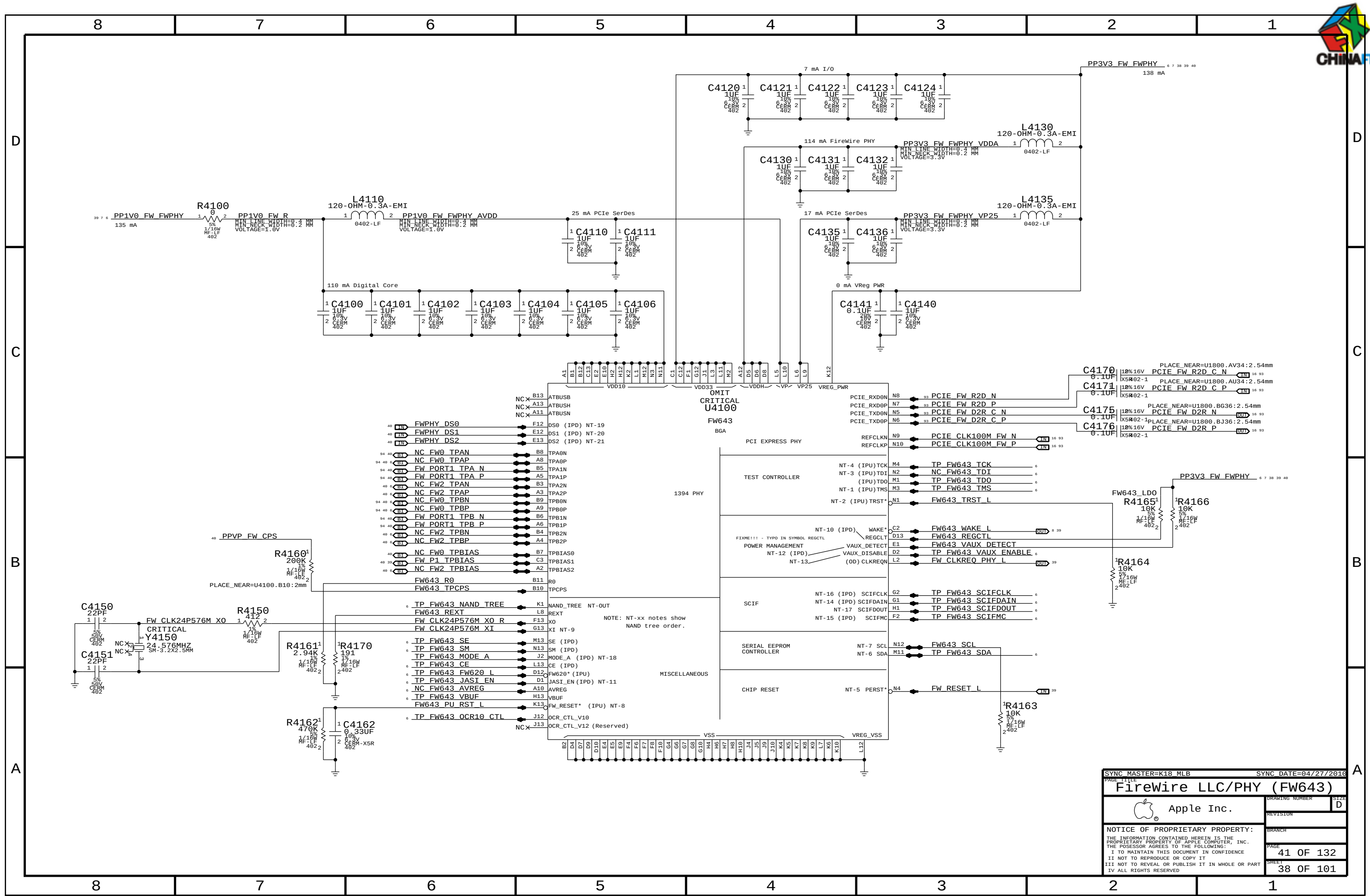
Power aliases required by this page:
(NONE)

Signal aliases required by this page:
(NONE)

BOM options provided by this page:
(NONE)



SYNC MASTER=K91 TRINHNT		SYNC DATE=05/26/2016	
PAGE TITLE		Ethernet Connector	
Apple Inc.		DRAWING NUMBER	SIZE
		REVISION	D
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE	40 OF 132
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		SHEET	37 OF 101
II NOT TO REPRODUCE OR COPY IT			
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			



SYNC MASTER=K18 MLB		SYNC DATE=04/27/2016	
PAGE 11111		FireWire LLC/PHY (FW643)	
Apple Inc.		DRAWING NUMBER	SIZE D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		REVISION	BRANCH
		PAGE	41 OF 132
		SHEET	38 OF 101



Page Notes

Power aliases required by this page:

- =PPBUS_S5_FWPWRSW (FW VP FET Input)
- =PPBUS_FW_FET (FW VP FET Output)
- =PP3V3_FW_P3V3FWFET (3.3V FET Input)
- =PP3V3_FW_FET (3.3V FET Output)
- =PP3V3_FW_FWPHY (PHY 3.3V Power)
- =PP3V3_S0_FWLATEVG
- =PP3V3_S0_FWPWRCTL
- =PP1V05_S0_FWPWRCTL (5KPD Bias Rail)
- =PP1V05_FW_P1V0FWFET (1.0V FET Input)
- =PP1V0_FW_FET_R (1.0V FET Output)
- =PP1V0_FW_FWPHY (PHY 1.0V)

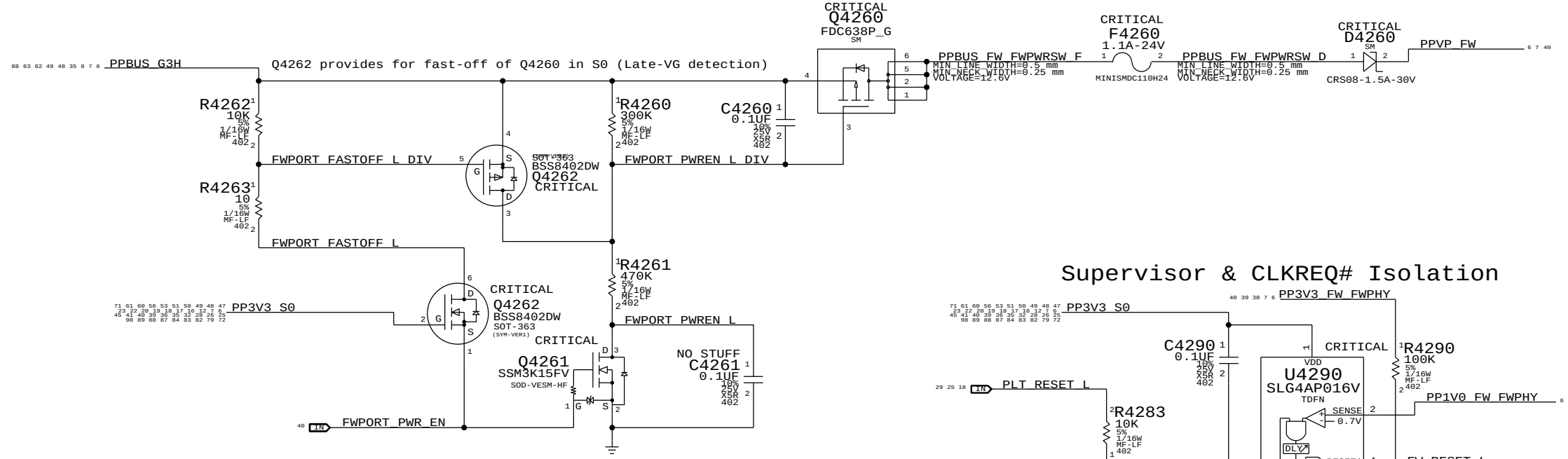
Signal aliases required by this page:

- =FW_CLKREQ_L
- =FW_PME_L

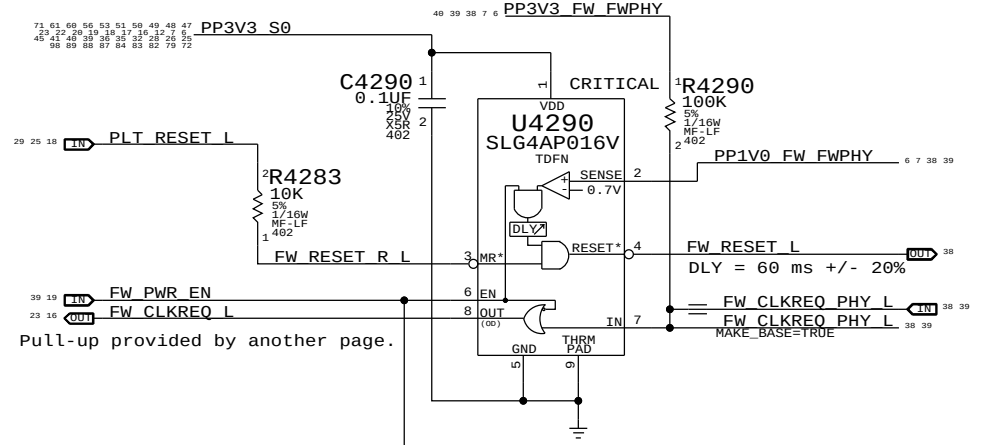
BOM options provided by this page:

(NONE)

FireWire Port Power Switch

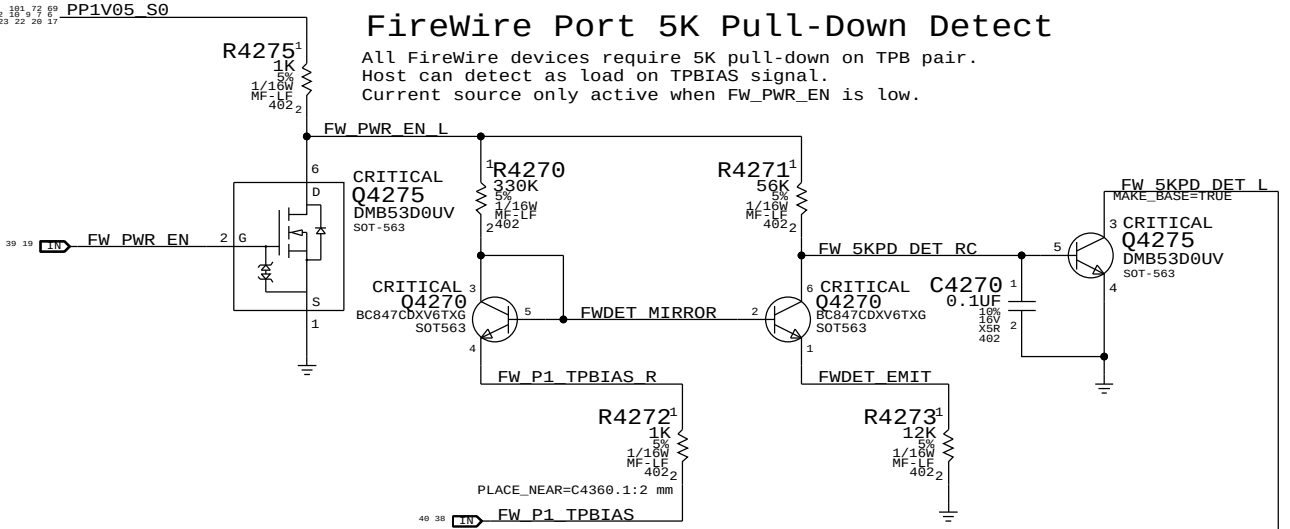


Supervisor & CLKREQ# Isolation



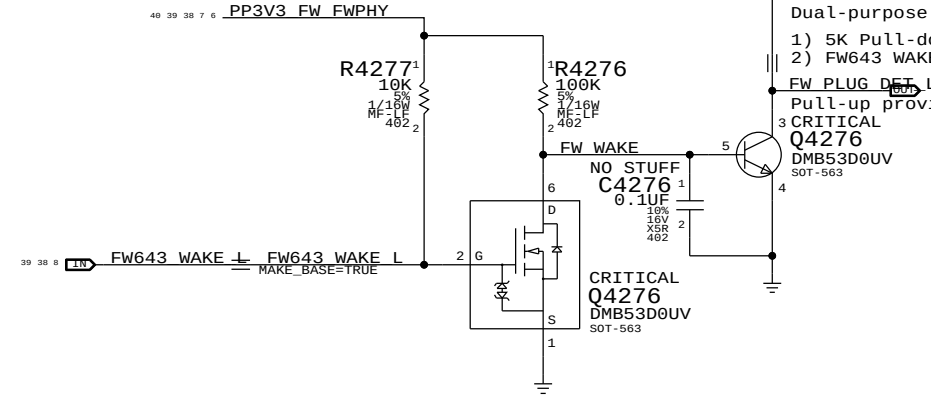
FireWire Port 5K Pull-Down Detect

All FireWire devices require 5K pull-down on TPB pair.
Host can detect as load on TPBIAS signal.
Current source only active when FW_PWR_EN is low.



FireWire PHY WAKE# Support

When PHY is powered, FW_5KPD_DET_L acts as legacy PME# signal.

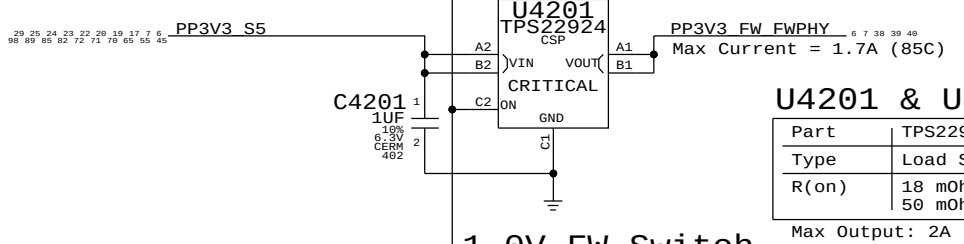


Dual-purpose output:

- 1) 5K Pull-down Detect when FW_PWR_EN is low.
- 2) FW643 WAKE# (PME#) when PHY is powered.

Pull-up provided on another page.

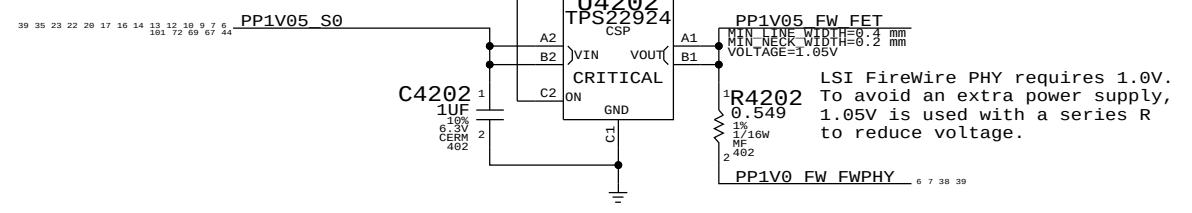
3.3V FW Switch



Part	TPS22924C
Type	Load Switch
R(on)	18 mOhm Typ 50 mOhm Max

Max Output: 2A

1.0V FW Switch



SYNC MASTER=T27 REF

SYNC DATE=06/10/2016

FireWire Port & PHY Power

Apple Inc.

NOTICE OF PROPRIETARY PROPERTY:

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:

I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE

II NOT TO REPRODUCE OR COPY IT

III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART

IV ALL RIGHTS RESERVED

DRAWING NUMBER

REVISION

BRANCH

PAGE

SHEET

SIZE

D

42 OF 132

39 OF 101



Page Notes

Power aliases required by this page:

- PPVP_FW_PORT1
- PPVP_FW_PHY_CPS_FET (From Port)
- PPVP_FW_PHY_CPS (To PHY)
- PP3V3_FW_FWPHY
- PP3V3_S0_FWLATEVG

Signal aliases required by this page:

- FW_PHY_DS0
- FW_PHY_DS1
- FW_PHY_DS2

NOTE: This page is expected to contain the necessary aliases to map the FireWire TPA/TPB pairs to their appropriate connectors and/or to properly terminate unused signals.

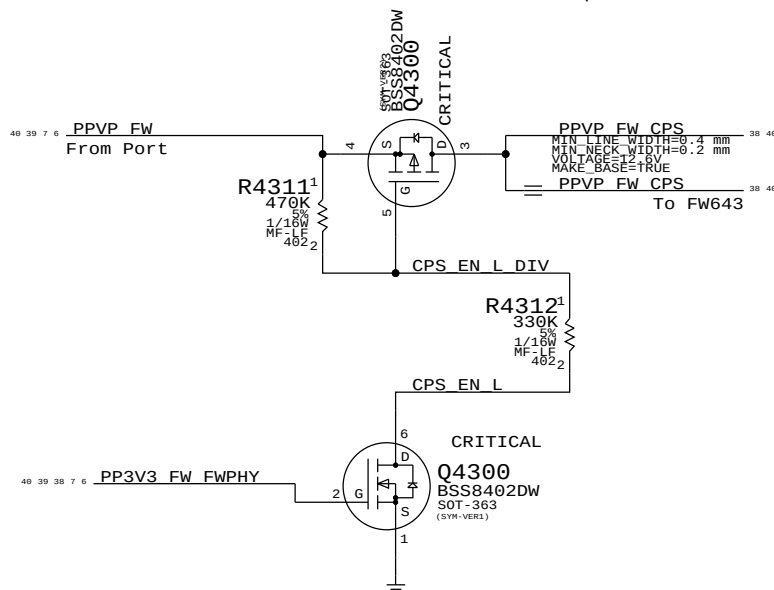
BOM options provided by this page:

(NONE)

1394b implementation based on Apple FireWire Design Guide (FWDG 0.6, 5/14/03)

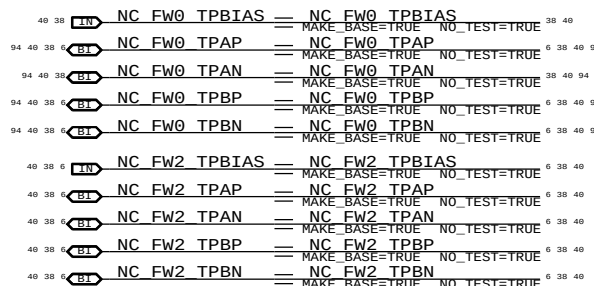
FW643 TPCPS Leakage Protection

FW643 has internal leakage path from TPCPS pin to VDD33.
FET blocks current to TPCPS until VDD33 is powered.



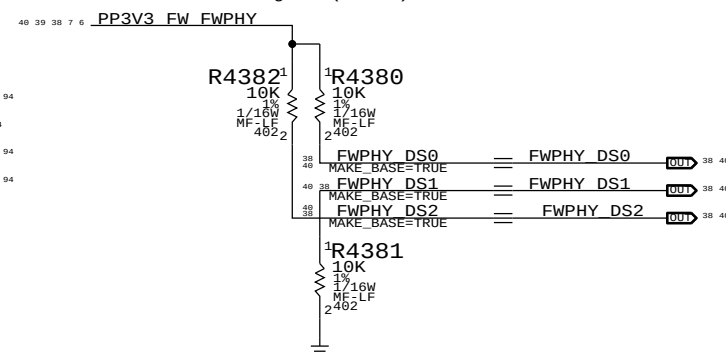
Unused FireWire Ports

Disabled per LSI instructions
(All unused port signals TP/NC)



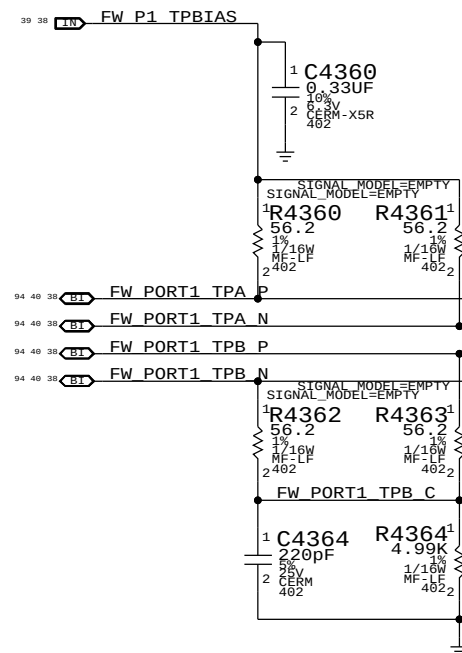
FireWire PHY Config Straps

Configures PHY for:
- Port "1" Bilingual (1394B)

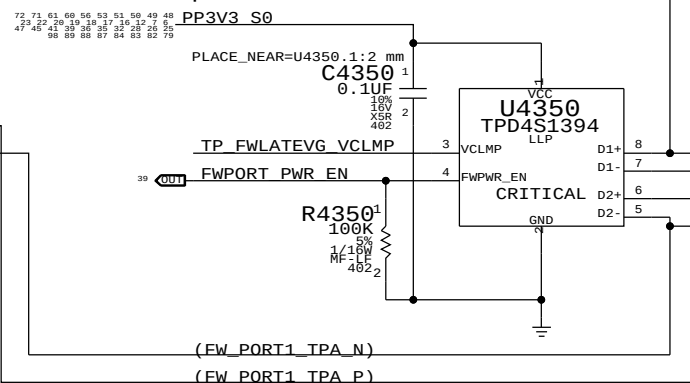


Termination

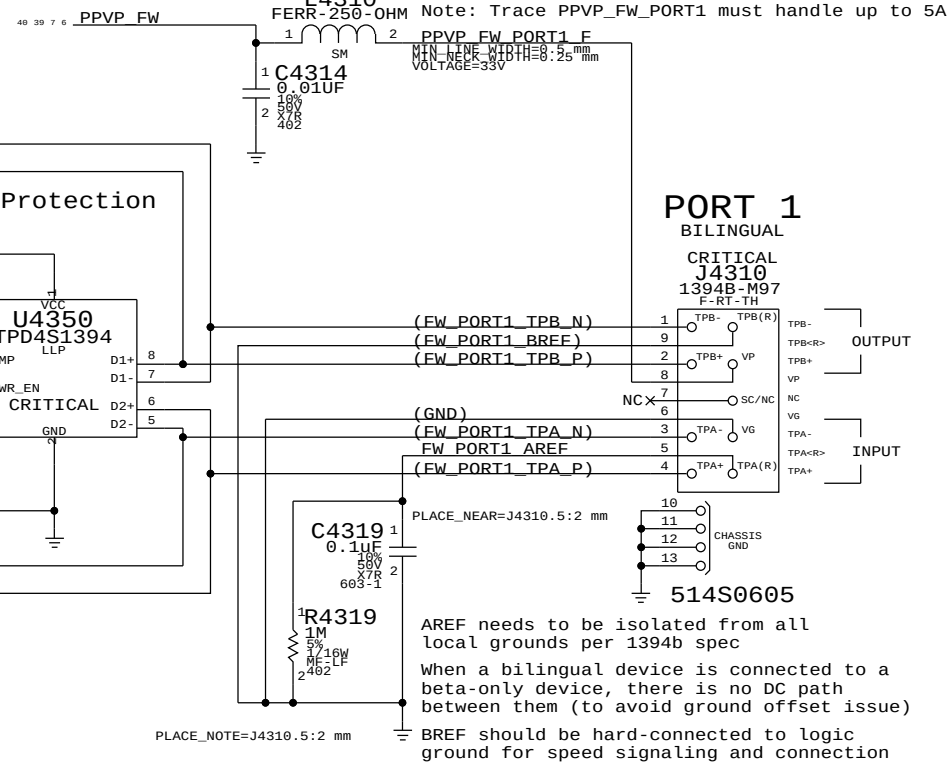
Place close to FireWire PHY



"Snapback" & "Late VG" Protection



Cable Power

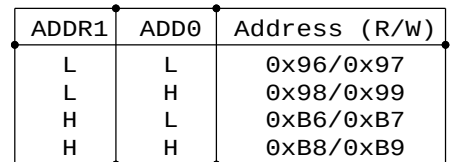


AREF needs to be isolated from all local grounds per 1394b spec
When a bilingual device is connected to a beta-only device, there is no DC path between them (to avoid ground offset issue)
BREF should be hard-connected to logic ground for speed signaling and connection

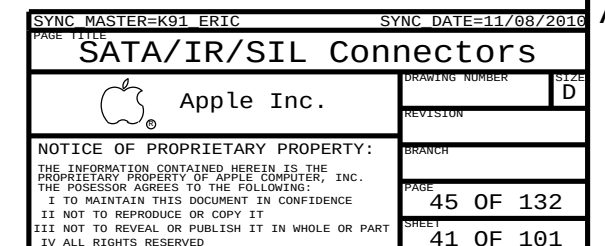
SYNC MASTER=T27 REF		SYNC DATE=06/10/2016	
PAGE TITLE		PAGE	
FireWire Connector		DRAWING NUMBER	
Apple Inc.		SIZE	
NOTICE OF PROPRIETARY PROPERTY:		REVISION	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:		BRANCH	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		PAGE	
II NOT TO REPRODUCE OR COPY IT		43 OF 132	
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART		SHEET	
IV ALL RIGHTS RESERVED		40 OF 101	



ODD Power Control

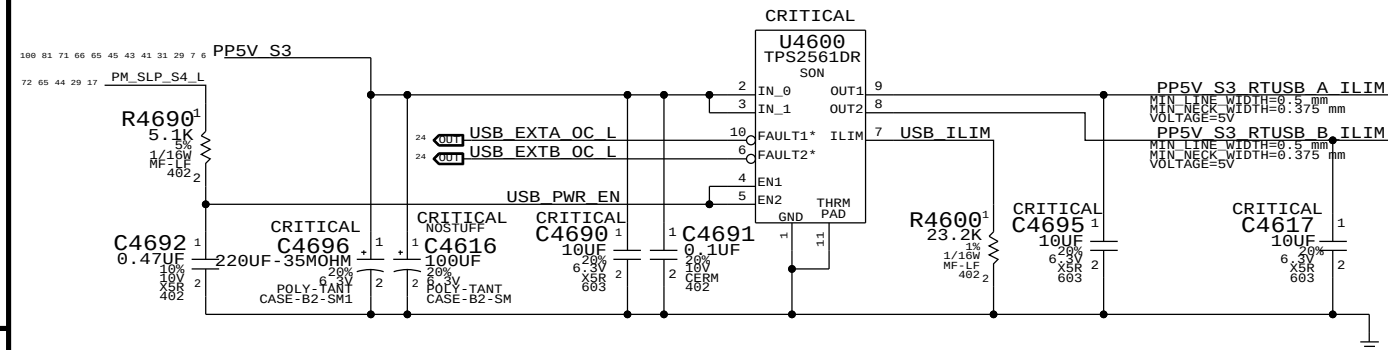


SATA HDD / IR / SIL Connector



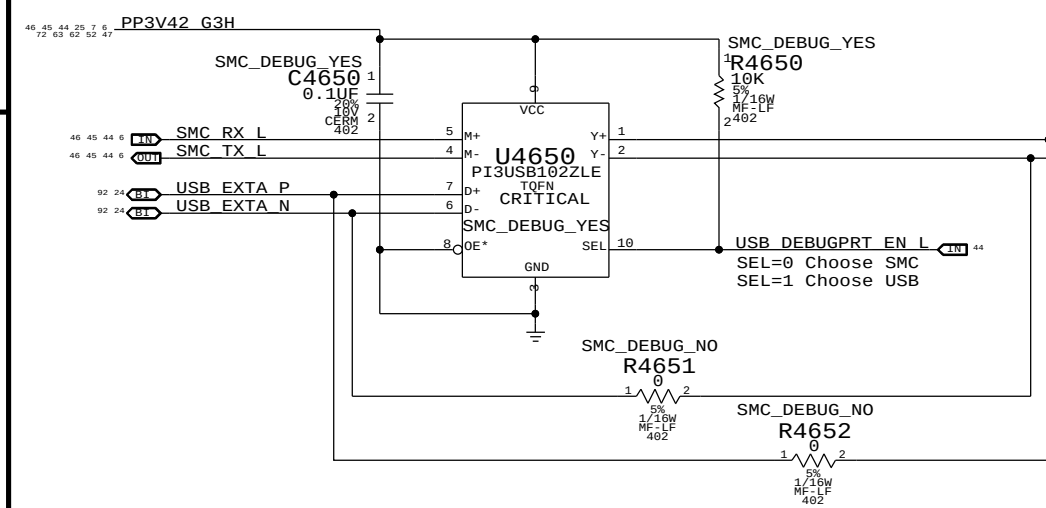


USB Port Power Switch

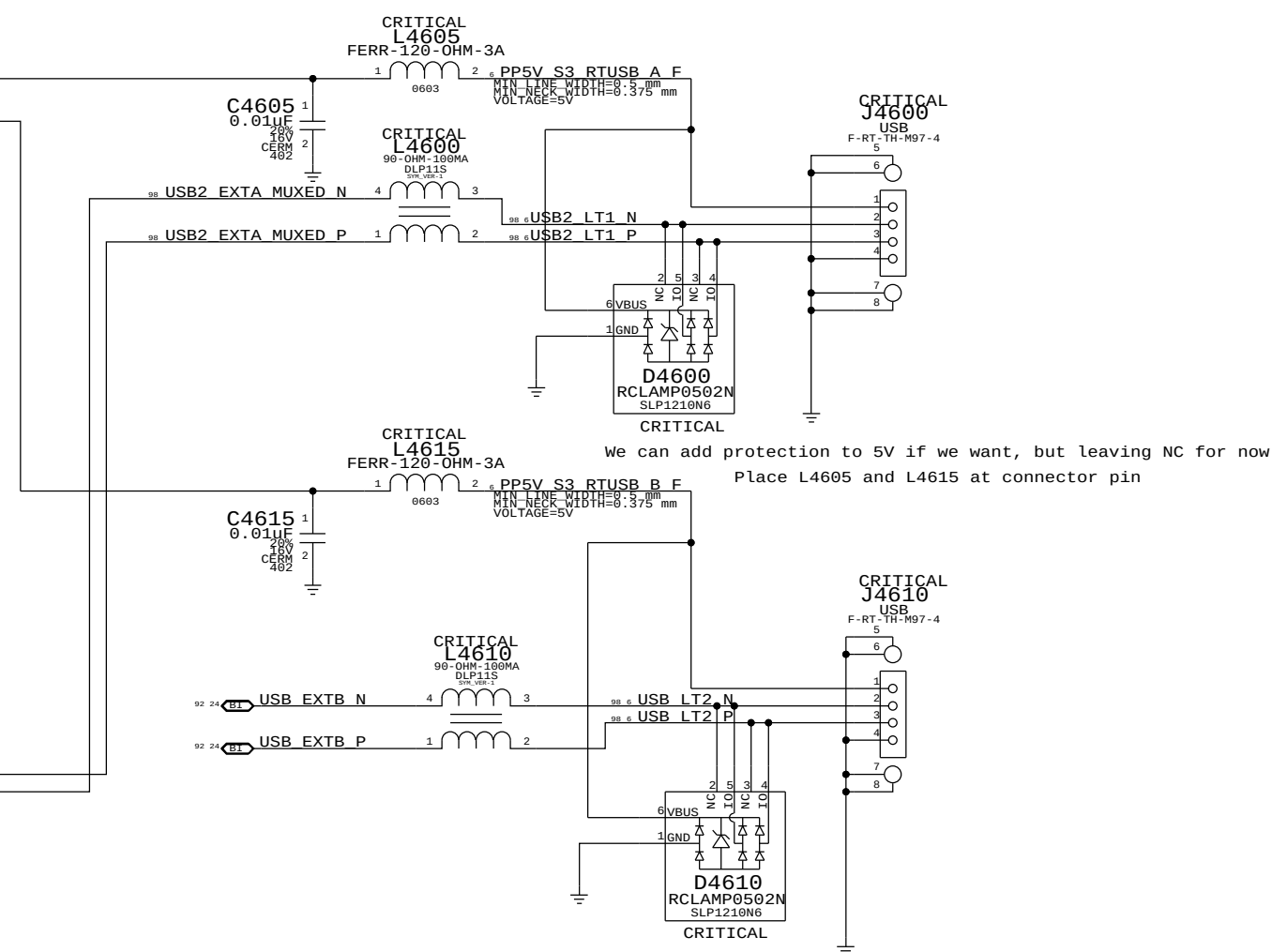


Current limit per port (R4600): 2.18A min / 2.63A max

USB/SMC Debug Mux



Left USB Port A

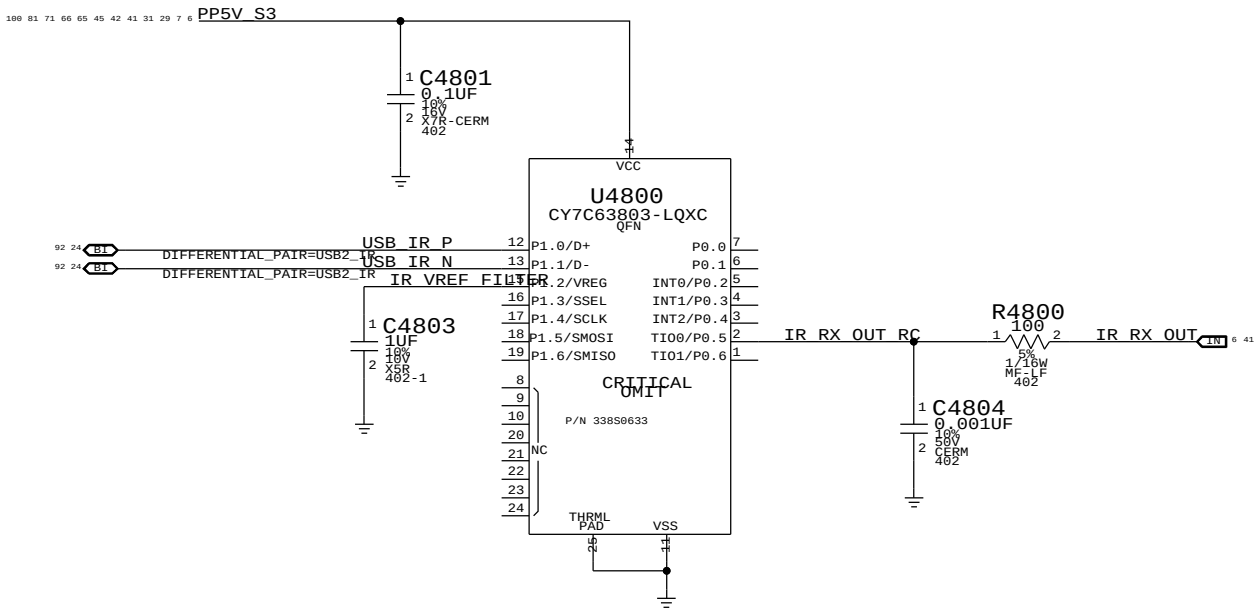


Left USB Port B

SYNC MASTER=K91 ERIC		SYNC DATE=10/08/2016	
External USB Connectors		DRAWING NUMBER	SIZE
Apple Inc.		REVISION	D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		BRANCH	PAGE
			46 OF 132
		SHEET	42 OF 101



IR SUPPORT





NOTE: Unused pins have "SMC_Pxx" names. Unused pins designed as outputs can be left floating, those designated as inputs require pull-ups.

D

D

C

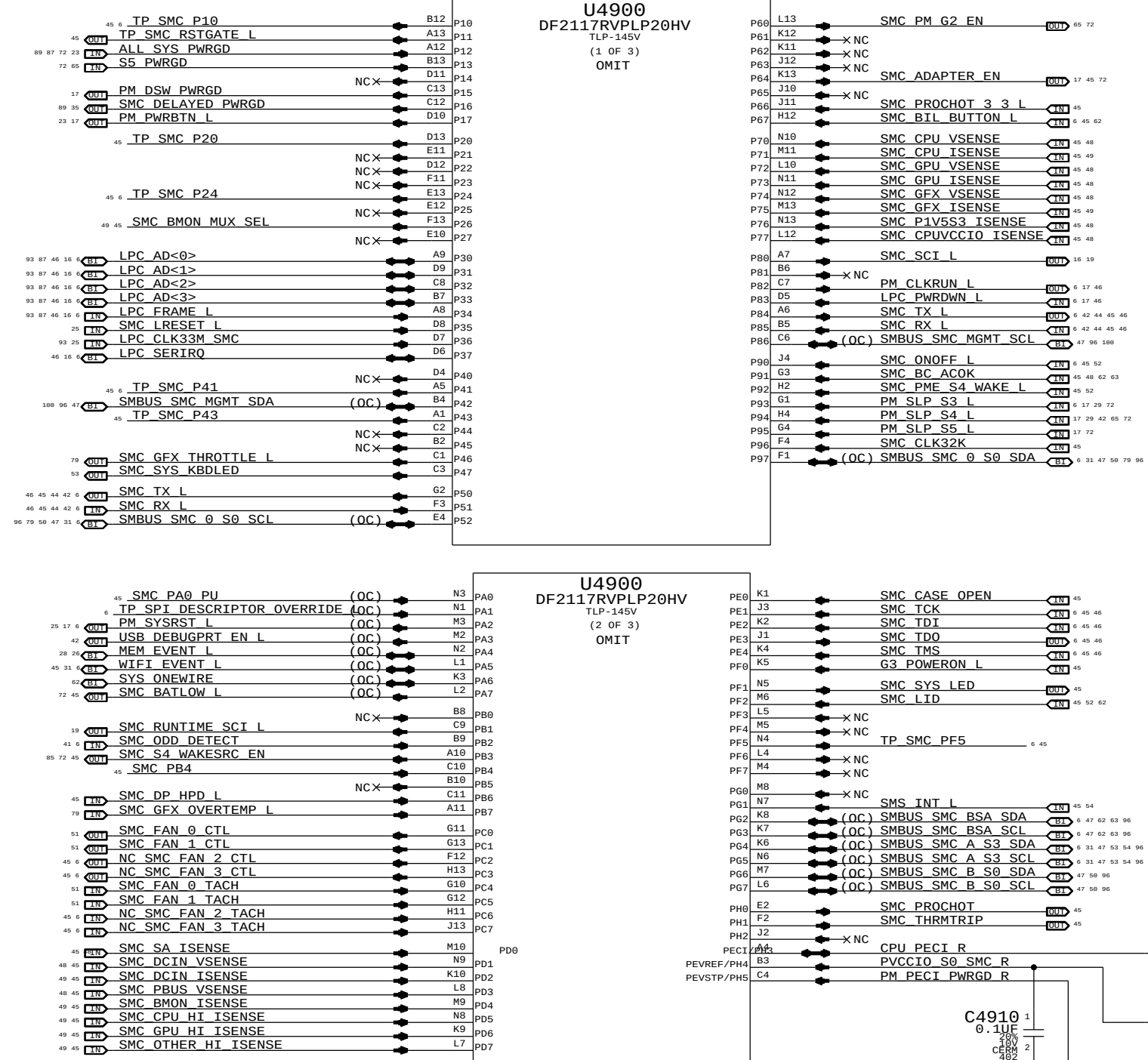
C

B

B

A

A

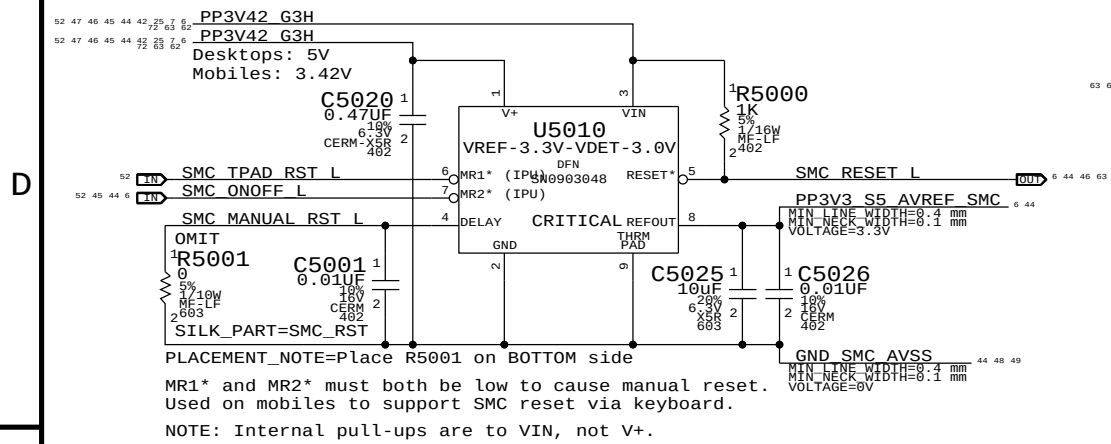


NOTE: SMS Interrupt can be active high or low, rename net accordingly. If SMS interrupt is not used, pull up to SMC rail.

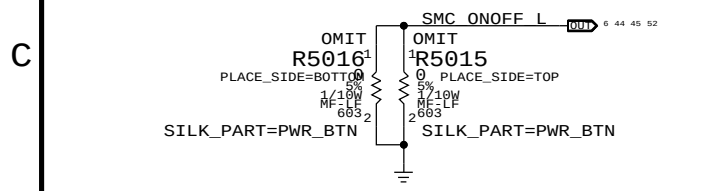
SYNC MASTER=K91 BEN		SYNC DATE=07/12/2016	
PAGE TITLE		SMC	
Apple Inc.		DRAWING NUMBER	SIZE D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		REVISION	
		BRANCH	
		PAGE	49 OF 132
		SHEET	44 OF 101



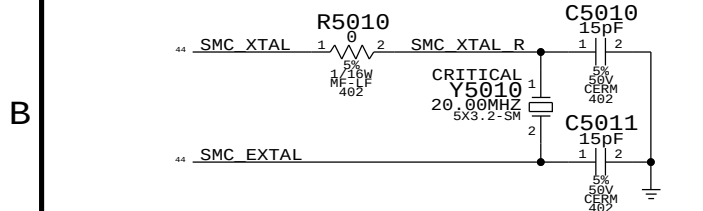
SMC Reset "Button", Supervisor & AVREF Supply



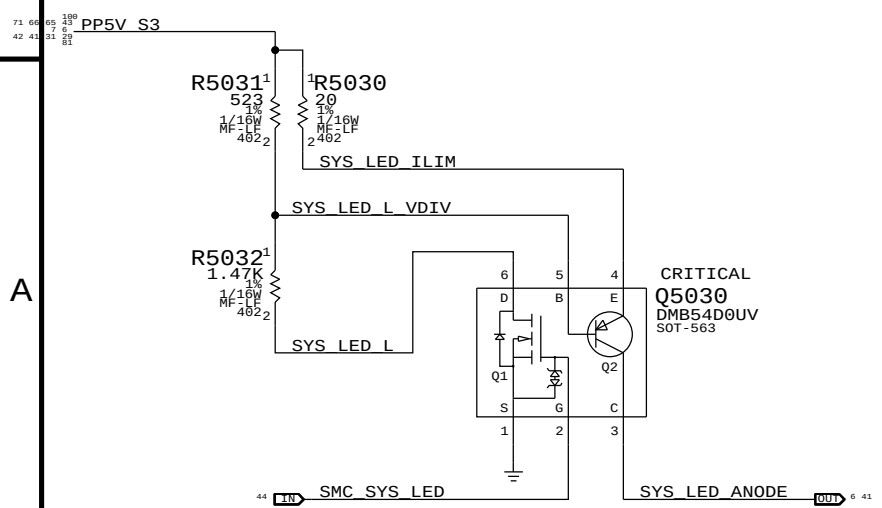
Debug Power "Buttons"



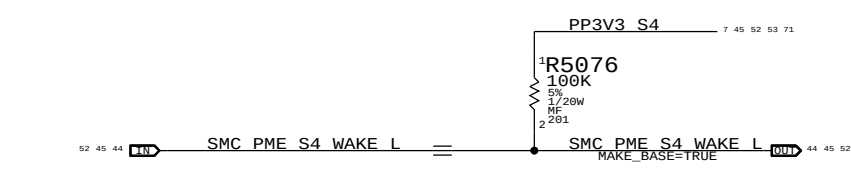
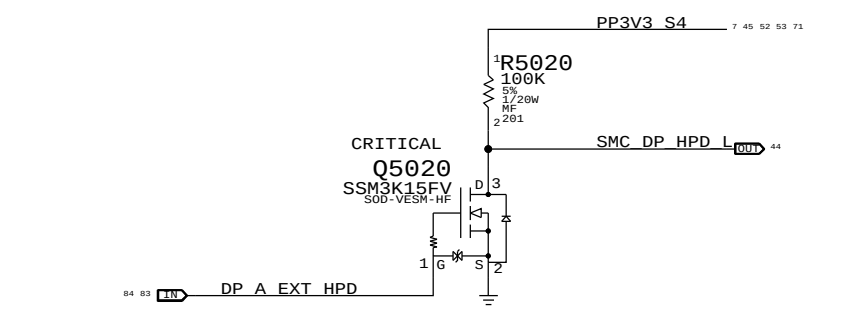
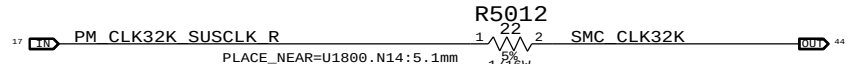
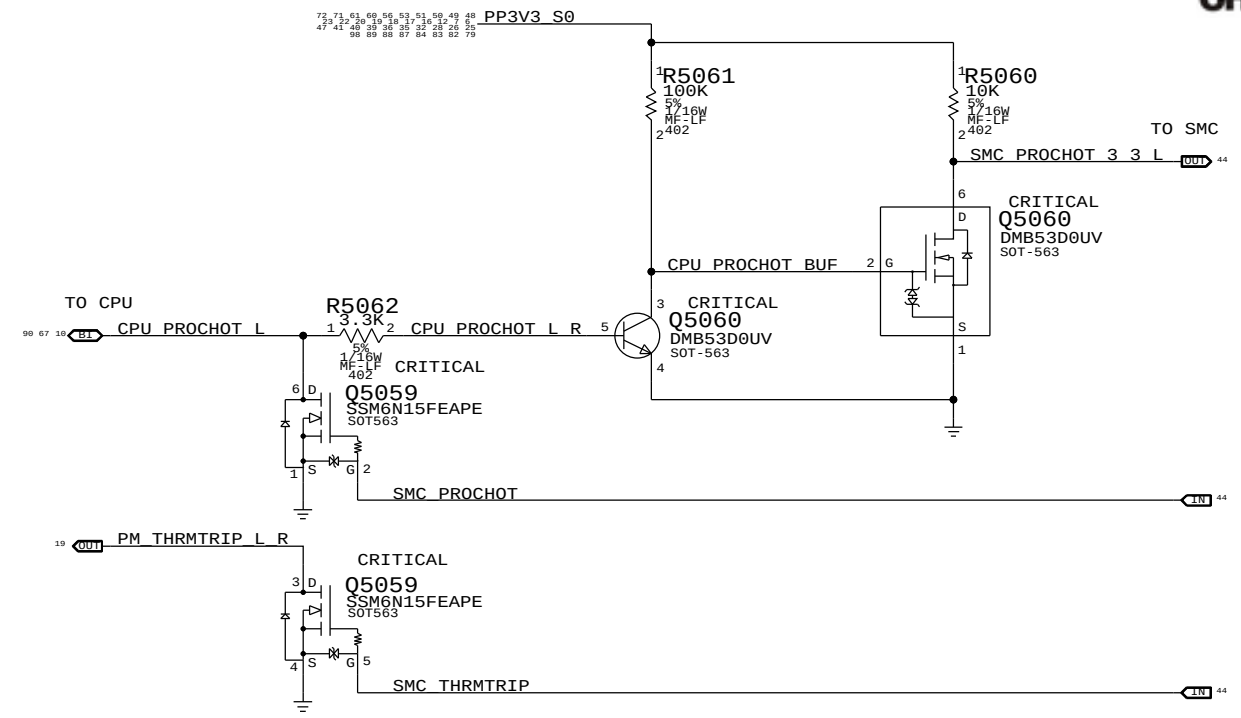
SMC Crystal Circuit



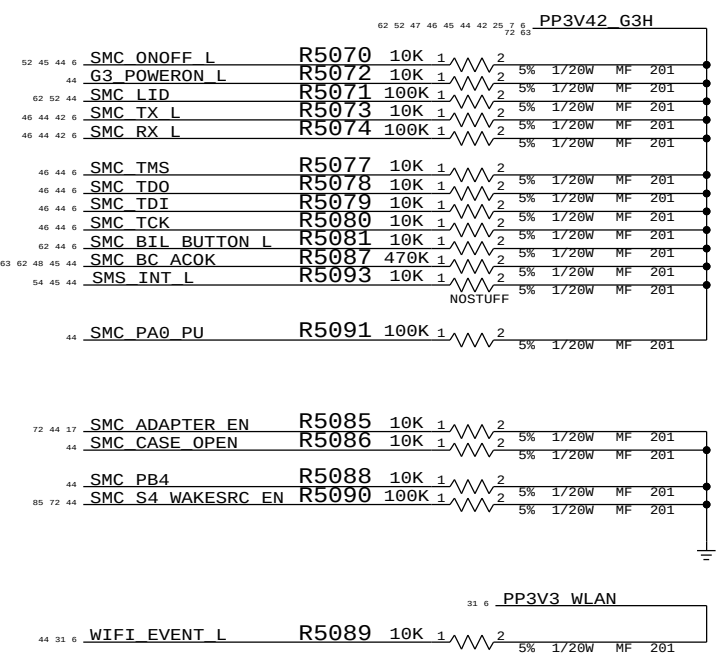
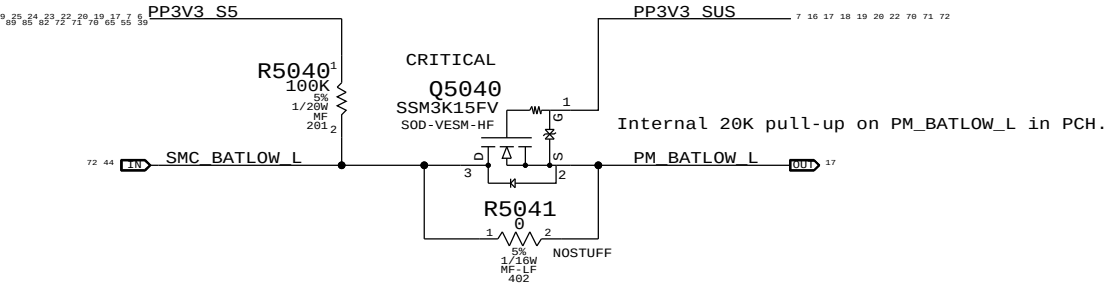
System (Sleep) LED Circuit



SMC FSB to 3.3V Level Shifting



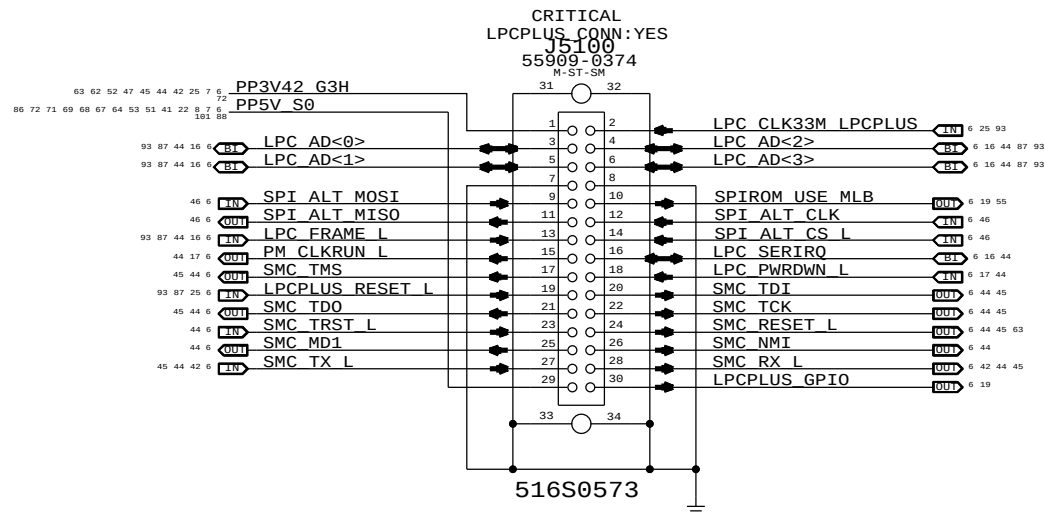
BATLOW# Isolation



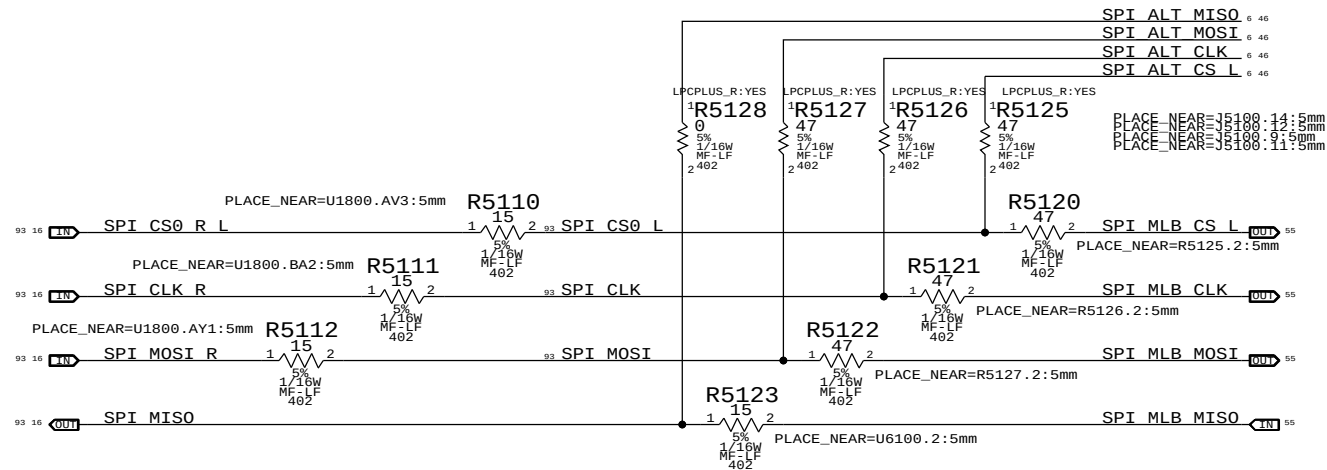
SYNC_MASTER=K91 BEN		SYNC_DATE=07/12/2016	
PAGE TITLE			
SMC Support		DRAWING NUMBER	SIZE D
Apple Inc.		REVISION	
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE	50 OF 132
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		SHEET	45 OF 101
II NOT TO REPRODUCE OR COPY IT			
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			



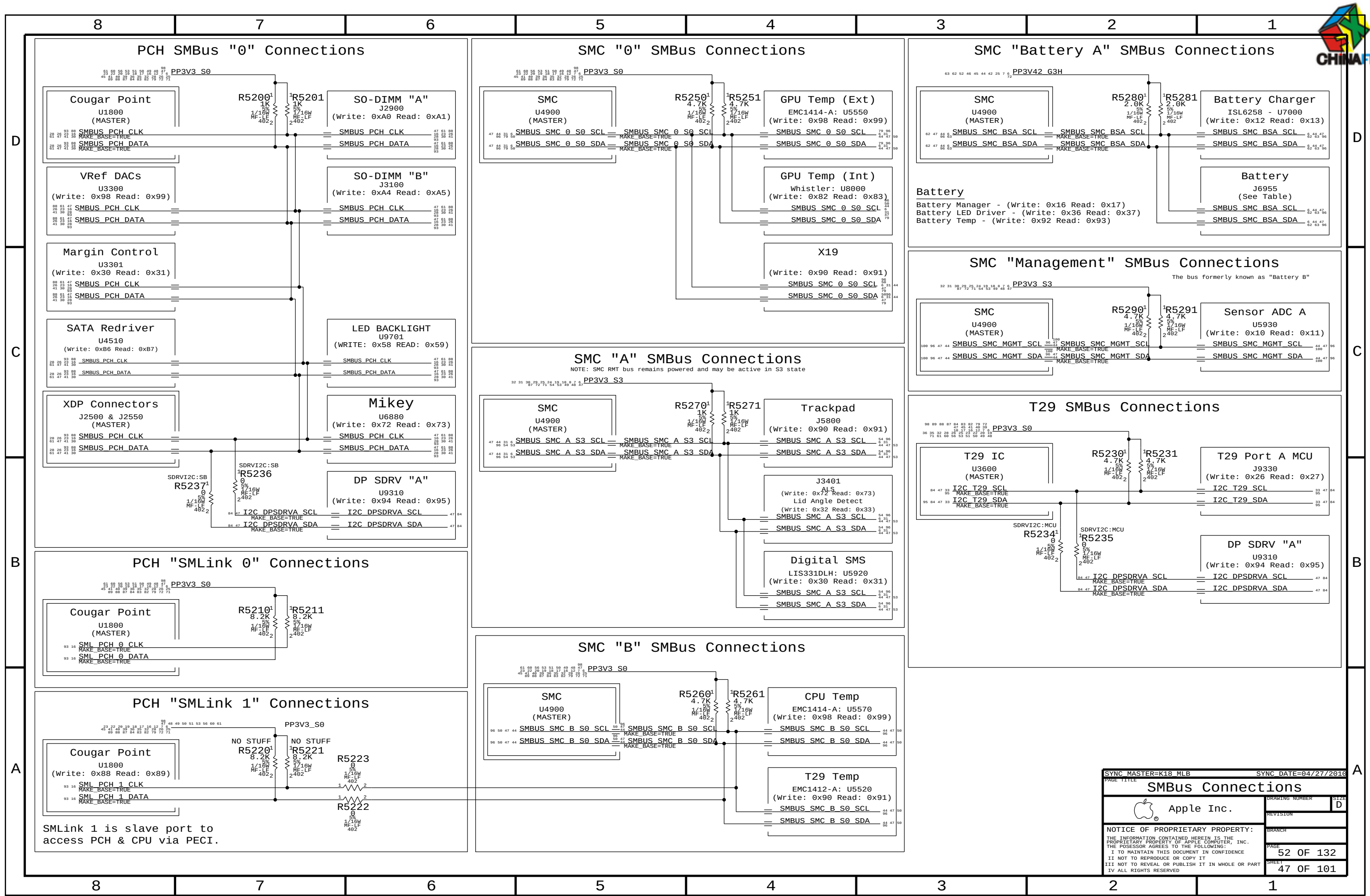
LPC+SPI Connector



SPI Bus Series Termination



SYNC_MASTER=K18_MLB		SYNC_DATE=04/27/2016	
PAGE TITLE			
LPC+SPI Debug Connector			
 Apple Inc.		DRAWING NUMBER	SIZE
		REVISION	D
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE	51 OF 132
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		SHEET	46 OF 101
II NOT TO REPRODUCE OR COPY IT			
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			



SYNC MASTER=K18 MLB		SYNC DATE=04/27/2016	
PAGE TITLE		DRAWING NUMBER	
SMBus Connections		SIZE	
Apple Inc.		D	
NOTICE OF PROPRIETARY PROPERTY:		REVISION	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:		BRANCH	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		PAGE	
II NOT TO REPRODUCE OR COPY IT		52 OF 132	
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART		SHEET	
IV ALL RIGHTS RESERVED		47 OF 101	



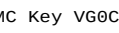
B



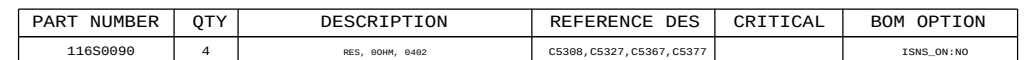
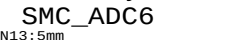
1



SMC_ADC4

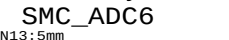


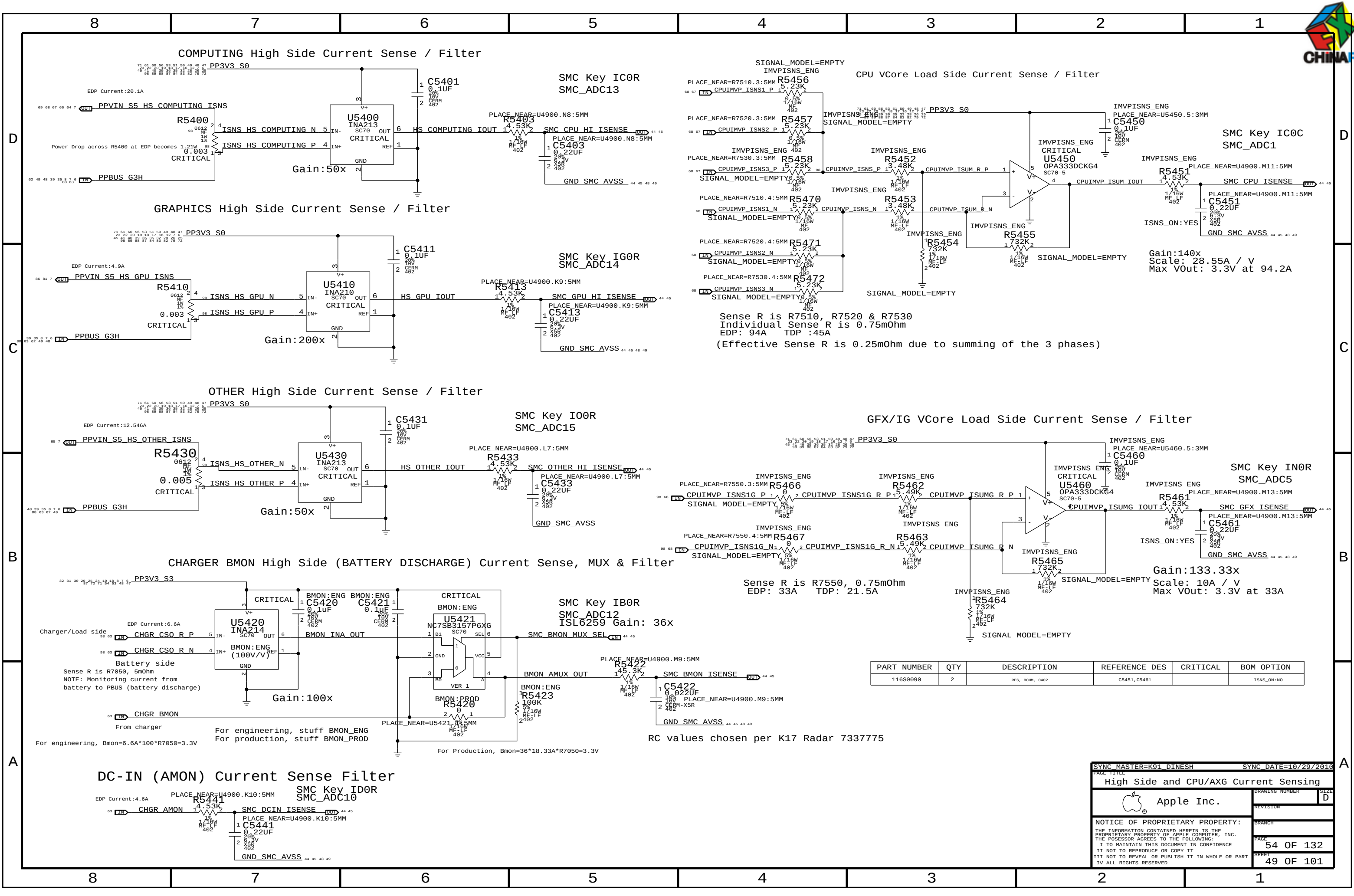
MC_ADC2



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
116S0090	4	RES, 60HM, 0402	C5308,C5327,C5367,C5377		ISNS_ON:NO

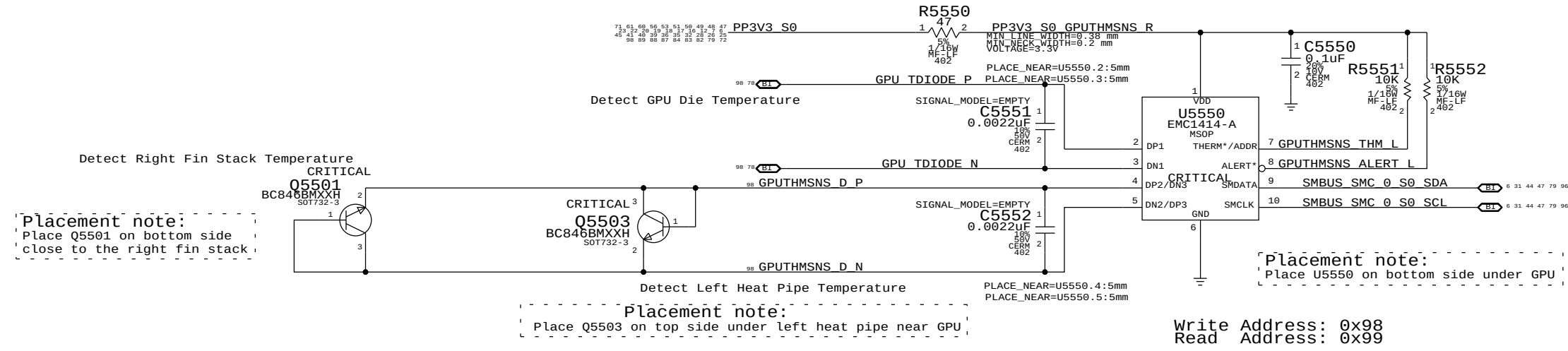
SMC Key IMOC

A |

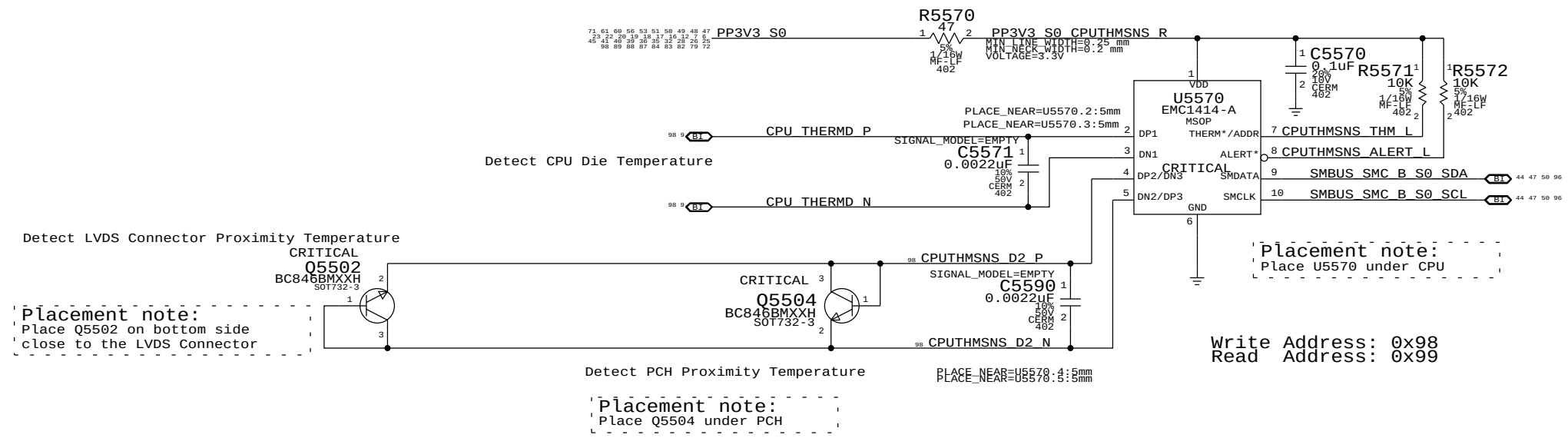




GPU Proximity/GPU Die/Left Heat Pipe/Right Fin Stack

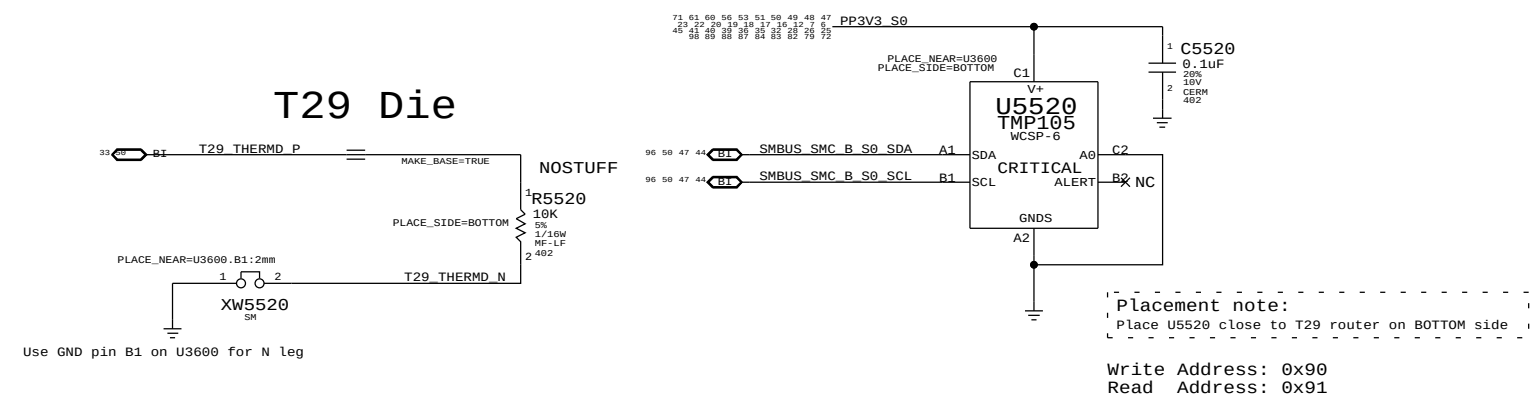


CPU Proximity/CPU Die/PCH Proximity/LVDS Connector Proximity



Note: EMC1414 can perform Beta Compensation for External Diode 1 only

T29 Proximity



SYNC MASTER=K91 DINESH		SYNC DATE=09/22/2016	
PAGE TITLE		Thermal Sensors	
Apple Inc.		DRAWING NUMBER	SIZE
		REVISION	D
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE	55 OF 132
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		SHEET	50 OF 101
II NOT TO REPRODUCE OR COPY IT			
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			



D

C

B

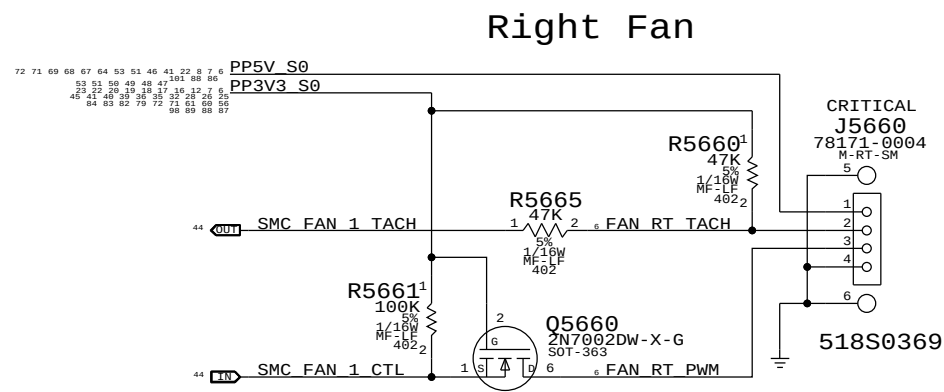
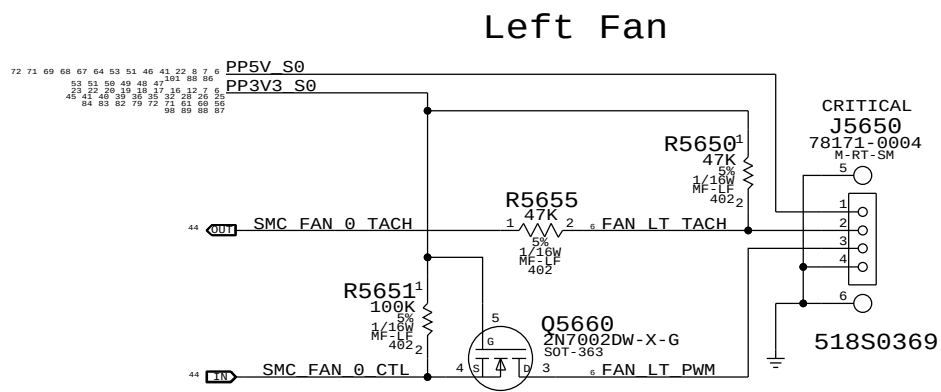
A

D

C

B

A



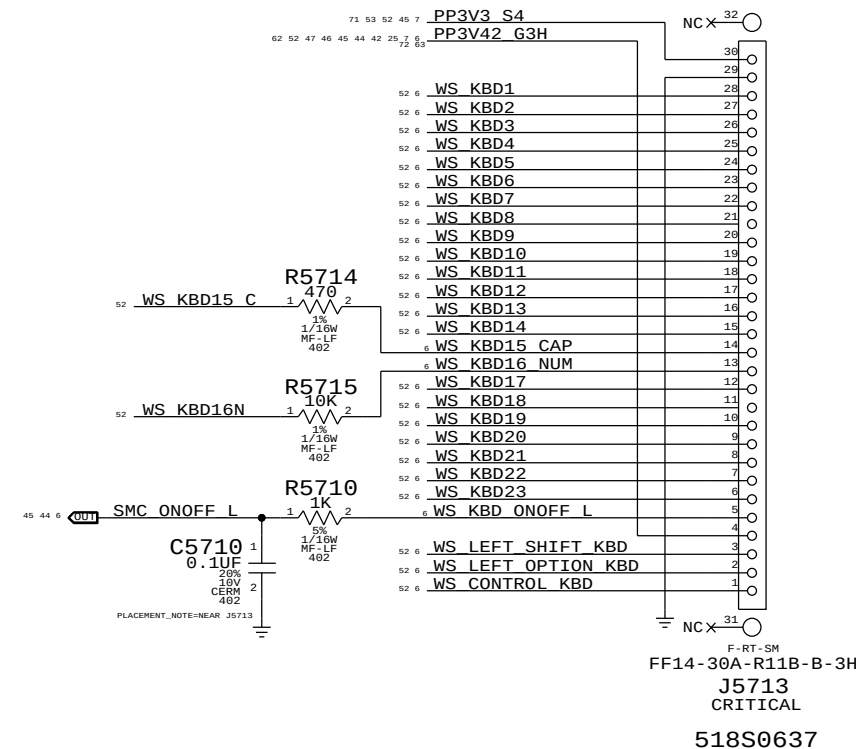
SYNC_MASTER=K18_MLB		SYNC_DATE=04/27/2016	
PAGE TITLE		Fan Connectors	
Apple Inc.		DRAWING NUMBER	SIZE
		REVISION	D
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE	56 OF 132
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		SHEET	51 OF 101
II NOT TO REPRODUCE OR COPY IT			
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			

PSOC USB CONTROLLER

- USB INTERFACES TO MLB
- SPI HOST TO Z2
- TRACKPAD PICK BUTTONS
- KEYBOARD SCANNER

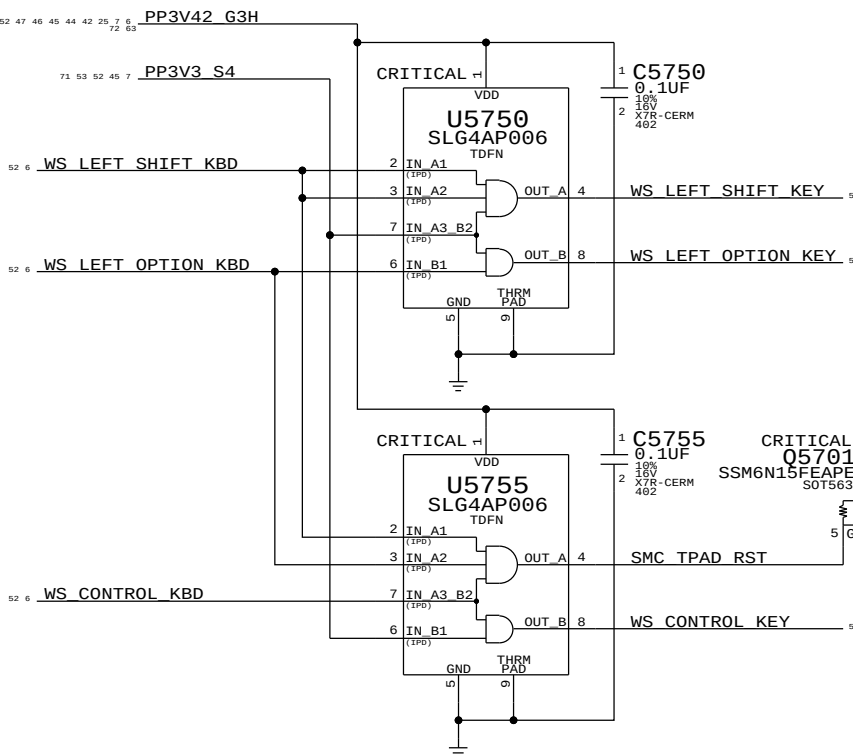
IC	PIN NAME	CURRENT	R_SNS	V_SNS	POWER
TMP102	V+	10UA 80UA	2.55 KOHM	0.0255 V 0.204 V	0.255E-6 W 16.32E-6 W
3V3 LDO	VDD VOUT	60MA (MAX) 60MA (MAX)	10 OHM 0.2 OHM	0.6 V 0.012 V	36E-3 W 0.72E-3 W
PSOC	VDD	8MA (TYP) 14MA (MAX)	1.5 OHM	0.012 V 0.021 V	96E-6 W 294E-6 W
18V BOOSTER	VIN	4MA (MAX)	4.7 OHM	0.0188 V	75.2E-6 W

Keyboard Connector

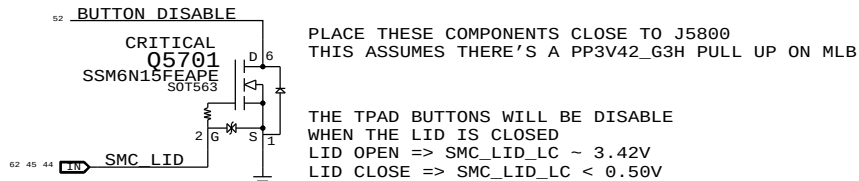


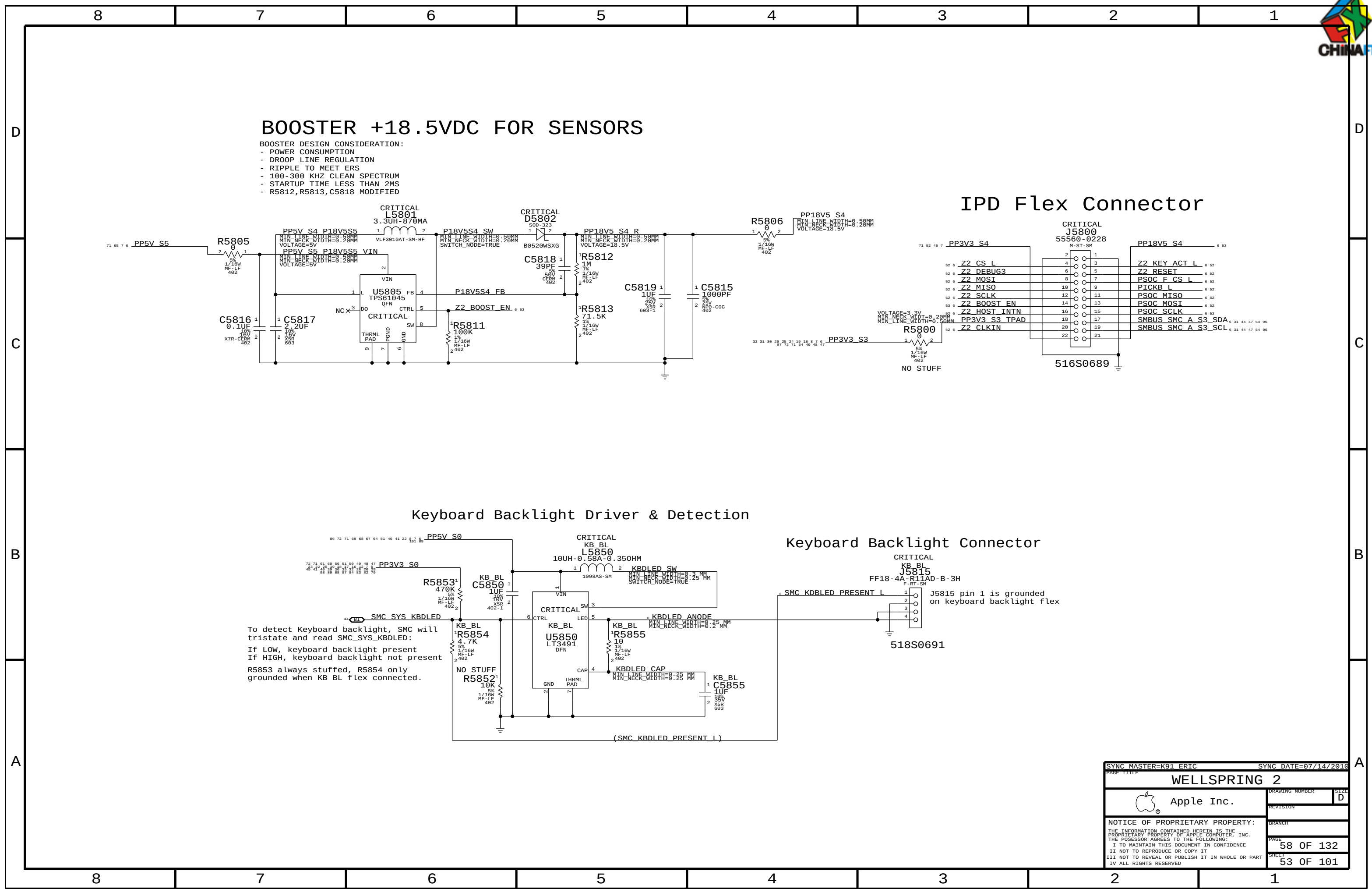
SMC Manual Reset & Isolation

Left shift, option & control keys combined with power button cause SMC RESET# assertion.
Keys ANDed with PSOC power to isolate when PSOC is not powered.



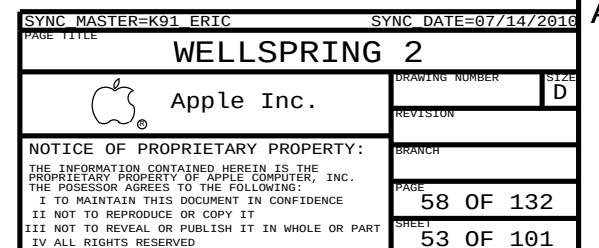
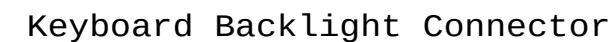
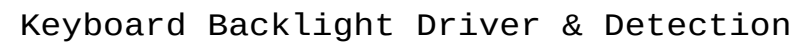
TPAD Buttons Disable

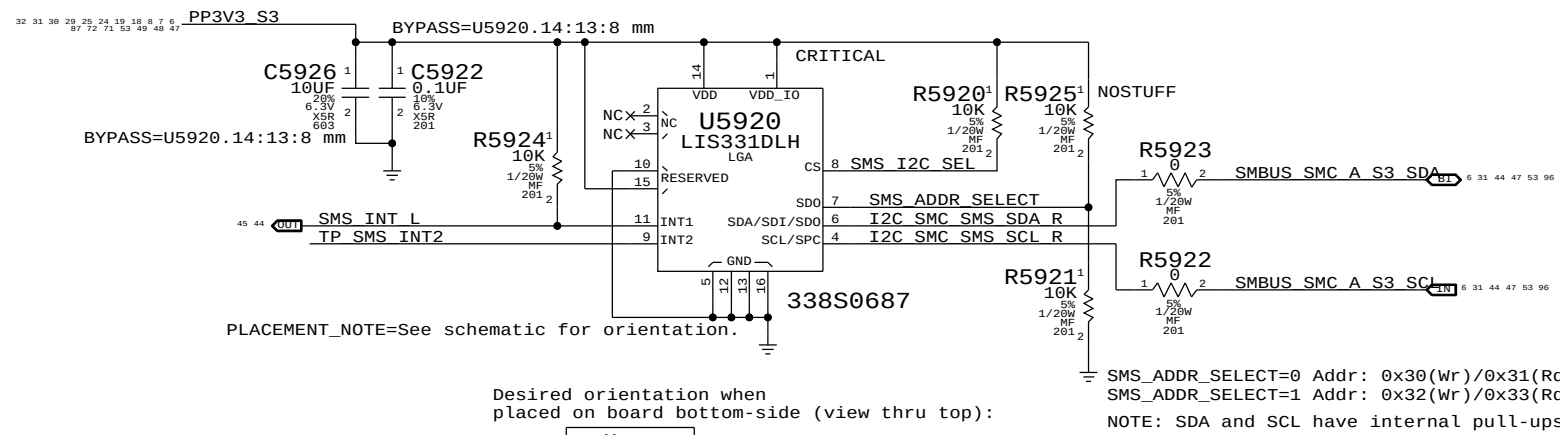




BOOSTER DESIGN CONSIDERATION:

- POWER CONSUMPTION
- DROOP LINE REGULATION
- RIPPLE TO MEET ERS
- 100-300 KHZ CLEAN SPECTRUM
- STARTUP TIME LESS THAN 2MS
- R5812, R5813, C5818 MODIFIED

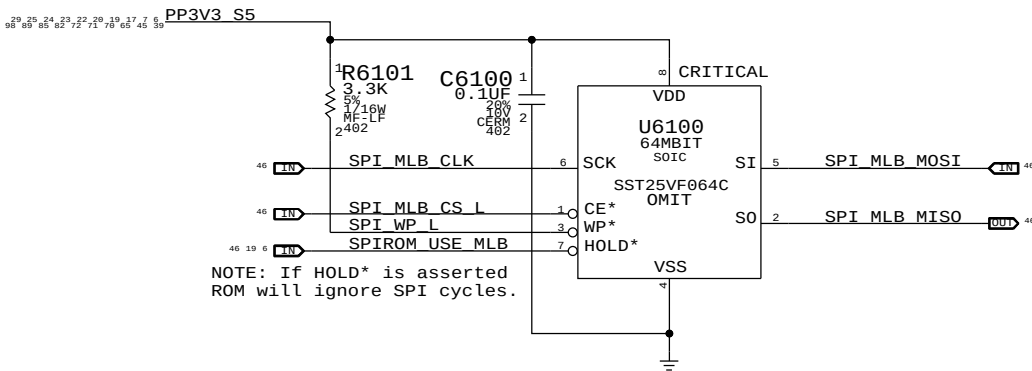




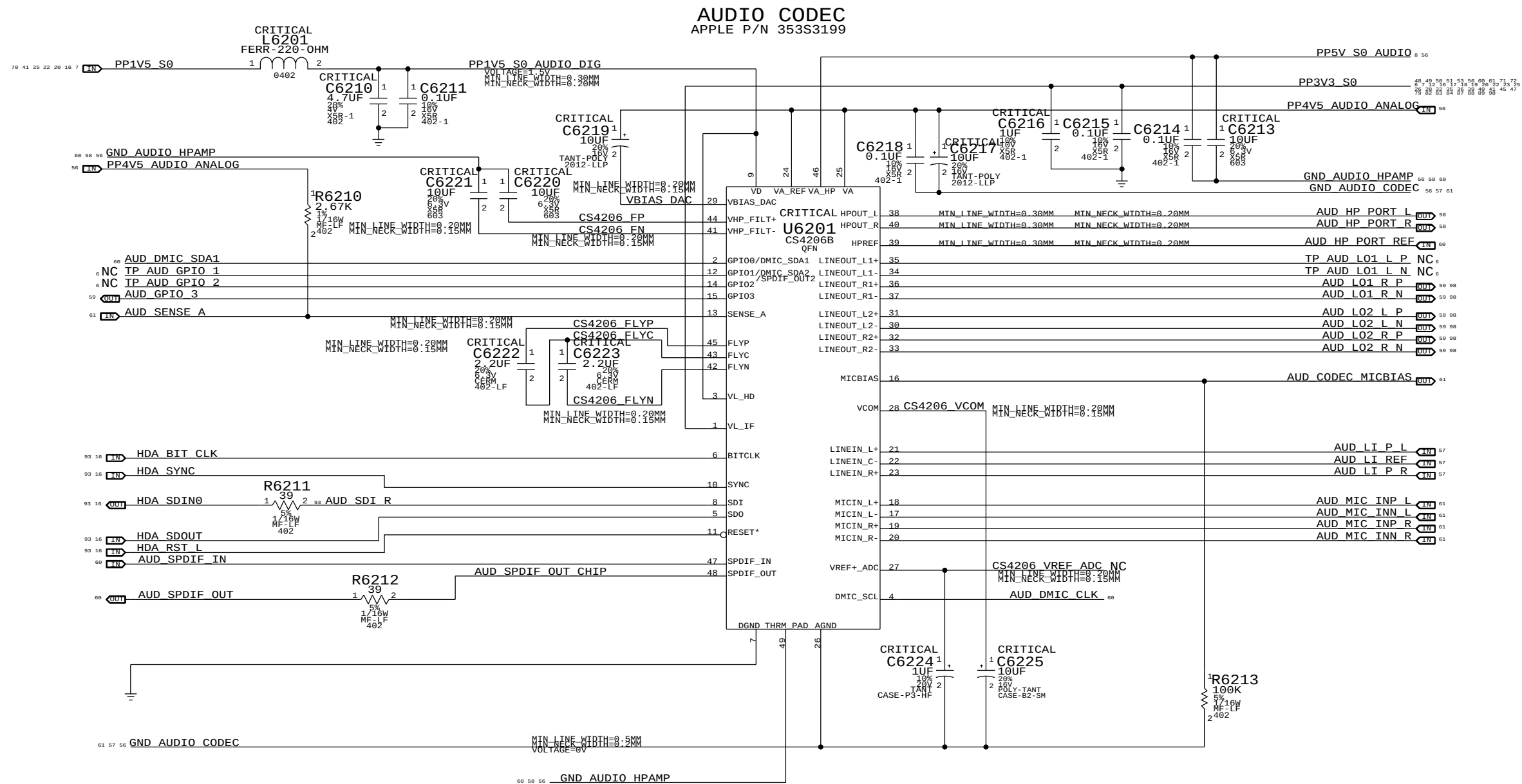
↓

Circle indicates pin 1 location when placed in correct orientation

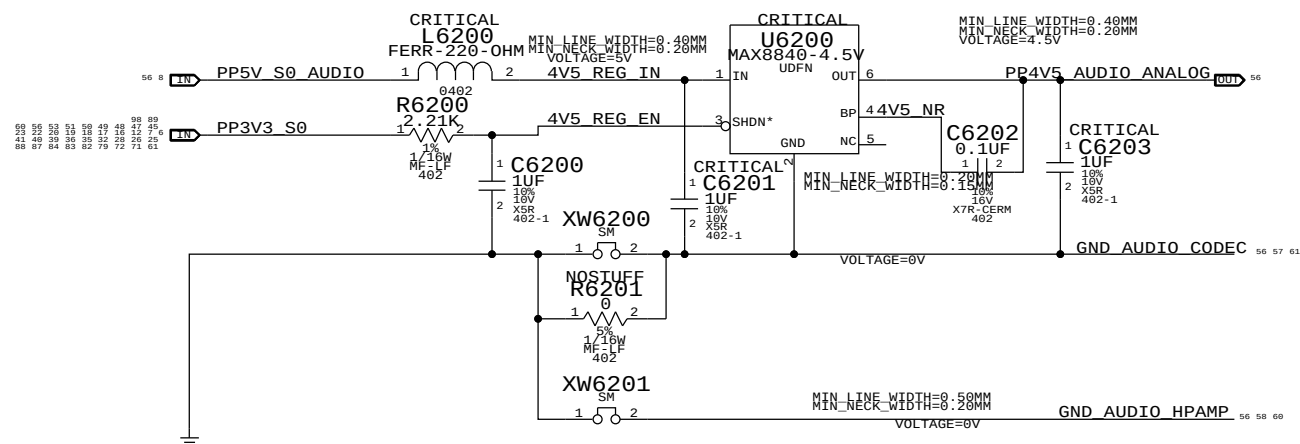
SYNCH MASTER-K91 DINESH		SYNCH DATE=08/06/2010	
PAGE TITLE			
Digital Accelerometer			
 Apple Inc.		DRAWING NUMBER	
		D	
		REVISION	
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		59 OF 132	
II NOT TO REPRODUCE OR COPY IT			
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART		SHEET	
IV ALL RIGHTS RESERVED		54 OF 101	



SYNC MASTER=K91 BEN		SYNC DATE=06/08/2016	
PAGE TITLE		SPI ROM	
		DRAWING NUMBER	SIZE
		REVISION	D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		BRANCH	PAGE
			61 OF 132
		SHEET	55 OF 101



4.5V POWER SUPPLY FOR CODEC
APPLE P/N 353S2234



NOTES ON CODEC I/O

DIFF FSINPUT= 2.45VRMS
SE FSINPUT= 1.22VRMS
DAC1 FSOUTPUT= 1.34VRMS
DAC2/3 FSOUTPUTDIFF= 2.67VRMS
DAC2/3 FSOUTPUTSE= 1.34VRMS

SYNC MASTER=K91 AUDIO		SYNC DATE=09/30/2016	
PAGE TITLE		AUDIO: CODEC/REGULATOR	
Apple Inc.		DRAWING NUMBER	SIZE D
NOTICE OF PROPRIETARY PROPERTY:		REVISION	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:		BRANCH	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		PAGE	62 OF 132
II NOT TO REPRODUCE OR COPY IT		SHEET	56 OF 101
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			



C

B

A

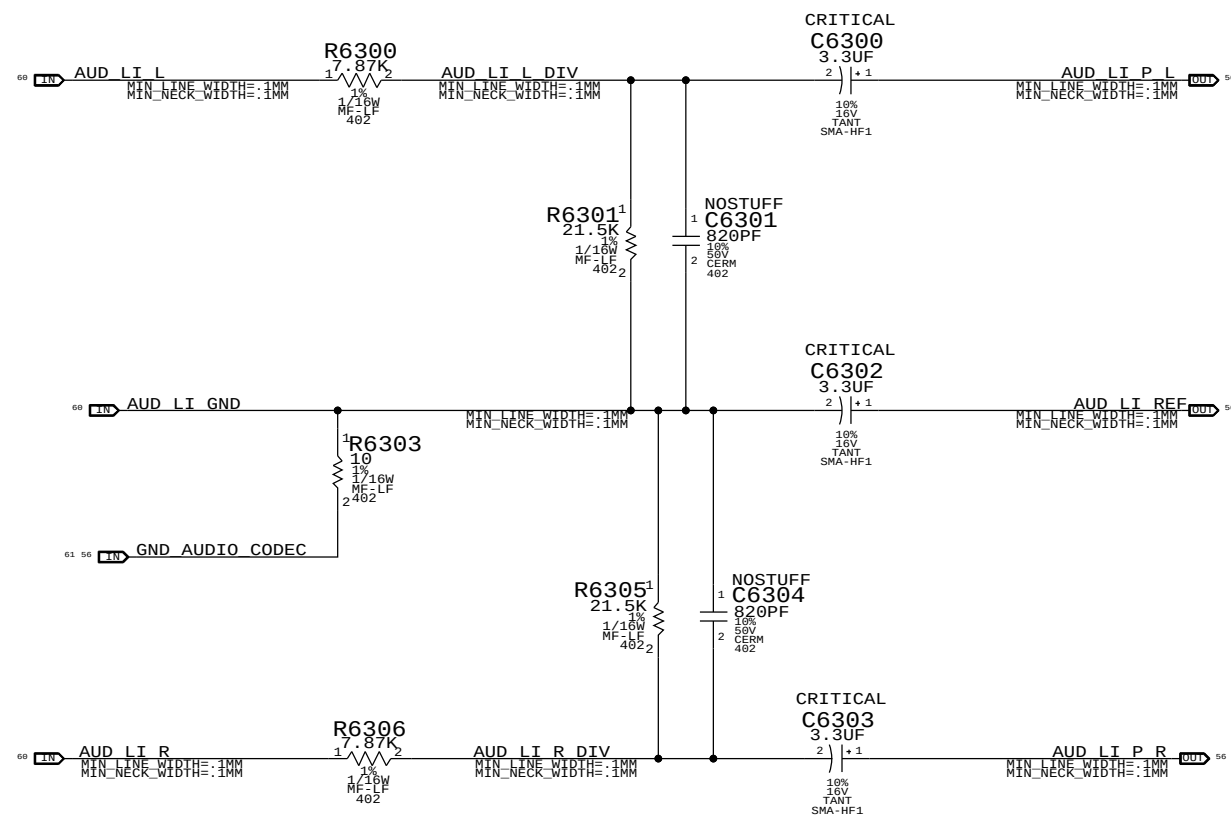
D |

C |

B

A |

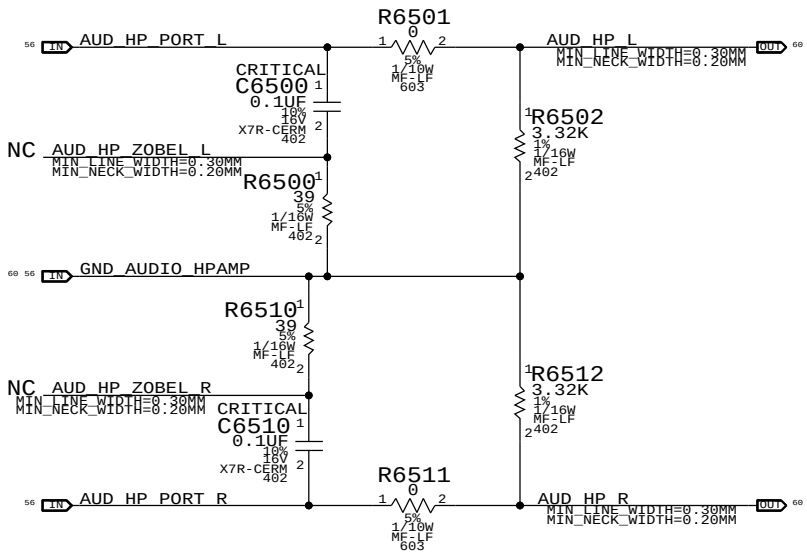
CODEC RIN = 20K OHMS
NET RIN = 18K OHMS
FC = 8 HZ
VIN = 2VRMS, CODEC VIN = 1.14 VRMS



SYNCH MASTER=K91 AUDIO		SYNCH DATE=07/12/2016	
PAGE 1 TITLE			
AUDIO: LINE INPUT FILTER			
 Apple Inc.		DRAWING NUMBER	
		SIZE D	
		REVISION	
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I. TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II. NOT TO REPRODUCE OR COPY IT III. NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV. ALL RIGHTS RESERVED		PAGE 63 OF 132	
		SHEET 57 OF 101	



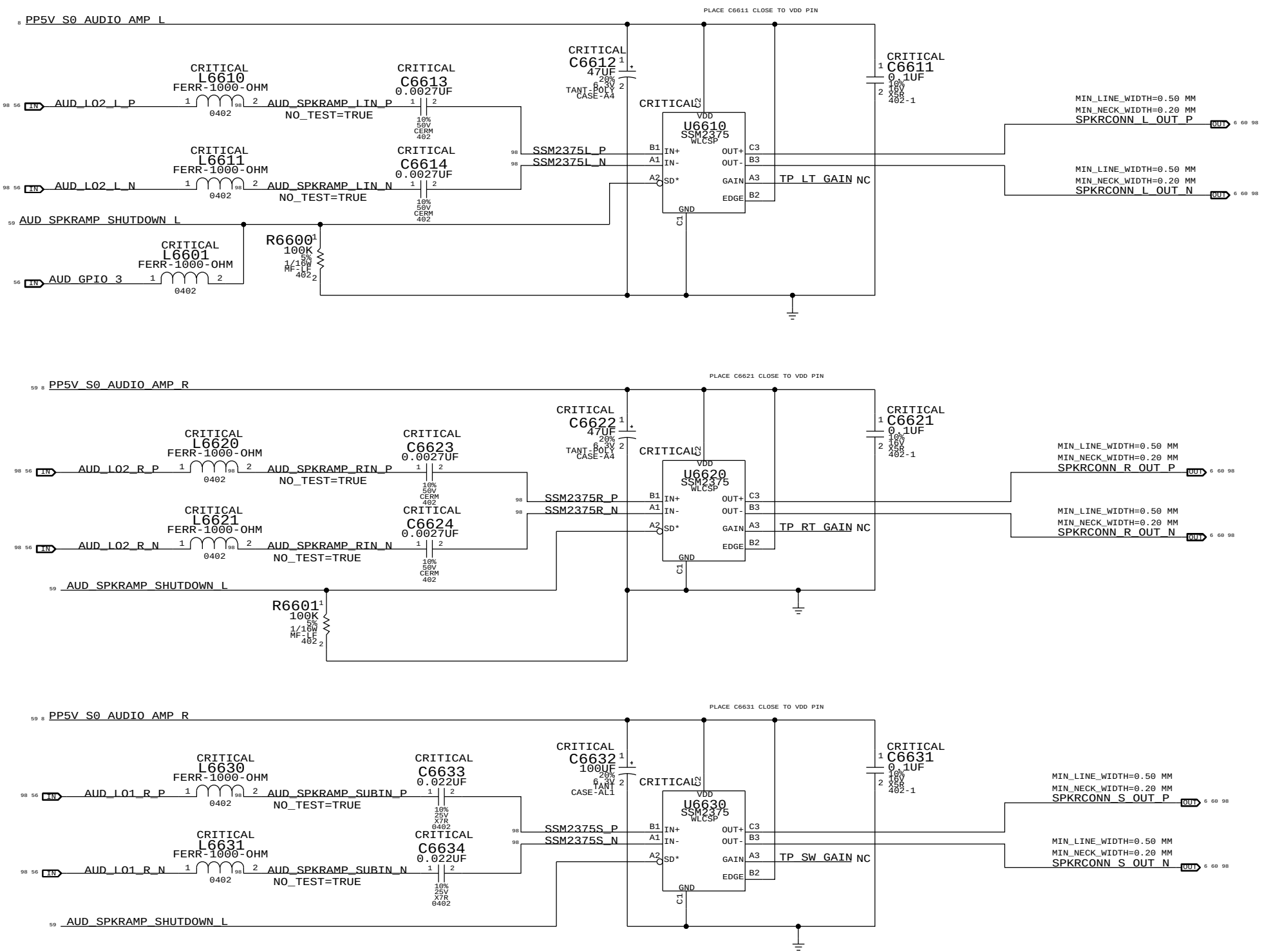
ZOBEL NETWORK & 1ST ORDER DAC FILTER PLACEHOLDER



SYNC_MASTER=K91_AUDIO		SYNC_DATE=07/12/2016	
PAGE_TITLE		AUDIO: HEADPHONE FILTER	
		DRAWING_NUMBER	SIZE
		REVISION	D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		BRANCH	PAGE
			65 OF 132
		SHEET	58 OF 101



3X MONO SPEAKER AMPLIFIERS (SSM2375)
APN: 353S2958
GAIN = +3 DB
1ST ORDER FC (L&R) = ~737 HZ
1ST ORDER FC (SUB) = ~90 HZ



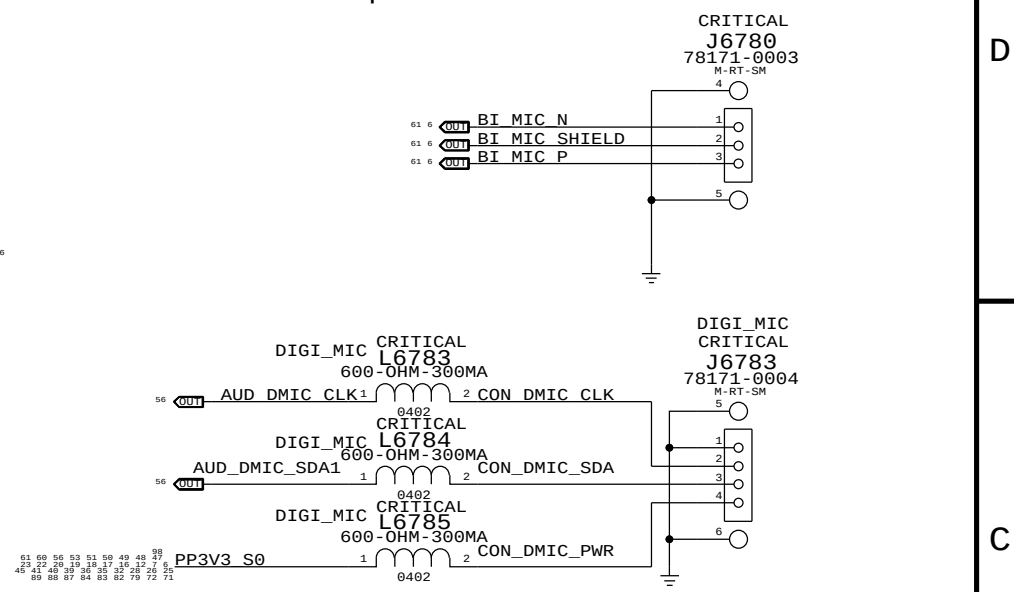
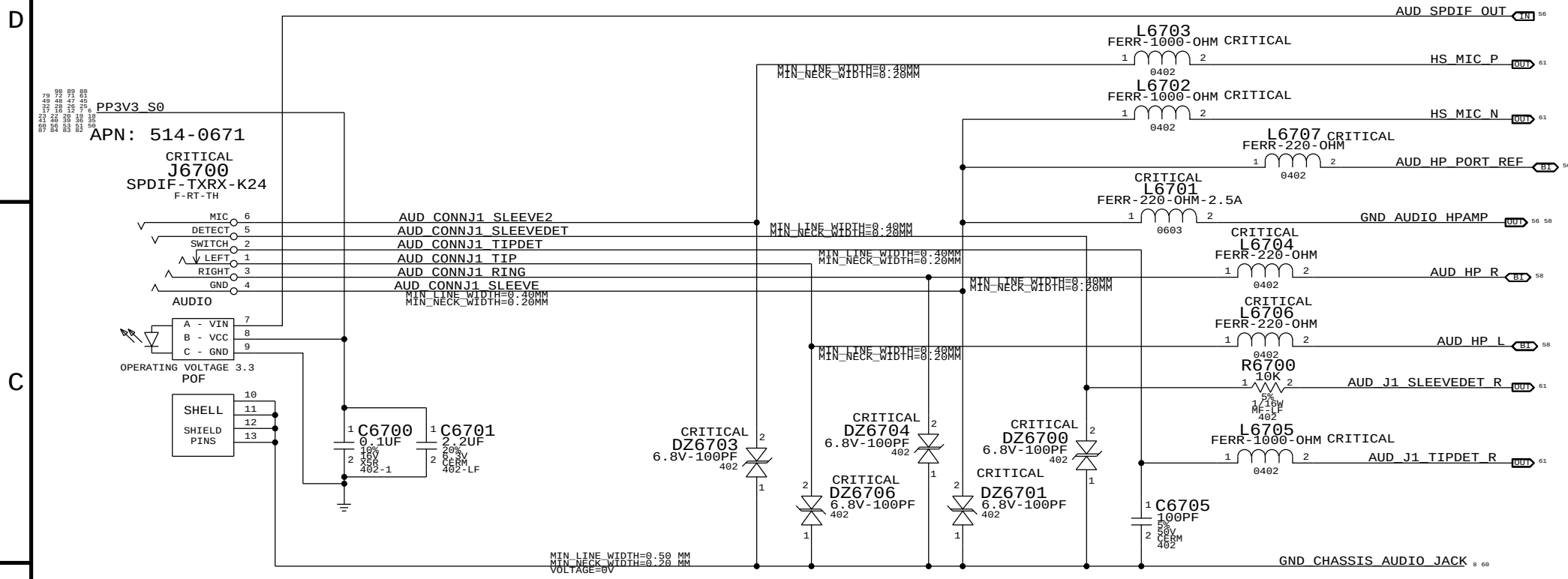
SYNC MASTER=K91 AUDIO		SYNC DATE=07/12/2016	
PAGE TITLE		AUDIO: SPEAKER AMP	
Apple Inc.		DRAWING NUMBER	SIZE D
NOTICE OF PROPRIETARY PROPERTY:		REVISION	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:		BRANCH	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		PAGE	66 OF 132
II NOT TO REPRODUCE OR COPY IT		SHEET	59 OF 101
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			



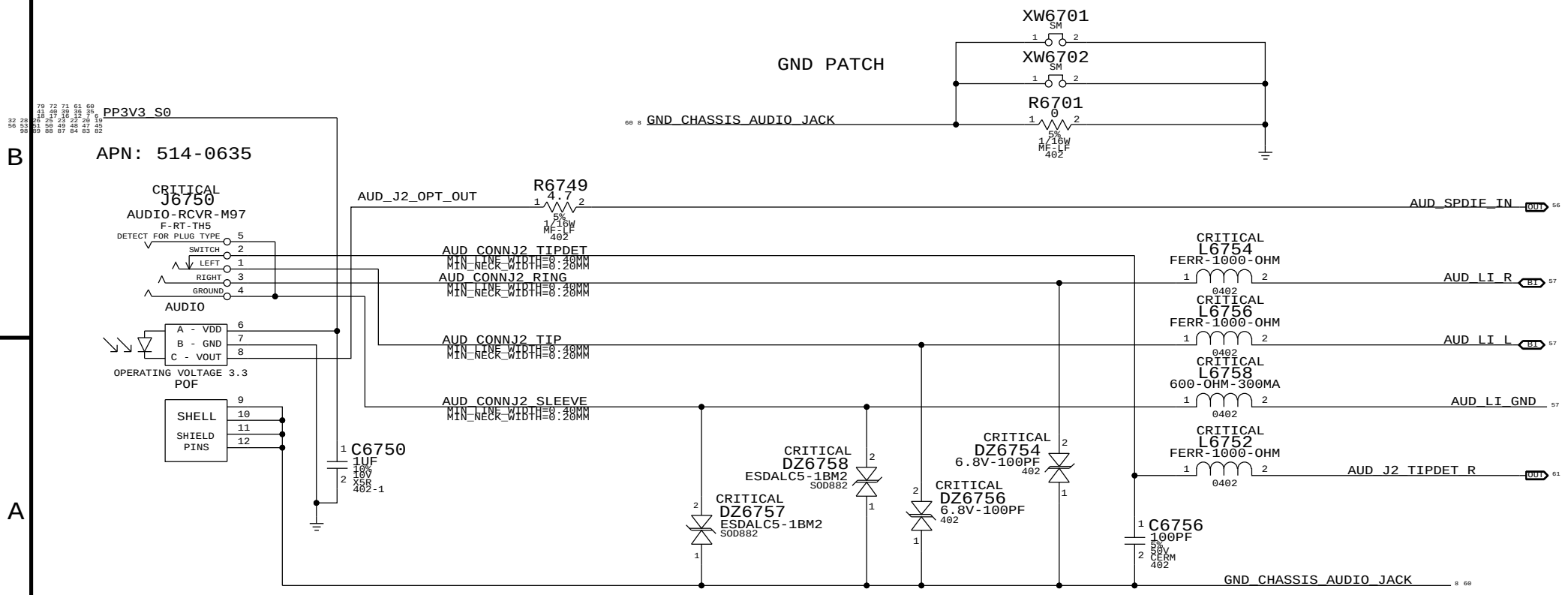
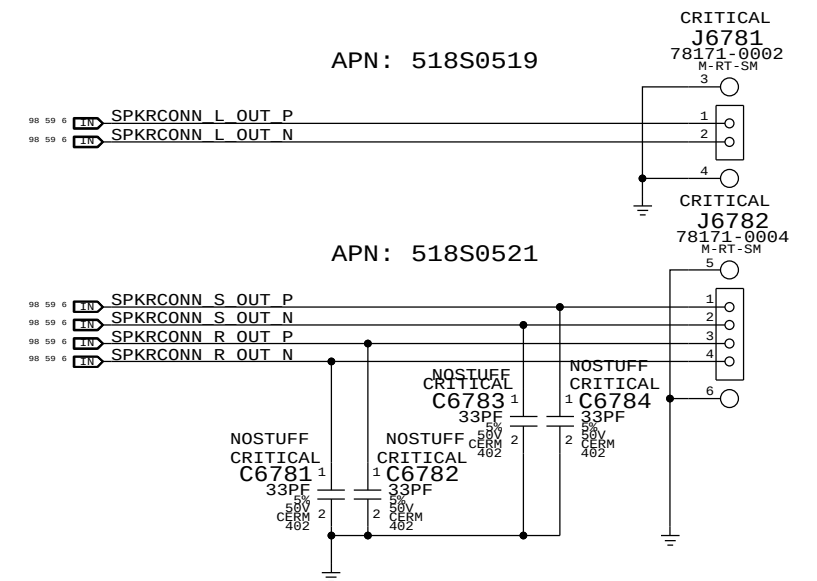
MIC CONNECTOR
Dual DMIC removed.Added single analog mic like K10
Sept 21st 2010

Place this in place of DMIC connector J6780

AUDIO JACK 1 LO/HP JACK, SPDIF TX



SPEAKER CONNECTOR



AUDIO JACK 2 LINE IN JACK, SPDIF RX

SYNC MASTER=K91 AUDIO		SYNC DATE=09/30/2010	
PAGE TITLE		AUDIO: JACKS	
Apple Inc.		DRAWING NUMBER	SIZE D
NOTICE OF PROPRIETARY PROPERTY:		REVISION	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:		BRANCH	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		PAGE	67 OF 132
II NOT TO REPRODUCE OR COPY IT		SHEET	60 OF 101
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			

CODEC OUTPUT SIGNAL PATHS

FUNCTION	VOLUME	CONVERTER	PIN COMPLEX	MUTE CONTROL	DET ASSIGNMENT
HP/LINE OUT	0X02 (2)	0X02 (2)	0X09 (9,A)	N/A	0X09 (A)
SATELLITES	0X04 (4)	0X04 (4)	0X0B (11)	GPIO_3	N/A
SUB	0X03 (3)	0X03 (03)	0X0A (10)	GPIO_3	N/A
SPDIF OUT	N/A	0X08 (8)	0X10 (16)	N/A	0X0C (B)

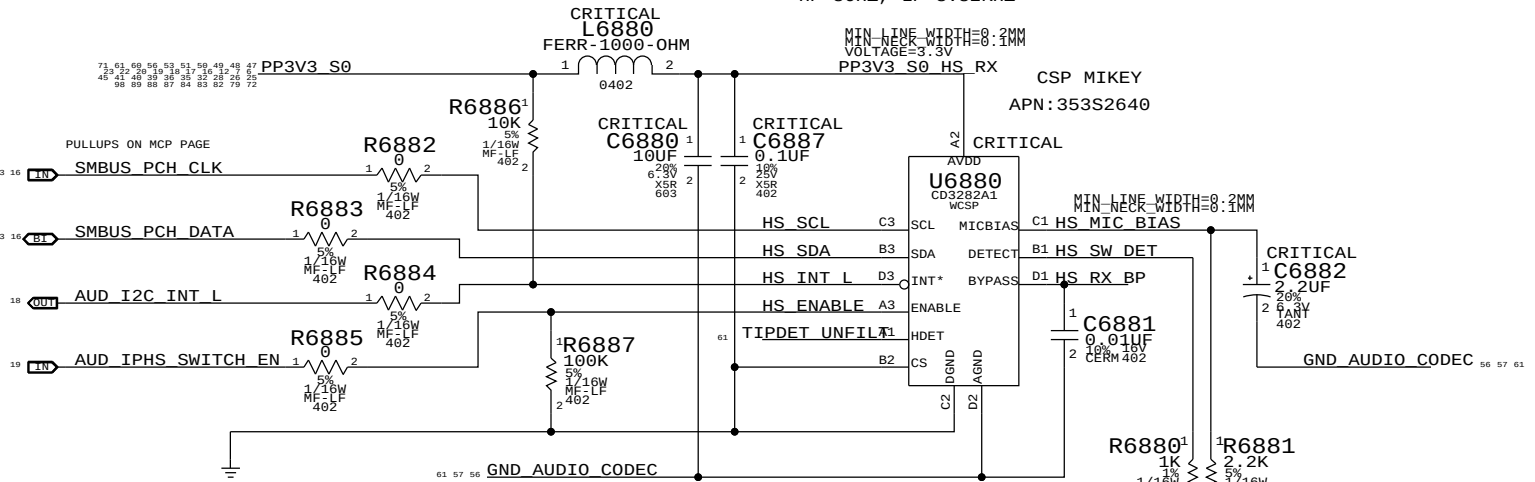
CODEC INPUT SIGNAL PATHS

FUNCTION	CONVERTER	PIN COMPLEX	VREF	DET ASSIGNMENT
LINE IN	0X05 (5)	0X0C (12,C)	N/A	0X0C (12,C)
SPDIF IN	0X07 (7)	0X0F (15)	N/A	N/A
BUILT-IN MIC	0X06 (6)	0X0D (13)	N/A	N/A
HEADSET MIC	0X06 (6)	0X0D (13,V22,B, LEFT)	MIKEY	MIKEY

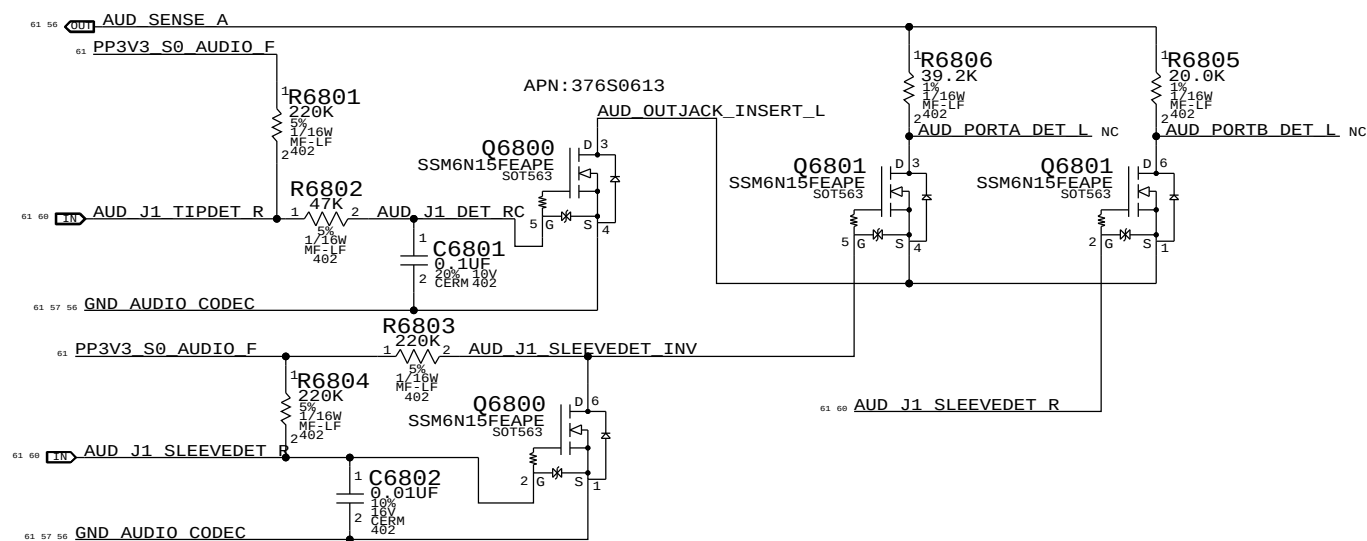
SYSTEM INT AND GPIO LINES

FUNCTION	INT	GPIO
MIKEY ENABLE		SATA4GP/GPIO 16
MIKEY INTERRUPT	PIRQ H	GPIO 5
PERIPHERAL DETECT	PIRQ F	GPIO 3

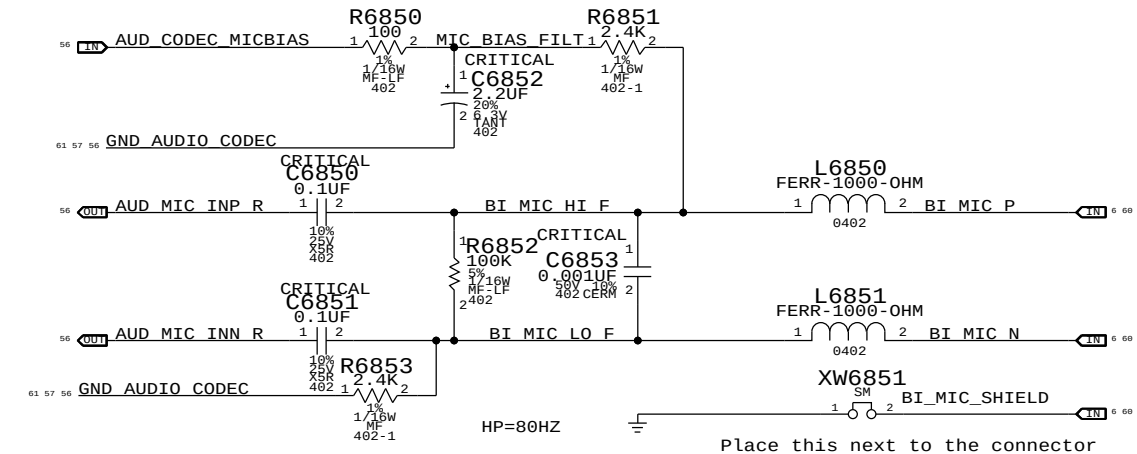
PORT B LEFT(HEADSET MIC)
HP=80HZ, LP=8.82KHZ



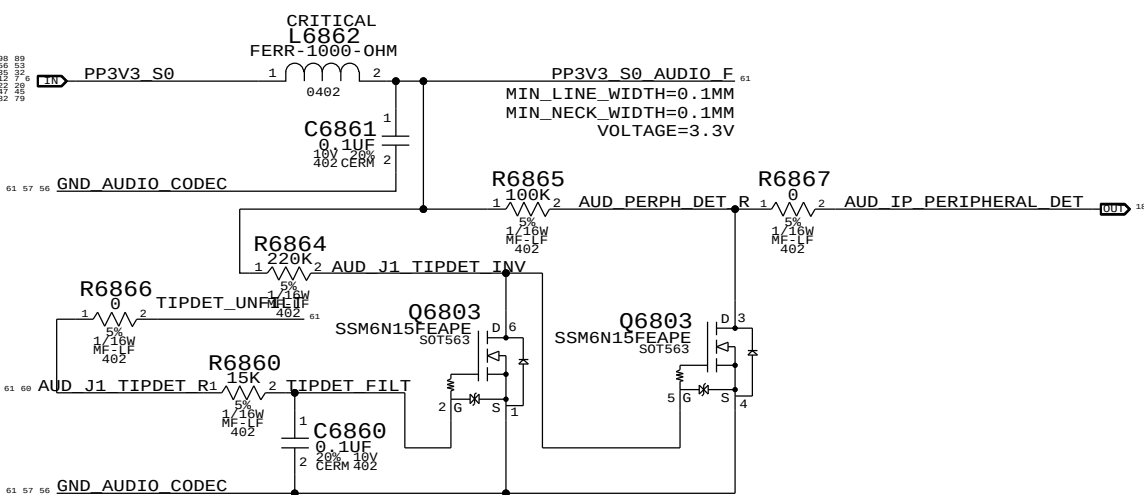
PORT A DETECT (HEADPHONES) PORT B DETECT(SPDIF DELEGATE)



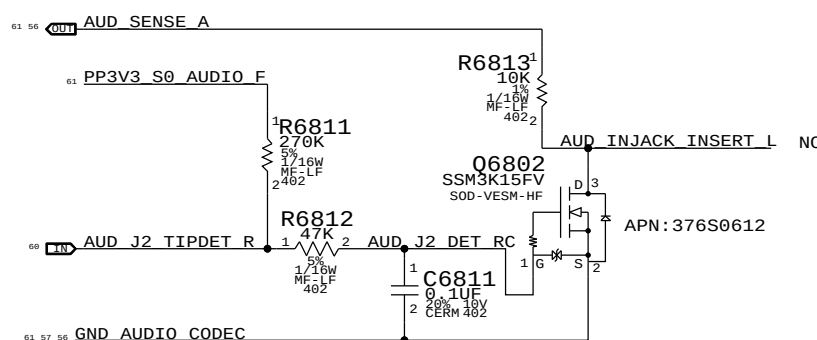
PORT B RIGHT(BUILT-IN MIC)



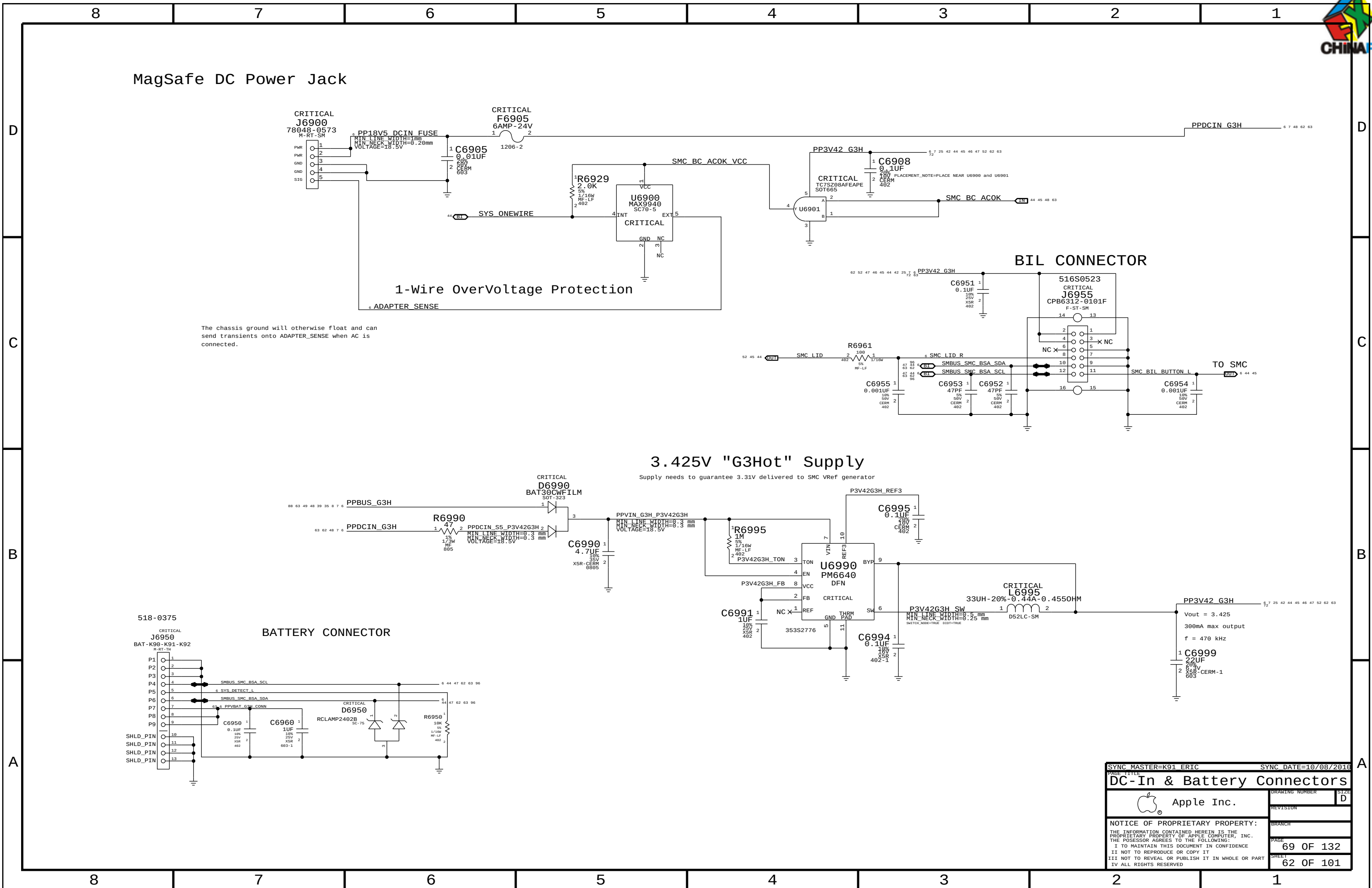
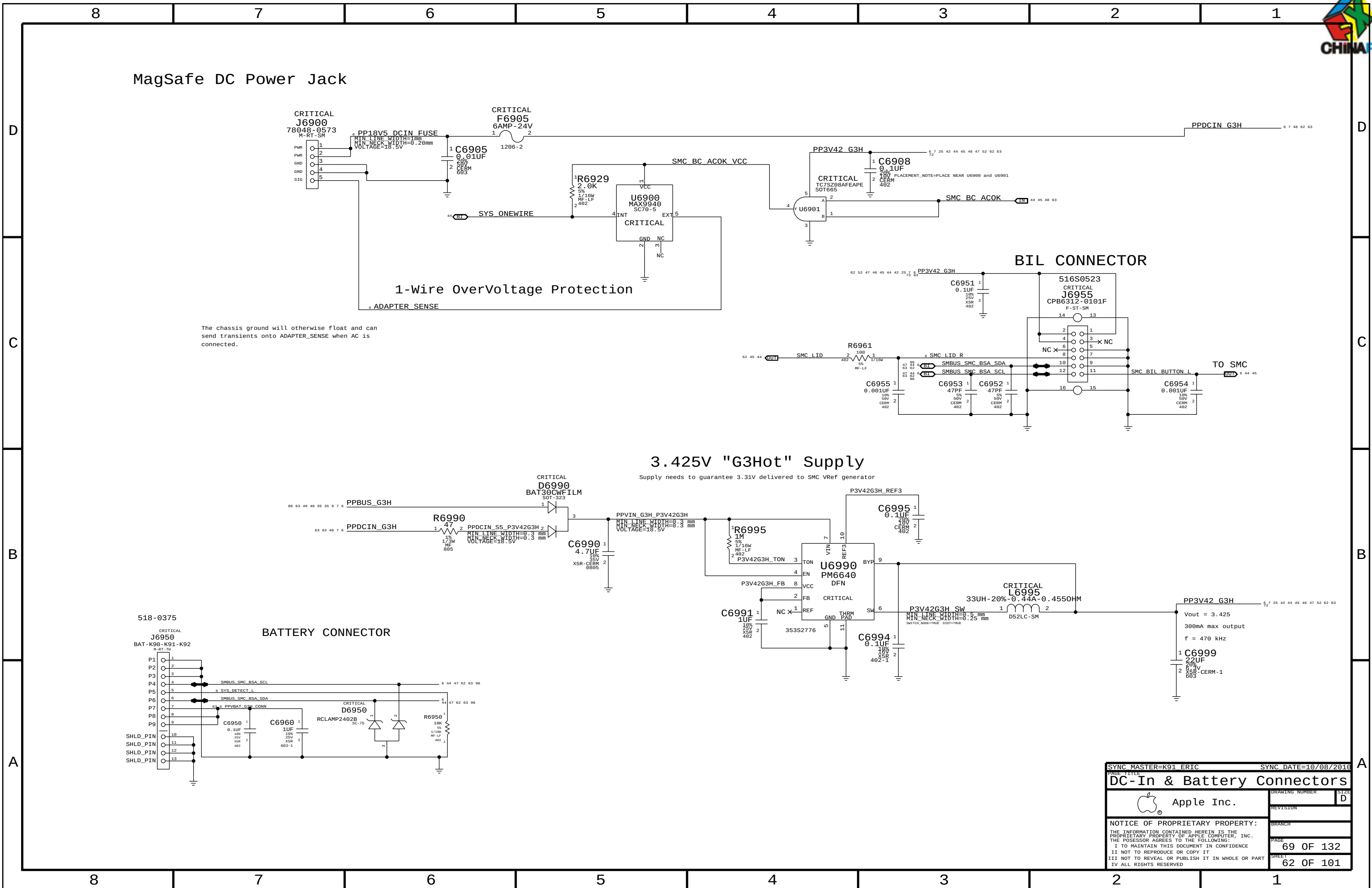
EXTRACTION NOTIFICATION



PORT C DETECT (LINE-IN)



PAGE TITLE		PAGE TITLE	
AUDIO: JACK TRANSLATORS		AUDIO: JACK TRANSLATORS	
Apple Inc.		Apple Inc.	
NOTICE OF PROPRIETARY PROPERTY:		NOTICE OF PROPRIETARY PROPERTY:	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:		THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE	
II NOT TO REPRODUCE OR COPY IT		II NOT TO REPRODUCE OR COPY IT	
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART		III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART	
IV ALL RIGHTS RESERVED		IV ALL RIGHTS RESERVED	

[illegible]

MagSafe DC Power Jack

CRITICAL J6900 78048-0573 M-RT-SM

PP18V5 DCIN FUSE

CRITICAL F6905 6AMP-24V

C6905 0.01UF

R6929 2.0K

U6900 MAX9940 SC70-5

SMC BC ACOK VCC

PP3V42 G3H

C6908 0.1UF

CRITICAL TC7SZ88AFAPE SOT665

SMC BC ACOK

1-Wire OverVoltage Protection

ADAPTER_SENSE

PPDCIN G3H

PP3V42 G3H

C6951 0.1UF

516S0523

CRITICAL J6955 CPB6312-0101F F-ST-SM

SMC BIL BUTTON L

TO SMC

C6954 0.001UF

3.425V "G3Hot" Supply

Supply needs to guarantee 3.31V delivered to SMC VRef generator

PPBUS_G3H

PPDCIN_G3H

R6990

PPVIN_G3H_P3V42G3H

C6990 4.7UF

U6990 PM6640 DFN

P3V42G3H_TON

P3V42G3H_FB

P3V42G3H_SW

C6991 1UF

C6994 0.1UF

PP3V42 G3H

Vout = 3.425

300mA max output

f = 470 kHz

C6999 22UF

CRITICAL D6950 BAT-K99-K91-K92 M-RT-TM

SHLD_PIN

SHLD_PIN

SHLD_PIN

SHLD_PIN

BATTERY CONNECTOR

CRITICAL D6950

RCLAMP2402B

R6950 10K

C6950 0.1UF

C6960 1UF

518-0375

SYNC MASTER=K91 ERIC	SYNC DATE=10/08/2010
DC-In & Battery Connectors	
Apple Inc.	DRAWING NUMBER D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED	
PAGE 69 OF 132	SHEET 62 OF 101

MagSafe DC Power Jack

CRITICAL J6900 78048-0573 M-RT-SM

PP18V5 DCIN FUSE

CRITICAL F6905 6AMP-24V

C6905 0.01UF

R6929 2.0K

U6900 MAX9940 SC70-5

SMC BC ACOK VCC

PP3V42 G3H

C6908 0.1UF

CRITICAL TC7SZ88AFAPE SOT665

SMC BC ACOK

1-Wire OverVoltage Protection

ADAPTER_SENSE

PPDCIN G3H

PP3V42 G3H

C6951 0.1UF

516S0523

CRITICAL J6955 CPB6312-0101F F-ST-SM

SMC BIL BUTTON L

TO SMC

C6954 0.001UF

3.425V "G3Hot" Supply

Supply needs to guarantee 3.31V delivered to SMC VRef generator

PPBUS_G3H

PPDCIN_G3H

R6990

PPVIN_G3H_P3V42G3H

C6990 4.7UF

U6990 PM6640 DFN

P3V42G3H_TON

P3V42G3H_FB

P3V42G3H_SW

C6991 1UF

C6994 0.1UF

PP3V42 G3H

Vout = 3.425

300mA max output

f = 470 kHz

C6999 22UF

CRITICAL D6950 BAT-K99-K91-K92 M-RT-TM

SHLD_PIN

SHLD_PIN

SHLD_PIN

SHLD_PIN

BATTERY CONNECTOR

CRITICAL D6950

RCLAMP2402B

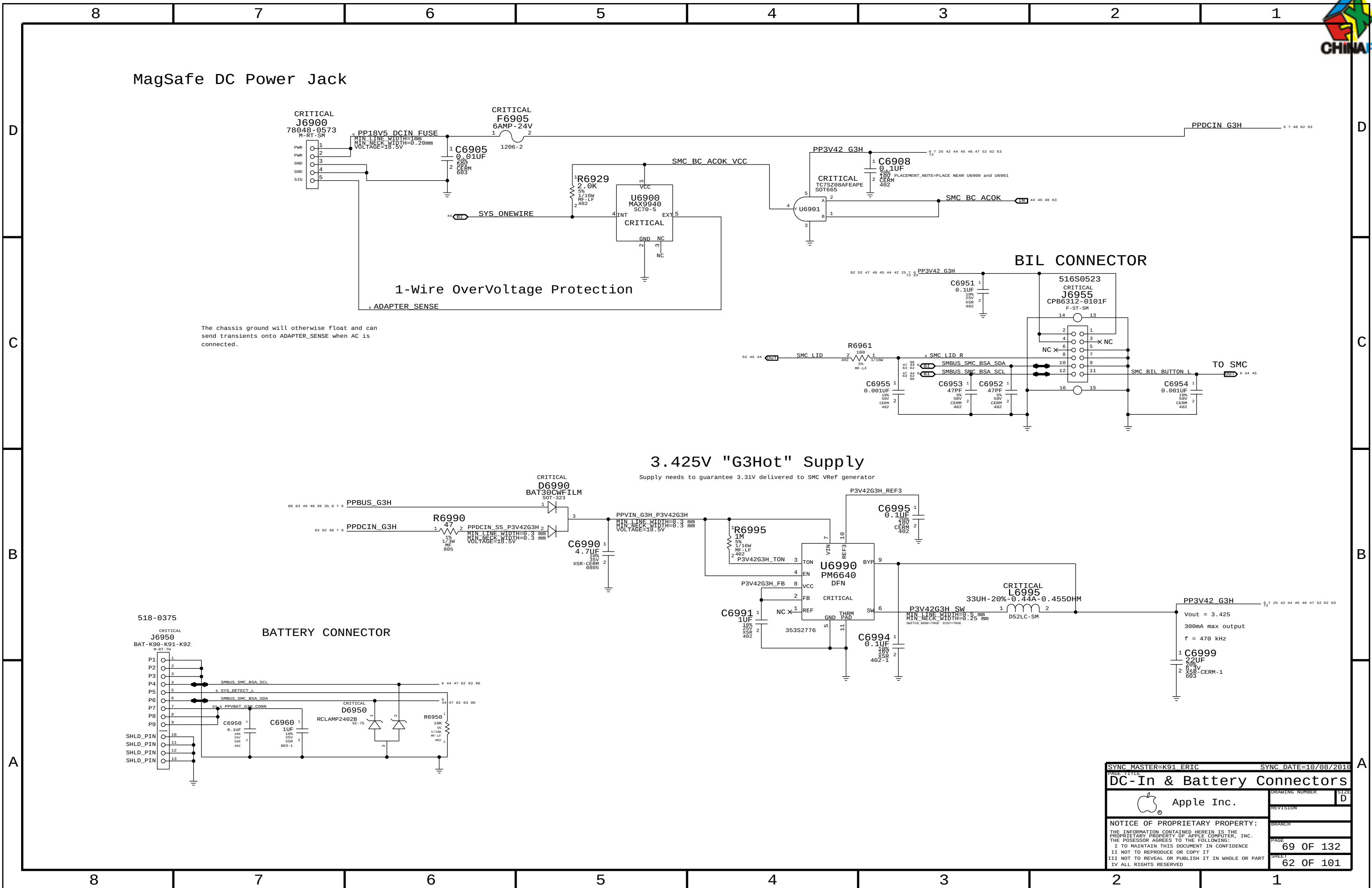
R6950 10K

C6950 0.1UF

C6960 1UF

518-0375

SYNC MASTER=K91 ERIC	SYNC DATE=10/08/2010
DC-In & Battery Connectors	
Apple Inc.	DRAWING NUMBER D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED	
PAGE 69 OF 132	SHEET 62 OF 101



MagSafe DC Power Jack

CRITICAL J6900 78048-0573 M-RT-SM

PP18V5 DCIN FUSE

CRITICAL F6905 6AMP-24V

C6905 0.01UF

R6929 2.0K

U6900 MAX9940 SC70-5

SMC BC ACOK VCC

PP3V42 G3H

C6908 0.1UF

CRITICAL TC7SZ88AFAPE SOT665

SMC BC ACOK

1-Wire OverVoltage Protection

ADAPTER_SENSE

PPDCIN G3H

PP3V42 G3H

C6951 0.1UF

516S0523

CRITICAL J6955 CPB6312-0101F F-ST-SM

SMC BIL BUTTON L

TO SMC

C6954 0.001UF

3.425V "G3Hot" Supply

Supply needs to guarantee 3.31V delivered to SMC VRef generator

PPBUS_G3H

PPDCIN_G3H

R6990

PPVIN_G3H_P3V42G3H

C6990 4.7UF

U6990 PM6640 DFN

P3V42G3H_TON

P3V42G3H_FB

P3V42G3H_SW

C6991 1UF

C6994 0.1UF

PP3V42 G3H

Vout = 3.425

300mA max output

f = 470 kHz

C6999 22UF

CRITICAL D6950 BAT-K99-K91-K92 M-RT-TM

SHLD_PIN

SHLD_PIN

SHLD_PIN

SHLD_PIN

BATTERY CONNECTOR

CRITICAL D6950

RCLAMP2402B

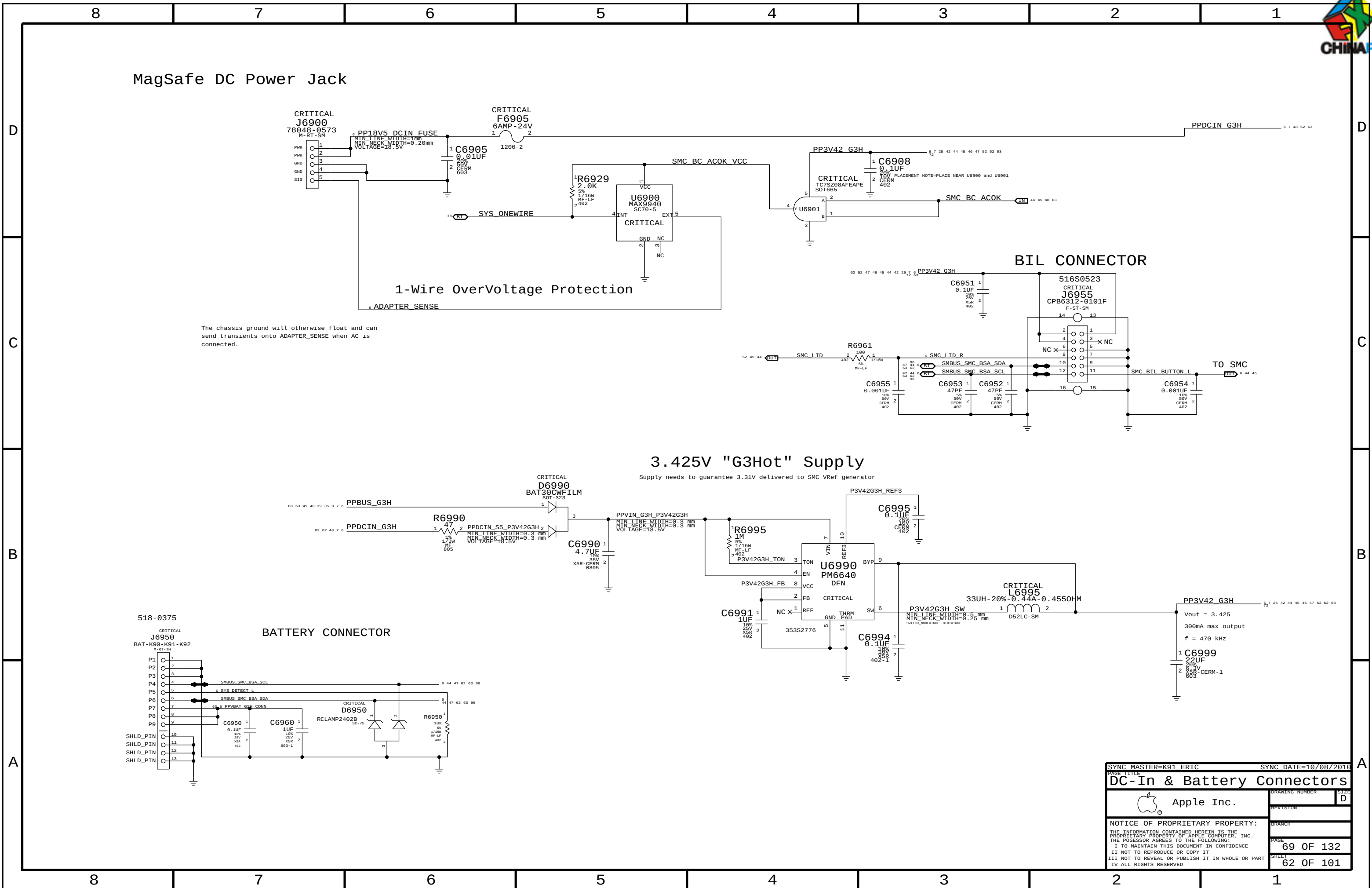
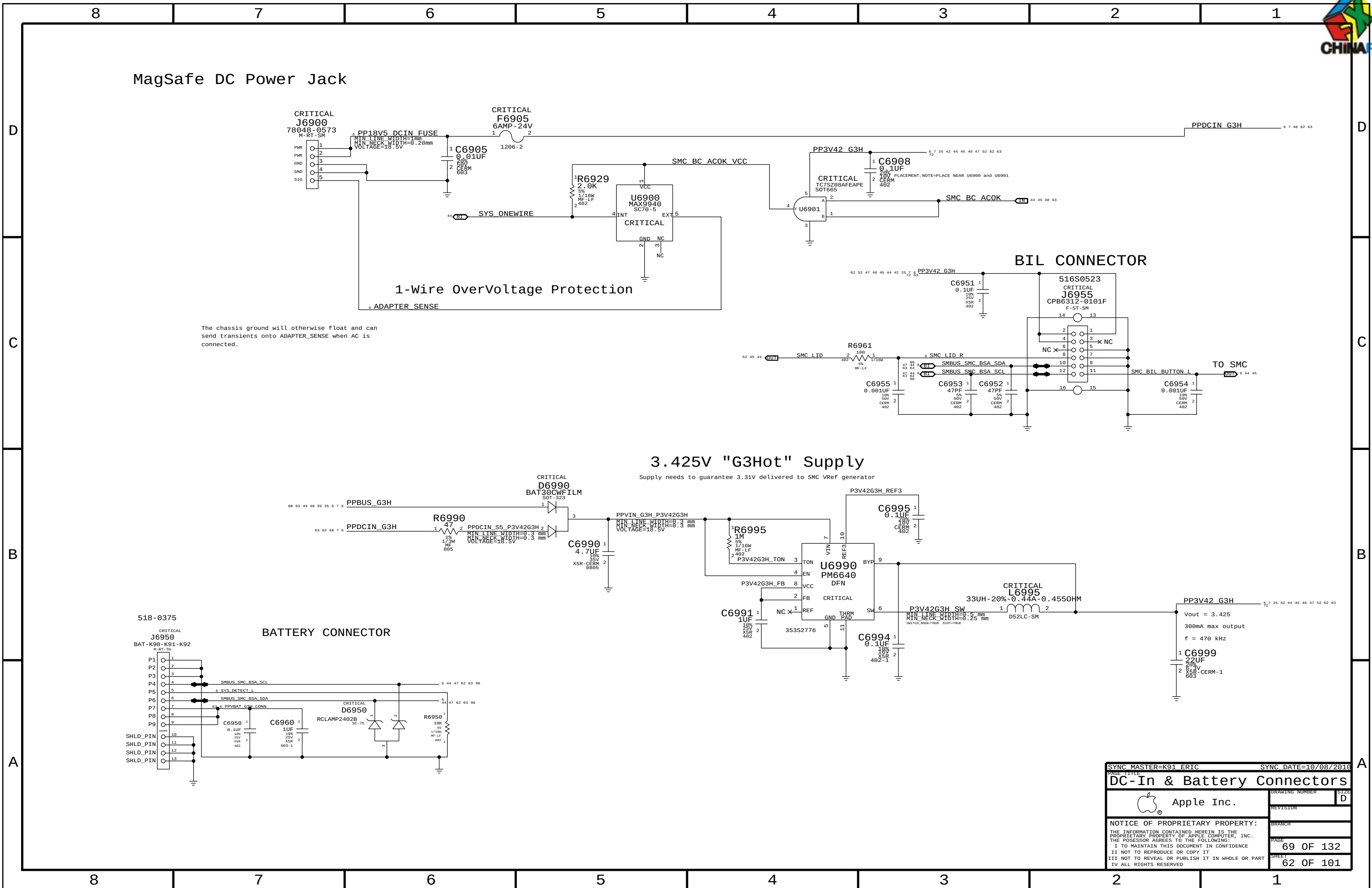
R6950 10K

C6950 0.1UF

C6960 1UF

518-0375

SYNC MASTER=K91 ERIC	SYNC DATE=10/08/2010
DC-In & Battery Connectors	
Apple Inc.	DRAWING NUMBER D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED	
PAGE 69 OF 132	SHEET 62 OF 101

[illegible][illegible]

MagSafe DC Power Jack

CRITICAL J6900 78048-0573 M-RT-SM

PP18V5 DCIN FUSE

CRITICAL F6905 6AMP-24V

C6905 0.01UF

R6929 2.0K

U6900 MAX9940 SC70-5

SMC BC ACOK VCC

PP3V42 G3H

C6908 0.1UF

CRITICAL TC7S288AFAPE SOT665

SMC BC ACOK

1-Wire OverVoltage Protection

ADAPTER_SENSE

PPDCIN G3H

PP3V42 G3H

C6951 0.1UF

516S0523

CRITICAL J6955 CPB6312-0101F F-ST-SM

SMC BIL BUTTON L

TO SMC

C6954 0.001UF

3.425V "G3Hot" Supply

Supply needs to guarantee 3.31V delivered to SMC VRef generator

PPBUS_G3H

PPDCIN_G3H

R6990

PPVIN_G3H_P3V42G3H

C6990 4.7UF

R6995

U6990 PM6640 DFN

P3V42G3H_TON

P3V42G3H_FB

P3V42G3H_SW

C6991 1UF

C6994 0.1UF

C6995 0.1UF

33UH-20%-0.44A-0.4550HM

PP3V42 G3H

Vout = 3.425

300mA max output

f = 470 kHz

C6999 22UF

BATTERY CONNECTOR

518-0375

CRITICAL J6950 BAT-K99-K91-K92 M-RT-TM

SHLD_PIN

SHLD_PIN

SHLD_PIN

SHLD_PIN

SMBUS_SMC_BSA_SCL

SYS_DETECT_L

SMBUS_SMC_BSA_SDA

PPVBAT_G3H_CONN

C6950 0.1UF

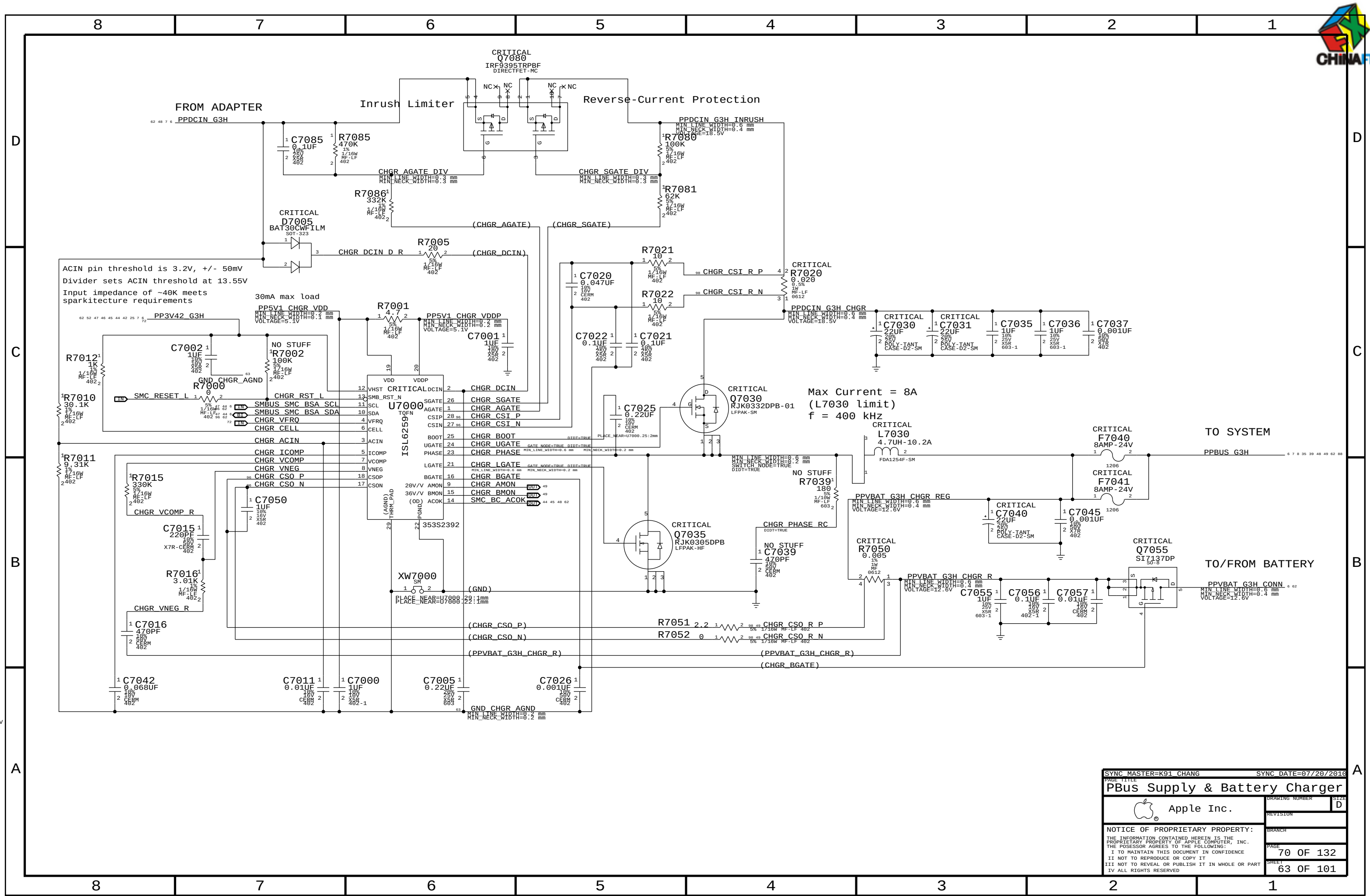
C6960 1UF

CRITICAL D6950 RCLAMP2402B SC-75

R6950 10K

518-0375

SYNC MASTER=K91 ERIC	SYNC DATE=10/08/2010
DC-In & Battery Connectors	
Apple Inc.	DRAWING NUMBER D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED	
PAGE 69 OF 132	SHEET 62 OF 101



SYNC MASTER=K91 CHANG		SYNC DATE=07/20/2016	
PAGE TITLE			
PBus Supply & Battery Charger			
 Apple Inc.		DRAWING NUMBER	SIZE
			D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		REVISION	
		BRANCH	
		PAGE	70 OF 132
		SHEET	63 OF 101



C

B

A



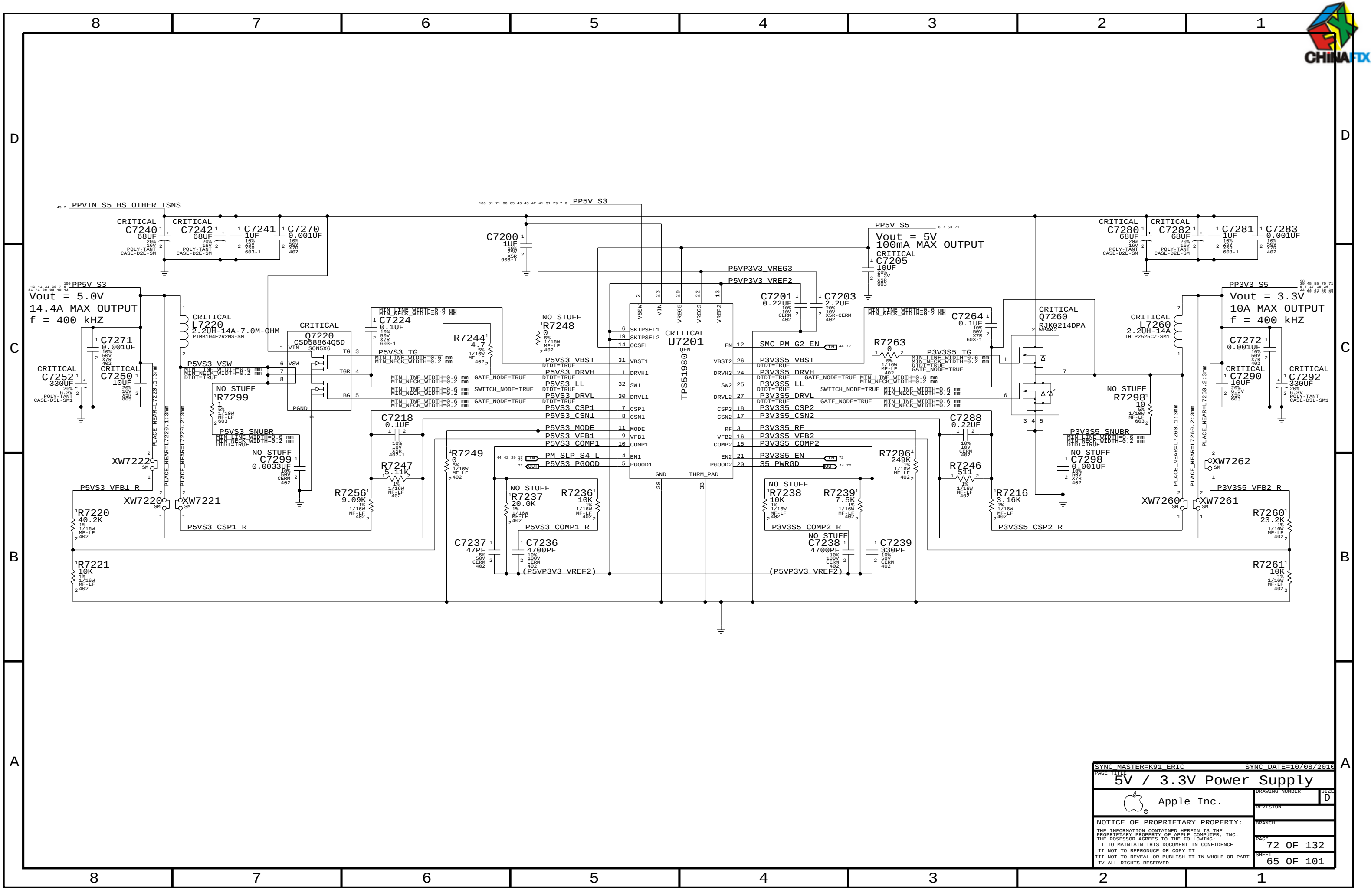
C

B

A |

1
$$\begin{aligned} \text{OCP} &= \text{R7141} \times 8.5\mu\text{A} / \text{R7140} \\ \text{OCP} &= 8.5\text{A} \end{aligned}$$

SYNC MASTER=K91 ERIC		SYNC DATE=10/08/2016	
PAGE TITLE			
System Agent Supply			
	Apple Inc.		DRAWING NUMBER
			SIZE
		REVISION	
NOTICE OF PROPRIETARY PROPERTY:			BRANCH
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:			PAGE
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE I NOT TO REPRODUCE OR COPY IT I NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART I ALL RIGHTS RESERVED			71 OF 132
			SHEET
			64 OF 101





C

B

A

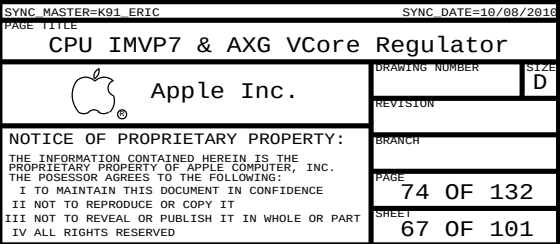


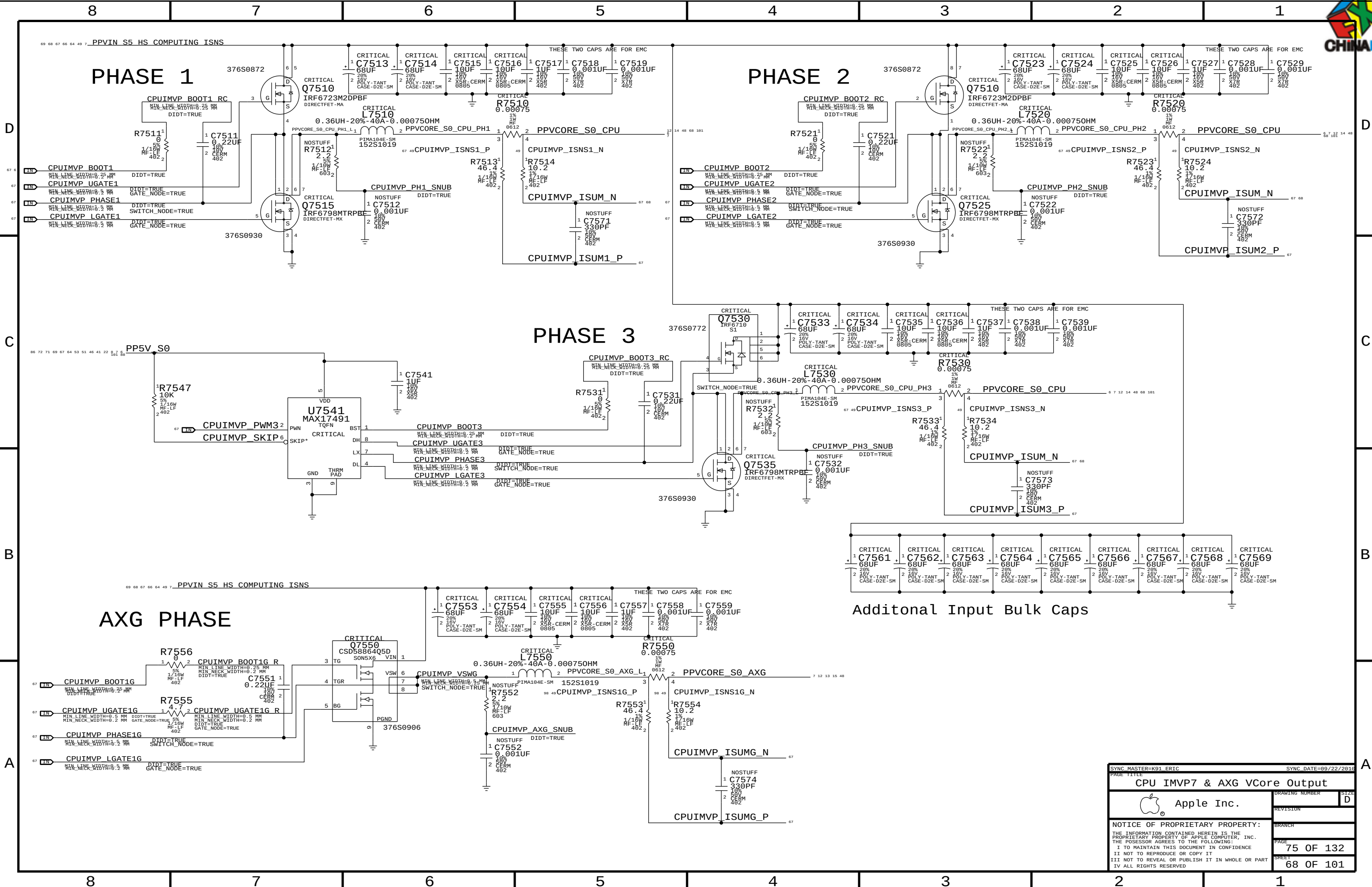
C

3

A |

SYNC_MASTER=K91_ERIC		SYNC_DATE=10/08/2010	
PAGE TITLE			
1.5V DDR3 Supply			
 Apple Inc.		DRAWING NUMBER	SIZE
		REVISION	D
BRANCH			
NOTICE OF PROPRIETARY PROPERTY:			
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:			
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE			
I TO NOT REPRODUCE OR COPY IT			
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			
PAGE		73 OF 132	
SHEET		66 OF 101	

A |

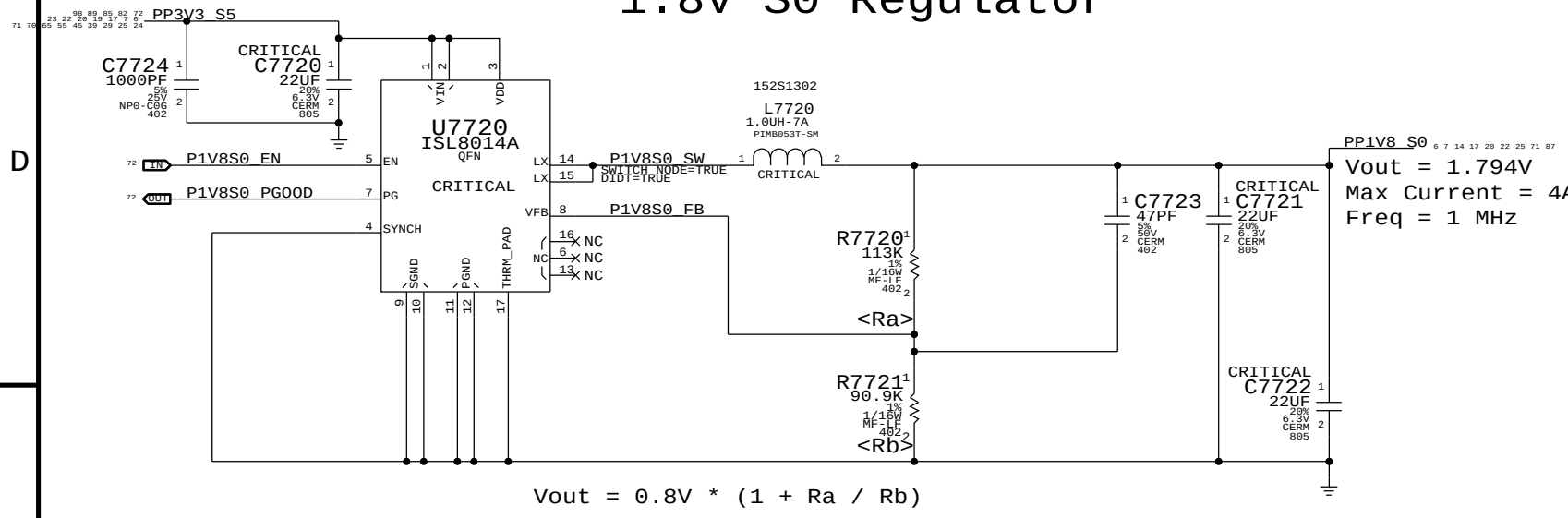


SYNC MASTER=K91 ERIC		SYNC DATE=89/22/2810	
PAGE TITLE		DRAWING NUMBER	
CPU IMVP7 & AXG VCore Output		SIZE	
Apple Inc.		REVISION	
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		75 OF 132	
II NOT TO REPRODUCE OR COPY IT		SHEET	
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART		68 OF 101	
IV ALL RIGHTS RESERVED			

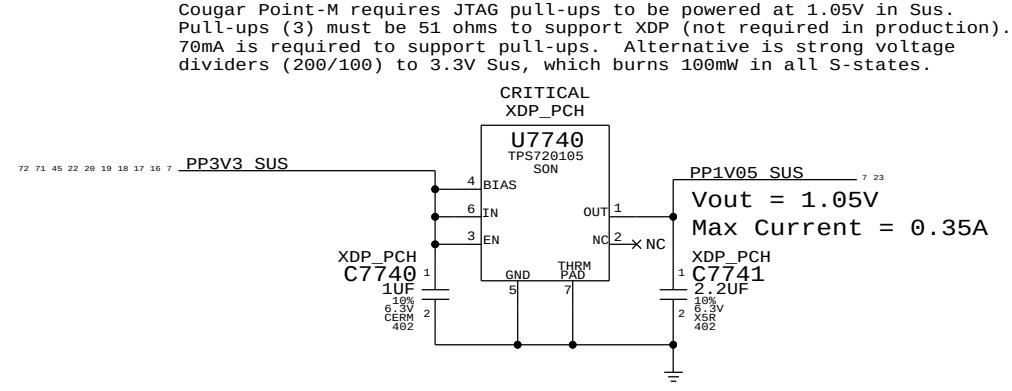
SYMC MASTER-K91 ERIC		SYNC DATE=10/08/2016	
PAGE TITLE			
CPU VCCIO (1.05V) Power Supply			
	Apple Inc.		DRAWING NUMBER
			REVISION
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE I NOT TO REPRODUCE OR COPY IT I NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART I ALL RIGHTS RESERVED		PAGE 76 OF 132	
		SHEET 69 OF 101	



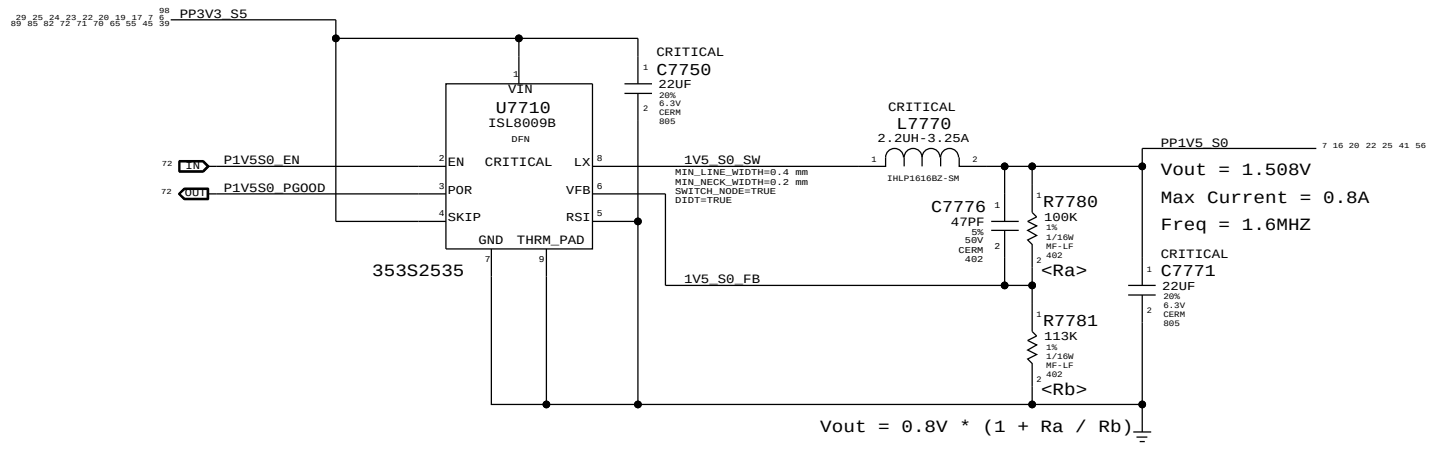
1.8V S0 Regulator



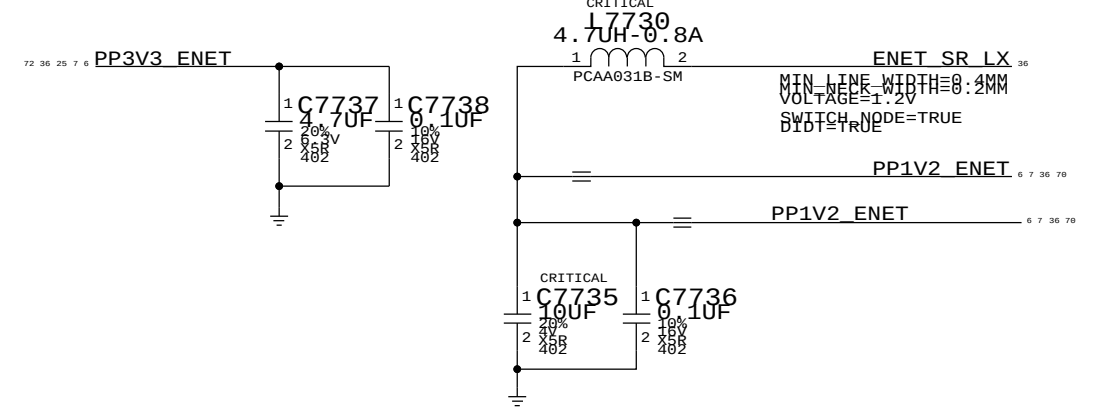
1.05V SUS LDO



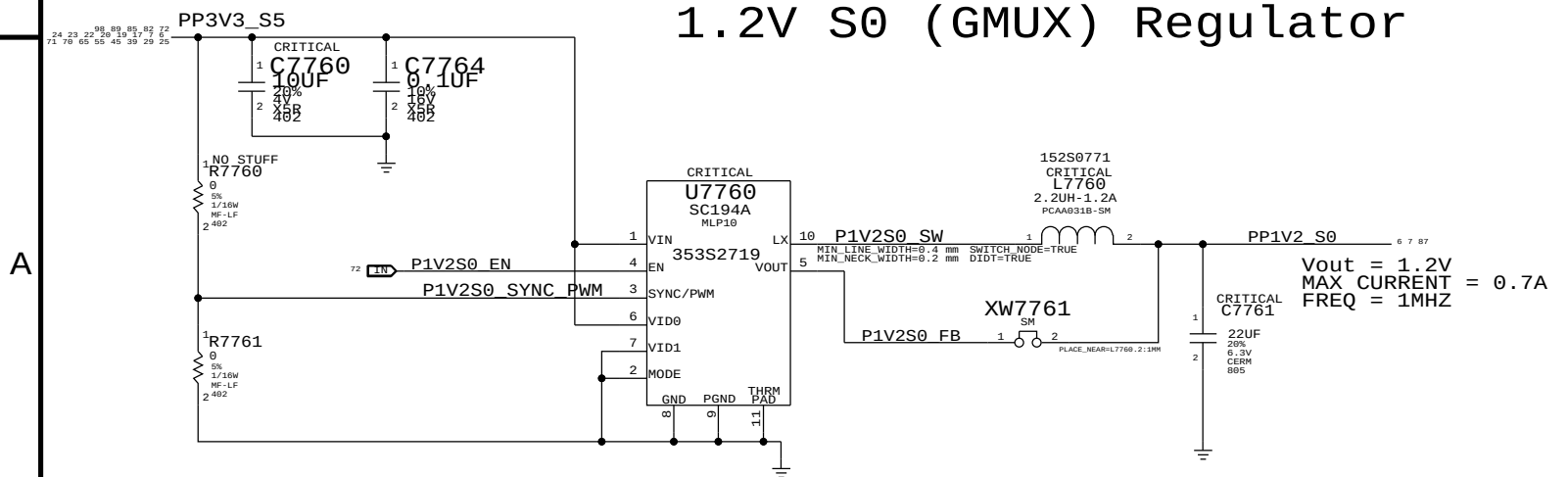
1.5V S0 Regulator



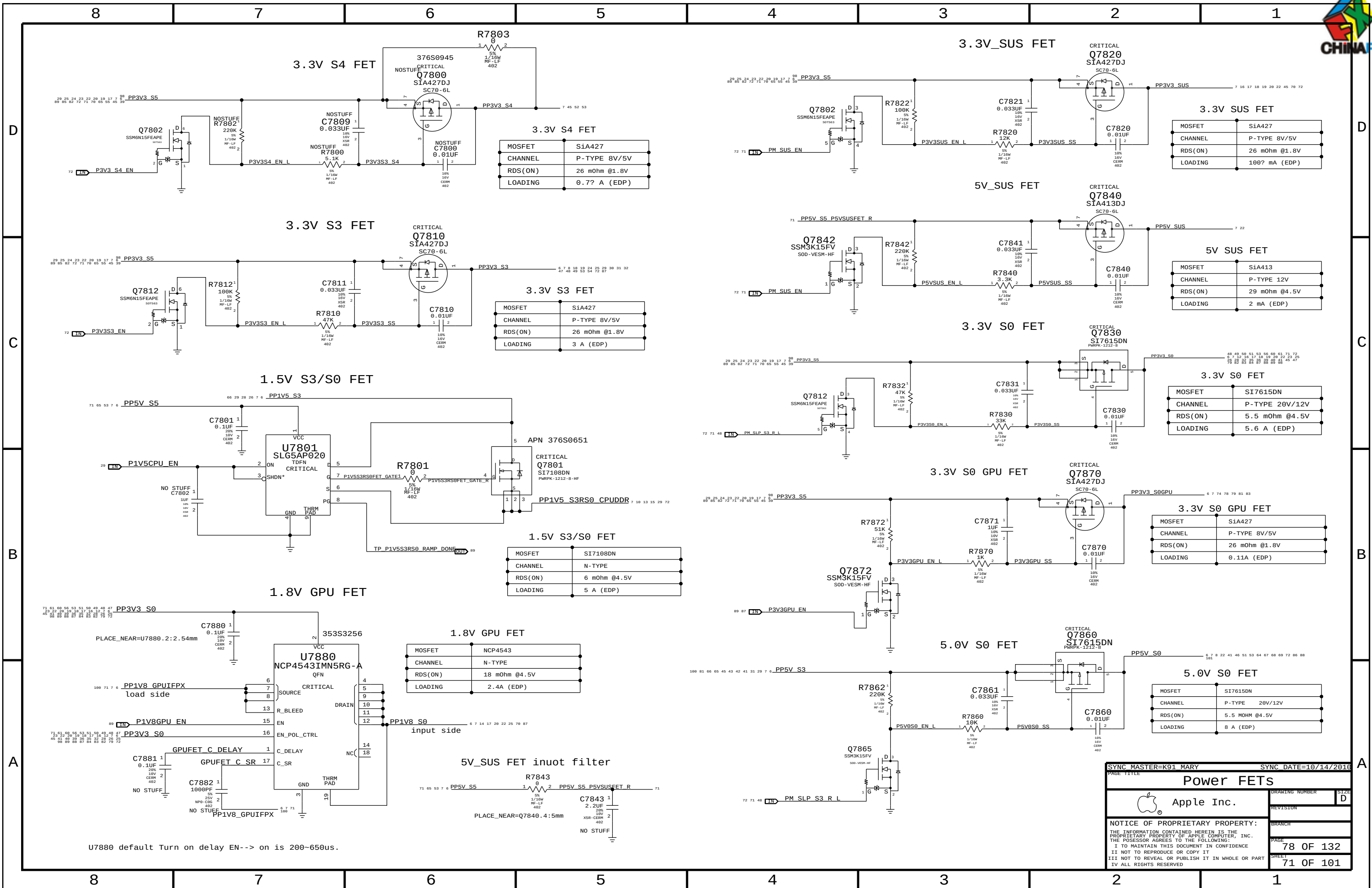
CAESAR IV 1.2V INT.VR CMPTS



1.2V S0 (GMUX) Regulator



SYNC MASTER=K91 ERIC		SYNC DATE=11/01/2016	
PAGE TITLE		Misc Power Supplies	
Apple Inc.		DRAWING NUMBER	SIZE
NOTICE OF PROPRIETARY PROPERTY:		REVISION	BRANCH
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE	77 OF 132
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		SHEET	70 OF 101
II NOT TO REPRODUCE OR COPY IT			
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			



MOSFET	SiA427
CHANNEL	P-TYPE 8V/5V
RDS(ON)	26 mOhm @1.8V
LOADING	0.7? A (EDP)

MOSFET	SiA427
CHANNEL	P-TYPE 8V/5V
RDS(ON)	26 mOhm @1.8V
LOADING	3 A (EDP)

MOSFET	SI7108DN
CHANNEL	N-TYPE
RDS(ON)	6 mOhm @4.5V
LOADING	5 A (EDP)

MOSFET	NCP4543
CHANNEL	N-TYPE
RDS(ON)	18 mOhm @4.5V
LOADING	2.4A (EDP)

3.3V_SUS FET

MOSFET	SiA427
CHANNEL	P-TYPE 8V/5V
RDS(ON)	26 mOhm @1.8V
LOADING	100? mA (EDP)

5V_SUS FET

MOSFET	SiA413
CHANNEL	P-TYPE 12V
RDS(ON)	29 mOhm @4.5V
LOADING	2 mA (EDP)

3.3V_S0 FET

MOSFET	SI7615DN
CHANNEL	P-TYPE 20V/12V
RDS(ON)	5.5 mOhm @4.5V
LOADING	5.6 A (EDP)

3.3V_S0 GPU FET

MOSFET	SiA427
CHANNEL	P-TYPE 8V/5V
RDS(ON)	26 mOhm @1.8V
LOADING	0.11A (EDP)

5.0V_S0 FET

MOSFET	SI7615DN
CHANNEL	P-TYPE 20V/12V
RDS(ON)	5.5 MOHM @4.5V
LOADING	8 A (EDP)

5V_SUS FET inuot filter

SYNC MASTER=K91 MARY

SYNC DATE=10/14/2016

Power FETs

Apple Inc.

NOTICE OF PROPRIETARY PROPERTY:

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:

I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE

II NOT TO REPRODUCE OR COPY IT

III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART

IV ALL RIGHTS RESERVED

DRAWING NUMBER

REVISION

BRANCH

PAGE

SHEET

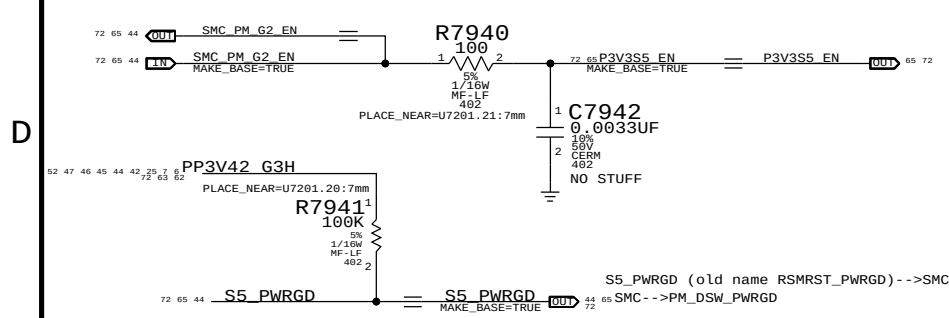
78 OF 132

71 OF 101

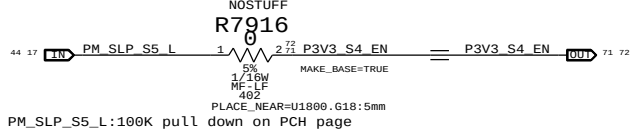


State	SMC_PM_G2_ENABLE	PM_SLP_S5_L	PM_SLP_S4_L	PM_SLP_S3_L
Run (S0)	1	1	1	1
Sleep (S3)	1	1	0	0
Deep Sleep (S4)	1	1	0	0
Deep Sleep (S5)	1	0	0	0
Battery Off (G3Hot)	0	0	0	0

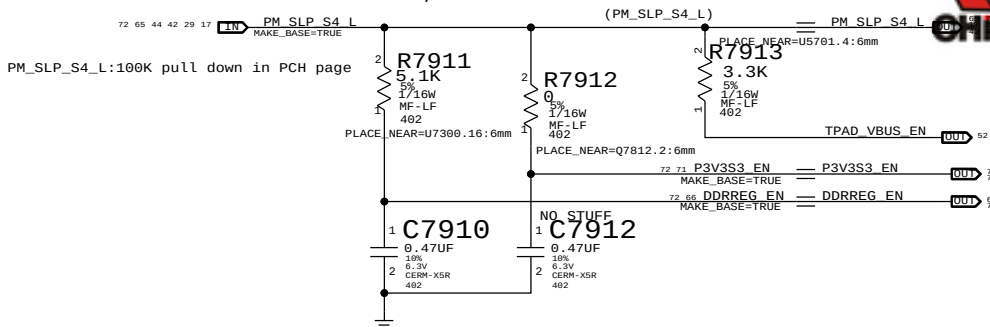
S5 Rail Enables & PG00D



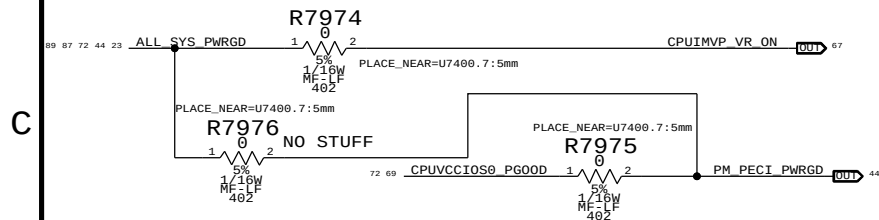
3.3V/5.0V S4 ENABLE



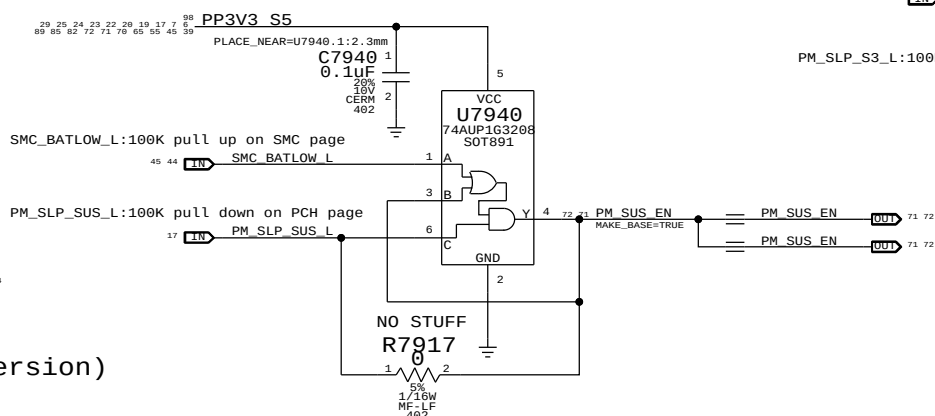
3.3V, 5V S3 ENABLE



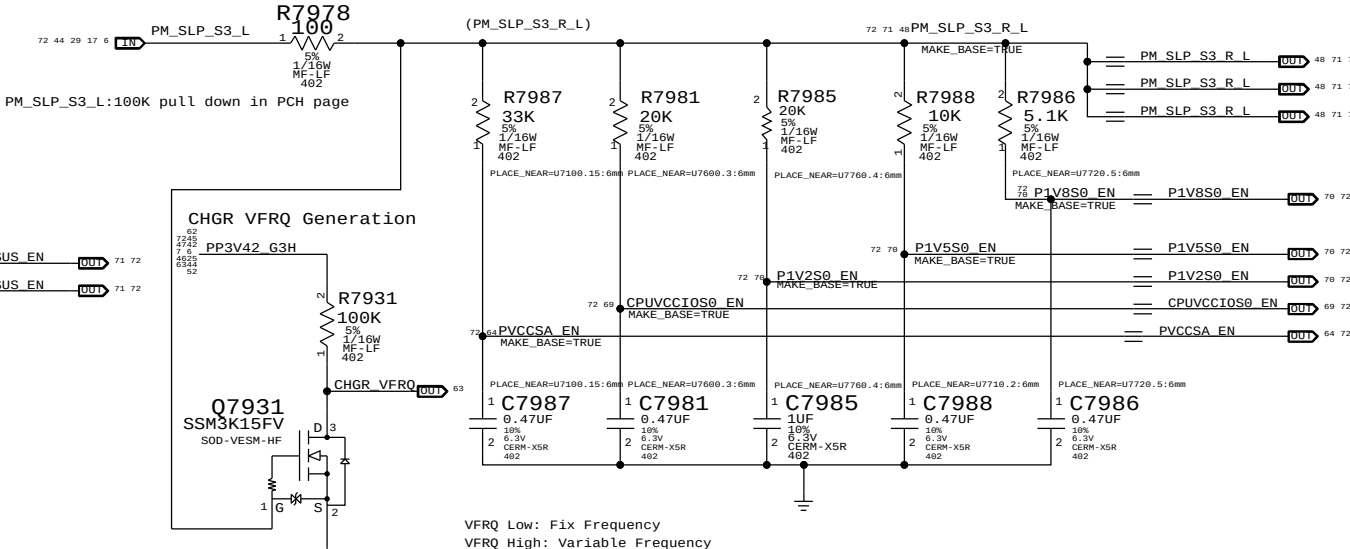
CPUVCORE ENABLE



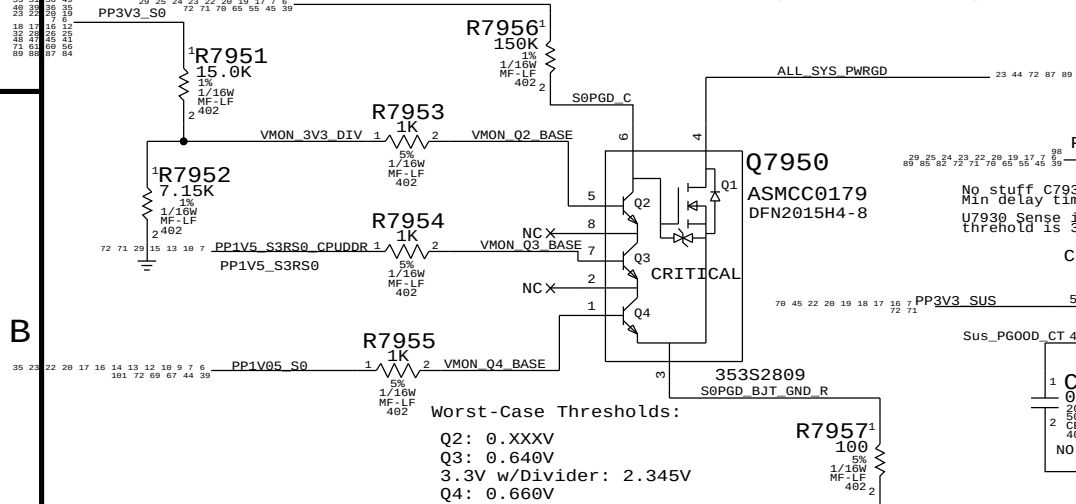
3.3V/5.0V Sus ENABLE



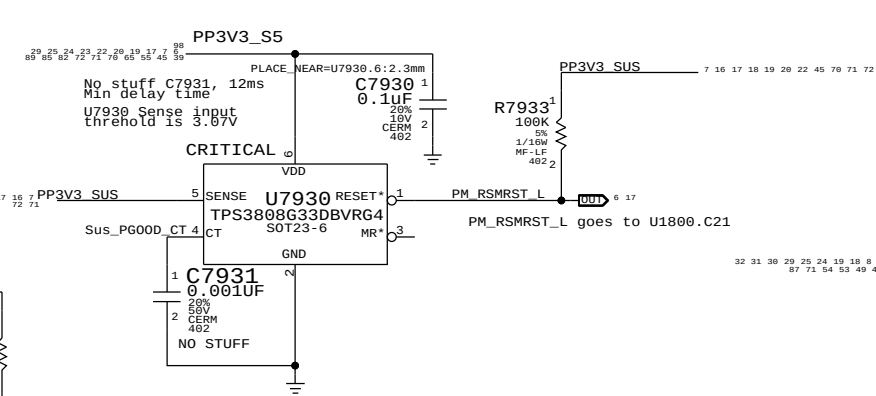
S0 ENABLE



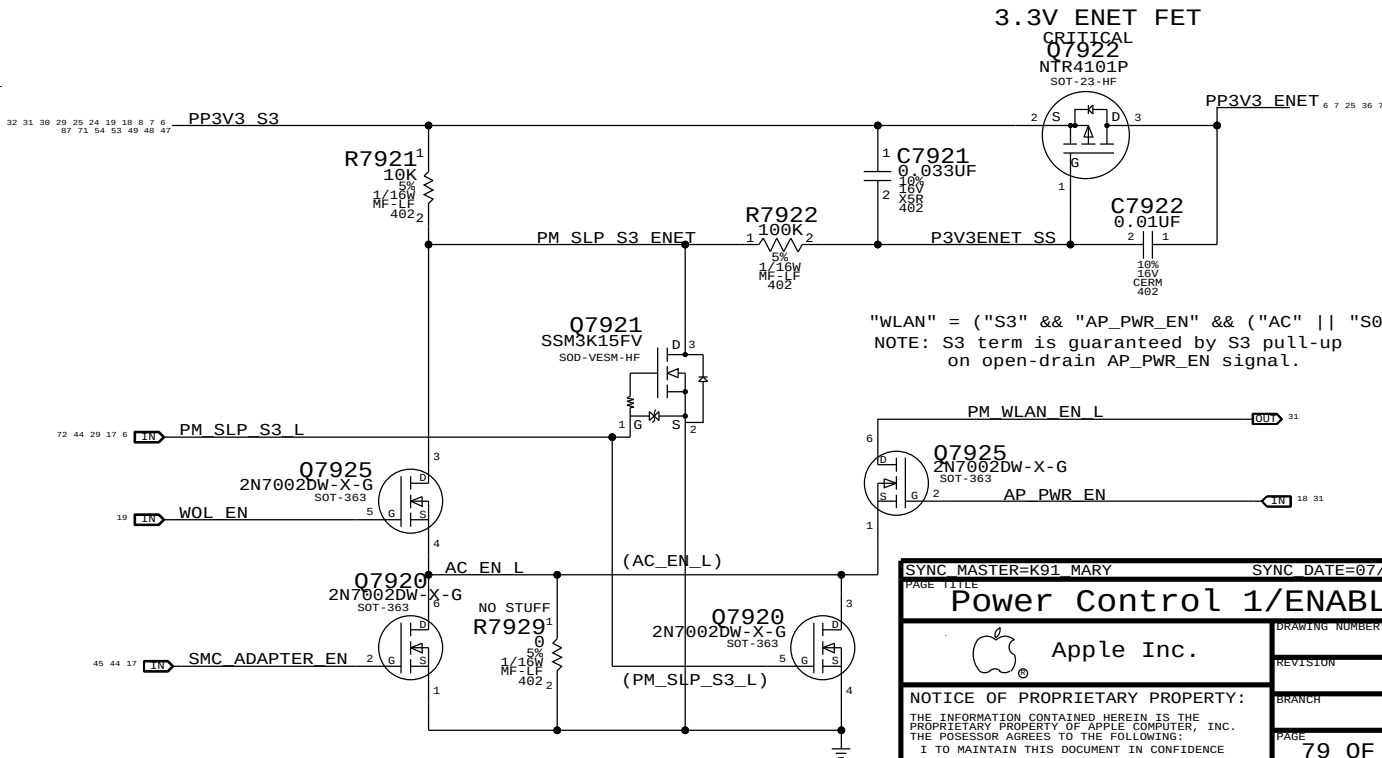
S0 Rail PG00D (BJT Version)



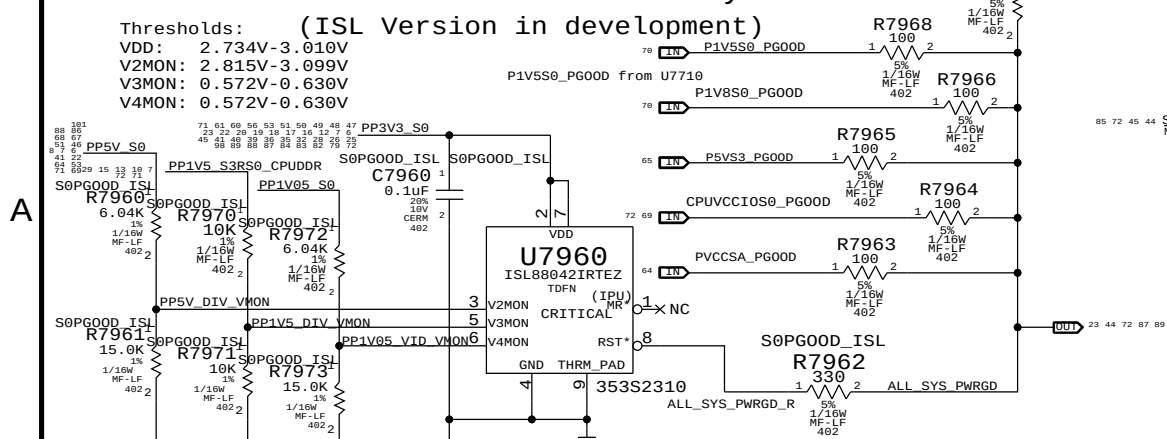
3.3V SUS Detect



ENET Enable Generation



S0 Rail PG00D Circuitry



"WLAN" = ("S3" && "AP_PWR_EN" && ("AC" || "S0"))
NOTE: S3 term is guaranteed by S3 pull-up on open-drain AP_PWR_EN signal.

SYNC MASTER=K91 MARY	SYNC DATE=07/22/2016
Power Control 1/ENABLE	
Apple Inc.	DRAWING NUMBER
NOTICE OF PROPRIETARY PROPERTY:	REVISION
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:	BRANCH
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE	PAGE
II NOT TO REPRODUCE OR COPY IT	79 OF 132
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART	SHEET
IV ALL RIGHTS RESERVED	72 OF 101



8

7

6

5

4

3

2

1

Page Notes

Power aliases required by this page:

- =PP1V2_GPU_PEX_PLLXVDD
- =PP1V2_GPU_PEX_IOVDDQ
- =PP1V2_GPU_PEX_IOVDD

Signal aliases required by this page:

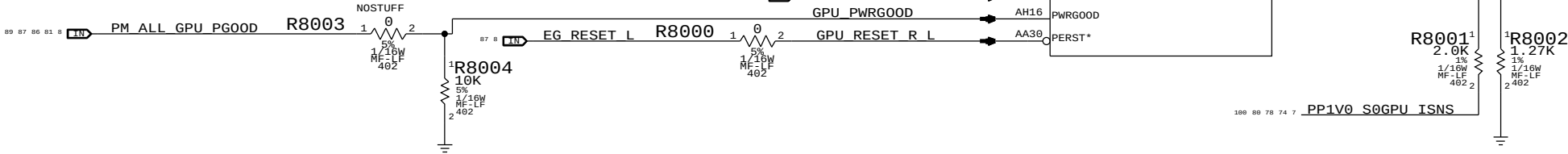
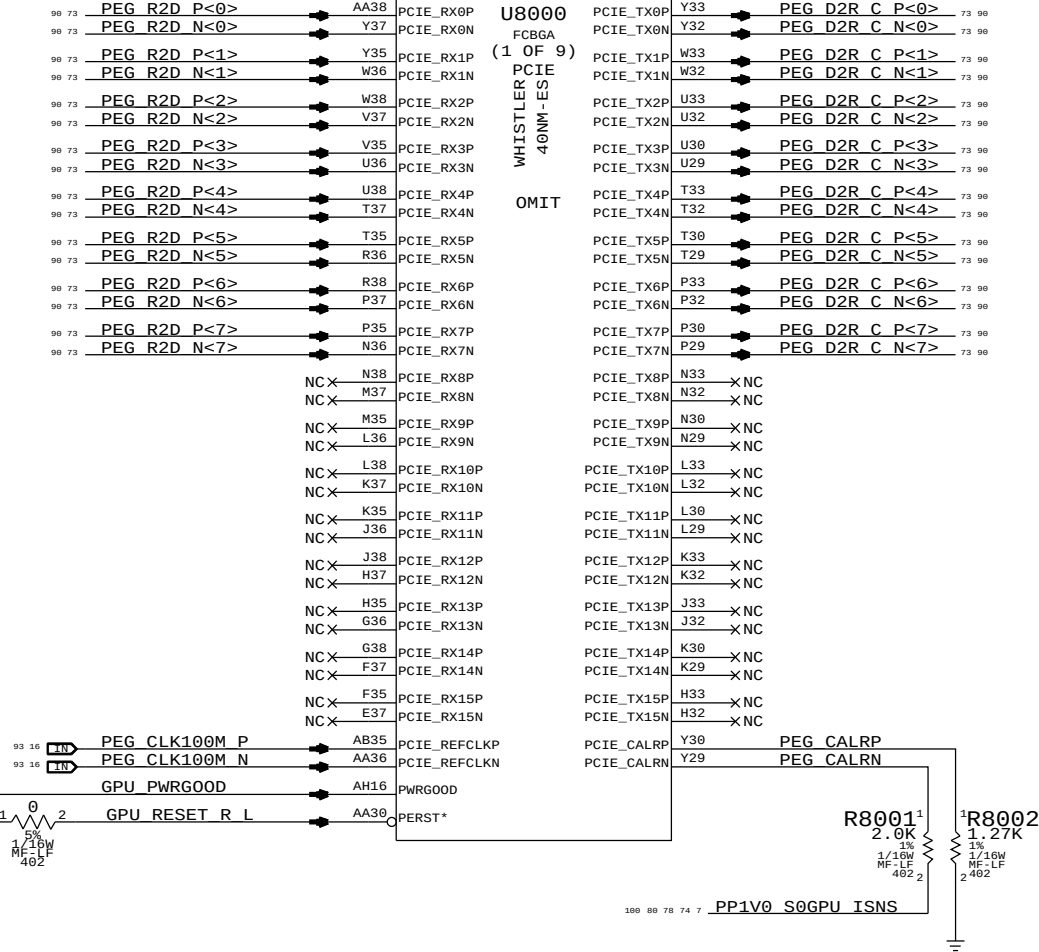
(NONE)

BOM options provided by this page:

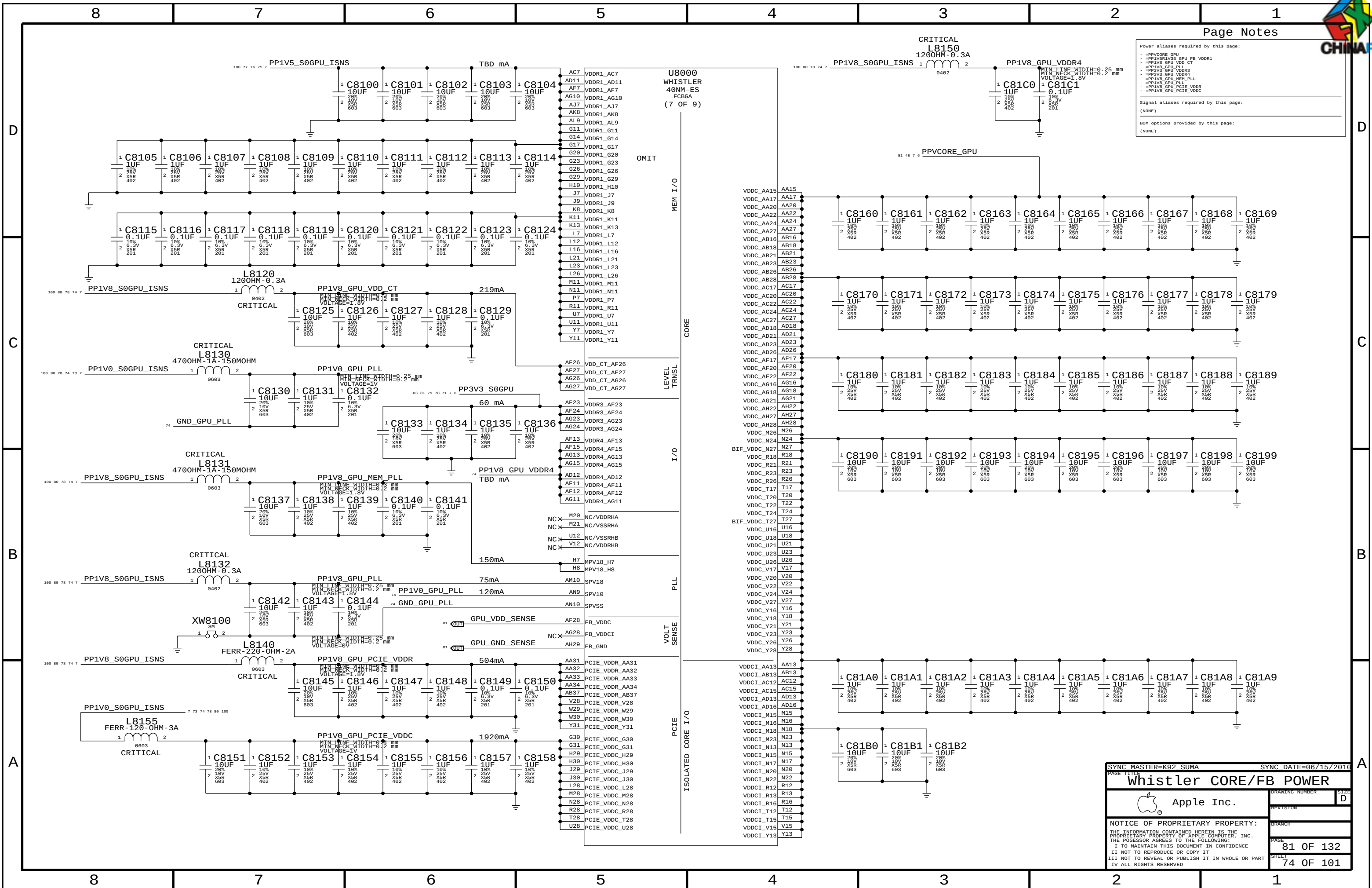
(NONE)

90 8	PEG R2D C P<0>	C8020 0.1UF	1	2	PEG R2D P<0>	73 90
				10% .3VX5R 201		
90 8	PEG R2D C N<0>	C8021 0.1UF	1	2	PEG R2D N<0>	73 90
				10% .3VX5R 201		
90 8	PEG R2D C P<1>	C8022 0.1UF	1	2	PEG R2D P<1>	73 90
				10% .3VX5R 201		
90 8	PEG R2D C N<1>	C8023 0.1UF	1	2	PEG R2D N<1>	73 90
				10% .3VX5R 201		
90 8	PEG R2D C P<2>	C8024 0.1UF	1	2	PEG R2D P<2>	73 90
				10% .3VX5R 201		
90 8	PEG R2D C N<2>	C8025 0.1UF	1	2	PEG R2D N<2>	73 90
				10% .3VX5R 201		
90 8	PEG R2D C P<3>	C8026 0.1UF	1	2	PEG R2D P<3>	73 90
				10% .3VX5R 201		
90 8	PEG R2D C N<3>	C8027 0.1UF	1	2	PEG R2D N<3>	73 90
				10% .3VX5R 201		
90 8	PEG R2D C P<4>	C8028 0.1UF	1	2	PEG R2D P<4>	73 90
				10% .3VX5R 201		
90 8	PEG R2D C N<4>	C8029 0.1UF	1	2	PEG R2D N<4>	73 90
				10% .3VX5R 201		
90 8	PEG R2D C P<5>	C8030 0.1UF	1	2	PEG R2D P<5>	73 90
				10% .3VX5R 201		
90 8	PEG R2D C N<5>	C8031 0.1UF	1	2	PEG R2D N<5>	73 90
				10% .3VX5R 201		
90 8	PEG R2D C P<6>	C8032 0.1UF	1	2	PEG R2D P<6>	73 90
				10% .3VX5R 201		
90 8	PEG R2D C N<6>	C8033 0.1UF	1	2	PEG R2D N<6>	73 90
				10% .3VX5R 201		
90 8	PEG R2D C P<7>	C8034 0.1UF	1	2	PEG R2D P<7>	73 90
				10% .3VX5R 201		
90 8	PEG R2D C N<7>	C8035 0.1UF	1	2	PEG R2D N<7>	73 90
				10% .3VX5R 201		

90 73	PEG D2R C P<0>	C8055 0.1UF	1	2	PEG D2R P<0>	8 90
				10% .3VX5R 201		
90 73	PEG D2R C N<0>	C8056 0.1UF	1	2	PEG D2R N<0>	8 90
				10% .3VX5R 201		
90 73	PEG D2R C P<1>	C8057 0.1UF	1	2	PEG D2R P<1>	8 90
				10% .3VX5R 201		
90 73	PEG D2R C N<1>	C8058 0.1UF	1	2	PEG D2R N<1>	8 90
				10% .3VX5R 201		
90 73	PEG D2R C P<2>	C8059 0.1UF	1	2	PEG D2R P<2>	8 90
				10% .3VX5R 201		
90 73	PEG D2R C N<2>	C8060 0.1UF	1	2	PEG D2R N<2>	8 90
				10% .3VX5R 201		
90 73	PEG D2R C P<3>	C8061 0.1UF	1	2	PEG D2R P<3>	8 90
				10% .3VX5R 201		
90 73	PEG D2R C N<3>	C8062 0.1UF	1	2	PEG D2R N<3>	8 90
				10% .3VX5R 201		
90 73	PEG D2R C P<4>	C8063 0.1UF	1	2	PEG D2R P<4>	8 90
				10% .3VX5R 201		
90 73	PEG D2R C N<4>	C8064 0.1UF	1	2	PEG D2R N<4>	8 90
				10% .3VX5R 201		
90 73	PEG D2R C P<5>	C8065 0.1UF	1	2	PEG D2R P<5>	8 90
				10% .3VX5R 201		
90 73	PEG D2R C N<5>	C8066 0.1UF	1	2	PEG D2R N<5>	8 90
				10% .3VX5R 201		
90 73	PEG D2R C P<6>	C8067 0.1UF	1	2	PEG D2R P<6>	8 90
				10% .3VX5R 201		
90 73	PEG D2R C N<6>	C8068 0.1UF	1	2	PEG D2R N<6>	8 90
				10% .3VX5R 201		
90 73	PEG D2R C P<7>	C8069 0.1UF	1	2	PEG D2R P<7>	8 90
				10% .3VX5R 201		
90 73	PEG D2R C N<7>	C8070 0.1UF	1	2	PEG D2R N<7>	8 90
				10% .3VX5R 201		



SYNC MASTER=K92 SUMA		SYNC DATE=06/15/2016	
PAGE TITLE			
whistler PCI-E			
Apple Inc.		DRAWING NUMBER	SIZE
		REVISION	D
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		80 OF 132	
II NOT TO REPRODUCE OR COPY IT		SHEET	
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART		73 OF 101	
IV ALL RIGHTS RESERVED			







Page Notes

Power aliases required by this page:
- #PP1V5_S0GPU_ISNS

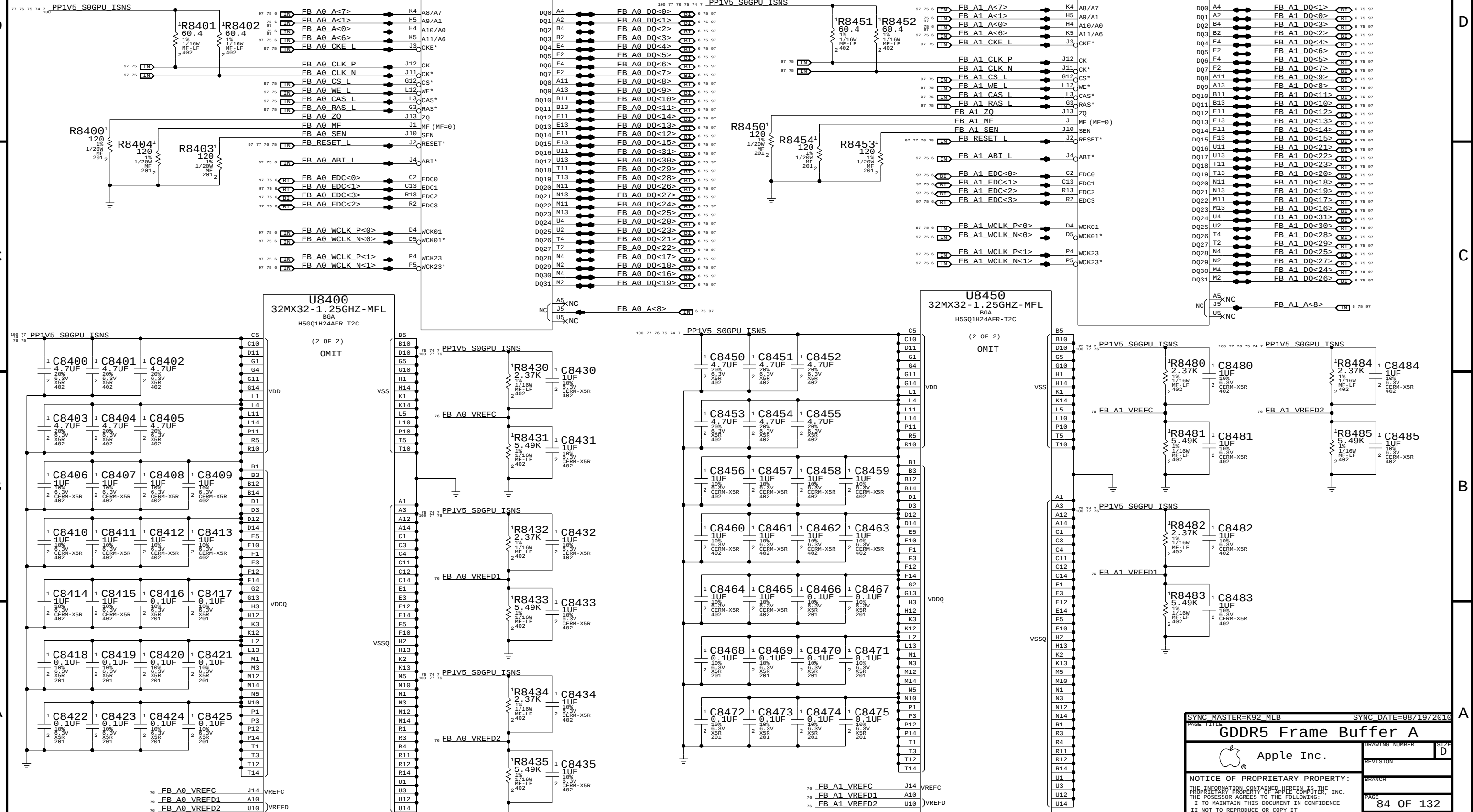
Signal aliases required by this page:
(NONE)

BOM options provided by this page:
GPU:WHISTLER is the BOM option called out on all Rs and Cs on this page

U8400
32MX32-1.25GHZ-MFL
BGA
H5GQ1H24AFR-T2C

GPU:WHISTLER is the BOM option called out on ALL Rs and Cs on this page!

U8450
32MX32-1.25GHZ-MFL
BGA
H5GQ1H24AFR-T2C



SYNC MASTER=K92 MLB		SYNC DATE=08/19/2016	
PAGE TITLE		SIZE	
GDDR5 Frame Buffer A		D	
Apple Inc.		REVISION	
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		84 OF 132	
II NOT TO REPRODUCE OR COPY IT		SHEET	
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART		76 OF 101	
IV ALL RIGHTS RESERVED			



Page Notes

Power aliases required by this page:

- PP1V5 S0GPU ISNS

Signal aliases required by this page:

(NONE)

BOM options provided by this page:

(NONE)

U8500
32MX32-1.25GHZ-MFL
BGA
H5GQ1H24AFR-T2C

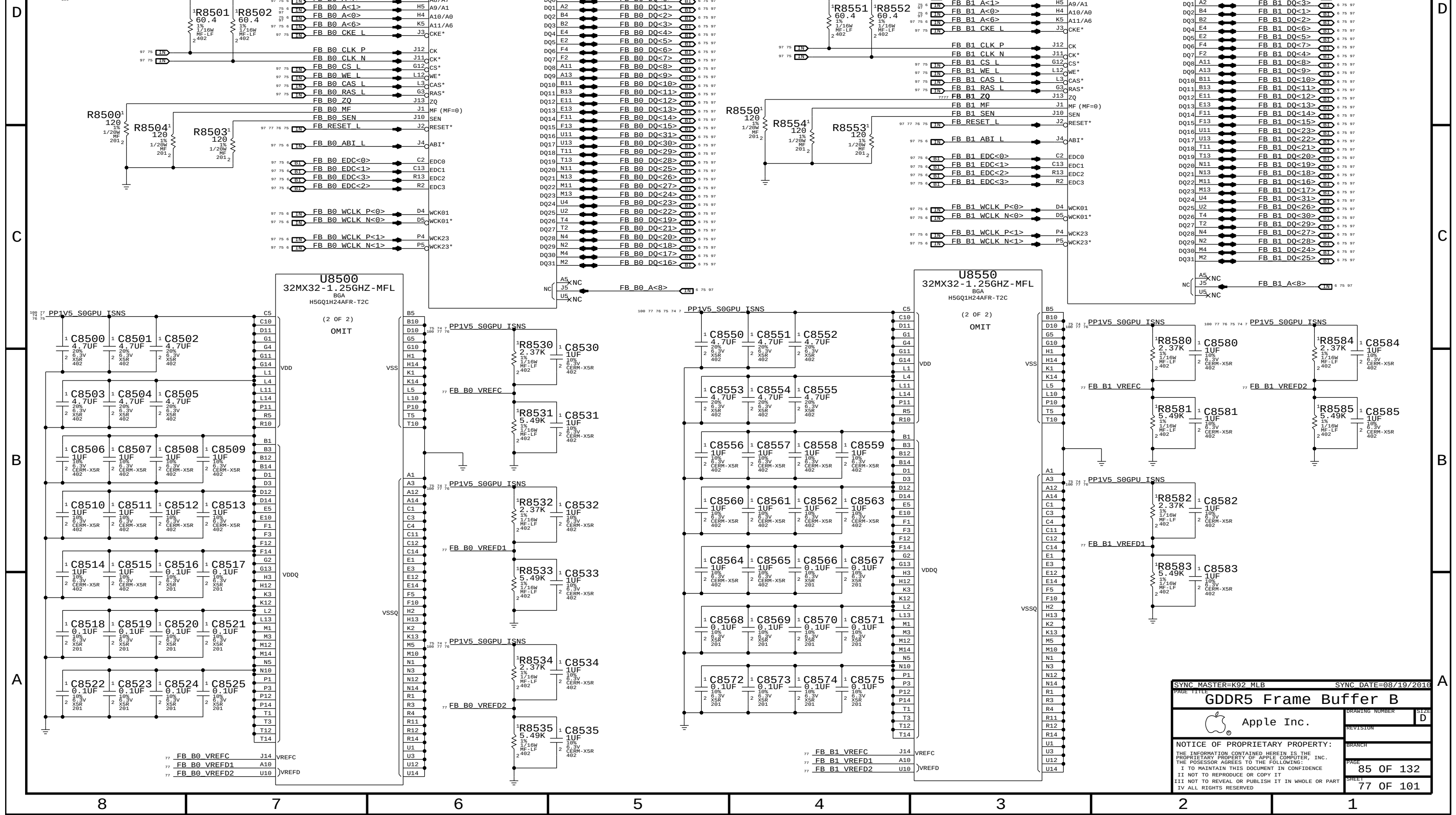
(1 OF 2)

OMIT

U8550
32MX32-1.25GHZ-MFL
BGA
H5GQ1H24AFR-T2C

(1 OF 2)

OMIT



PAGE TITLE		SYNC MASTER=K92 MLB		SYNC DATE=08/19/2016	
GDDR5 Frame Buffer B					
Apple Inc.				DRAWING NUMBER	SIZE
NOTICE OF PROPRIETARY PROPERTY:				REVISION	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:				BRANCH	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE				PAGE	
II NOT TO REPRODUCE OR COPY IT				85 OF 132	
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART				SHEET	
IV ALL RIGHTS RESERVED				77 OF 101	



Power aliases required by this page:
- =PP3V3_GPU_I2C
- =PP1V8_GPU_VREFG
- =PP1V8_GPU_DPLL
- =PP1V8_GPU_DPLL
- =PP1V8_GPU_TS

Signal aliases required by this page:
(NONE)

BOM options provided by this page:
(NONE)

VRAM BOM OPTION TABLE

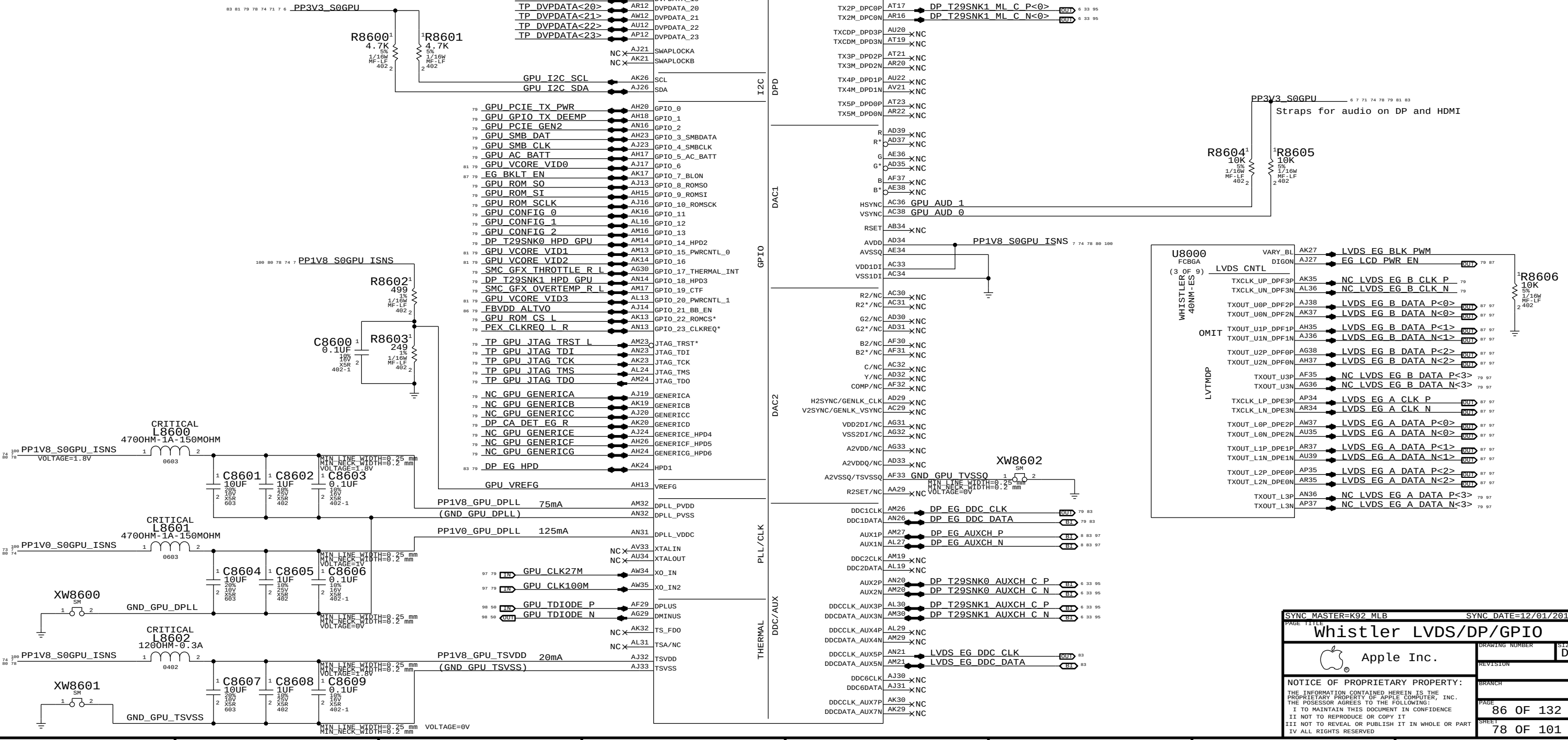
	VRAM_DVP0 (STUFF R86137)	VRAM_DVP1 (STUFF R86127)	VRAM_DVP2 (STUFF R86117)
K92 SAMSUNG 1G	NO	NO	YES
K92 HYNIX 1G	YES	NO	YES
K91F SAMSUNG 512M	NO	NO	NO
K91F HYNIX 512M	NO	YES	NO
K91F SAMSUNG 1G	YES	NO	NO
K91F HYNIX 1G	YES	YES	NO

NOTE: AMD STRAPS FOR IDENTIFYING VRAM VENDOR & SIZE FOR WHISTLER

K92 Samsung 1G - NOSTUFF R8613, NOSTUFF R8612, STUFF R8611
K92 Hynix 1G - STUFF R8613, NOSTUFF R8612, STUFF R8611

K91FSamsung 512M - NOSTUFF R8613, NOSTUFF R8612, NOSTUFF R8611
K91FHynix 512M - NOSTUFF R8613, STUFF R8612, NOSTUFF R8611
K91FSamsung 1G - STUFF R8613, NOSTUFF R8612, NOSTUFF R8611
K91FHynix 1G - STUFF R8613, STUFF R8612, NOSTUFF R8611

NO STRAP CHANGES ARE REQUIRED FOR SEYMOUR BASED SYSTEMS



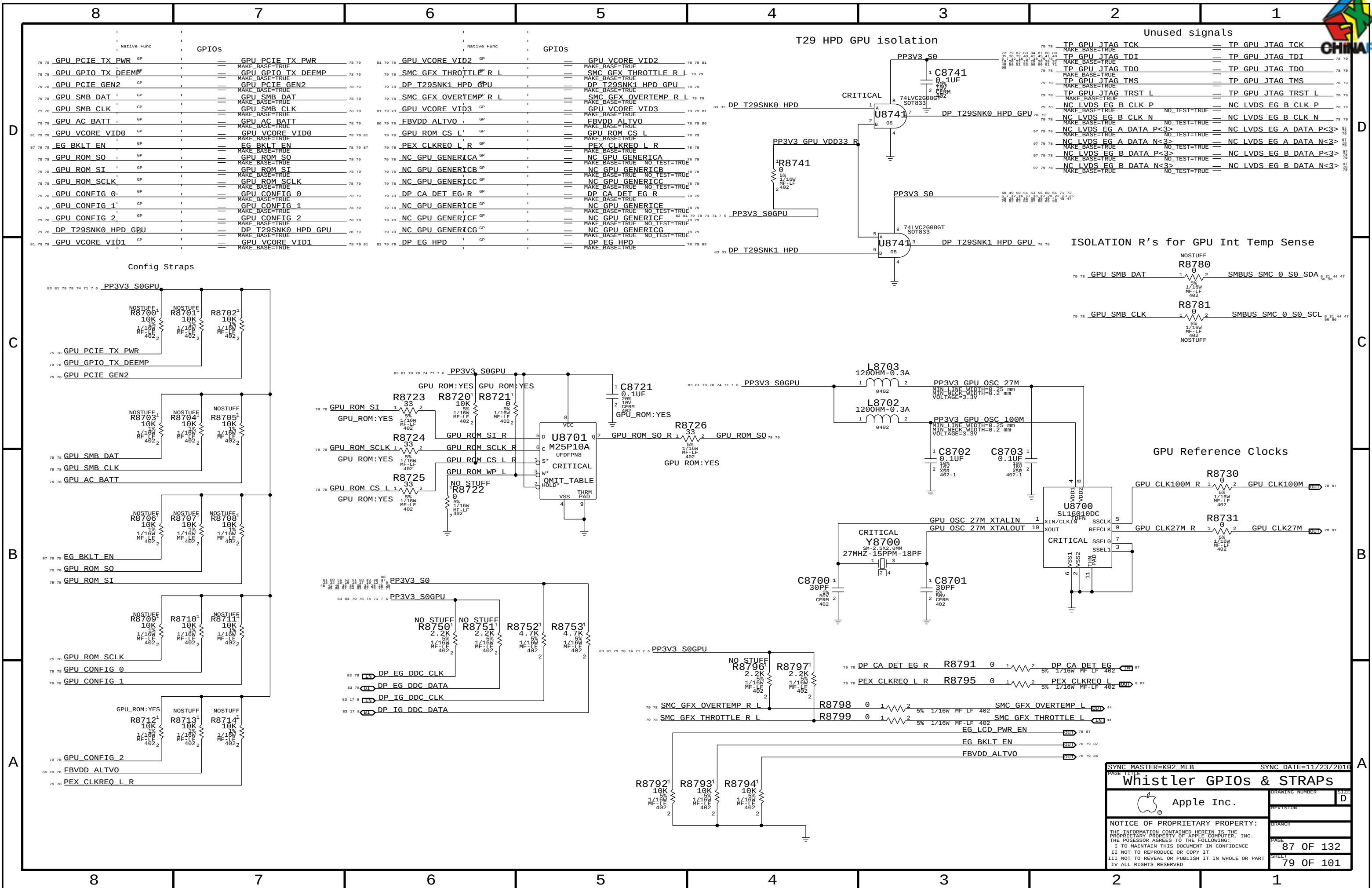
SYNC MASTER=K92 MLB SYNC DATE=12/01/2016

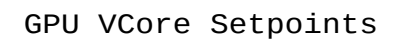
Whistler LVDS/DP/GPIO

Apple Inc.

NOTICE OF PROPRIETARY PROPERTY:
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

DRAWING NUMBER: D
REVISION:
BRANCH:
PAGE: 86 OF 132
SHEET: 78 OF 101





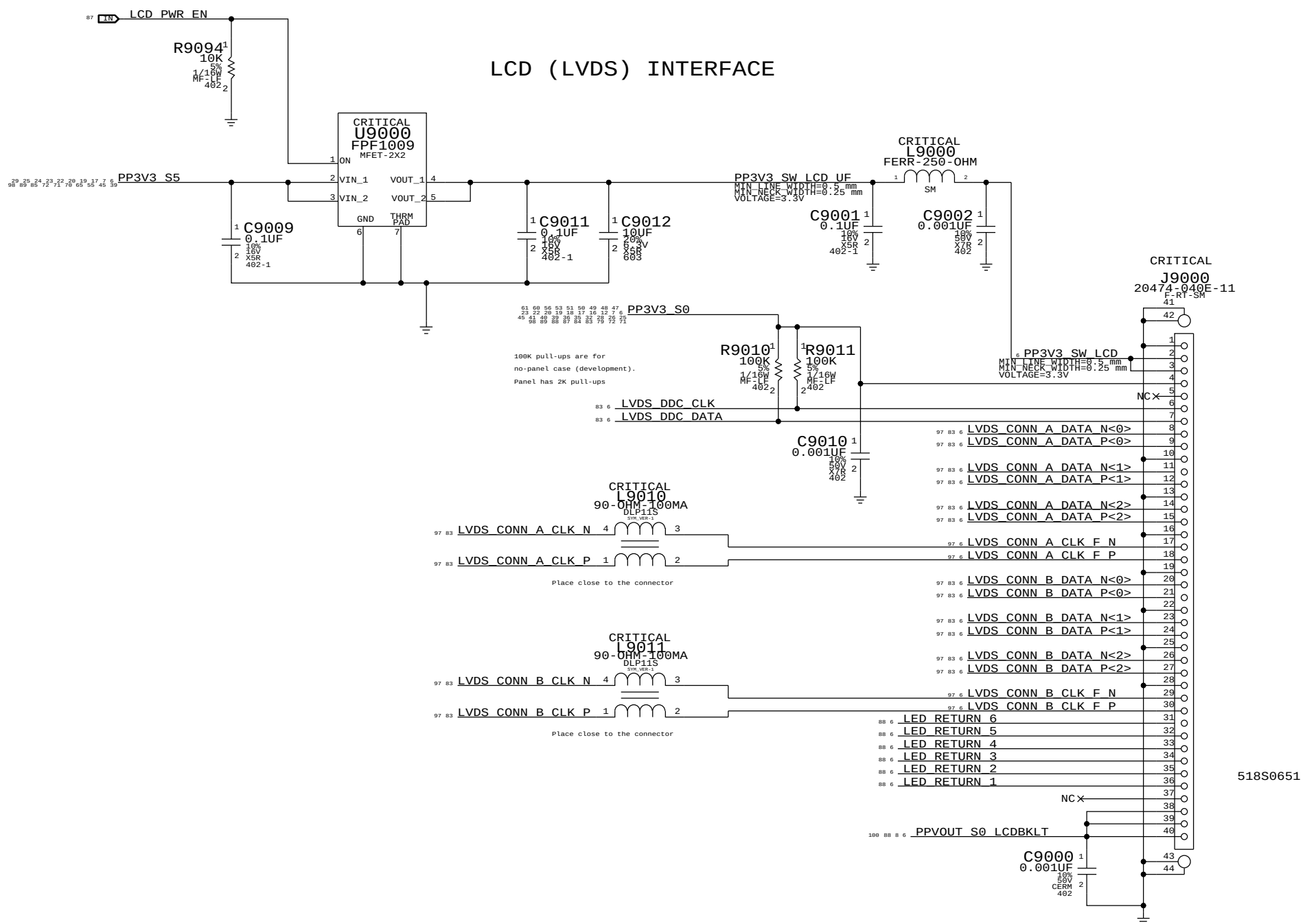
GPU VCore Setpoints not up-to-date!

K91F Default Vcore Setpoints

BOM_GROUP	BOM_OPTIONS
GPUVID_1P11V	GPUVID4_0, GPUVID3_0, GPUVID2_1, GPUVID1_1, GPUVID0_1

VIMON = 31 x Io x Rsns x (1 + Ris2/Ris1)
In K9x, the equation will be:
VIMON = 31 x Io x R8940 x (1 + R8902/R8901)
VIMON = 31x Io x 1m0hm x 2.074 = Io x 64.29 mV/A

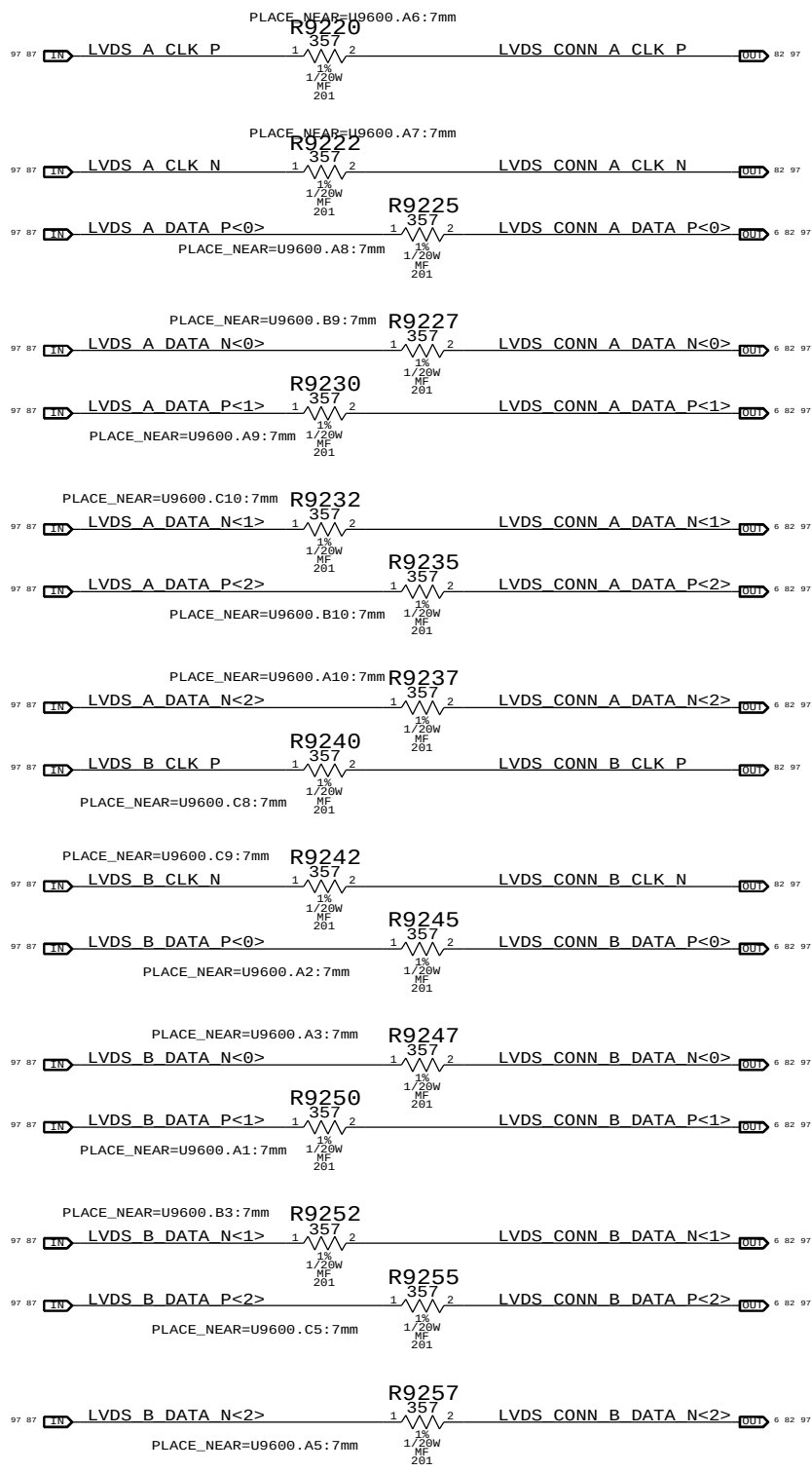
SYNC MASTER-K91 ERIC		SYNC DATE=12/21/2010	
PAGE 1 FILE			
GPU (Whistler)		CORE SUPPLY	
 Apple Inc.		DRAWING NUMBER	SIZE
		REVISION	D
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		PAGE	89 OF 132
		SHEET	81 OF 101



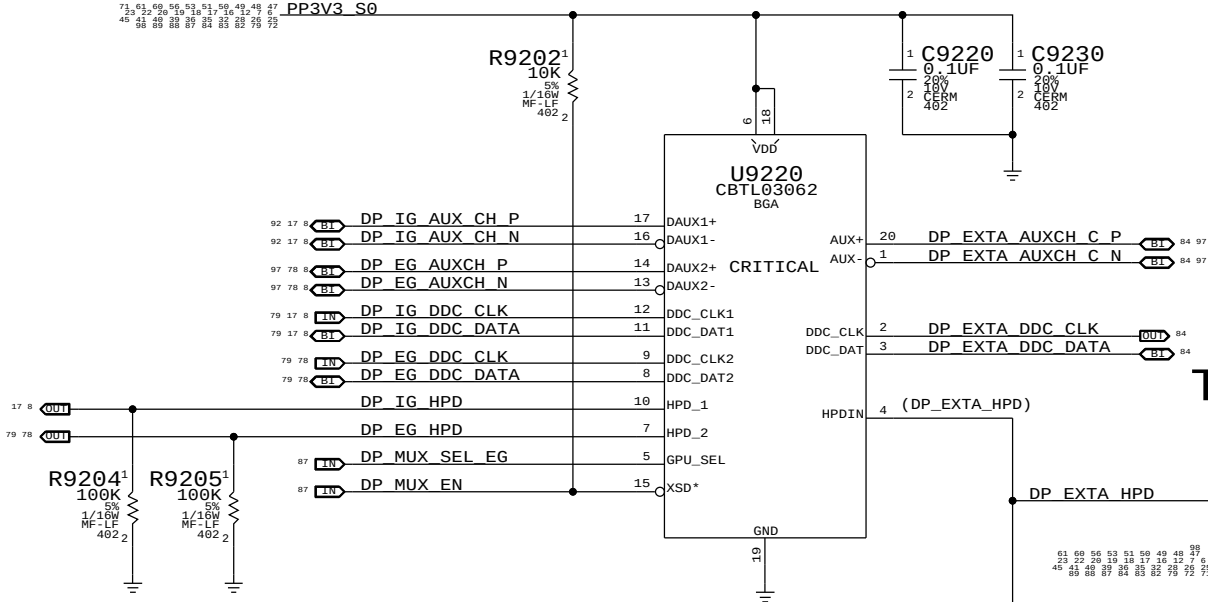


LVDS Transmitter Termination

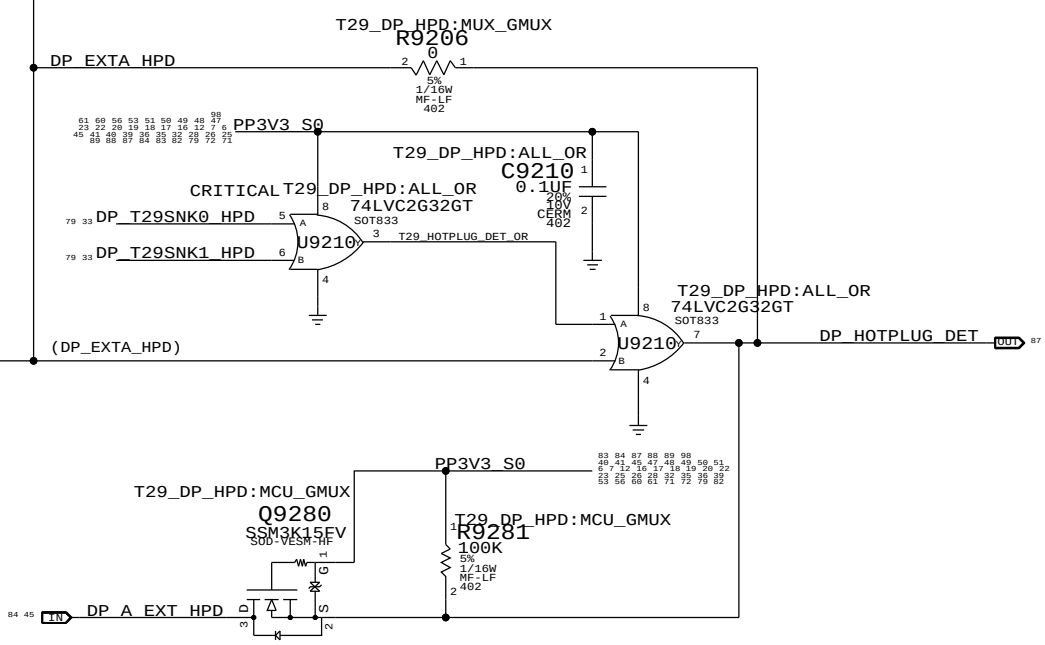
All emulated LVDS outputs require this termination



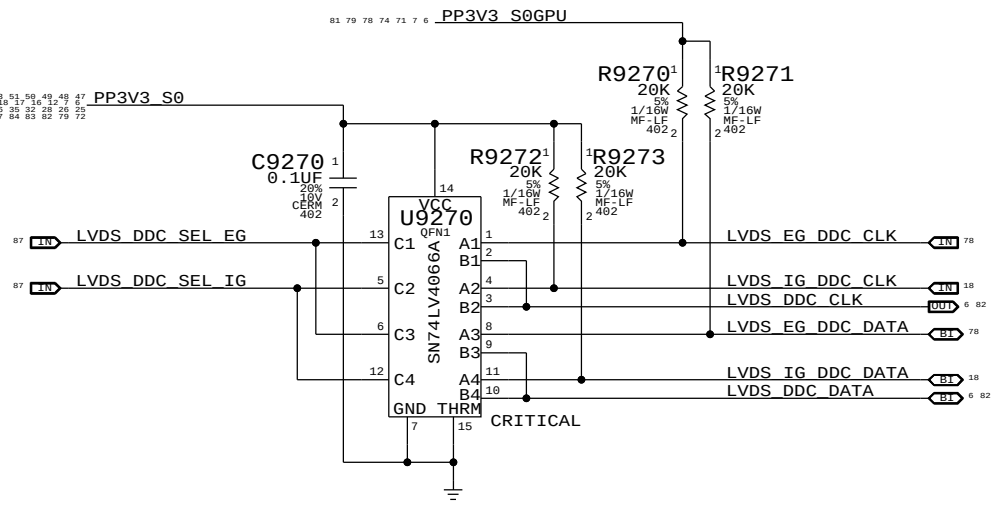
DP AUX, DDC, & HPD muxing to IG/EG



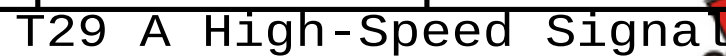
T29/DP HOT PLUG IN



LVDS DDC MUX



PAGE TITLE		PAGE NUMBER	
Muxed Graphics Support		92 OF 132	
Apple Inc.		83 OF 101	
NOTICE OF PROPRIETARY PROPERTY:		THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		II NOT TO REPRODUCE OR COPY IT	
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART		IV ALL RIGHTS RESERVED	



T29 signals are
P/N-swapped after AC
caps to improve layout.

If GPU uses common pins for AUX_CH and DDC, alias nets together at GPU.

R9308/R9309 maintain bias on C9308/C9309 to prevent spikes when U9310 AUXDDC_OFF transitions from high to low.

PS8301 I2C Addresses:

A1	A0	Addr (W/R)
0	0	0x96/0x97
0	1	0xB6/0xB7
1	0	0x94/0x95
1	1	0xB4/0xB5

Note: Other Parade devices use 96/B6, so only 94/B4 are used for this part.

DP_EXT_A ML_P<0>	1	IN_D0P	CRITICAL	OUT_D0P
DP_EXT_A ML_N<0>	2	IN_D0N		OUT_D0N
DP_EXT_A ML_P<1>	4	IN_D1P		OUT_D1P
DP_EXT_A ML_N<1>	5	IN_D1N		OUT_D1N
DP_EXT_A ML_P<2>	7	IN_D2P		OUT_D2P
DP_EXT_A ML_N<2>	8	IN_D2N		OUT_D2N
DP_EXT_A ML_P<3>	9	IN_D3P		OUT_D3P
DP_EXT_A ML_N<3>	10	IN_D3N		OUT_D3N
DP_EXT_A DDC_CLK	14	IN_SCL		AC_AUXP
DP_EXT_A DDC_DATA	13	IN_SDA		AC_AUXN
DP_EXT_A AUXCH_P	16	IN_AUXP		OUT_AUXP_SCL
DP_EXT_A AUXCH_N	15	IN_AUXN		OUT_AUXN_SDA
DP_EXT_A HPD	3	IN_HPD		(IPD) OUT_HPD
DPSDRVA_I2C_CTL_EN	26	I2C_CTL_EN (IPU)		CA_DET
DPSDRVA_I2C_ADDR0	36	I2C_ADDR0 (IPD)		
DPSDRVA_I2C_ADDR1	35	I2C_ADDR1 (IPD)		CEXT
I2C_DPSDRVA_SCL	38	SCL_CTL		
I2C_DPSDRVA_SDA	37	SDA_CTL		
DPSDRVA_REXT	12	REXT		
DP_AUXCH_ISOL	39	AUXDDC_OFF (IPD)		
DP_A_PWRDWN_R	34	PD (IPD)		

PS8301 has internal ~150K pull-down on PD pin. Okay to drive this pin even when VCC=0V per Parade (pin is 5V-tolerant).

Port A MCU

(DP_SDRVA_AUXCH_P)
(DP_SDRVA_AUXCH_N)

IC supports input high while $V_{CC} = 0V$

DP/T29 A Low-Speed MUX

Must be 3.3V DP A port power

CBTL04DP081 (353S3151) and PI3vEDP212 (353S3055) are footprint-compatible parts with similar pinouts. NXP uses pin 10 for ML and HPD, Pericom uses pin 10 for ML and pin 11 for HP

Note: U9390 ML/HPD defaults to T29 mode so that DP/T29 Display can detect host T29 support using I2C pull-ups on ML<>>. U9390 AUX defaults to DP mode because 100-ohm pull-downs would defeat DP Sink's detection of DP source.

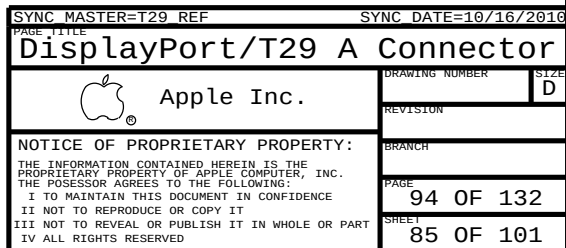
```
=T29_WAKE_L:  
use PCIe WAKE#
```

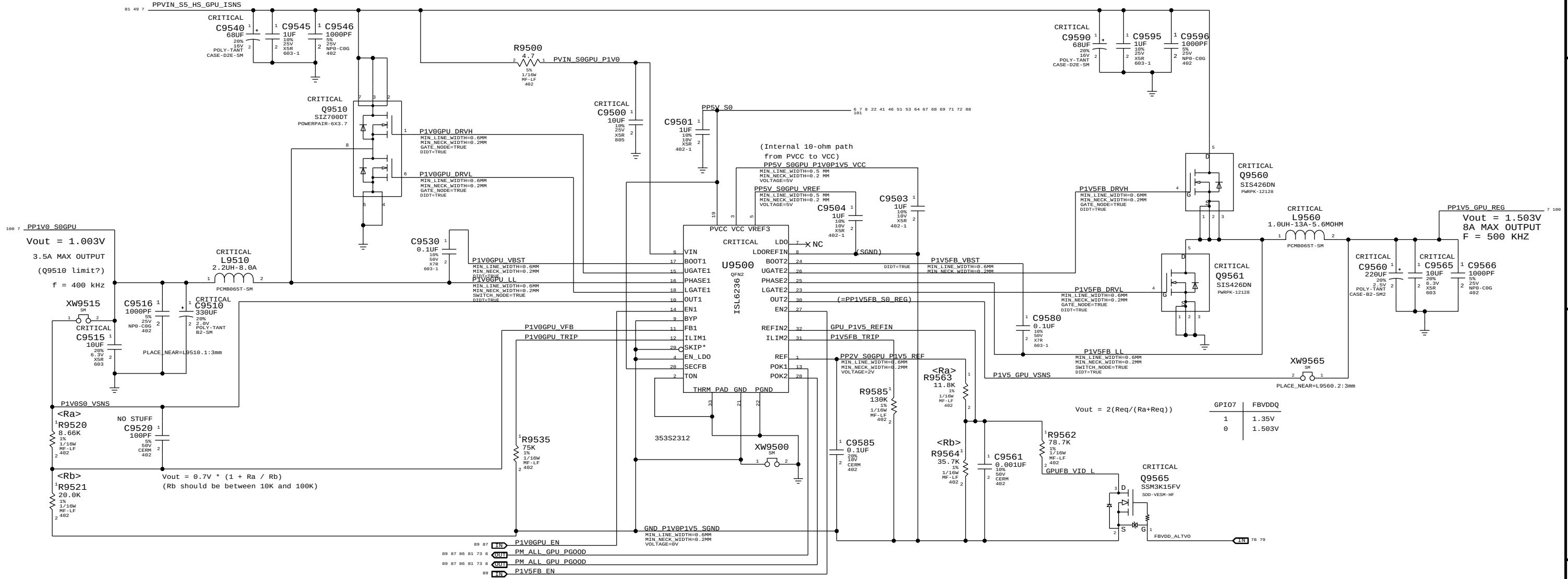
R9330 provides pads for programming/debug of MCU, please make accessible.
If project has space for 10-pin programming header it should be used.

SYNC MASTER=T29 REF		SYNC DATE=10/16/2010	
PAGE TITLE			
DisplayPort/T29 A MUXing			
 Apple Inc.		DRAWING NUMBER	
		D	
		REVISION	
		BRANCH	
NOTICE OF PROPRIETARY PROPERTY:			
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED			
		PAGE	
		93 OF 132	
		SHEET	
		84 OF 101	

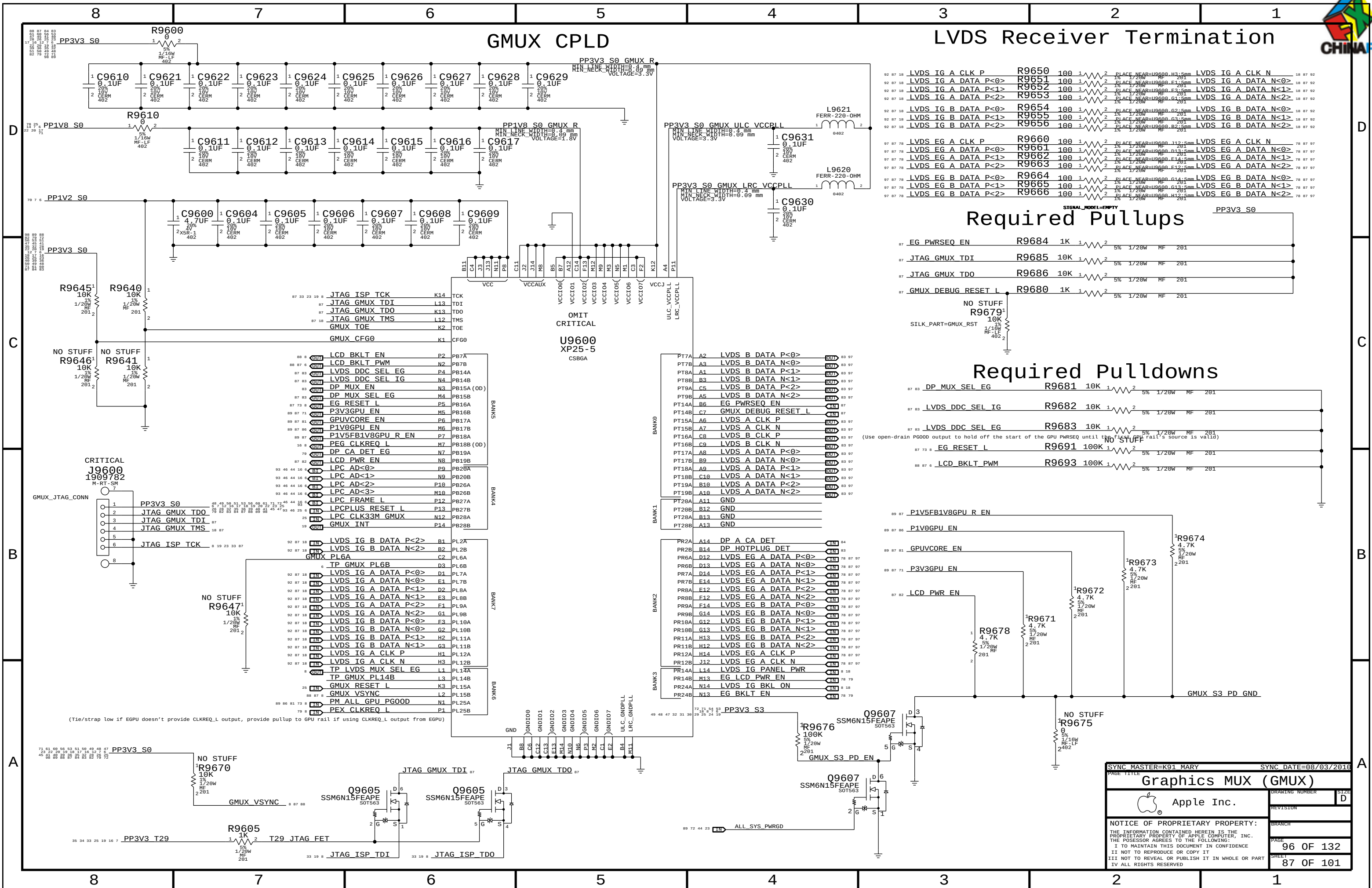


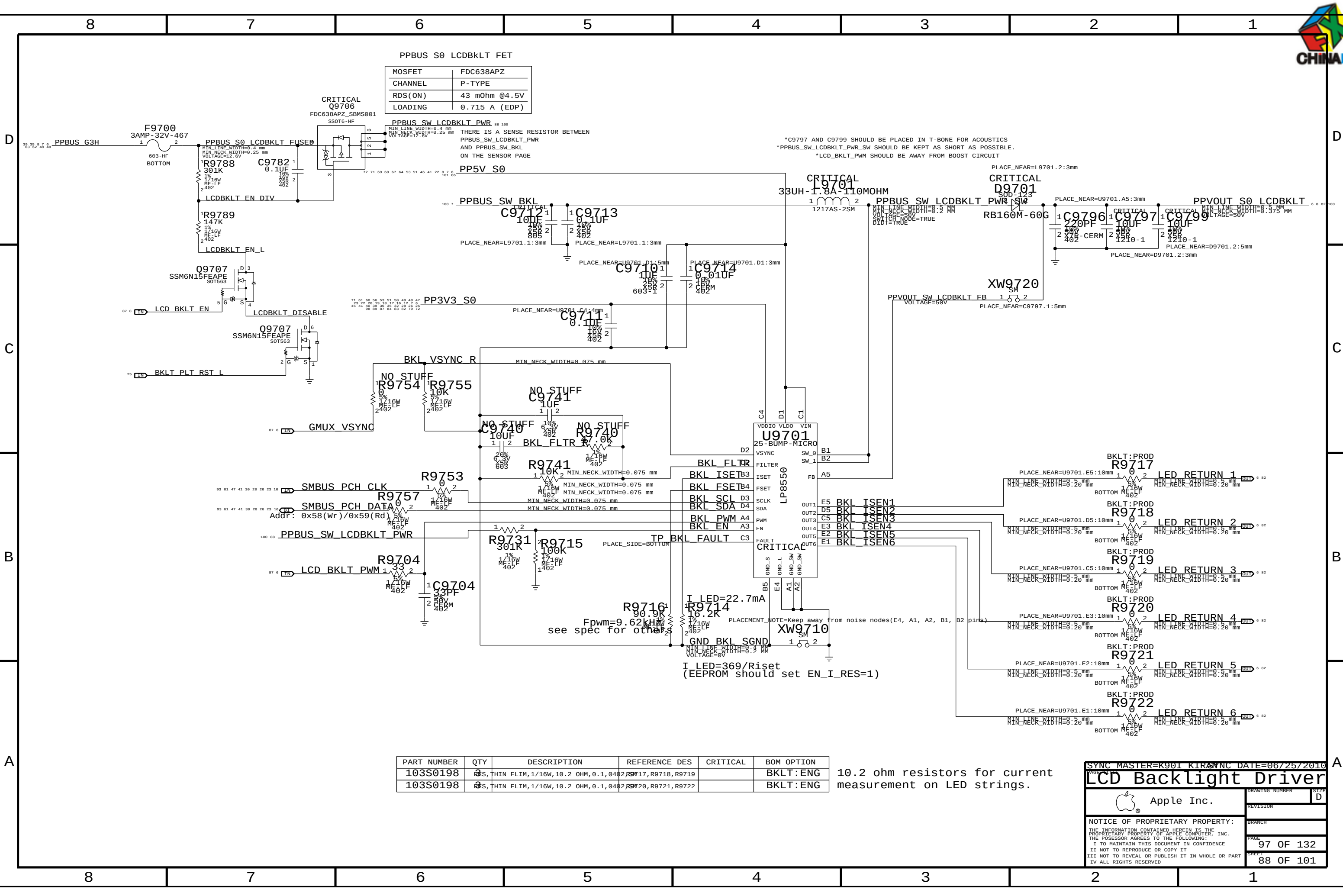
Port A 3.3V Power Switch





SYNC MASTER=K91 ERIC		SYNC DATE=10/08/2016	
PAGE TITLE		1V0 GPU / 1V5 FB Power Supply	
Apple Inc.		DRAWING NUMBER	SIZE
NOTICE OF PROPRIETARY PROPERTY:		REVISION	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:		BRANCH	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		PAGE	95 OF 132
II NOT TO REPRODUCE OR COPY IT		SHEET	86 OF 101
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			





PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
103S0198	2	RES, THIN FLIM, 1/16W, 10.2 OHM, 0.1, 0402	R9717, R9718, R9719		BKLT:ENG
103S0198	2	RES, THIN FLIM, 1/16W, 10.2 OHM, 0.1, 0402	R9720, R9721, R9722		BKLT:ENG

10.2 ohm resistors for current measurement on LED strings.

SYNC MASTER=K901 KIRAS

DATE=06/25/2010

LCD Backlight Driver

Apple Inc.

NOTICE OF PROPRIETARY PROPERTY:
THE INFORMATION CONTAINED HEREIN IS THE
PROPERTY OF APPLE COMPUTER, INC.
THE POSSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

DRAWING NUMBER

REVISION

BRANCH

PAGE

SHEET

SIZE

D

97 OF 132

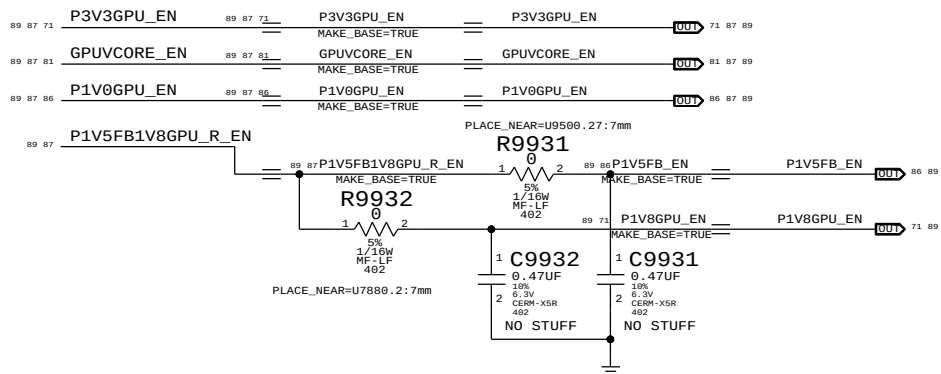
88 OF 101



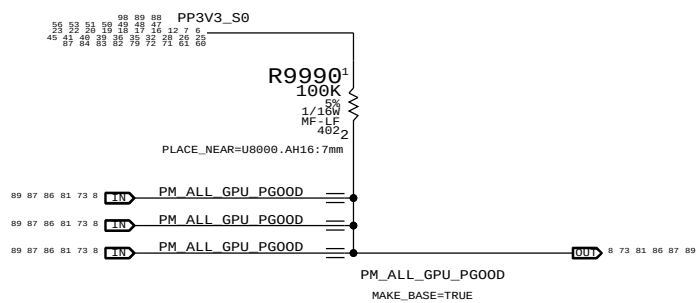
GPU Rail Sequencing

Whistler GPU requires rails to come up in the following order:

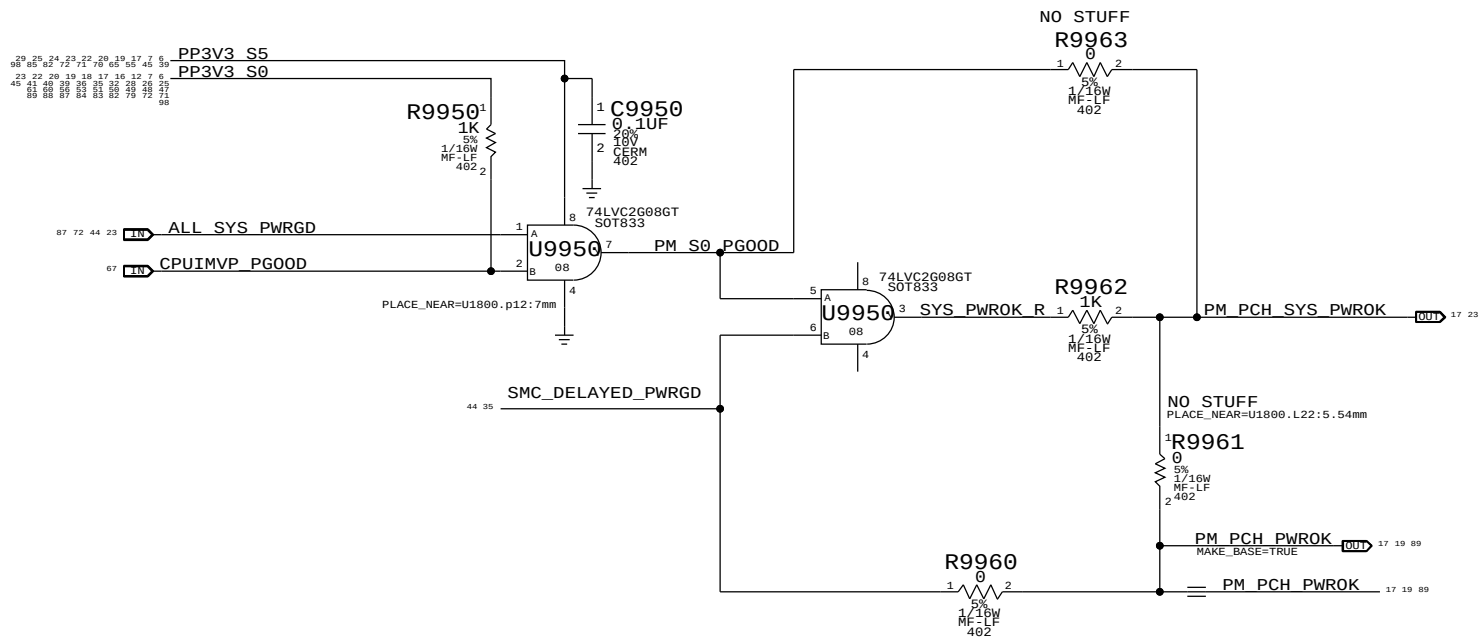
- 1) GPU_3.3V
- 2) GPUVcore
- 3) GPU_1.0V
- 4) GPU_1.8V; GDDR5 1.5/1.35V



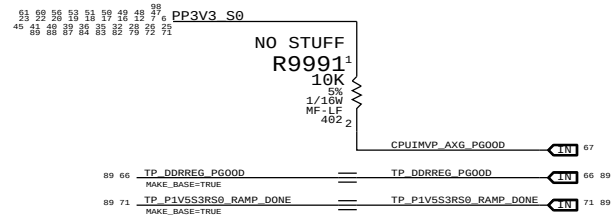
EXT GPU PWRGD Pullup



PCH S0 PWRGD



Unused PG00D signal



SYNC MASTER=K91 MARY		SYNC DATE=08/03/2016	
PAGE TITLE		Power Sequencing EG/PCH S0	
Apple Inc.		DRAWING NUMBER	SIZE D
NOTICE OF PROPRIETARY PROPERTY:		REVISION	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:		BRANCH	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		PAGE	99 OF 132
II NOT TO REPRODUCE OR COPY IT		SHEET	89 OF 101
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			



CPU Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CPU_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
CPU_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
CPU_27P4S	*	=27P4_OHM_SE	=27P4_OHM_SE	=27P4_OHM_SE	=27P4_OHM_SE	7 MIL	7 MIL

NOTE: 7 mil gap is for VCCSense pair, which Intel says to route with 7 mil spacing without specifying a target impedance.

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CPU_AGTL	*	=STANDARD	?
CPU_8MIL	*	8 MIL	?
CPU_COMP	*	20 MIL	?
CPU_ITP	*	=2:1_SPACING	?
CPU_VCCSENSE	*	25 MIL	?

Most CPU signals with impedance requirements are 50-ohm single-ended.
Some signals require 27.4-ohm single-ended impedance.

SOURCE: Calpella SFF DG (DG-407364_v1.5), Section 2.8

PCI-Express

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCIE_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
CLK_PCIE_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCIE	*	=3X_DIELECTRIC	?
CLK_PCIE	*	20 MIL	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CPU_AGTL	TOP, BOTTOM	=2X_DIELECTRIC	?
CPU_VID	*	0.457 MM	?

CPU Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE			
		PHYSICAL	SPACING		
	DMI_S2N	PCIE_85D	PCIE	DMI_S2N P<3:0>	6 9 17
	DMI_S2N	PCIE_85D	PCIE	DMI_S2N N<3:0>	6 9 17
	DMI_N2S	PCIE_85D	PCIE	DMI_N2S P<3:0>	9 17
	DMI_N2S	PCIE_85D	PCIE	DMI_N2S N<3:0>	9 17
	FDI_DATA	PCIE_85D	PCIE	FDI_DATA P<7:0>	6 9 17
	FDI_DATA	PCIE_85D	PCIE	FDI_DATA N<7:0>	6 9 17
		CPU_50S	CPU_AGTL	FDI_FSYNC<1..0>	6 9 17
		CPU_50S	CPU_AGTL	FDI_LSYNC<1..0>	6 9 17
	DMI_CLK100M	CLK_PCIE_90D	CLK_PCIE	DMI_CLK100M_CPU_P	10 16
		CLK_PCIE_90D	CLK_PCIE	DMI_CLK100M_CPU_N	10 16
		CPU_50S	CPU_AGTL	FDI_INT	6 9 17
	CPU_PECT	CPU_50S	PCIE	CPU_PECT	10 19 44
	PM_SYNC	CPU_50S	CPU_AGTL	PM_SYNC	10 17
	PM_MEM_PWRGD	CPU_50S	CPU_AGTL	PM_MEM_PWRGD	10 17 29
	XDP_CPU_PWRG00D	CPU_50S	CPU_ITP	XDP_CPU_PWRGD	23
	XDP_DBRESET_I	CPU_50S	CPU_ITP	XDP_DBRESET_L	10 23 25
	XDP_PRDY_I	CPU_50S	CPU_ITP	XDP_CPU_PRDY_L	10 23
	XDP_PREQ_I	CPU_50S	CPU_ITP	XDP_CPU_PREQ_L	10 23
	CPU_SM_RCOMP	CPU_27P4S	CPU_COMP	CPU_SM_RCOMP0	
	CPU_SM_RCOMP	CPU_27P4S	CPU_COMP	CPU_SM_RCOMP1	
	CPU_SM_RCOMP	CPU_27P4S	CPU_COMP	CPU_SM_RCOMP2	
	CPU_CFG	CPU_50S	CPU_ITP	CPU_CFG<11..0>	9 23
	CPU_CFG	CPU_50S	CPU_ITP	CPU_CFG<17..16>	9 23
	CPU_CATERR_I	CPU_50S	CPU_AGTL	CPU_CATERR_L	10
		CPU_50S	CPU_AGTL	CPU_PROC_SEL_L	10 17
		CPU_50S	CPU_AGTL	TP_CPU_VTT_SELECT	8
	CPU_PROCHOT_I	CPU_50S	CPU_AGTL	CPU_PROCHOT_L	10 45 67
	CPU_PWRGD	CPU_50S	CPU_AGTL	CPU_PWRGD	10 19 23
	PM_THRMTRIP_I	CPU_50S	CPU_8MIL	PM_THRMTRIP_L	10 19
	XDP_CLK_CPU	CLK_PCIE_90D	CLK_PCIE	ITPCPU_CLK100M_P	10 16
	XDP_CLK_CPU	CLK_PCIE_90D	CLK_PCIE	ITPCPU_CLK100M_N	10 16
	XDP_CLK_PCH	CLK_PCIE_90D	CLK_PCIE	ITPXDPCLK100M_P	16 23
	XDP_CLK_PCH	CLK_PCIE_90D	CLK_PCIE	ITPXDPCLK100M_N	16 23
	XDP_CLK_ITP	CLK_PCIE_90D	CLK_PCIE	XDP_CPU_CLK100M_P	23
	XDP_CLK_ITP	CLK_PCIE_90D	CLK_PCIE	XDP_CPU_CLK100M_N	23
		CPU_55S	CPU_8MTI	CPU_PSI_L	
	PM_DPRSPLVR	CPU_50S	CPU_AGTL	PM_DPRSPLVR	
		CPU_27P4S	CPU_COMP	CPU_PEG_COMP	9
		CPU_27P4S	CPU_COMP	CPU_PEG_RBIAS	
	CPU_COMP	CPU_27P4S	CPU_COMP	CPU_COMP3	
	CPU_COMP	CPU_27P4S	CPU_COMP	CPU_COMP2	
	CPU_COMP	CPU_27P4S	CPU_COMP	CPU_COMP1	
	CPU_COMP	CPU_27P4S	CPU_COMP	CPU_COMP0	
	XDP_TDI	CPU_50S	CPU_ITP	XDP_CPU_TDI	10 23
	XDP_TDO	CPU_50S	CPU_ITP	XDP_CPU_TDO	10 23
	XDP_TMS	CPU_50S	CPU_ITP	XDP_CPU_TMS	10 23
	XDP_TCK	CPU_50S	CPU_ITP	XDP_CPU_TCK	10 23
	XDP_TRST_I	CPU_50S	CPU_ITP	XDP_CPU_TRST_L	10 23
	XDP_BPM	CPU_50S	CPU_ITP	XDP_BPM_L<3..0>	10 23
	XDP_BPM_I	CPU_50S	CPU_ITP	XDP_BPM_L<7..4>	10 23
	(FSB_CPURST_I)	CPU_50S	CPU_ITP	XDP_CPURST_L	23
		CPU_55S	CPU_8MTI	CPU_VID<6..0>	8
		CPU_50S	CPU_AGTL	CPUIMVP_IMON	
	CPU_VCCSENSE	CPU_27P4S	CPU_VCCSENSE	CPU_VCCSENSE_P	12 67
	CPU_VCCSENSE	CPU_27P4S	CPU_VCCSENSE	CPU_VCCSENSE_N	12 67
	CPU_VCCSENSE	CPU_27P4S	CPU_VCCSENSE	CPU_VCCIOSENSE_P	12 69
	CPU_VCCSENSE	CPU_27P4S	CPU_VCCSENSE	CPU_VCCIOSENSE_N	12 69
	CPU_VCCSENSE	CPU_27P4S	CPU_VCCSENSE	CPU_AXG_SENSE_P	12 67
	CPU_VCCSENSE	CPU_27P4S	CPU_VCCSENSE	CPU_AXG_SENSE_N	12 67
		CPU_27P4S	CPU_VCCSENSE	CPU_VCC_VALSENSE_P	12
		CPU_27P4S	CPU_VCCSENSE	CPU_VCC_VALSENSE_N	12
		CPU_27P4S	CPU_VCCSENSE	CPU_AXG_VALSENSE_P	12
		CPU_27P4S	CPU_VCCSENSE	CPU_AXG_VALSENSE_N	12
		CPU_55S	CPU_8MTI	GFX_VID<6..0>	8
	PM_DPRSPLVR	CPU_50S	CPU_AGTL	GFX_DPRSPLVR	
		CPU_50S	CPU_AGTL	GFX_VR_EN	
		CPU_50S	CPU_AGTL	GFXIMVP_IMON	
		PCIE_85D	PCIE	PEG_R2D_P<7..0>	73
		PCIE_85D	PCIE	PEG_R2D_N<7..0>	73
	PEG_R2D	PCIE_85D	PCIE	PEG_R2D_C_P<7..0>	8 73
		PCIE_85D	PCIE	PEG_R2D_C_N<7..0>	8 73
	PEG_D2R	PCIE_85D	PCIE	PEG_D2R_P<7..0>	8 73
		PCIE_85D	PCIE	PEG_D2R_N<7..0>	8 73
		PCIE_85D	PCIE	PEG_D2R_C_P<7..0>	73
		PCIE_85D	PCIE	PEG_D2R_C_N<7..0>	73
		CPU_50S	CPU_VID	CPU_VIDSOUT	12 67
		CPU_50S	CPU_VID	CPU_VIDSCLK	12 67
		CPU_50S	CPU_VID	CPU_VIDALERT_L	12 67

SYNC_MASTER=K92_MLB		SYNC_DATE=08/09/2016	
PAGE TITLE		CPU Constraints	
Apple Inc.		DRAWING NUMBER	SIZE D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		REVISION	
		BRANCH	
		PAGE	100 OF 132
		SHEET	90 OF 101



Memory Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MEM_37S	*	=37_OHM_SE	=37_OHM_SE	=37_OHM_SE	=37_OHM_SE	=STANDARD	=STANDARD
MEM_40S	*	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=STANDARD	=STANDARD
MEM_72D	*	=72_OHM_DIFF	=72_OHM_DIFF	=72_OHM_DIFF	=72_OHM_DIFF	=72_OHM_DIFF	=72_OHM_DIFF
MEM_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
MEM_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MEM_CLK2MEM	*	=4:1_SPACING	?
MEM_CTRL2CTRL	*	=3:1_SPACING	?
MEM_CTRL2MEM	*	=2.5:1_SPACING	?
MEM_CMD2CMD	*	=1.5:1_SPACING	?
MEM_CMD2MEM	*	=3:1_SPACING	?
MEM_DATA2DATA	*	=1.5:1_SPACING	?
MEM_DATA2MEM	*	=3:1_SPACING	?
MEM_DQS2MEM	*	=3:1_SPACING	?
MEM_20THER	*	25 MILS	?

Memory Bus Spacing Group Assignments

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CLK	MEM_CLK	*	MEM_CLK2MEM
MEM_CLK	MEM_CTRL	*	MEM_CLK2MEM
MEM_CLK	MEM_CMD	*	MEM_CLK2MEM
MEM_CLK	MEM_DATA	*	MEM_CLK2MEM
MEM_CLK	MEM_DQS	*	MEM_CLK2MEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CMD	MEM_CLK	*	MEM_CMD2MEM
MEM_CMD	MEM_CTRL	*	MEM_CMD2MEM
MEM_CMD	MEM_CMD	*	MEM_CMD2CMD
MEM_CMD	MEM_DATA	*	MEM_CMD2MEM
MEM_CMD	MEM_DQS	*	MEM_CMD2MEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CTRL	MEM_CLK	*	MEM_CTRL2MEM
MEM_CTRL	MEM_CTRL	*	MEM_CTRL2CTRL
MEM_CTRL	MEM_CMD	*	MEM_CTRL2MEM
MEM_CTRL	MEM_DATA	*	MEM_CTRL2MEM
MEM_CTRL	MEM_DQS	*	MEM_CTRL2MEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_DATA	MEM_CLK	*	MEM_DATA2MEM
MEM_DATA	MEM_CTRL	*	MEM_DATA2MEM
MEM_DATA	MEM_CMD	*	MEM_DATA2MEM
MEM_DATA	MEM_DATA	*	MEM_DATA2DATA
MEM_DATA	MEM_DQS	*	MEM_DATA2MEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_DQS	MEM_CLK	*	MEM_DQS2MEM
MEM_DQS	MEM_CTRL	*	MEM_DQS2MEM
MEM_DQS	MEM_CMD	*	MEM_DQS2MEM
MEM_DQS	MEM_DATA	*	MEM_DQS2MEM
MEM_DQS	MEM_DQS	*	MEM_DQS2MEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CLK	*	*	MEM_20THER
MEM_CTRL	*	*	MEM_20THER
MEM_CMD	*	*	MEM_20THER
MEM_DATA	*	*	MEM_20THER
MEM_DQS	*	*	MEM_20THER

DDR3:

DQ/DM signals should be matched within 0.508mm of associated DQS pair.
DQS intra-pair matching should be within 0.127mm, no inter-pair matching requirement.
DQS to clock matching should be within [CLK-12.7mm] and [CLK+25.4mm].
CLK intra-pair matching should be within 0.127mm, inter-pair matching should be within 0.508mm.
CONTROL signals should be matched within [CLK-12.7mm] to [CLK+0.0mm] of CLK pairs.
A/BA/CMD signals should be matched within [CLK-12.7mm] to [CLK+12.7mm] of CLK pairs.
DQ/DQS/A/BA/cmd signal spacing is 4x dielectric, CLK is 5x dielectric.
Maximum length of any signal from die pad to SODIMM pad is 139.7mm, from processor ball to SODIMM pad is 114.3mm.

SOURCE: Calpella SFF Platform DG, Rev 1.5 (#407364), Section 2.2

Memory Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE			
	PHYSICAL	SPACING		
MEM_A_CLK	MEM_72D	MEM_CLK	MEM A CLK P<5..0>	6 11 26
MEM_A_CLK	MEM_72D	MEM_CLK	MEM A CLK N<5..0>	6 11 26
MEM_A_CNTL	MEM_37S	MEM_CTRL	MEM A CKE<3..0>	6 11 26
MEM_A_CNTL	MEM_37S	MEM_CTRL	MEM A CS L<3..0>	6 11 26
MEM_A_CNTL	MEM_37S	MEM_CTRL	MEM A ODT<3..0>	6 11 26
MEM_A_CMD	MEM_40S	MEM_CMD	MEM A A<15..0>	6 11 26
MEM_A_CMD	MEM_40S	MEM_CMD	MEM A BA<2..0>	6 11 26
MEM_A_CMD	MEM_40S	MEM_CMD	MEM A RAS L	6 11 26
MEM_A_CMD	MEM_40S	MEM_CMD	MEM A CAS L	6 11 26
MEM_A_CMD	MEM_40S	MEM_CMD	MEM A WE L	6 11 26
MEM_A_DQ_BYTE0	MEM_50S	MEM_DATA	MEM A DQ<7..0>	6 11 27
MEM_A_DQ_BYTE1	MEM_50S	MEM_DATA	MEM A DQ<15..8>	6 11 27
MEM_A_DQ_BYTE2	MEM_50S	MEM_DATA	MEM A DQ<23..16>	6 11 27
MEM_A_DQ_BYTE3	MEM_50S	MEM_DATA	MEM A DQ<31..24>	6 11 27
MEM_A_DQ_BYTE4	MEM_50S	MEM_DATA	MEM A DQ<39..32>	6 11 26 27
MEM_A_DQ_BYTE5	MEM_50S	MEM_DATA	MEM A DQ<47..40>	6 11 27
MEM_A_DQ_BYTE6	MEM_50S	MEM_DATA	MEM A DQ<55..48>	6 11 27
MEM_A_DQ_BYTE7	MEM_50S	MEM_DATA	MEM A DQ<63..56>	6 11 27
MEM_A_DQS0	MEM_85D	MEM_DQS	MEM A DQS P<0>	6 11 27
MEM_A_DQS0	MEM_85D	MEM_DQS	MEM A DQS N<0>	6 11 27
MEM_A_DQS1	MEM_85D	MEM_DQS	MEM A DQS P<1>	6 11 27
MEM_A_DQS1	MEM_85D	MEM_DQS	MEM A DQS N<1>	6 11 27
MEM_A_DQS2	MEM_85D	MEM_DQS	MEM A DQS P<2>	6 11 27
MEM_A_DQS2	MEM_85D	MEM_DQS	MEM A DQS N<2>	6 11 27
MEM_A_DQS3	MEM_85D	MEM_DQS	MEM A DQS P<3>	6 11 27
MEM_A_DQS3	MEM_85D	MEM_DQS	MEM A DQS N<3>	6 11 27
MEM_A_DQS4	MEM_85D	MEM_DQS	MEM A DQS P<4>	6 11 27
MEM_A_DQS4	MEM_85D	MEM_DQS	MEM A DQS N<4>	6 11 27
MEM_A_DQS5	MEM_85D	MEM_DQS	MEM A DQS P<5>	6 11 27
MEM_A_DQS5	MEM_85D	MEM_DQS	MEM A DQS N<5>	6 11 27
MEM_A_DQS6	MEM_85D	MEM_DQS	MEM A DQS P<6>	6 11 26 27
MEM_A_DQS6	MEM_85D	MEM_DQS	MEM A DQS N<6>	6 11 26 27
MEM_A_DQS7	MEM_85D	MEM_DQS	MEM A DQS P<7>	6 11 27
MEM_A_DQS7	MEM_85D	MEM_DQS	MEM A DQS N<7>	6 11 27
MEM_B_CLK	MEM_72D	MEM_CLK	MEM B CLK P<5..0>	6 11 28
MEM_B_CLK	MEM_72D	MEM_CLK	MEM B CLK N<5..0>	6 11 28
MEM_B_CNTL	MEM_37S	MEM_CTRL	MEM B CKE<3..0>	6 11 28
MEM_B_CNTL	MEM_37S	MEM_CTRL	MEM B CS L<3..0>	6 11 28
MEM_B_CNTL	MEM_37S	MEM_CTRL	MEM B ODT<3..0>	6 11 28
MEM_B_CMD	MEM_40S	MEM_CMD	MEM B A<15..0>	6 11 28
MEM_B_CMD	MEM_40S	MEM_CMD	MEM B BA<2..0>	6 11 28
MEM_B_CMD	MEM_40S	MEM_CMD	MEM B RAS L	6 11 28
MEM_B_CMD	MEM_40S	MEM_CMD	MEM B CAS L	6 11 28
MEM_B_CMD	MEM_40S	MEM_CMD	MEM B WE L	6 11 28
MEM_B_DQ_BYTE0	MEM_50S	MEM_DATA	MEM B DQ<7..0>	6 11 27
MEM_B_DQ_BYTE1	MEM_50S	MEM_DATA	MEM B DQ<15..8>	6 11 27
MEM_B_DQ_BYTE2	MEM_50S	MEM_DATA	MEM B DQ<23..16>	6 11 27
MEM_B_DQ_BYTE3	MEM_50S	MEM_DATA	MEM B DQ<31..24>	6 11 27
MEM_B_DQ_BYTE4	MEM_50S	MEM_DATA	MEM B DQ<39..32>	6 11 27 28
MEM_B_DQ_BYTE5	MEM_50S	MEM_DATA	MEM B DQ<47..40>	6 11 27
MEM_B_DQ_BYTE6	MEM_50S	MEM_DATA	MEM B DQ<55..48>	6 11 27
MEM_B_DQ_BYTE7	MEM_50S	MEM_DATA	MEM B DQ<63..56>	6 11 27
MEM_B_DQS0	MEM_85D	MEM_DQS	MEM B DQS P<0>	6 11 27
MEM_B_DQS0	MEM_85D	MEM_DQS	MEM B DQS N<0>	6 11 27
MEM_B_DQS1	MEM_85D	MEM_DQS	MEM B DQS P<1>	6 11 27
MEM_B_DQS1	MEM_85D	MEM_DQS	MEM B DQS N<1>	6 11 27
MEM_B_DQS2	MEM_85D	MEM_DQS	MEM B DQS P<2>	6 11 27
MEM_B_DQS2	MEM_85D	MEM_DQS	MEM B DQS N<2>	6 11 27
MEM_B_DQS3	MEM_85D	MEM_DQS	MEM B DQS P<3>	6 11 27
MEM_B_DQS3	MEM_85D	MEM_DQS	MEM B DQS N<3>	6 11 27
MEM_B_DQS4	MEM_85D	MEM_DQS	MEM B DQS P<4>	6 11 27
MEM_B_DQS4	MEM_85D	MEM_DQS	MEM B DQS N<4>	6 11 27
MEM_B_DQS5	MEM_85D	MEM_DQS	MEM B DQS P<5>	6 11 27
MEM_B_DQS5	MEM_85D	MEM_DQS	MEM B DQS N<5>	6 11 27
MEM_B_DQS6	MEM_85D	MEM_DQS	MEM B DQS P<6>	6 11 27 28
MEM_B_DQS6	MEM_85D	MEM_DQS	MEM B DQS N<6>	6 11 27 28
MEM_B_DQS7	MEM_85D	MEM_DQS	MEM B DQS P<7>	6 11 27
MEM_B_DQS7	MEM_85D	MEM_DQS	MEM B DQS N<7>	6 11 27

SYNC_MASTER=K18 MLB		SYNC_DATE=04/27/2016	
PAGE TITLE			
Memory Constraints		DRAWING NUMBER	SIZE D
 Apple Inc.		REVISION	
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE	101 OF 132
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		SHEET	91 OF 101
II NOT TO REPRODUCE OR COPY IT			
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			



Digital Video Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
DP_85D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF
LVDS_85D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
DISPLAYPORT	TS13, TS14, TS19, TS110	=4:1_SPACING	?	DISPLAYPORT	TOP, BOTTOM	=4:1_SPACING	?
LVDS	TS13, TS14, TS19, TS110	=4:1_SPACING	?	LVDS	TOP, BOTTOM	=4:1_SPACING	?

SOURCE: HR PLATFORM DESIGN GUIDE, TABLES 191,193

SATA Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
SATA_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF
SATA_37SE	*	=37_OHM_SE	=37_OHM_SE	=37_OHM_SE	=37_OHM_SE	=37_OHM_SE	=37_OHM_SE
SATA_50SE	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
SATA	TS13, TS14, TS19, TS110	=5:1_SPACING	?	SATA	TOP, BOTTOM	=5:1_SPACING	?
SATA_ICOMP	*	15 MIL	?				

SOURCE: HR PLATFORM DESIGN GUIDE, TABLES 191,193

USB 2.0 Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
PCH_USB_RBIAIS	*	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
USB_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

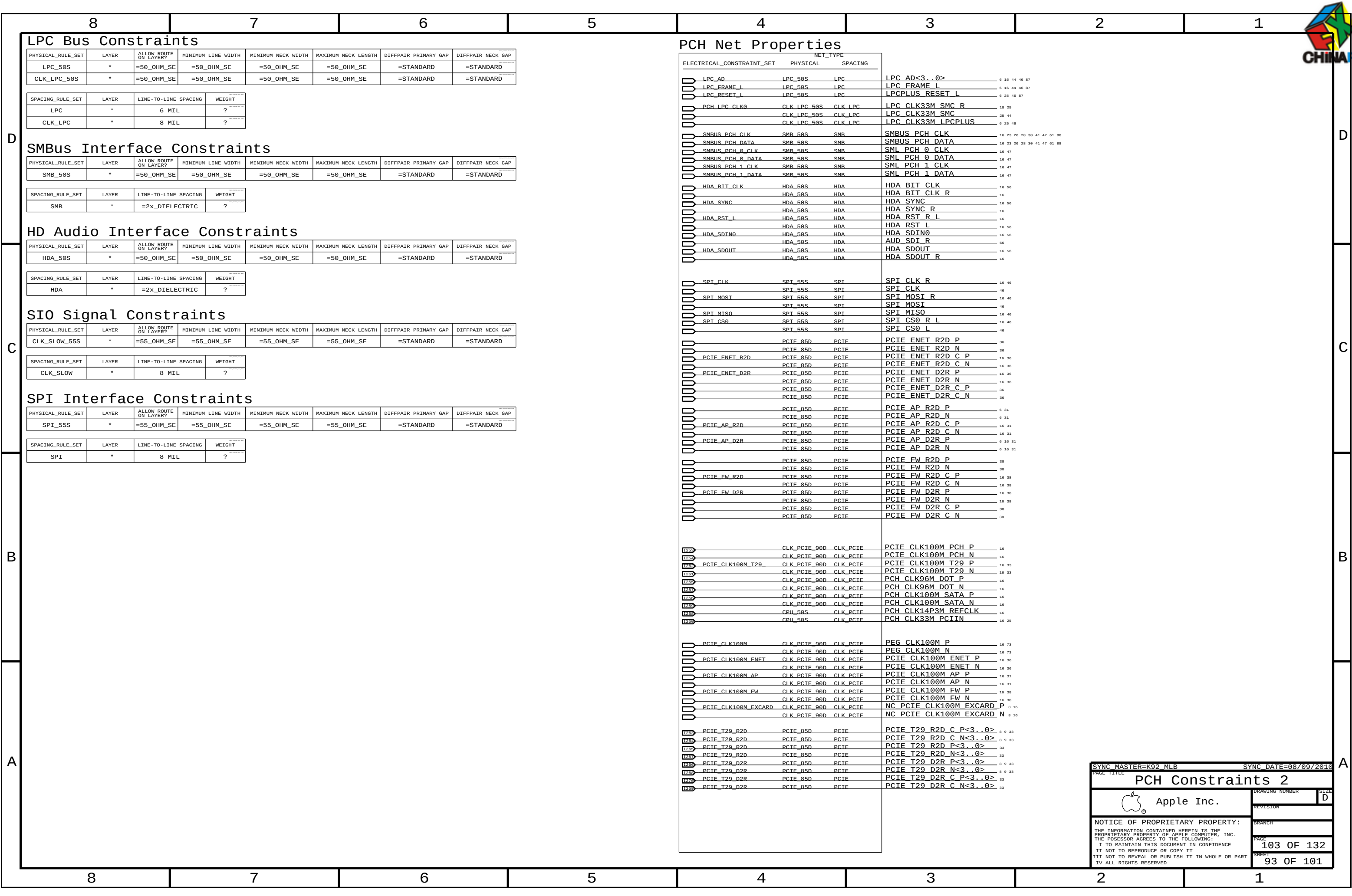
SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
USB	TS13, TS14, TS19, TS110	=4:1_SPACING	?	USB	TOP, BOTTOM	=4:1_SPACING	?
USB_RBIAIS	*	15 MIL	?				

SOURCE: HR PLATFORM DESIGN GUIDE, TABLES 191,193

PCH Net Properties

NET_TYPE			
ELECTRICAL_CONSTRAINT_SET	PHYSICAL	SPACING	
DP_AUX_CH	DP_85D	DISPLAYPORT	DP_IG_AUX_CH_P 8 17 83
DP_AUX_CH	DP_85D	DISPLAYPORT	DP_IG_AUX_CH_N 8 17 83
LVDS_IG_A_CLK	LVDS_85D	LVDS	LVDS_IG_A_CLK_P 18 87
LVDS_IG_A_CLK	LVDS_85D	LVDS	LVDS_IG_A_CLK_N 18 87
LVDS_IG_A_DATA	LVDS_85D	LVDS	LVDS_IG_A_DATA_P<2..0> 18 87
LVDS_IG_A_DATA	LVDS_85D	LVDS	LVDS_IG_A_DATA_N<2..0> 18 87
LVDS_IG_A_DATA3	LVDS_85D	LVDS	NC_LVDS_IG_A_DATAP<3> 8 18
LVDS_IG_A_DATA3	LVDS_85D	LVDS	NC_LVDS_IG_A_DATAN<3> 8 18
LVDS_IG_B_DATA	LVDS_85D	LVDS	LVDS_IG_B_DATA_P<2..0> 18 87
LVDS_IG_B_DATA	LVDS_85D	LVDS	LVDS_IG_B_DATA_N<2..0> 18 87
SATA_HDD_R2D	SATA_90D	SATA	SATA_HDD_R2D_C_P 16 41
SATA_HDD_R2D	SATA_90D	SATA	SATA_HDD_R2D_C_N 16 41
SATA_HDD_R2D	SATA_90D	SATA	SATA_HDD_R2D_P 6 41
SATA_HDD_R2D	SATA_90D	SATA	SATA_HDD_R2D_N 6 41
SATA_HDD_R2D	SATA_90D	SATA	SATA_HDD_R2D_UF_P 41
SATA_HDD_R2D	SATA_90D	SATA	SATA_HDD_R2D_UF_N 41
SATA_HDD_D2R	SATA_90D	SATA	SATA_HDD_D2R_P 16 41
SATA_HDD_D2R	SATA_90D	SATA	SATA_HDD_D2R_N 16 41
SATA_HDD_D2R	SATA_90D	SATA	SATA_HDD_D2R_C_P 6 41
SATA_HDD_D2R	SATA_90D	SATA	SATA_HDD_D2R_C_N 6 41
SATA_ODD_R2D	SATA_90D	SATA	SATA_ODD_R2D_C_P 16 41
SATA_ODD_R2D	SATA_90D	SATA	SATA_ODD_R2D_C_N 16 41
SATA_ODD_R2D	SATA_90D	SATA	SATA_ODD_R2D_P 6 41
SATA_ODD_R2D	SATA_90D	SATA	SATA_ODD_R2D_N 6 41
SATA_ODD_D2R	SATA_90D	SATA	SATA_ODD_D2R_P 16 41
SATA_ODD_D2R	SATA_90D	SATA	SATA_ODD_D2R_N 16 41
SATA_ODD_D2R	SATA_90D	SATA	SATA_ODD_D2R_UF_P 6 41
SATA_ODD_D2R	SATA_90D	SATA	SATA_ODD_D2R_UF_N 6 41
PCH_SATA3_ICOMP	SATA_50SE	SATA_ICOMP	PCH_SATA3COMP 16
PCH_SATA_ICOMP	SATA_37SE	SATA_ICOMP	PCH_SATAICOMP 16
USB_HUB1_UP	USB_85D	USB	USB_HUB1_UP_P 18 24
USB_HUB1_UP	USB_85D	USB	USB_HUB1_UP_N 18 24
USB_HUB2_UP	USB_85D	USB	USB_HUB2_UP_P 18 24
USB_HUB2_UP	USB_85D	USB	USB_HUB2_UP_N 18 24
USB_EXT_A	USB_85D	USB	USB_EXT_A_P 24 42
USB_EXT_A	USB_85D	USB	USB_EXT_A_N 24 42
USB_EXT_B	USB_85D	USB	USB_EXT_B_P 24 42
USB_EXT_B	USB_85D	USB	USB_EXT_B_N 24 42
USB_EXT_C	USB_85D	USB	USB_EXT_C_P 8 24
USB_EXT_C	USB_85D	USB	USB_EXT_C_N 8 24
USB_CAMERA	USB_85D	USB	USB_CAMERA_CONN_P 6 31
USB_CAMERA	USB_85D	USB	USB_CAMERA_CONN_N 6 31
USB_BT	USB_85D	USB	USB_BT_P 24 31
USB_BT	USB_85D	USB	USB_BT_N 24 31
USB_TPAD	USB_85D	USB	USB_TPAD_P 24 52
USB_TPAD	USB_85D	USB	USB_TPAD_N 24 52
USB_IR	USB_85D	USB	USB_IR_P 24 43
USB_IR	USB_85D	USB	USB_IR_N 24 43
PCH_USB_RBIAIS	PCH_USB_RBIAIS	USB_RBIAIS	PCH_USB_RBIAIS 18
USB_T29A	USB_85D	USB	USB_T29A_P 8 24
USB_T29A	USB_85D	USB	USB_T29A_N 8 24

SYNC_MASTER=K92_MLB		SYNC_DATE=08/09/2016	
PAGE TITLE			
PCH Constraints 1			
 Apple Inc.		DRAWING NUMBER	SIZE
			D
NOTICE OF PROPRIETARY PROPERTY:		REVISION	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:		BRANCH	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		PAGE	
II NOT TO REPRODUCE OR COPY IT		102 OF 132	
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART		SHEET	
IV ALL RIGHTS RESERVED		92 OF 101	





8

7

6

5

4

3

2

1

CAESAR IV (Ethernet) Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
ENET_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
ENET_3X	*	=3:1_SPACING	?

SOURCE: Broadcom 5764-DS04-RDS Page 38

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
ENET_CR	*	=3X_DIELECTRIC	?

SOURCE: Attila Farkas Email - 8/2/10

CAESAR IV (Ethernet PHY) Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
ENET_100D	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
ENET_MDI	*	0.6 MM	?

SOURCE: Broadcom 5764-DS04-RDS Page 38

Ethernet Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE		
	PHYSICAL	SPACING	
	ENET_50S	ENET_3X	BCM5764_CLK25M_XTALI
	ENET_50S	ENET_3X	BCM5764_CLK25M_XTALO
	ENET_50S	ENET_3X	ENET_RESET_L
ENET_MDT	ENET_100D	ENET_MDT	ENET_MDI_P<3..0>
	ENET_100D	ENET_MDI	ENET_MDI_N<3..0>
EEED CR_DATA_A0	ENET_50S	ENET_CR	SDCONN_DATA_R<7..0>
EEED CR_DATA_A0	ENET_50S	ENET_CR	SDCONN_CMD_R
EEED CR_CLK	ENET_50S	ENET_CR	SDCONN_CLK_R
EEED CR_DATA_A0	ENET_50S	ENET_CR	SDCONN_DATA<7..0>
EEED CR_DATA_A0	ENET_50S	ENET_CR	SDCONN_CMD
EEED CR_CLK	ENET_50S	ENET_CR	SDCONN_CLK
EEED CR_CLK	ENET_50S	ENET_CR	SDCONN_CLK_R_L_32

D

D

C

FireWire Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
FW_110D	*	=110_OHM_DIFF	=110_OHM_DIFF	=110_OHM_DIFF	=110_OHM_DIFF	=110_OHM_DIFF	=110_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
FW_TP	*	=3:1_SPACING	?

FireWire Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE		
	PHYSICAL	SPACING	
EEED FW_P0_TPA	FW_110D	FW_TP	NC_FW0_TPAP
EEED FW_P0_TPA	FW_110D	FW_TP	NC_FW0_TPAN
EEED FW_P0_TPB	FW_110D	FW_TP	NC_FW0_TBPB
EEED FW_P0_TPB	FW_110D	FW_TP	NC_FW0_TPBH
EEED FW_P1_TPA	FW_110D	FW_TP	FW_PORT1_TPA_P
EEED FW_P1_TPA	FW_110D	FW_TP	FW_PORT1_TPA_N
EEED FW_P1_TPB	FW_110D	FW_TP	FW_PORT1_TPB_P
EEED FW_P1_TPB	FW_110D	FW_TP	FW_PORT1_TPB_N
Port 2 Not Used			

C

B

B

A

A

8

7

6

5

4

3

2

1

SYNC_MASTER=K91_ERIC		SYNC_DATE=08/03/2010	
PAGE TITLE			
Ethernet/FW Constraints		DRAWING NUMBER	SIZE
 Apple Inc.		REVISION	D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		BRANCH	
		PAGE	104 OF 132
		SHEET	94 OF 101



D

C

B

A

1

1

1

1

D

C

B

A

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
1T01_DIFFPAIR	*	=STANDARD	=STANDARD	=STANDARD	=STANDARD	0.1 MM	0.1 MM

SMC SMBus Net Properties

NET_TYPE			
ELECTRICAL_CONSTRAINT_SET	PHYSICAL	SPACING	
 SMBUS_SMC_A_S3_SCL	SMB_50S	SMB	SMBUS_SMC_A_S3_SCL 6 31 44 47 53 54
 SMBUS_SMC_A_S3_SDA	SMB_50S	SMB	SMBUS_SMC_A_S3_SDA 6 31 44 47 53 54
 SMBUS_SMC_B_S0_SCL	SMB_50S	SMB	SMBUS_SMC_B_S0_SCL 44 47 50
 SMBUS_SMC_B_S0_SDA	SMB_50S	SMB	SMBUS_SMC_B_S0_SDA 44 47 50
 SMBUS_SMC_0_S0_SCL	SMB_50S	SMB	SMBUS_SMC_0_S0_SCL 6 31 44 47 50 79
 SMBUS_SMC_0_S0_SDA	SMB_50S	SMB	SMBUS_SMC_0_S0_SDA 6 31 44 47 50 79
 SMBUS_SMC_BSA_SCL	SMB_50S	SMB	SMBUS_SMC_BSA_SCL 6 44 47 62 63
 SMBUS_SMC_BSA_SDA	SMB_50S	SMB	SMBUS_SMC_BSA_SDA 6 44 47 62 63
 SMBUS_SMC_MGMT_SCL	SMB_50S	SMB	SMBUS_SMC_MGMT_SCL 44 47 100
 SMBUS_SMC_MGMT_SDA	SMB_50S	SMB	SMBUS_SMC_MGMT_SDA 44 47 100

SMBus Charger Net Properties

NET_TYPE			
ELECTRICAL_CONSTRAINT_SET	PHYSICAL	SPACING	
 CHGR_CSI	1T01_DIEFPATR		CHGR_CSI_P 63
 CHGR_CSI	1T01_DIEFPATR		CHGR_CSI_N 63
 CHGR_CS0	1T01_DIEFPATR		CHGR_CS0_P 63
 CHGR_CS0	1T01_DIEFPATR		CHGR_CS0_N 63



D

C

B

A

SYNC_MASTER=K18_MLB

SYNC_DATE=04/27/2016

PAGE_TITLE

SMC Constraints

 Apple Inc.

NOTICE OF PROPRIETARY PROPERTY:
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

DRAWING_NUMBER

REVISION

BRANCH

PAGE

SHEET

SIZE

D

106 OF 132

96 OF 101



GDDR5 Frame Buffer Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
GDDR5_45R50SE	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	12.7 MM	=STANDARD	=STANDARD
GDDR5_45SE	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
GDDR5_80D	*	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GDDR5_CLK	*	=5x_DIELECTRIC	?
GDDR5_CMD	*	=2x_DIELECTRIC	?
GDDR5_DATA	*	=3x_DIELECTRIC	?
GDDR5_EDC	*	=7x_DIELECTRIC	?

Digital Video Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DP_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
LVDS_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DISPLAYPORT	*	=3x_DIELECTRIC	?	DISPLAYPORT	TOP, BOTTOM	=4x_DIELECTRIC	?
LVDS	*	=3x_DIELECTRIC	?	LVDS	TOP, BOTTOM	=4x_DIELECTRIC	?

LVDS intra-pair matching should be 0.127 mm. Pairs should be within 0.508mm of entire channel.
DisplayPort/TMDS intra-pair matching should be 0.127mm. Inter-pair matching should be within 2.54cm. Max Length 241.3mm.
DisplayPort AUX CH intra-pair matching should be 0.127mm. Max length 330.2mm.
Max length of LVDS/DisplayPort/TMDS traces: 13 inches.
SOURCE: Calpella SFF DG Rev 1.5 (497364) and Family GPU DG-04202-001-v04.

GDDR5 FB A Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE			
	PHYSICAL	SPACING		
FB_A0_CLK	GDDR5_80D	GDDR5_CLK	FB A0 CLK P	75 76
FB_A0_CLK	GDDR5_80D	GDDR5_CLK	FB A0 CLK N	75 76
FB_A1_CLK	GDDR5_80D	GDDR5_CLK	FB A1 CLK P	75 76
FB_A1_CLK	GDDR5_80D	GDDR5_CLK	FB A1 CLK N	75 76
FB_A0_CMD	GDDR5_45R50SE	GDDR5_CMD	FB A0 A<8..0>	6 75 76
FB_A1_CMD	GDDR5_45R50SE	GDDR5_CMD	FB A1 A<8..0>	6 75 76
FB_A0_CMD	GDDR5_45R50SE	GDDR5_CMD	FB A0 ABI L	6 75 76
FB_A1_CMD	GDDR5_45R50SE	GDDR5_CMD	FB A1 ABI L	6 75 76
FB_A0_CMD	GDDR5_45R50SE	GDDR5_CMD	FB A0 RAS L	75 76
FB_A1_CMD	GDDR5_45R50SE	GDDR5_CMD	FB A1 RAS L	75 76
FB_A0_CMD	GDDR5_45R50SE	GDDR5_CMD	FB A0 CAS L	75 76
FB_A1_CMD	GDDR5_45R50SE	GDDR5_CMD	FB A1 CAS L	75 76
FB_A0_CMD	GDDR5_45R50SE	GDDR5_CMD	FB A0 WE L	75 76
FB_A1_CMD	GDDR5_45R50SE	GDDR5_CMD	FB A1 WE L	75 76
FB_A0_CMD	GDDR5_45R50SE	GDDR5_CMD	FB A0 CKE L	75 76
FB_A1_CMD	GDDR5_45R50SE	GDDR5_CMD	FB A1 CKE L	75 76
FB_A0_CMD	GDDR5_45R50SE	GDDR5_CMD	FB A0 CS L	75 76
FB_A1_CMD	GDDR5_45R50SE	GDDR5_CMD	FB A1 CS L	75 76
FB_A0_EDC0	GDDR5_45SE	GDDR5_EDC	FB A0 EDC<0>	6 75 76
FB_A0_EDC1	GDDR5_45SE	GDDR5_EDC	FB A0 EDC<1>	6 75 76
FB_A0_EDC2	GDDR5_45SE	GDDR5_EDC	FB A0 EDC<2>	6 75 76
FB_A0_EDC3	GDDR5_45SE	GDDR5_EDC	FB A0 EDC<3>	6 75 76
FB_A1_EDC0	GDDR5_45SE	GDDR5_EDC	FB A1 EDC<0>	6 75 76
FB_A1_EDC1	GDDR5_45SE	GDDR5_EDC	FB A1 EDC<1>	6 75 76
FB_A1_EDC2	GDDR5_45SE	GDDR5_EDC	FB A1 EDC<2>	6 75 76
FB_A1_EDC3	GDDR5_45SE	GDDR5_EDC	FB A1 EDC<3>	6 75 76
FB_A0_DBI_L0	GDDR5_45SE	GDDR5_DATA	FB A0 DBI L<0>	6 75 76
FB_A0_DBI_L1	GDDR5_45SE	GDDR5_DATA	FB A0 DBI L<1>	6 75 76
FB_A0_DBI_L2	GDDR5_45SE	GDDR5_DATA	FB A0 DBI L<2>	6 75 76
FB_A0_DBI_L3	GDDR5_45SE	GDDR5_DATA	FB A0 DBI L<3>	6 75 76
FB_A1_DBI_L0	GDDR5_45SE	GDDR5_DATA	FB A1 DBI L<0>	6 75 76
FB_A1_DBI_L1	GDDR5_45SE	GDDR5_DATA	FB A1 DBI L<1>	6 75 76
FB_A1_DBI_L2	GDDR5_45SE	GDDR5_DATA	FB A1 DBI L<2>	6 75 76
FB_A1_DBI_L3	GDDR5_45SE	GDDR5_DATA	FB A1 DBI L<3>	6 75 76
FB_A0_WCLK0	GDDR5_80D	GDDR5_CMD	FB A0 WCLK P<0>	6 75 76
FB_A0_WCLK0	GDDR5_80D	GDDR5_CMD	FB A0 WCLK N<0>	6 75 76
FB_A0_WCLK1	GDDR5_80D	GDDR5_CMD	FB A0 WCLK P<1>	6 75 76
FB_A0_WCLK1	GDDR5_80D	GDDR5_CMD	FB A0 WCLK N<1>	6 75 76
FB_A1_WCLK0	GDDR5_80D	GDDR5_CMD	FB A1 WCLK P<0>	6 75 76
FB_A1_WCLK0	GDDR5_80D	GDDR5_CMD	FB A1 WCLK N<0>	6 75 76
FB_A1_WCLK1	GDDR5_80D	GDDR5_CMD	FB A1 WCLK P<1>	6 75 76
FB_A1_WCLK1	GDDR5_80D	GDDR5_CMD	FB A1 WCLK N<1>	6 75 76
FB_A0_DQ_BYTE0	GDDR5_45SE	GDDR5_DATA	FB A0 DQ<7..0>	6 75 76
FB_A0_DQ_BYTE1	GDDR5_45SE	GDDR5_DATA	FB A0 DQ<15..8>	6 75 76
FB_A0_DQ_BYTE2	GDDR5_45SE	GDDR5_DATA	FB A0 DQ<23..16>	6 75 76
FB_A0_DQ_BYTE3	GDDR5_45SE	GDDR5_DATA	FB A0 DQ<31..24>	6 75 76
FB_A1_DQ_BYTE0	GDDR5_45SE	GDDR5_DATA	FB A1 DQ<7..0>	6 75 76
FB_A1_DQ_BYTE1	GDDR5_45SE	GDDR5_DATA	FB A1 DQ<15..8>	6 75 76
FB_A1_DQ_BYTE2	GDDR5_45SE	GDDR5_DATA	FB A1 DQ<23..16>	6 75 76
FB_A1_DQ_BYTE3	GDDR5_45SE	GDDR5_DATA	FB A1 DQ<31..24>	6 75 76
FB_AB_RESET	GDDR5_45R50SE	GDDR5_CMD	FB RESET L	75 76 77

GDDR5 FB B Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE			
	PHYSICAL	SPACING		
FB_B0_CLK	GDDR5_80D	GDDR5_CLK	FB B0 CLK P	75 77
FB_B0_CLK	GDDR5_80D	GDDR5_CLK	FB B0 CLK N	75 77
FB_B1_CLK	GDDR5_80D	GDDR5_CLK	FB B1 CLK P	75 77
FB_B1_CLK	GDDR5_80D	GDDR5_CLK	FB B1 CLK N	75 77
FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD	FB B0 A<8..0>	6 75 77
FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD	FB B1 A<8..0>	6 75 77
FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD	FB B0 ABI L	6 75 77
FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD	FB B1 ABI L	6 75 77
FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD	FB B0 RAS L	75 77
FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD	FB B1 RAS L	75 77
FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD	FB B0 CAS L	75 77
FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD	FB B1 CAS L	75 77
FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD	FB B0 WE L	75 77
FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD	FB B1 WE L	75 77
FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD	FB B0 CKE L	75 77
FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD	FB B1 CKE L	75 77
FB_B0_CMD	GDDR5_45R50SE	GDDR5_CMD	FB B0 CS L	75 77
FB_B1_CMD	GDDR5_45R50SE	GDDR5_CMD	FB B1 CS L	75 77
FB_B0_EDC0	GDDR5_45SE	GDDR5_EDC	FB B0 EDC<0>	6 75 77
FB_B0_EDC1	GDDR5_45SE	GDDR5_EDC	FB B0 EDC<1>	6 75 77
FB_B0_EDC2	GDDR5_45SE	GDDR5_EDC	FB B0 EDC<2>	6 75 77
FB_B0_EDC3	GDDR5_45SE	GDDR5_EDC	FB B0 EDC<3>	6 75 77
FB_B1_EDC0	GDDR5_45SE	GDDR5_EDC	FB B1 EDC<0>	6 75 77
FB_B1_EDC1	GDDR5_45SE	GDDR5_EDC	FB B1 EDC<1>	6 75 77
FB_B1_EDC2	GDDR5_45SE	GDDR5_EDC	FB B1 EDC<2>	6 75 77
FB_B1_EDC3	GDDR5_45SE	GDDR5_EDC	FB B1 EDC<3>	6 75 77
FB_B0_DBI_L0	GDDR5_45SE	GDDR5_DATA	FB B0 DBI L<0>	6 75 77
FB_B0_DBI_L1	GDDR5_45SE	GDDR5_DATA	FB B0 DBI L<1>	6 75 77
FB_B0_DBI_L2	GDDR5_45SE	GDDR5_DATA	FB B0 DBI L<2>	6 75 77
FB_B0_DBI_L3	GDDR5_45SE	GDDR5_DATA	FB B0 DBI L<3>	6 75 77
FB_B1_DBI_L0	GDDR5_45SE	GDDR5_DATA	FB B1 DBI L<0>	6 75 77
FB_B1_DBI_L1	GDDR5_45SE	GDDR5_DATA	FB B1 DBI L<1>	6 75 77
FB_B1_DBI_L2	GDDR5_45SE	GDDR5_DATA	FB B1 DBI L<2>	6 75 77
FB_B1_DBI_L3	GDDR5_45SE	GDDR5_DATA	FB B1 DBI L<3>	6 75 77
FB_B0_WCLK0	GDDR5_80D	GDDR5_CMD	FB B0 WCLK P<0>	6 75 77
FB_B0_WCLK0	GDDR5_80D	GDDR5_CMD	FB B0 WCLK N<0>	6 75 77
FB_B0_WCLK1	GDDR5_80D	GDDR5_CMD	FB B0 WCLK P<1>	6 75 77
FB_B0_WCLK1	GDDR5_80D	GDDR5_CMD	FB B0 WCLK N<1>	6 75 77
FB_B1_WCLK0	GDDR5_80D	GDDR5_CMD	FB B1 WCLK P<0>	6 75 77
FB_B1_WCLK0	GDDR5_80D	GDDR5_CMD	FB B1 WCLK N<0>	6 75 77
FB_B1_WCLK1	GDDR5_80D	GDDR5_CMD	FB B1 WCLK P<1>	6 75 77
FB_B1_WCLK1	GDDR5_80D	GDDR5_CMD	FB B1 WCLK N<1>	6 75 77
FB_B0_DQ_BYTE0	GDDR5_45SE	GDDR5_DATA	FB B0 DQ<7..0>	6 75 77
FB_B0_DQ_BYTE1	GDDR5_45SE	GDDR5_DATA	FB B0 DQ<15..8>	6 75 77
FB_B0_DQ_BYTE2	GDDR5_45SE	GDDR5_DATA	FB B0 DQ<23..16>	6 75 77
FB_B0_DQ_BYTE3	GDDR5_45SE	GDDR5_DATA	FB B0 DQ<31..24>	6 75 77
FB_B1_DQ_BYTE0	GDDR5_45SE	GDDR5_DATA	FB B1 DQ<7..0>	6 75 77
FB_B1_DQ_BYTE1	GDDR5_45SE	GDDR5_DATA	FB B1 DQ<15..8>	6 75 77
FB_B1_DQ_BYTE2	GDDR5_45SE	GDDR5_DATA	FB B1 DQ<23..16>	6 75 77
FB_B1_DQ_BYTE3	GDDR5_45SE	GDDR5_DATA	FB B1 DQ<31..24>	6 75 77

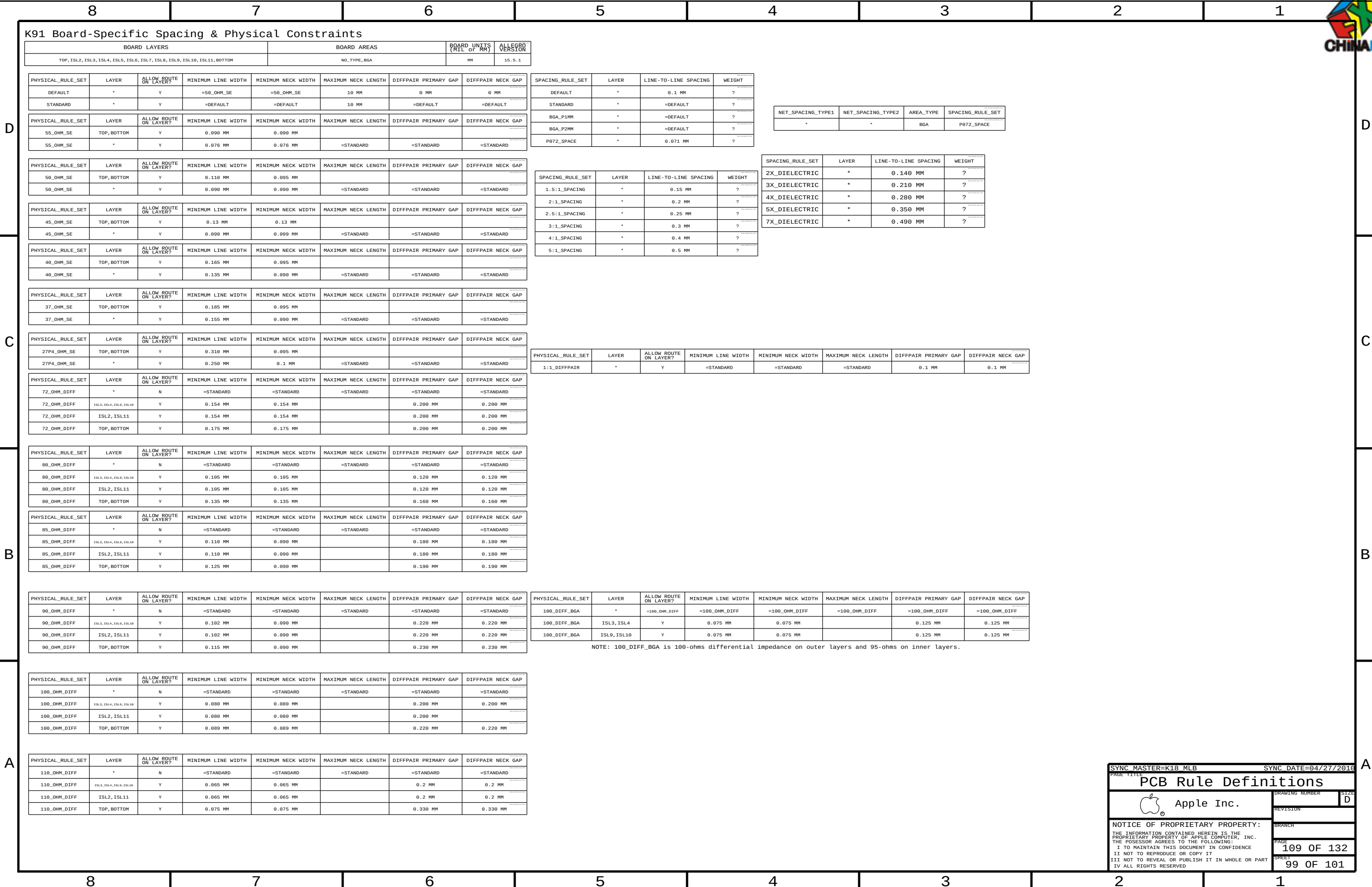
MUXGFX Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE			
	PHYSICAL	SPACING		
LVDS_A_CLK	LVDS_85D	LVDS	LVDS A CLK P	83 87
LVDS_A_CLK	LVDS_85D	LVDS	LVDS A CLK N	83 87
LVDS_A_DATA	LVDS_85D	LVDS	LVDS A DATA P<2..0>	83 87
LVDS_A_DATA	LVDS_85D	LVDS	LVDS A DATA N<2..0>	83 87
LVDS_B_CLK	LVDS_85D	LVDS	LVDS B CLK P	83 87
LVDS_B_CLK	LVDS_85D	LVDS	LVDS B CLK N	83 87
LVDS_B_DATA	LVDS_85D	LVDS	LVDS B DATA P<2..0>	83 87
LVDS_B_DATA	LVDS_85D	LVDS	LVDS B DATA N<2..0>	83 87
	LVDS_85D	LVDS	LVDS CONN A CLK F P	6 82
	LVDS_85D	LVDS	LVDS CONN A CLK F N	6 82
	LVDS_85D	LVDS	LVDS CONN B CLK F P	6 82
	LVDS_85D	LVDS	LVDS CONN B CLK F N	6 82
	LVDS_85D	LVDS	LVDS CONN A CLK P	82 83
	LVDS_85D	LVDS	LVDS CONN A CLK N	82 83
	LVDS_85D	LVDS	LVDS CONN A DATA P<2..0>	6 82 83
	LVDS_85D	LVDS	LVDS CONN A DATA N<2..0>	6 82 83
	LVDS_85D	LVDS	LVDS CONN B CLK P	82 83
	LVDS_85D	LVDS	LVDS CONN B CLK N	82 83
	LVDS_85D	LVDS	LVDS CONN B DATA P<2..0>	6 82 83
	LVDS_85D	LVDS	LVDS CONN B DATA N<2..0>	6 82 83

Whistler Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE			
	PHYSICAL	SPACING		
GPU_CLK27M	CLK_SLOW_55S	CLK_SLOW	GPU_CLK27M	78 79
GPU_CLK100M	CLK_SLOW_55S	CLK_SLOW	GPU_CLK100M	78 79
LVDS_EG_A_CLK	LVDS_85D	LVDS	LVDS EG A CLK P	78 87
LVDS_EG_A_CLK	LVDS_85D	LVDS	LVDS EG A CLK N	78 87
LVDS_EG_A_DATA	LVDS_85D	LVDS	LVDS EG A DATA P<2..0>	78 87
LVDS_EG_A_DATA	LVDS_85D	LVDS	LVDS EG A DATA N<2..0>	78 87
LVDS_EG_A_DATA3	LVDS_85D	LVDS	NC LVDS EG A DATA P<3>	78 79
LVDS_EG_A_DATA3	LVDS_85D	LVDS	NC LVDS EG A DATA N<3>	78 79
LVDS_EG_B_DATA	LVDS_85D	LVDS	LVDS EG B DATA P<2..0>	78 87
LVDS_EG_B_DATA	LVDS_85D	LVDS	LVDS EG B DATA N<2..0>	78 87
LVDS_EG_B_DATA3	LVDS_85D	LVDS	NC LVDS EG B DATA P<3>	78 79
LVDS_EG_B_DATA3	LVDS_85D	LVDS	NC LVDS EG B DATA N<3>	78 79
DP_MI	DP_85D	DISPLAYPORT	DP EXTA ML C P<3..0>	78 84
DP_MI	DP_85D	DISPLAYPORT	DP EXTA ML C N<3..0>	78 84
DP_AUX_CH	DP_85D	DISPLAYPORT	DP EXTA AUXCH C P	83 84
DP_AUX_CH	DP_85D	DISPLAYPORT	DP EXTA AUXCH C N	83 84
DP_AUX_CH	DP_85D	DISPLAYPORT	DP EG AUXCH P	8 78 83
DP_AUX_CH	DP_85D	DISPLAYPORT	DP EG AUXCH N	8 78 83

SYNC MASTER=K92 MLB		SYNC DATE=08/09/2016	
GPU (Whistler) CONSTRAINTS			
Apple Inc.		DRAWING NUMBER	SIZE D
NOTICE OF PROPRIETARY PROPERTY:		REVISION	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:		BRANCH	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		PAGE	107 OF 132
II NOT TO REPRODUCE OR COPY IT		SHEET	97 OF 101
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			



K91 Board-Specific Spacing & Physical Constraints

BOARD LAYERS				BOARD AREAS			BOARD UNITS (ALL OF MM)	ALLEGRO VERSION
TOP, ISL2, ISL3, ISL4, ISL5, ISL6, ISL7, ISL8, ISL9, ISL10, ISL11, BOTTOM				NO_TYPE, BGA			MM	15.5.1

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DEFAULT	*	Y	=50_OHM_SE	=50_OHM_SE	10 MM	0 MM	0 MM
STANDARD	*	Y	=DEFAULT	=DEFAULT	10 MM	=DEFAULT	=DEFAULT

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
55_OHM_SE	TOP,BOTTOM	Y	0.090 MM	0.090 MM			
55_OHM_SE	*	Y	0.076 MM	0.076 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
50_OHM_SE	TOP,BOTTOM	Y	0.110 MM	0.095 MM			
50_OHM_SE	*	Y	0.090 MM	0.090 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
45_OHM_SE	TOP,BOTTOM	Y	0.13 MM	0.13 MM			
45_OHM_SE	*	Y	0.090 MM	0.090 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
40_OHM_SE	TOP,BOTTOM	Y	0.165 MM	0.095 MM			
40_OHM_SE	*	Y	0.135 MM	0.090 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
37_OHM_SE	TOP,BOTTOM	Y	0.185 MM	0.095 MM			
37_OHM_SE	*	Y	0.155 MM	0.090 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
27P4_OHM_SE	TOP,BOTTOM	Y	0.310 MM	0.095 MM			
27P4_OHM_SE	*	Y	0.250 MM	0.1 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
72_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
72_OHM_DIFF	ISL3, ISL4, ISL9, ISL10	Y	0.154 MM	0.154 MM		0.200 MM	0.200 MM
72_OHM_DIFF	ISL2, ISL11	Y	0.154 MM	0.154 MM		0.200 MM	0.200 MM
72_OHM_DIFF	TOP,BOTTOM	Y	0.175 MM	0.175 MM		0.200 MM	0.200 MM

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
80_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
80_OHM_DIFF	ISL3, ISL4, ISL9, ISL10	Y	0.105 MM	0.105 MM		0.120 MM	0.120 MM
80_OHM_DIFF	ISL2, ISL11	Y	0.105 MM	0.105 MM		0.120 MM	0.120 MM
80_OHM_DIFF	TOP,BOTTOM	Y	0.135 MM	0.135 MM		0.160 MM	0.160 MM

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
85_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
85_OHM_DIFF	ISL3, ISL4, ISL9, ISL10	Y	0.110 MM	0.090 MM		0.180 MM	0.180 MM
85_OHM_DIFF	ISL2, ISL11	Y	0.110 MM	0.090 MM		0.180 MM	0.180 MM
85_OHM_DIFF	TOP,BOTTOM	Y	0.125 MM	0.090 MM		0.190 MM	0.190 MM

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
90_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
90_OHM_DIFF	ISL3, ISL4, ISL9, ISL10	Y	0.102 MM	0.090 MM		0.220 MM	0.220 MM
90_OHM_DIFF	ISL2, ISL11	Y	0.102 MM	0.090 MM		0.220 MM	0.220 MM
90_OHM_DIFF	TOP,BOTTOM	Y	0.115 MM	0.090 MM		0.230 MM	0.230 MM

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
100_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
100_OHM_DIFF	ISL3, ISL4, ISL9, ISL10	Y	0.080 MM	0.080 MM		0.200 MM	0.200 MM
100_OHM_DIFF	ISL2, ISL11	Y	0.080 MM	0.080 MM		0.200 MM	0.200 MM
100_OHM_DIFF	TOP,BOTTOM	Y	0.080 MM	0.080 MM		0.220 MM	0.220 MM

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
110_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
110_OHM_DIFF	ISL3, ISL4, ISL9, ISL10	Y	0.065 MM	0.065 MM		0.2 MM	0.2 MM
110_OHM_DIFF	ISL2, ISL11	Y	0.065 MM	0.065 MM		0.2 MM	0.2 MM
110_OHM_DIFF	TOP,BOTTOM	Y	0.075 MM	0.075 MM		0.330 MM	0.330 MM

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DEFAULT	*	0.1 MM	?
STANDARD	*	=DEFAULT	?
BGA_P1MM	*	=DEFAULT	?
BGA_P2MM	*	=DEFAULT	?
P072_SPACE	*	0.071 MM	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
1.5:1_SPACING	*	0.15 MM	?
2:1_SPACING	*	0.2 MM	?
2.5:1_SPACING	*	0.25 MM	?
3:1_SPACING	*	0.3 MM	?
4:1_SPACING	*	0.4 MM	?
5:1_SPACING	*	0.5 MM	?

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
1:1_DIFFPAIR	*	Y	=STANDARD	=STANDARD	=STANDARD	0.1 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
100_DIFF_BGA	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF
100_DIFF_BGA	ISL3, ISL4	Y	0.075 MM	0.075 MM		0.125 MM	0.125 MM
100_DIFF_BGA	ISL9, ISL10	Y	0.075 MM	0.075 MM		0.125 MM	0.125 MM

NOTE: 100_DIFF_BGA is 100-ohms differential impedance on outer layers and 95-ohms on inner layers.

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
*	*	BGA	P072_SPACE

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
2X_DIELECTRIC	*	0.140 MM	?
3X_DIELECTRIC	*	0.210 MM	?
4X_DIELECTRIC	*	0.280 MM	?
5X_DIELECTRIC	*	0.350 MM	?
7X_DIELECTRIC	*	0.490 MM	?

SYNC_MASTER=K18_MLB

SYNC_DATE=04/27/2016

PCB Rule Definitions

Apple Inc.

NOTICE OF PROPRIETARY PROPERTY:
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

DRAWING NUMBER

REVISION

BRANCH

PAGE

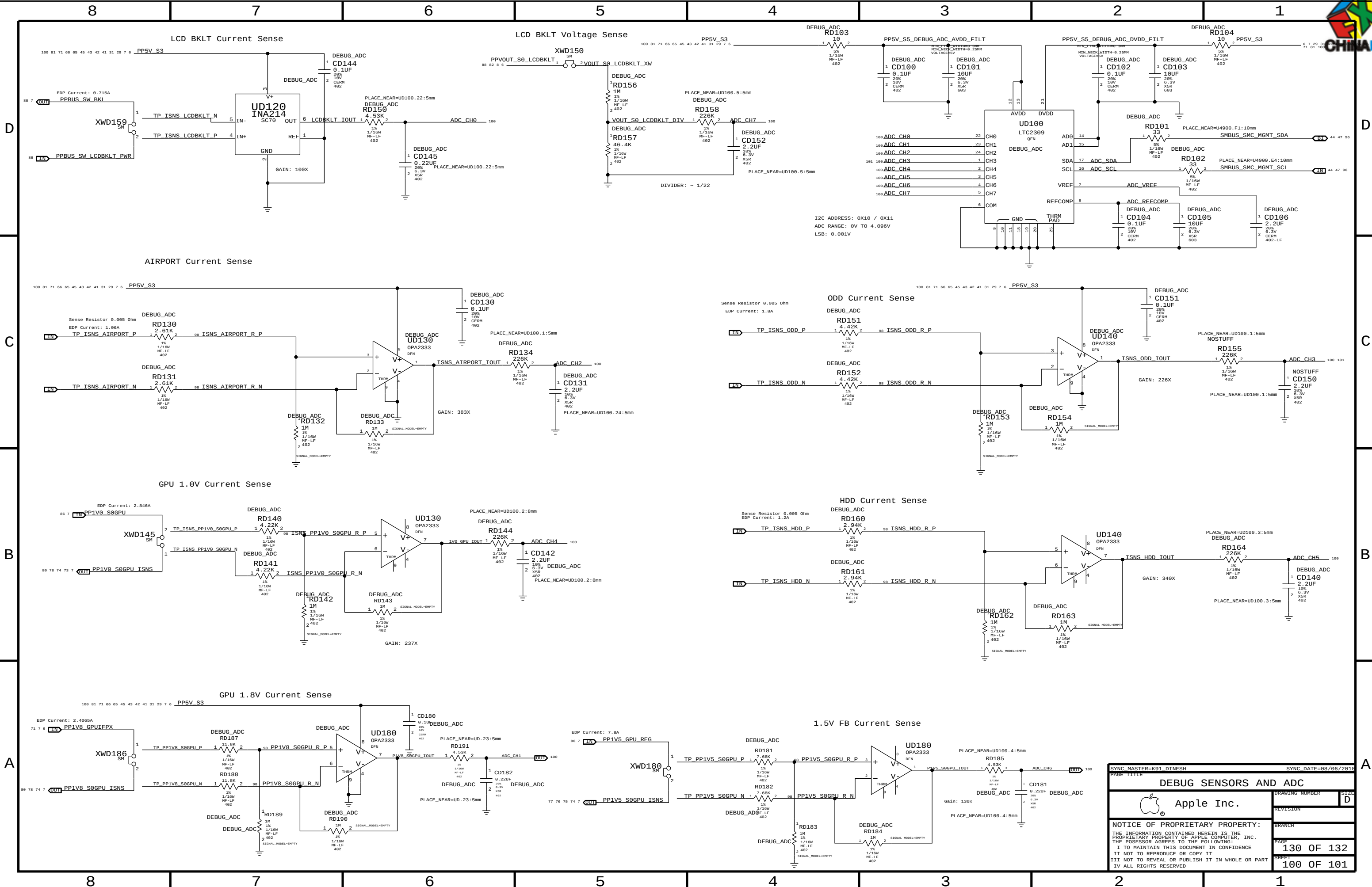
SHEET

SIZE

D

109 OF 132

99 OF 101



SYNC MASTER=K91 DINESH		SYNC DATE=08/06/2010	
PAGE TITLE		DRAWING NUMBER	
DEB		SIZE	
Apple Inc.		REVISION	
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE COMPUTER, INC. THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		130 OF 132	
II NOT TO REPRODUCE OR COPY IT		SHEET	
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART		100 OF 101	
IV ALL RIGHTS RESERVED			

A